



**GREEN
CLIMATE
FUND**

Meeting of the Board

17 – 20 February 2025

Songdo, Incheon, Republic of Korea

Provisional agenda item 11

GCF/B.41/02/Add.05

27 January 2025

Consideration of funding proposals – Addendum V

Funding proposal package for FP257

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "RE-GAIN: Scaling solutions for food loss in Africa";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the project-specific assessment approach applicant to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title: RE-GAIN: Scaling solutions for food loss in Africa

Countries: Burkina Faso
Ethiopia
Kenya
Malawi
Tanzania
Uganda
Zambia

Accredited Entity: AGRA

Date of first submission: [2024/09/27]

Date of current submission [2024/12/02]

Version number [V.1.4]



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NOTE TO ACCREDITED ENTITIES ON THE USE OF THE FUNDING PROPOSAL TEMPLATE

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Programme	A.2. Public or private sector	Public	
A.3. Request for Proposals (RFP)	Not applicable			
A.4. Result area(s)			GCF contribution	Co-financers' contribution ¹
	Mitigation total			
	☐ Energy generation and access			
	☐ Low-emission transport			
	☐ Buildings, cities, industries and appliances			
	☐ Forestry and land use			
	Adaptation total			
	☒ Most vulnerable people and communities		50%	50%
	☒ Health and well-being, and food and water security		50%	50%
	☐ Infrastructure and built environment			
☐ Ecosystems and ecosystem services				
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	N/A	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	Direct Total: 2,632,553 0.73% of the population across the 7 countries	Indirect Total: 2,735,621 0.75% of the population across the 7 countries
A.7. Total financing (GCF + co-finance²)	105,000,000 USD	A.8. Project size	Medium (Upto USD 250 million)	
A.9. Total GCF funding requested	75,623,754 USD			
A.10. Financial instrument(s) requested for the GCF funding	☒ Grant 75,623,754 USD ☐ Equity <u>Enter number</u> ☐ Loan <u>Enter number</u> ☐ Results-based payment <u>Enter number</u> ☐ Guarantee <u>Enter number</u>			
A.11. Implementation period	5 years	A.12. Total lifespan	10 years	
A.13. Expected date of AE internal approval	2/9/2024	A.14. ESS category	B	

1 Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

2 Refer to the Policy of Co-financing of the GCF.

A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☐ No ☐
A.19. Complementarity and coherence	Yes ☒ No ☐		
A.20. Executing Entity information	The project will be executed directly by AGRA through their Programme Operational Unit – which falls under the same legal entity as the PSAA Applicant. As a continental institution in agriculture and food systems, AGRA is well placed to undertake the planned activities through its widespread infrastructure and partnerships on the continent, having strong AGRA country teams and infrastructure in the target countries. Using AGRA's continental forums, the Africa Food Systems Forum (AFSF) and Agribusiness Deal Room, AGRA brings on board close working relationships with governments, partners and other stakeholders to enhance ownership and coordinate key actors, knowledge and resources, which are fundamental to take food loss innovation under changing climates to scale. The executing entity role of AGRA will be fulfilled both through the Nairobi-based headquarters, and their country offices, all of which are the same legal entity as the PSAA Applicant.		

A.21. Executive summary

4. Sub-Saharan Africa stands at a crossroads with an unprecedented opportunity and challenges for agricultural transformation, driven by the demands of a rapidly growing population of 1.5 billion and the pressures of a changing climate. The vicious cycle of poverty and environmental degradation in Africa is evident in low crop productivity, deforestation, land degradation, conflict, migration, and vulnerability to climate shocks, which perpetuate persistent food insecurity and poverty. The effects of climate change are expected to be severe in Africa, where the capacity to adapt and respond to a changing climate is weak — exacerbating the continent's food insecurity and climate vulnerability challenges.
5. The impacts of climate change have increased over the past decades in Africa, manifesting in more frequent, intense, and prolonged extreme weather events, such as floods, droughts, heatwaves, pest outbreaks, landslides, desertification, and sandstorms. These extreme weather events have resulted in increased temperatures and humidity, shifts in precipitation patterns, water stress, and soil erosion. Most African countries already face recurrent droughts that affect growing seasons, often leading to short growing periods reducing the viability of farming in marginal agricultural areas. Projected reductions in yield in some countries could reach as much as 50% by 2030, and crop net revenues may fall by up to 90% by 2100, with smallholder farmers being the most affected. Food security, however, is affected not only by changes in crop production but also by changes throughout the crop value chain, including post-harvest. There is, therefore, a crucial need to address the impacts of climate change on the full agriculture value chain, including production, aggregation, storage, processing, and distribution.
6. While there is a wealth of information on the observed and projected impacts of climate change on crop production pre-harvest (i.e., germination, fruiting, cultivation, maturation, seed or grain quality, and most of all yields), relatively little attention has been paid to how climate change variability and hazards affect harvesting, post-harvest handling and storage (e.g., threshing, sorting, drying, warehousing), and processing, transport and logistics (grading, packaging, storage, transport, distribution, and marketing).
7. Of the range of **climate hazards** that affect harvest and post-harvest value chains of cereal crops and legumes, scholarly publications and country experts pointed to **variable, erratic, and unpredictable rainfall (including an increase in heavy rainfall events), high temperatures (extreme heat), and flooding** as the priority hazards of concern. Literature and stakeholders also highlighted **drought** as an

important hazard. To a lesser degree, sources also acknowledged the role of wildfires and landslides. Consistently, stakeholder input coalesced around the several stage-specific risks and impacts:

- **Harvesting Processes:** shattering of the grain/seed due to untimely heavy rains ; challenges to field drying due to excessive rainfall; rotting, mold, and mycotoxin (principally aflatoxin) contamination in the field due to high humidity; damage to stored harvest from flooding; pest and rodents.
- **Post-Harvest Handling and Storage:** damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mold, and mycotoxin (aflatoxin) contamination due to damage to storage infrastructure from flooding;
- **Processing, Transport, and Logistics:** damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.

8. In addition to the impacts of climate change on post-harvest food losses, such losses also contribute to greenhouse gas emissions.
9. In response to the critical gaps noted above, the RE-GAIN programme will support the innovative adaptation of the harvest and post-harvest value chain in seven African countries — Burkina Faso, Ethiopia, Kenya, Malawi, Tanzania, Uganda, and Zambia; strengthening food security by addressing the climate change impacts on post-harvest food losses through the wide-scale adoption of Food Loss-Reduction Solutions (FL-RS) among smallholder farmers. In addition to increasing the amount of food available for consumption due to reduced losses, this approach will also improve the quality of produce leaving the farms, thereby increasing nutritional value, marketability and income potential, while simultaneously reducing the GHG emissions associated with food loss. To achieve the project goal, actions will be taken across three primary pathways — farmer-centric, supplier-centric, and institutional support — which together will form the foundation of a sustainable and scalable market for FL-RS in the target countries. These interconnected solutions will address food loss under smallholder settings of Africa.
10. The first development pathway focuses on increasing demand and absorption capacity for FL-RS among smallholder farming communities. This will involve raising awareness among farmers of the impacts of climate change on post-harvest food loss, as well as on the availability of FL-RS options on the market and how FL-RS will help reduce food loss under changing climate conditions. This awareness raising will be coupled with training on the use of specific FL-RS that have been identified for key crops in each country as well as on the effective use of weather information and advisories, while supporting the initial transfer of suitable technologies to the farmers to catalyse uptake and demonstrate the effectiveness of FL-RS. These activities will be further supported through actions to improve market linkages between farmers and agri-value chain actors such as processors and aggregators.
11. The second development pathway complements the demand-side activities by enhancing the supply and affordability of FL-RS. The first output of this will support business development linked to the local supply of FL-RS, targeting both existing suppliers, as well as entrepreneurs seeking to enter the market — particularly MSMEs, local cooperatives and youth groups. Support to local businesses will increase the supply of FL-RS in local markets, which in turn will enable cost reduction, underpinning the outcome of improved affordability. This will be complemented by the establishment of innovative financial mechanisms for smallholders and MSMEs to support the adoption of FL-RS. In particular, the project will bring together key stakeholders — including suppliers, agro-dealers, financial institutions, MSMEs and local farmers — to establish finance models that stimulate the FL-RS market, using catalytic co-payments and conditional procurement to provide added security to the transactions.
12. Combined, the increased demand and the enhanced supply and affordability of FL-RS stimulate the market, enabling suppliers to develop products and services that are affordable to smallholder farmers, thereby facilitating a balance between supply and demand, and establishing a self-sufficient market that will operate beyond the project support. The integrated nature of these Outputs and the causal pathways reinforcing the market development are depicted in Figure 3 below.

These first two development pathways will be underpinned and sustained by the third, which seeks to create an enabling environment for the uptake of FL-RS; which will not only mobilise increased investment in the sector but also provide wider policy interventions in strengthening the market systems and advisory services. This will involve policy revisions and put in place enforcement mechanisms in each of the target countries in support of enabling FL-RS — including stimulating tax incentives and promoting standards for FL-RS quality — as well as creating platforms for the scaling and replication of successful FL-RS business models.

13. Overall, the resulting uptake of FL-RS, at scale, will reduce food loss, improve food quality and enhance local capacity, thereby enabling the long-term outcome of improved food and nutritional security. This will have the co-benefit of increasing food safety, while the increased investment in FL-RS resulting will create additional employment opportunities in the market. Moreover, the project will reduce GHG emission, both directly – through the reduced emissions from the wasted food itself – and indirectly by reducing land use change for agricultural expansion driven by the need to compensate for food losses.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context

14. Sub-Saharan Africa stands at a crossroads with an unprecedented opportunity and challenges for agricultural transformation, driven by the demands of a rapidly growing population of 1.5 billion and the pressures of a changing climate³. The vicious cycle of poverty and environmental degradation in Africa is evident in low crop productivity, deforestation, land degradation, conflict, migration, and vulnerability to climate shocks, which perpetuate persistent food insecurity and poverty. The effects of climate change are expected to be severe in Africa, where the capacity to adapt and respond to a changing climate is weak — exacerbating the continent's food insecurity challenge.
15. A great deal of attention has been paid to the impacts of climate change on crop cultivation, with extensive research on how climate variability affects crop yields, and management strategies to address these challenges⁴. However, food security is influenced not only by changes in crop production but also by the entire crop value chain, including post-harvest stages⁵. There is, therefore, a crucial need to consider how climate change impacts the full agriculture value chain, including production, aggregation, storage, processing, and distribution.

Harvesting and Post Harvesting Value Chains

16. The RE-GAIN programme will support the adaptation of the harvest and post-harvest management innovations in seven African countries — Burkina Faso, Ethiopia, Kenya, Malawi, Tanzania, Uganda, and Zambia; strengthening food security by addressing the climate change impacts on post-harvest food losses. For the purpose of the RE-GAIN Programme proposal, the post-harvest value chain is considered to start at the point of harvest, and spans across three key stages, as shown in Figure 1. Strategic⁶



Figure 1. Strategic value chain stages included in the RE-GAIN Programme.

Climate Risks for Post-Harvest Value Chains

17. Climate change-driven (or exacerbated) food spoilage and contamination in post-production stages of the value chains compound the climate change-driven productivity and yield losses during cultivation. This results in a double-blow to farming communities from climatic factors such as rising temperatures, higher moisture levels (warmer air holds more moisture), increased precipitation, more frequent or intense heavy rainfall events, flooding.
18. Climate change is thus a threat-multiplier to crop value chains. Climate change impacts in the post-production value chain, which result in substantial post-harvest losses and reduce the output that farmers have left to sell, depress incomes and jeopardize livelihoods of small-holder farmers (IFPRI, 2022). This perpetuates vulnerability to climatic shocks and stresses and prolongs the cycle of persistent poverty.
19. In the face of climate change-driven post-harvest losses, farmers have an impetus to keep less of the reduced output for their domestic consumption and to offset the volume losses (during harvesting, field-

³ World Bank, 2023. The World Bank in Africa

⁴ CGIAR. (2023). A food system approach to climate action. CGIAR.

⁵ Akoth, A. S. (2020). Impact of climate change on post-harvest value chain. International Journal of Agriculture Extension and Social Development, 3(2), 76-80.

⁶ Details on how each of these are defined and the actions entailed within each are presented in Annex 2.

drying, on-farm storage) by selling more of what remains in the market. This worsens household food insecurity in countries that already have markedly high levels of undernourishment.

Climate change impact on food loss

20. In all seven countries, historic trends in temperature (such as annual average/mean surface temperature, and the number of days with extreme heat – i.e., temperatures over 35°C) have displayed an increasing trend over time, with the rate of increase accelerating in recent decades. Historic rainfall trends display far greater variability, and for some countries a measure of uncertainty regarding the direction of the trend.
21. By and large, future projections of rainfall (such as annual average/mean precipitation, and the number of consecutive wet days, and – as a corollary – the number of consecutive dry days) suggest a projected rise in rainfall levels and rainfall days across most of East Africa, with the exception of Zambia (where projections display a slight drying signal, consistent with regional trends for Southern Africa).

Table 1. Climate change trends⁷

	Climatic variable/ hazard	Historic change (only statistically significant trends)	Historic Baseline (1950 – 2014)	Projected change by 2040
Burkina Faso	Mean Annual Surface Temperature (°C)	Statistically significant increasing trend of 0.16 °C per decade since 1951 to 2020 (0.23 °C per decade from 1971-2020; 0.48°C per decade from 1991-2020)	28.38 °C	• SSP2-4.5 = 29.73 °C (1.35 °C rise) • SSP5-8.5 = 30.02 °C (1.64 °C rise)
	Mean Annual Precipitation (mm)	Statistically significant decreasing trend of -19.6 per decade since 1951 to 2020	627.85 mm	• SSP2-4.5 = 691.85 mm (64 mm rise) • SSP5-8.5 = 710.59 mm (82.74 rise)
	No. of hot days (>35°C)	Statistically significant increasing trend of 5.6 such days per decade since 1951 to 2020 (+6.16 days per decade from 1971-2020; +9.53 days per decade from 1991-2020)	151.72 days	• SSP2-4.5 = 199.44 days (47.72 days rise) • SSP5-8.5 = 210.82 days (140.28 days rise)
	Maximum no. of consecutive dry days (linked to drought)	Statistically significant increasing trend of 2.43 more such days per decade since 1971 to 2020	101.92days	• SSP2-4.5 = 101.41 days (-0.51 days decrease) • SSP5-8.5 = 99.68 days (-2.24 days decrease)
	Maximum no. of consecutive wet days (linked to flooding)	Statistically significant decreasing trend of -1.01 less such days per decade since 1951 to 2020	22.30 days	• SSP2-4.5 = 23.08 days (0.78 days increase) • SSP5-8.5 = 21.84 days (-0.46 days decrease)
Ethiopia	Mean Annual Surface Temperature (°C)	Statistically significant increasing trend of 0.28 °C per decade since 1951 to 2020 (0.29 °C per decade from 1971-2020; 0.30°C per decade from 1991-2020)	23.17 °C	• SSP2-4.5 = 24.81 oC (1.64 oC rise) • SSP5-8.5 = 25.28 oC (2.11 oC rise)
	Mean Annual Precipitation (mm)	Statistically significant decreasing trend of -34.37 mm per decade since 1951 to 2020 (-31.88 mm from 1971-2020)	1039.72 mm	• SSP2-4.5 = 1099.63 mm (59.91 mm rise) • SSP5-8.5 = 1151.29 mm (111.57 mm rise)
	No. of hot days (>35°C)	Statistically significant increasing trend of 0.29 such days per decade since 1951 to 2020 (+0.30 days per decade from 1971-2020; +0.32 days per decade from 1991-2020)	33.77 days	• SSP2-4.5 = 60.38 days (26.61 days rise) • SSP5-8.5 = 70.26 days (36.49 days rise)
	Maximum no. of consecutive dry days (linked to drought)	Statistically significant increasing trend of 2.24 more such days per decade since 1951 to 2020 (+2.62 days from 1971-2020)	70.11 days	• SSP2-4.5 = 69.94 days (-0.17 days decrease) • SSP5-8.5 = 67.02 days (-3.09 days decrease)
	Maximum no. of consecutive wet	Statistically significant decreasing trend of -1.74 less such days per decade since 1951 to 2020 (-1.46 days from 1971-2020)	58.83 days	• SSP2-4.5 = 58.64 days (-0.19 days decrease)

⁷ The analysis uses data from the [World Bank's Climate Change Knowledge Portal](#), and includes two distinct future climate change emissions scenarios, to allow a study of the range of potential climate change impacts in 2040 regardless of which of the two emissions pathways the world is more likely to reflect in 2040. It is important for the design and operationalization of investments under RE-GAIN to plan and prepare for where emissions pathways are already heading, as well as a 'worst case scenario' so that investments are effective in addressing potential risks and do not fall short of the protections they are intended to afford.

	days (linked to flooding)			• SSP5-8.5 = 57.92 days (-0.91 days decrease)
Kenya	Mean Annual Surface Temperature (°C)	Statistically significant increasing trend of 0.27 °C per decade since 1951 to 2020 (0.29 °C per decade from 1971-2020; 0.35°C per decade from 1991-2020)	25.17 °C	• SSP2-4.5 = 26.33 oC (1.16 oC rise) • SSP5-8.5 = 26.73 oC (1.56 oC rise)
	Mean Annual Precipitation (mm)	Statistically significant decreasing trend of -27.37 mm per decade since 1951 to 2020	739.09 mm	• SSP2-4.5 = 773.48 mm (34.39 mm rise) • SSP5-8.5 = 786.04 mm (46.95 mm rise)
	No. of hot days (>35°C)	Statistically significant increasing trend of 0.04 such days per decade since 1951 to 2020 (+0.08 days per decade from 1971-2020; +0.32 days per decade from 1991-2020)	33.79 days	• SSP2-4.5 = 64.14 days (30.35 days rise) • SSP5-8.5 = 76.72 days (42.93 days rise)
	Maximum no. of consecutive dry days (linked to drought)	Statistically significant increasing trend of 2.31 more such days per decade since 1971 to 2020	76.98 days	• SSP2-4.5 = 79.32 days (2.32 days increase) • SSP5-8.5 = 73.01 days (-3.97 days decrease)
	Maximum no. of consecutive wet days (linked to flooding)	Statistically significant decreasing trend of -1.56 less such days per decade since 1951 to 2020 (-1.53 days from 1971-2020)	28.06 days	• SSP2-4.5 = 29.88 days (1.82 days increase) • SSP5-8.5 = 31.25 days (3.19 days increase)
Malawi	Mean Annual Surface Temperature (°C)	Statistically significant increasing trend of 0.16 °C per decade since 1951 to 2020 (0.21 °C per decade from 1971-2020; 0.18°C per decade from 1991-2020)	21.98 °C	• SSP2-4.5 = 23.27 oC (1.16 oC rise) • SSP5-8.5 = 23.85 oC (1.56 oC rise)
	Mean Annual Precipitation (mm)	No statistically significant trends; high variability and uncertainty (marginal decreasing signal from 1951-2020)	1382.45 mm	• SSP2-4.5 = 1402.34 mm (19.89 mm rise) • SSP5-8.5 = 1320.05 mm (62.40 mm rise)
	No. of hot days (>35°C)	Statistically significant increasing trend of 0.01 such days per decade since 1951 to 2020 (+0.02 days per decade from 1971-2020; +0.05 days per decade from 1991-2020)	2.52 days	• SSP2-4.5 = 6.29 days (3.77 days rise) • SSP5-8.5 = 9.28 days (6.76 days rise)
	Maximum no. of consecutive dry days (linked to drought)	Statistically significant increasing trend of 3.09 more such days per decade since 1971 to 2020	124.42 days	• SSP2-4.5 = 126.95 days (2.53 days increase) • SSP5-8.5 = 129.31 days (4.89 days increase)
	Maximum no. of consecutive wet days (linked to flooding)	Statistically significant decreasing trend of -1.56 less such days per decade since 1951 to 2020	61.20 days	• SSP2-4.5 = 62.52 days (1.32 days increase) • SSP5-8.5 = 63.36 days (2.16 days increase)
Tanzania	Mean Annual Surface Temperature (°C)	Statistically significant increasing trend of 0.22 °C per decade since 1951 to 2020 (0.27 °C per decade from 1971-2020; 0.27°C per decade from 1991-2020)	22.69 °C	• SSP2-4.5 = 24.32 oC (1.63 oC rise) • SSP5-8.5 = 24.79 oC (2.10 oC rise)
	Mean Annual Precipitation (mm)	Statistically significant decreasing trend of -45.89 mm per decade since 1951 to 2020 (-51.99 from 1971-2020)	1151.98 mm	• SSP2-4.5 = 1161.14 mm (9.16 mm rise) • SSP5-8.5 = 1177.43 mm (25.45 mm rise)
	No. of hot days (>35°C)	Statistically significant but slight increasing trend of 0.00 such days per decade since 1951 to 2020 (+0.00 days per decade from 1971-2020; +0.01 days per decade from 1991-2020)	0.95 days	• SSP2-4.5 = 7.18 days (6.23 days rise) • SSP5-8.5 = 11.42 days (10.47 days rise)
	Maximum no. of consecutive dry days (linked to drought)	Statistically significant increasing trend of 2.95 more such days per decade since 1971 to 2020	108.14 days	• SSP2-4.5 = 112.19 days (4.05 days increase) • SSP5-8.5 = 108.99 days (0.85 days increase)
	Maximum no. of consecutive wet days (linked to flooding)	Statistically significant decreasing trend of -2.23 less such days per decade since 1951 to 2020 (-2.13 days from 1971-2020)	45.39 days	• SSP2-4.5 = 47.60 days (2.21 days increase) • SSP5-8.5 = 44.04 days (-1.35 days decrease)

Uganda	Mean Annual Surface Temperature (°C)	Statistically significant increasing trend of 0.30 °C per decade since 1951 to 2020 (0.31 °C per decade from 1971-2020; 0.26°C per decade from 1991-2020)	22.89 °C	• SSP2-4.5 = 24.16 oC (1.27 oC rise) • SSP5-8.5 = 24.55 oC (1.66 oC rise)
	Mean Annual Precipitation (mm)	Statistically significant decreasing trend of -102.3 mm per decade since 1951 to 2020 (-111.984 from 1971-2020)	1765.05 mm	• SSP2-4.5 = 1791.29 mm (26.24 mm rise) • SSP5-8.5 = 1863.93 mm (98.88 mm rise)
	No. of hot days (>35°C)	No statistically significant trends (the no. of such days has remained relatively stable per observational records)	3.21 days	• SSP2-4.5 = 7.50 days (4.29 days rise) • SSP5-8.5 = 11.16 days (7.95 days rise)
	Maximum no. of consecutive dry days (linked to drought)	Statistically significant increasing trend of 1.40 more such days per decade since 1951 to 2020 (+1.67 days from 1971-2020)	16.00 days	• SSP2-4.5 = 14.67 days (-1.33 days decrease) • SSP5-8.5 = 14.85 days (-1.15 days decrease)
	Maximum no. of consecutive wet days (linked to flooding)	Statistically significant decreasing trend of -4.36 less such days per decade since 1951 to 2020 (-4.74 days from 1971-2020; -3.84 days from 1991-2020)	81.00 days	• SSP2-4.5 = 77.90 days (-3.1 days decrease) • SSP5-8.5 = 79.08 days (-1.92 days decrease)
Zambia	Mean Annual Surface Temperature (°C)	Statistically significant increasing trend of 0.18 °C per decade since 1951 to 2020 (0.23 °C per decade from 1971-2020)	22.01 °C	• SSP2-4.5 = 23.33 oC (1.23 oC rise) • SSP5-8.5 = 24.00 oC (1.99 oC rise)
	Mean Annual Precipitation (mm)	No statistically significant trends (there is a great deal of variability and uncertainty; initially a marginal decreasing signal from 1951-2020 in the observational records, followed by a marginal increasing signal in the 1991-2020 trend)	1112.50 mm	• SSP2-4.5 = 1113.05 mm (0.55 mm increase) • SSP5-8.5 = 1057.41 mm (-55.09 mm decrease)
	No. of hot days (>35°C)	Statistically significant trend of 0 additional such days per decade from 1951-2020; 0 from 1971-2020; and 0 from 1991-2020 (the number of such days has remained relatively stable)	6.11 days	• SSP2-4.5 = 18.76days (12.65 days rise) • SSP5-8.5 = 25.77 days (19.66 days rise)
	Maximum no. of consecutive dry days (linked to drought)	Statistically significant increasing trend of 3.57 more such days per decade since 1971 to 2020	175.07 days	• SSP2-4.5 = 177.35 days (2.28 days increase) • SSP5-8.5 = 177.48 days (2.41 days increase)
	Maximum no. of consecutive wet days (linked to flooding)	Statistically significant decreasing trend of -1.62 less such days per decade since 1951 to 2020	50.87 days	• SSP2-4.5 = 52.04 days (1.17 days increase) • SSP5-8.5 = 44.01 days (-6.86 days decrease)

Impact of climate change on food loss

22. The impact of climate change on post-harvest food loss has not been well studied, either in terms of observed impacts (i.e., research on historic food losses in the post-harvest value chain, with some attribution to climatic factors), or in terms of projected impacts (i.e., research on future anticipated food losses in the post-harvest value chain, with plausible attribution or linkage to rise in temperature, increased rainfall variability, and a climate change-induced exacerbation of hazards such as flooding and drought).
23. Thus, in the table below the observed impacts of climate change in the post-harvest value chain are captured simply as the level of climate-sensitivity of the grain/seed (drawn from peer reviewed literature that focused on yield and productivity impacts of climate change). The column on projected impacts of climate change in the post-harvest value chain is populated based on country (national and local) stakeholder guidance (including farmers and agricultural decision-makers) plus some additional sectoral expert judgment.

Table 2. Observed and projected impact of climate change on food loss ⁸

	Crop (s)	Climate change hazard	Observed impact (Sensitivity)	Projected climate change impact on food loss
Burkina Faso	Cowpea	Flooding; Excessive and/or erratic rainfall (including heavy rainfall events); High temperatures (extreme heat)	Moderate sensitivity to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and mycotoxin (aflatoxin) contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and mycotoxin (aflatoxin) contamination due to moisture content damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
	Rice	High temperatures (extreme heat) Flooding; Excessive and/or erratic rainfall (including heavy rainfall events)	Moderate sensitivity to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and aflatoxin contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and aflatoxin contamination due to moisture content damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
Ethiopia	Teff	Excessive and/or erratic rainfall (including heavy rainfall events); Flooding; High temperatures (extreme heat)	Moderate sensitivity to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and aflatoxin contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and aflatoxin contamination due to moisture content damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
	Wheat	Excessive and/or erratic rainfall (including heavy rainfall events); Flooding;	High sensitivity to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and aflatoxin contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and aflatoxin contamination due to moisture content

⁸ Note: due to the similarity of crops/grains across multiple countries – and, more crucially, due to the similarities in the way that temperature and moisture impact the grain/seed in the post-harvest stages – the projected impacts emphasized by stakeholders and experts are very homogeneous and highly consistent across countries and crops

		High temperatures (extreme heat)		damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
Kenya	Maize	Excessive and/or erratic rainfall (including heavy rainfall events); High temperatures (extreme heat); Climate-driven increase in pests and diseases.	High sensitivity to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and aflatoxin contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and aflatoxin contamination due to moisture content damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
	Beans	Climate-driven increase in pests and diseases; Excessive and/or erratic rainfall (including heavy rainfall events); High temperatures (extreme heat).	Moderate sensitivity to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and aflatoxin contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and aflatoxin contamination due to moisture content damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
Malawi	Maize	Excessive and/or erratic rainfall (including heavy rainfall events); Flooding; High temperatures (extreme heat)	High sensitivity to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and aflatoxin contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and aflatoxin contamination due to moisture content damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
	Groundnut	Drought / dry spells; Flooding (including as a result of including heavy rainfall events);	Moderate sensitivity to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and aflatoxin contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and aflatoxin contamination due to moisture content

		High temperatures (extreme heat)		damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
Tanzania	Maize	Flooding (including due to erratic and/or heavy rainfall events); Drought / dry spells; High temperatures (extreme heat)	High sensitivity to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and aflatoxin contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and aflatoxin contamination due to moisture content damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
	Rice	Rainfall variability (including erratic, unpredictable, or heavy rainfall events); Flooding; High temperatures (extreme heat)	Moderate sensitivity to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and aflatoxin contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and aflatoxin contamination due to moisture content damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
Uganda	Maize	Excessive and/or erratic rainfall (including heavy rainfall events); Flooding; High temperatures (extreme heat)	High sensitivity to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and aflatoxin contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and aflatoxin contamination due to moisture content damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
	Beans	Excessive and/or erratic rainfall (including heavy rainfall events); Flooding;	Moderate sensitivity to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and mycotoxin (aflatoxin) contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and mycotoxin (aflatoxin) contamination due to moisture

		High temperatures (extreme heat)		content damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
Zambia	Maize	Rainfall variability (including erratic, unpredictable, or heavy rainfall events that cause flooding); Drought / dry spells; High temperatures (extreme heat)	<i>High sensitivity</i> to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and mycotoxin (aflatoxin) contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and mycotoxin (aflatoxin) contamination due to moisture content damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.
	Soybeans	Rainfall variability (including erratic, unpredictable, or heavy rainfall events that cause flooding); Drought / dry spells; High temperatures (extreme heat)	<i>Moderate sensitivity</i> to climatic factors.	<i>Harvesting Processes:</i> shattering of the grain/seed due to heat; challenges to field drying due to excessive rainfall; rotting, mould, and mycotoxin (aflatoxin) contamination due to high moisture content; damage to stored harvest from flooding. <i>Post-Harvest Handling and Storage:</i> damage to or decay of grain/seed quality from high temperatures and humidity; rotting, mould, and mycotoxin (aflatoxin) contamination due to moisture content damage to storage infrastructure from flooding.; <i>Processing, Transport, and Logistics:</i> damaged or inaccessible road networks and transport infrastructure due to flooding; damage to processing and storage facilities and infrastructure from excessive rainfall and flooding; damage to or decay of grain/seed quality from high temperatures and humidity, and lower shelf life; disruption of mobility and access to markets due to extreme weather, impeding distribution and marketing.

Adaptive capacity – current capacity to anticipate or respond to the impact of climate change on food loss

24. All seven RE-GAIN countries are recognized as having low adaptive capacity and low readiness for climate change impacts, in terms of their relative global standing. Per the ND-GAIN index⁹, based on scores for 2021, the countries' ranking on adaptive capacity¹⁰ and readiness¹¹ – out of 192 countries – is presented in Table 3. This economy-wide low level of adaptive capacity also applies to the agricultural sector, including post-harvest management of value chains. This was affirmed and validated through inputs in national and sub-national stakeholder workshops, at which stakeholders identified several factors that contribute to low adaptive capacity against climate-induced risks and impacts. Stakeholder inputs across all 14 crops in the seven countries points to consistent, common factors affecting adaptive capacity in all fourteen value chains (of eight different crops). This consistency is both understandable (given common or similar barriers in all seven countries' agricultural value chains) and valuable (as it offers opportunity for RE-GAIN's

⁹ Notre Dame Global Adaptation Initiative, n.d

¹⁰ Under the ND GAIN Index, 'Adaptive Capacity' is the ability of society and its supporting sectors to adjust to reduce potential damage and to respond to the negative consequences of climate events. This captures a collection of means, readily deployable to deal with sector-specific climate change impacts.

¹¹ Under the ND GAIN Index, 'Readiness' is the ability to make effective use of investments for adaptation actions thanks to a safe and efficient business environment. Readiness has three components: economic readiness, governance readiness, and social readiness.

responses to align and harmonize at a cross-national scale, enabling programmatic efficiencies and economies of scale).

Table 3 Applicable ND-GAIN Adaptive Capacity Rankings for RE-GAIN countries

Country	Adaptive Capacity Rank (out of 192)	Readiness Rank (out of 192)
Burkina Faso	154 (low adaptive capacity)	158 (low readiness)
Ethiopia	159 (low adaptive capacity)	156 (low readiness)
Kenya	133 (low adaptive capacity)	152 (low readiness)
Malawi	150 (low adaptive capacity)	157 (low readiness)
Tanzania	141 (low adaptive capacity)	151 (low readiness)
Uganda	147 (low adaptive capacity)	163 (low readiness)
Zambia	103 (low adaptive capacity)	141 (low readiness)

25. Across the board, the major factors of vulnerability that impede adaptive capacity, highlighted by stakeholders in all seven countries, are summarised as follows:
 - **Infrastructure:** Reliance on climate-vulnerable (non-climate-robust) storage and transport infrastructure leads to greater vulnerability and exposure to ambient temperature and moisture, as well as during extreme weather conditions such as flooding and heavy rainfall events.
 - **Finance:** Lack of access or limited access to credit reduces the ability for risk-management, and limits financial capacity to invest in climate-adaptive technology, equipment, facilities, and tools. Limited information about markets reduces the ability to plan for volumes of storage, transport, and sale, with the risk of more of the harvested crop being exposed to temperature and humidity than optimal.
 - **Technology:** Reliance on traditional methods and practices rather than greater mechanization and specialized tools (for harvesting, threshing, sorting, grading, packaging) reduces ability to protect the harvested crop from the impacts of high temperatures, moisture, humidity, and exposure to flooding. Limited information about pest control also increases the harvested crop's exposure to climate-sensitive pests (including insects). The lack of / limited access to climate information services (such as early warning systems and weather alerts) also straitjackets adaptive capacity against hazards like extreme weather events.
 - **Capacity:** Lack of/limited access to technology, equipment, facilities, and tools (for threshing, drying, storage, and transport) reduces ability to protect harvested crop from the impacts of high temperatures, moisture, humidity, and increases exposure to flooding and heavy rains.
26. Climate change-driven (or exacerbated) food spoilage and contamination in post-production stages of the value chains compound the climate change-driven productivity and yield losses during cultivation. This results in a double-blow to farming communities from climatic factors such as rising temperatures, higher moisture levels (warmer air holds more moisture), increased precipitation, more frequent or intense heavy rainfall events, flooding. Climate change is thus a threat-multiplier to crop value chains. The impacts from climate change in the post-production value chain stages, which result in substantial post-harvest losses and reduce the output that farmers have left to sell, depress incomes and jeopardize livelihoods of smallholder farmers. This perpetuates vulnerability to climatic shocks and stresses and prolongs the cycle of persistent poverty.
27. Furthermore, in the face of climate change-driven post-harvest losses, farmers have an impetus to keep less of the reduced output for their own domestic consumption and to offset the volume losses (during harvesting, field-drying, on-farm storage) by selling more of what remains in the market. This worsens household food insecurity in countries that already have markedly high levels of undernourishment (per FAO figures). All of these reduce adaptive capacity and must be ameliorated to ensure resilience of smallholder farmers in the seven countries.
28. In light of these findings, it is imperative for RE-GAIN to support adaptation to climate change through harvest and post-harvest loss-reduction solutions, and enable climate resilience in crops' post-harvest value chains of smallholder farmers and communities.

Adaptation Needs and Climate Change Solutions

29. The impact of climate change on post-harvest food value chain stage is felt largely through the exposure of the grain/seed to high temperatures, high moisture levels, direct contact with water (such as during heavy rainfall events), and damage caused by extreme weather events such as flooding. Additionally, high temperature and moisture levels also foster the growth of pests and diseases, including fungal and parasitic agents. Thus, the most effective solutions for reducing climate-change induced food losses in the post-harvest value chain is to reduce the harvested seed's/grain's exposure to the climatic variable or hazard. This involves interventions that improve efficiency and speed of harvest and related actions (such as threshing, sorting, grading etc. – to reduce the time period of exposure), as well as interventions that offer higher protection from climatic elements (barriers between the grain/seed and the external environment), plus interventions that provide higher levels of actionable information on the levels of or prevalence of the climatic variables (e.g., measurement tools).
30. The mere provision of such FL-RS solutions by itself is not enough if this is not accompanied by capacity-building, training, awareness-generation of how best to use such solutions, and how to increase financial wherewithal to maintain and augment such solutions in the future. To address the pressing need for broader implementation of solutions aimed at reducing climate-related harvest and post-harvest food loss, it is necessary to raise awareness and build capacity to promote the adoption of Food Loss Reduction Solutions (FL-RS) in the light of expected climate impacts. This needs to be complemented by strengthened multisectoral institutional capacities, facilitating the uptake of FL-RS by end users and service providers, increasing options of solutions' availability, and enabling practical application through policy interventions. There is also an urgent need for enhanced financial access for farmers and Micro, Small, and Medium Enterprises (MSMEs), empowering them to invest in climate-friendly FL-RS and incentivising vendors, manufacturers, and suppliers of climate-adapted FL-RS, fostering a robust end-to-end market ecosystem.
31. For this programme, a key focus of adaptation support needs to be on strengthening the capabilities of countries to develop climate-resilient post-harvest infrastructure, through providing physical solutions alongside technical and institutional capacity building for actors across harvest and postharvest value chain stages – from processing through to market distribution. This includes investing in strategic approaches that leverage finance and build markets for food loss reduction strategies. By enhancing access to markets of existing postharvest solutions, the program will encourage farmers to adopt FL-RS products and services, thereby boosting their climate and economic resilience.

Table 4. Solutions to reducing/preventing food losses induced by climate change

Typology of Solution	Climate change solution	Climate change hazard being addressed (enhanced protection from):	Applicable crops
Physical Food-Loss Reduction Solutions (FL-RS) (infrastructure, equipment, tools, machinery interventions)	Mechanical multi-crop threshers and shellers	Increased average temperatures	Maize
	Tarpaulins and plastic sheets	Extreme heat and heatwaves	Wheat
	Hermetic bags	Increase in number of hot days over 35°C	Rice
	Metal and plastic silos	Higher moisture levels and flooding from heavy rains (days with rainfall > 20mm)	Teff
	Moisture meters	Higher moisture levels and flooding from large 1-day rains and large 5-day rains)	Beans
	Improved storage structures	River floods and/or urban floods (from increased in consecutive wet days)	Soybeans
	Storage protectants and control agents		
	Harvesting machinery		
Non-Physical Food-Loss Reduction Solutions (FL-RS) (awareness-generation, capacity-)	Gender-responsive awareness campaign on the impacts of CC on post-harvest food losses and the availability of FL-RS.	(Through reduction of vulnerability and enhancement of adaptive capacity against the following hazards):	Cowpeas
	Demonstration, training and tech. transfer for the use of weather/ climate information, FL-RS and related practices	Increased average temperatures Extreme heat and heatwaves	Groundnuts

building, training)	Capacity development of extension services and agro-dealers	Increase in number of hot days over 35°C Higher moisture levels and flooding from heavy rains (days with rainfall > 20mm) Higher moisture levels and flooding from large 1-day rains and large 5-day rains) River floods and/or urban floods (from increased in consecutive wet days)	
	Facilitate market linkages between institutional markets & other buyers & smallholders, Support to structuring of value chains & coordination between market actors		
	Provide business development support & market intelligence for FL-RS manufacturers		
	Capacity and market development for all market actors		
	Training of new FL-RS providers (MSMEs, cooperatives, incl. women- and youth -led initiatives)		
	Facilitate access to finance for FL-RS providers through innovative de-risking schemes		
	Support inclusion of FL-RS in climate-resilient input packages		
	Structure prefinancing partnership arrangements that include FL-RS		
	Facilitate the development and deployment of smart subsidy and catalytic grant models, as well as 'lease-to-own models for FL-RS focussing on women and youth as key beneficiaries.		
	Support the revision of policies that enable FL-RS investments, including tax exemptions, certification and standards for FL-RS quality		
	Promote successful FL-RS business models for scaling-up & replication		

32. According to the World Bank estimates, a one percent reduction in postharvest losses in Sub-Saharan Africa could lead to economic gains of \$40 million each year, and most of the benefits would go directly to smallholder farmers¹². Moreover, food loss and waste depict an extremely inefficient use of resources and account for about 3.3 gigatonnes of greenhouse gas emissions globally¹³. In addition, large amounts of water and fertilizer go into the production of this food that never reaches human mouths. Recovering the food that is lost during harvest and postharvest handling can help close that calorie gap in Africa while strengthening livelihoods and improving food security— without requiring any additional environmental cost.

Complementarity with ongoing and planned initiatives

The RE-GAIN Programme operates in a complex environment of ongoing initiatives, and has been designed with this in mind. While many of these initiatives are focused on the production side of the agriculture value chain, there are also initiatives in each of the target countries targeting various aspects of the post-harvest value chain. Through their ongoing operations, AGRA has developed a strong understanding of the landscape of initiatives in the target countries and has developed working relations with many of the organisations operating these initiatives. The design of this programme has closely considered the ongoing work in each country and seeks to complement and collaborate with these initiatives to maximise the effectiveness of the programme¹⁴. A list of the key initiatives in each of the target countries is presented in Table 10-2 of Annex 2.

¹² World bank 2011. MISSING FOOD: The Case of Postharvest Grain Losses in Sub-Saharan Africa

¹³ FAO 2013. Food wastage footprint: Impacts on natural resources - Summary report

¹⁴ It should be noted that although the other initiatives listed in Annex 2 are complementary, the expected benefits of RE-GAIN are not contingent on the success of such initiatives; as FL-RS will have a positive impact on food security and GHG emission reduction, even in isolation.

B.2. (a). Theory of change narrative and diagram

Goal Statement

33. **IF** the capacity of the target countries and communities to respond to climate-triggered food losses is strengthened through improved and inclusive access to financing, promotion of context-specific and gender-responsive innovations to reduce food losses, and better enabling conditions for public and private investments, **THEN** smallholder farmers will have enhanced food security and livelihood resilience, **BECAUSE** the widespread use of food loss-reduction technologies will reduce food loss and reduce the carbon footprint of food systems, while increasing household income and building the resilience of smallholder farmers, MSMEs and rural communities to climate shocks.

Barriers

34. Several barriers currently impede the uptake of food loss-reducing solutions in Africa. While the specifics of each barrier might vary between countries and farming systems, six common barriers have been identified, cutting across several themes, including limited knowledge of the options available and capacities to use what is available, access to and supply of affordable harvesting and post harvesting equipment and technologies, and access to sustainable and inclusive finance opportunities, all encompassed by an overarching lack of an enabling environment. An outline of these core barriers is presented below, with deeper country-specific analysis provided in the appendices to the Feasibility Study (Annex 2).

Barrier 1: Inadequate access to information and solutions for smallholder farmers

35. While most smallholder farmers are acutely aware of the impacts of post-harvest losses on their food and income security, many do not have access to the information needed to adopt improved post-harvest processing and storage practices needed reduce such losses. The resulting knowledge gaps extend across the full range of pre- and post-harvest processes — including harvesting, handling, storage, transporting, and packaging of agricultural commodities – and are typically more significant for women-managed farms, contributing to notable gender gaps in farm productivity and profitability. For some farmers, the knowledge gaps are fundamental, in that they lack the basic understanding of resilient post-harvest processes and the options available to reduce food loss. For others, there is a general understanding of the need for improved practices, or even the availability of food loss-reducing solutions (FL-RS) but lack the understanding of where and how to access such solutions, or how to properly implement them. These knowledge gaps are systemic in the agriculture sector of the target countries, with limited integration of FL-RS information into sectoral policies and agricultural extension services. As a result, farmers lack access to the required information to adopt FL-RS. Overcoming this barrier requires a combination of targeted and inclusive awareness raising among smallholder farmers, as well as the establishment of an enabling environment in which information on FL-RS is mainstreamed into agriculture sectoral policies and support services.

Barrier 2: Limited awareness of climate change and climate risks

36. As noted above, climate change impacts the agriculture sector across its entire value chain – from production, through harvest, storage, processing and distribution. Despite this significant impact, many smallholder farmers do not fully understand climate change and the associated climate risks and the impacts they have on food security, and consequently are not adequately motivated to adopt resilient solutions. Given that the impacts of climate change on the agricultural sector are expected to increase in the future, the consequences of inaction stemming from limited awareness will become increasingly severe. Although there is growing understanding among farmers of the impacts of climate change on production – particularly as observed during increasingly frequency extreme climate events – such awareness is less prevalent for post-harvest losses. This is partly due to the links being less obvious to the observer, for example where a changing climate influences the prevalence of pests or disease during storage. This lack of awareness perpetuates the information gaps noted in Barrier 1. Specifically, when a farmer does not understand the links between climate change and food loss, efforts to seek resilient solutions are often subdued. Moreover, a lack of awareness of potential future climate trajectories

exacerbates the problem, as farmers are less likely to act early if they don't understand how the problem of climate change-induced food loss will increase over time.

Barrier 3: Limited availability of appropriate harvest and post-harvest technology

37. While a wide range of FL-RS options have been developed globally, penetration of such solutions into smallholder farming communities in Africa is limited due to limited access to technologies and practices, low community engagement and insufficient demand and perceptions of risk from FL-RS providers. This lack of market penetration has been attributed to several closely linked factors. First, because of the limited awareness noted in Barriers 1 and 2, there is currently limited demand for FL-RS in rural communities. The lack of demand is further exacerbated by unavailability of working capital or access to affordable finance (see Barrier 4 below), with many farmers therefore considering it unfeasible even if they wanted to adopt improved FL-RS.
38. Second, as a result of the limited demand, suppliers have not demonstrated sufficient interest to develop the market for FL-RS. In particular, without the economies of scale driven by sufficient demand, FL-RS providers are not incentivised to develop localised production, distribution networks and markets that would bring the cost of FL-RS down. This also inhibits research into improving FL-RS solutions that could improve efficiency or affordability of the products.
39. Third, the development of local markets for FL-RS is further constrained by the perceived risk for suppliers of engaging with smallholder farmers. As with financial institutions, FL-RS providers need to balance risk in setting up operations within, and providing services to, smallholder farming communities. Creating a market for smallholder farmers – for whom working capital is limited – requires innovative market tools to be developed, many of which require the sharing of risk between different stakeholders – including providers, users and financial operators. However, such systems ultimately rely on the ability of smallholder farmers to secure sufficient income from their produce. Consequently, FL-RS suppliers often perceive smallholder farmers as high risk because of their general limited capacity to sustainably produce, process and market agricultural goods.
40. The ultimate result of these challenges is that smallholder farmers do not have easy access to affordable FL-RS. To overcome this barrier, a catalyst is required to initiate demand and establish a sustainable local market that would provide the necessary incentives to attract FL-RS providers.

Barrier 4: Limited access to finance for FL-RS uptake

41. As noted above, the market for FL-RS for smallholder farmers in Africa is influenced by the limited access to markets for smallholder farmers, working capital for MSMEs and access to finance for the purchasing of equipment. While there is a diverse financial market across all target countries — ranging from commercial banks to micro-finance institutions (MFIs) — several inter-related factors impede access to finance. First, MSMEs are perceived as high risk, largely because of the uncertainty of production (particularly for rainfed agriculture under changing climate conditions) and limited access to smallholder produce. The feasibility of agricultural loans is dependent on MSMEs being able to develop a profitable business case for the FL-RS equipment to be able to repay the loan at the end of the season. However, the vast array of factors that impact the business case creates a high level of uncertainty, while inadequate on-farm post-harvest storage and processing can lead to further food losses before a farmer is ever able to use the FL-RS services. This uncertainty is being exacerbated by climate change, with growing risk that entire harvest can be lost in a single extreme climate event. Consequently, FIs are often hesitant to provide agricultural loan products that target agrifood MSMEs. In instances where financial products have been developed, the terms of the credit are often untenable for agrifood MSMEs.
42. Second, and exacerbating the risk perceptions mentioned above, is the lack of or limited collateral and track record among smallholder farmers and the companies serving them, including youth groups, agro-dealers and cooperatives. To offset the risk and comply with Central bank regulation, FIs generally require collateral from the borrowers, generally in the form of an immovable asset, such as land or property. However, most small and medium sized enterprises especially women and youth led businesses lack suitable collateral or have already pledged their assets, restricting access to available finance products. For example, land tenure arrangements in rural areas of Africa are generally dominated by communal land

rights, in which land is not owned by the individual, but rather an individual is given access to land under customary law. In such instances, the land cannot be used as collateral, and in the absence of other valuable assets, entrepreneurs cannot meet the collateral requirements. This challenge is generally amplified for women and other vulnerable groups who have less access to and control of assets due to sociocultural norms and practices that override the implementation of gender equitable laws (which now exist in many African countries) relating to land and inheritance. Another factor commonly used to offset risk is the presence of a strong track record; the business case for FL-RS service provision is not proven in many countries, hence the hesitation for financial institutions to invest in productive assets that will be rented out or used by small scale farmers. To create sustainable and more equitable markets for FL-RS, it is essential to improve access to affordable finance for agripreneurs. This requires innovation in the market, working with various stakeholders to develop innovative finance products that share the risk and maximise opportunities for the sustainable adoption of FL-RS. Such innovations need to consider – and address – gender disparities in access to collateral, farm productivity, and control over use of proceeds, to ensure such financial support is accessible to both women and men. This includes enabling activities, such as the sensitization of local communities on women's land rights, as a basis for promoting women's greater ownership and control over land and other productive assets, all of which are key enablers for securing credit

Barrier 5: Inadequate policy and regulatory frameworks to enable FL-RS investments

43. Development of national policies, strategies, frameworks and implementation plans aiming to address agricultural sector and climate change challenges has been a priority for many of the African governments in the recent years. Despite this, there are still considerable gaps in the policy and regulatory environment across the target countries that impede the mainstreaming of sustainable FL-RS in the agricultural sector. This gap is underpinned by several factors. First, there is limited or no use of the scientific evidence base in the formulation of policies and strategies on post-harvest loss reduction. This challenge is exacerbated by limited technical and institutional capacities and systems for institution-level data collection, monitoring, and evaluation of implementation and effectiveness policies and programmes.
44. Second, policies generally lack coordination and harmonisation across sectors, limiting opportunities for integrated approaches to building resilience in the agriculture sector. For example, the volatility and vulnerability of food systems, and associated food loss, is often exacerbated by poor infrastructure and/or uneven and deteriorating power or energy access; and the lack of harmonisation of policies across these different sectors prevents collaborative action, perpetuating the challenges.
45. Third, where policies or regulatory frameworks are in place, they are often constrained to agricultural input subsidies — mainly for fertilizer and seeds — with no subsidy on post-harvest technologies. This further extends to the focus of information and extension services, which are generally geared towards primary production, and less so to post-harvest processes. Similarly, existing taxation regimes do not favour or incentivise investments in storage and processing technologies for food in the smallholder system.
46. The underdeveloped private sector in these countries require policy incentive mechanisms, for example import tax exemption that would enable reduction of cost of import and distribution of FS equipment and services. Further, tax exemptions on key components that can be used in the manufacturing or assembly of FL-RS should also be a key consideration in these countries.
47. The lack of access to finance as described above is also restricted by Central bank regulation and prudential guidelines on loan loss provisioning, consideration of movable assets as collateral and value of guarantees for banks in reducing their reserve requirements. Addressing these policy barriers can unlock financing for equipment (movable assets), storage facilities (fixed assets) and other FL-RS identified. The existence of warehouse receipt legislation and authorities are also an important factor to consider enhancing the uptake and usage of storage solutions. Without cash in hand at harvest, farmers will be hesitant to store their produce.

Barrier 6: Insufficient public sector capacity to steer investments in FL-RS

48. Actions to overcome the full range of barriers described above require a strong public sector to create the enabling environment and steer investments into FL-RS. The public sector plays an important role in

steering investment, not only by creating an enabling policy environment (see Barrier 5), providing context-specific extension and advisory services to smallholder farmers but also through its role in coordination and as a facilitator of public-private-community partnerships. However, public sector actors across the target countries currently have limited technical and institutional capacity to create the required enabling environment. This barrier is partly rooted in human capacity constraints, with insufficient staff available in key public institutions stretching available human resources; but also, in the technical capacity of those staff that are available to facilitate investments, technology adoption and partnerships. This includes public sector staff having limited understanding and awareness of FL-RS and the innovative models available to develop sustainable business and market opportunities.

Proposed Solution

49. The proposed project will contribute to food security in seven African countries by promoting the wide-scale adoption of Food Loss-Reduction Solutions (FL-RS) among smallholder farmers. In addition to increasing the amount of food available for consumption due to reduced losses, this approach will also improve the quality of produce leaving the farms, thereby increasing nutritional value, marketability and income potential, while simultaneously reducing the GHG emissions associated with food loss. To achieve the project goal, and actions, will be taken across three primary pathways — farmer-centric, supplier-centric, and institutional support — which together will form the foundation of a sustainable and scalable market for FL-RS in the target countries. Interconnected solutions will be addressing food loss under smallholder setting of Africa.

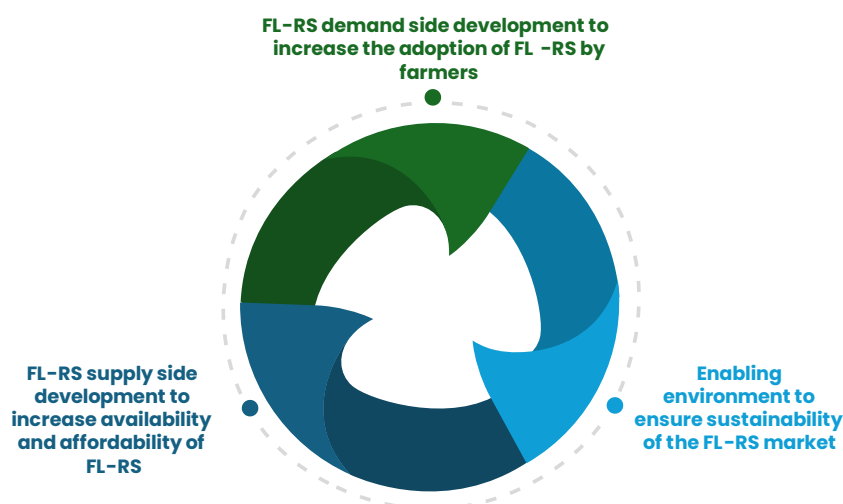


Figure 2. Interconnected solutions addressing food loss under smallholder setting of Africa.

50. The first development pathway focuses on increasing demand and absorption capacity for FL-RS among smallholder farming communities. Activities under this pathway will target overcoming Barriers 1 and 2, raising awareness among farmers of the impacts of climate change on post-harvest food loss, as well as on the availability of FL-RS options on the market and how FL-RS will help reduce food loss under changing climate conditions. This awareness raising will be coupled with training on the use of specific FL-RS that have been identified for key crops in each country as well as on the effective use of weather information and advisories, while supporting the initial transfer of suitable technologies to the farmers to catalyse uptake and demonstrate the effectiveness of FL-RS. These combined activities will be catalysing the demand and build the absorption capacity needed to achieve short-term Outcome 1: FL-RS adopted by smallholder farmers. Contributions to Outcome 1 are further enhanced through Output 1.2, which will improve market linkages between farmers and agri-value chain actors such as processors and aggregators. This will enhance the value-add of FL-RS, ensuring that the increased quality produce resulting from improved harvesting methods, post-harvest handling, storage, and processing is able to make it to suitable markets to maximize income; creating sustainability in the whole value chain that will enable future scaling of and maintain demand for FL-RS.

51. The second development pathway complements the demand-side activities by enhancing the supply and affordability of FL-RS (Outcome 2). This component involves two Outputs, one focused on business development for the improved provision of FL-RS on local markets, and the second focused on complementary financial mechanisms for smallholders and MSMEs to enable the adoption of FL-RS. Activities under Output 2.1 will support business development linked to the local supply of FL-RS, targeting both existing suppliers, as well as entrepreneurs seeking to enter the market – particularly MSMEs, local cooperatives and youth groups. The business development support will include providing access to market intelligence, as well as facilitating access to existing credit/guarantee schemes as well as any new financial mechanisms established under Output 2.2. Support to local businesses will increase the supply of FL-RS in local markets, which in turn will enable cost reduction, underpinning the outcome of improved affordability. This will be complemented by activities under Output 2.2 aimed at establishing innovative financial mechanisms for smallholders and MSMEs to support the adoption of FL-RS. In particular, the project will establish partnerships across the FL-RS value chain to create accessible finance opportunities based on the foundations of risk sharing and cooperation. These partnerships will bring together key stakeholders — including suppliers, agro-dealers, financial institutions, MSMEs and local farmers — to establish finance models that stimulate the FL-RS market. Catalytic co-payments and conditional procurements under the system using GCF grants will provide added security to the transactions, further enhancing the risk-sharing nature of the approach. Finance models will be adapted to fit the local context of each target community, considering the nature of the targeted crop and the required solutions, the needs of the local farmers, and the capacity of stakeholders to absorb risk. Where available, the development of financial modalities will integrate existing credit lines available from local FIs, as well as supporting the inclusion of FL-RS in climate-resilient input packages.
52. Combined, the increased demand from Outcome 1 and the enhanced supply and affordability of FL-RS from Output 2 will be catalytic in the market, overcoming Barriers 3 and 4. Specifically, the transactions facilitated by the project will stimulate the market, enabling suppliers to develop products and services that are affordable to smallholder farmers, thereby facilitating a balance between supply and demand, and establishing a self-sufficient market that will operate beyond the project support. The integrated nature of these Outputs and the causal pathways reinforcing the market development are depicted in Figure 3 below.

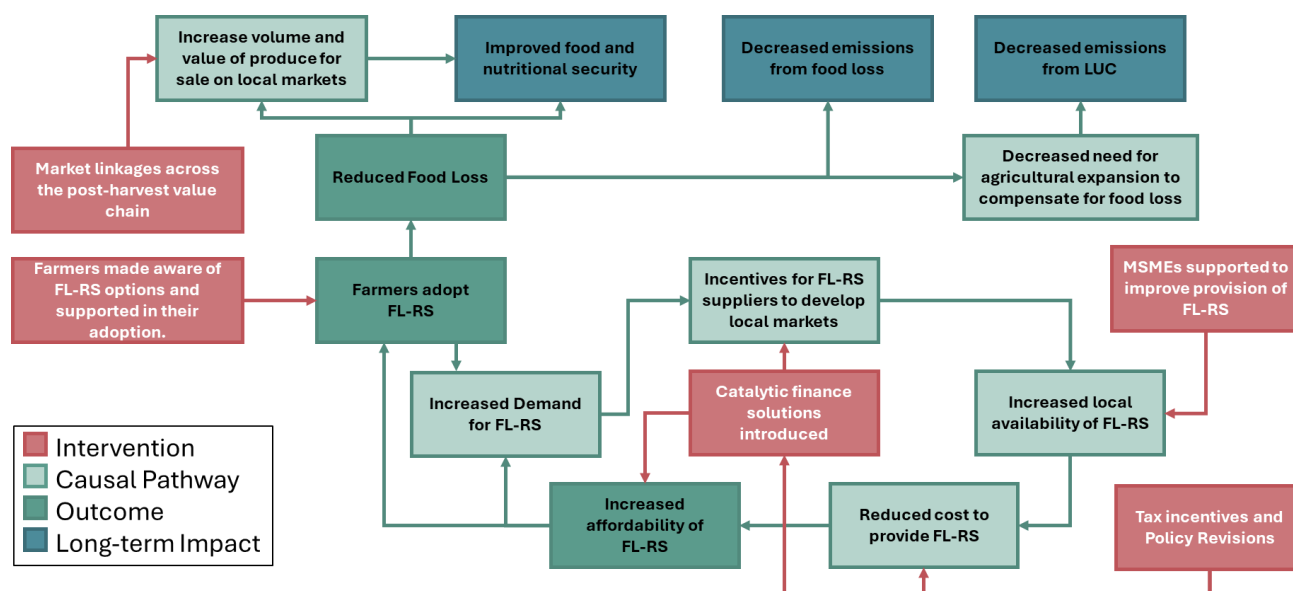


Figure 3: Causal pathways underpinning the Theory of Change.

53. These first two development pathways will be underpinned and sustained by the third, which seeks to create an enabling environment for the uptake of FL-RS (Outcome 3); which together with Outcome 2 will not only mobilise increased investment in the sector but also provide wider policy interventions in strengthening the market systems and advisory services. Through Output 3.1, the capacity of national institutions will be enhanced to enable investments in FL-RS – specifically tackling Barriers 5 and 6. This will involve policy revisions and put in place enforcement mechanisms in each of the target countries in support of enabling FL-RS — including stimulating tax incentives and promoting standards for FL-RS

quality — as well as creating platforms for the scaling and replication of successful FL-RS business models.

54. Overall, the resulting uptake of FL-RS, at scale, will reduce food loss, improve food quality and enhance local capacity, thereby enabling the long-term outcome of improved food and nutritional security. This will have the co-benefit of increasing food safety, while the increased investment in FL-RS resulting from Outcome 3 will create additional employment opportunities in the market. Moreover, the project will reduce GHG emission, both directly – through the reduced emissions from the wasted food itself – and indirectly by reducing land use change for agricultural expansion driven by the need to compensate for food losses.

Assumptions

55. The Theory of Change described above and illustrated in Figure 4, is underscored by several assumptions. These assumptions are outlined below, and mapped against specific Outcomes and Outputs in Section E.
- The proposed FL-RS technologies are market-ready in Africa or globally (i.e. transfer and adoption ready).
 - MSMEs and smallholder farmers are receptive to capacity development and adopt alternative post-harvest management technologies, with focus on managing contamination by mycotoxins and adaptation practices for key crops and value chains, including the use of renewable energy.
 - MSMEs and smallholder farmers, mainly youth and women, are willing to adopt climate-smart postharvest handling techniques, land-use practices and RETs and are motivated to sustain them over the programmes duration (5 years) and lifetime (10 years).
 - Market actors (input suppliers and off-takers) are interested in providing the FL-RS solutions and in buying improved FL-RS at better prices.
 - Existing financial schemes and local financial institutions are willing to participate in the proposed interventions.
 - Accurate weather information and climate advisory is readily available to be disseminated to farmers to enhance FL-RS effectiveness.

B.2. (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☒	☐	☐
Outcome 2	☐	☐	☐	☐	☒	☒	☐	☐
Outcome 3	☐	☐	☐	☐	☒	☒	☐	☐

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Increased food quality and safety	☐	☒	☐	☐	☒	☐
Co-benefit 2: Additional employment opportunities for women and youth	☐	☒	☒	☒	☒	☐
Co-benefit 3: Reduced GHG emissions including from avoided food loss, deforestation and land degradation	☐	☐	☐	☐	☐	☒

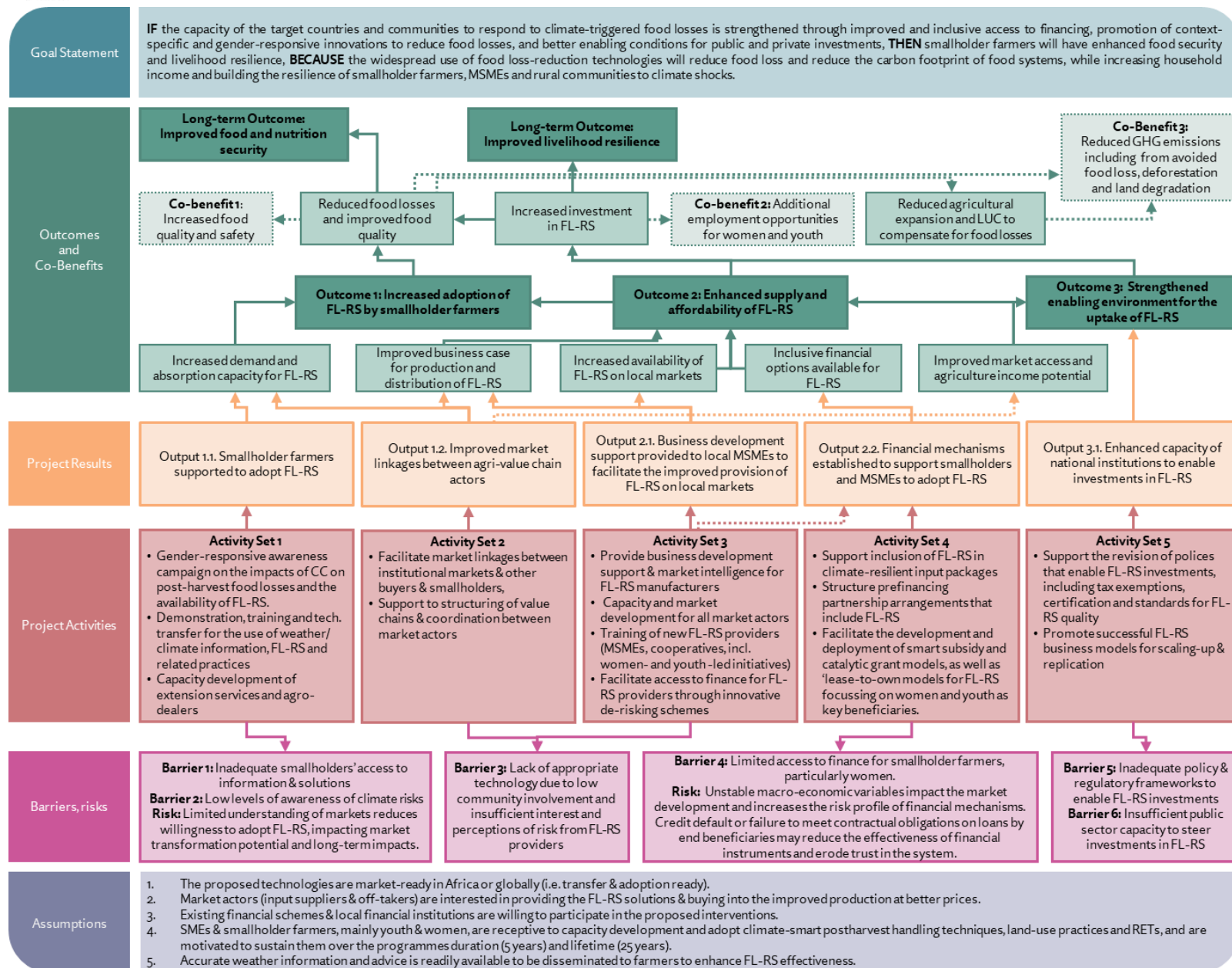


Figure 4: Theory of Change

B.3. Programme description

58. The programme description below outlines the overarching design of the programme's approach as it applies across all seven countries. The activities are presented thematically as they apply across the focal countries as a whole, with specific details on the nature of interventions for each individual country presented in the country-specific Appendixes to Annex 2.

Crop and Site Selection

59. Key crops were identified by expert assessment, supported by AGRA and the National Designated Authority (NDA) of each target country through consultative processes. Two crops per target country under AGRA's portfolio were selected to ensure that sufficient resources would be available for the crafting and execution of targeted solutions (as depicted in Figure 10). Selected crops are representative of the agricultural dynamics of each country and aligned with the specific needs and strategic agricultural goals of the nation. In addition, these crops hold substantial importance to the country's food security and/or experience particularly high rates of loss within the value chain. Finally, these crops are produced in large parts of the country through a significant number of smallholder farmers engaged in the value chain. The key crops therefore reflect the agronomic and economic realities of each country and provide opportunities for targeted enhancement of food security and sustainable agricultural practices and improved management of these crops is also expected to reduce GHG emissions given the aggregated contributions of these value chains to country's GHG emissions, contributing to the NDCs of the target country. Please see Annex 2: Section 3 and 4 for details on the climate vulnerability of selected crops.

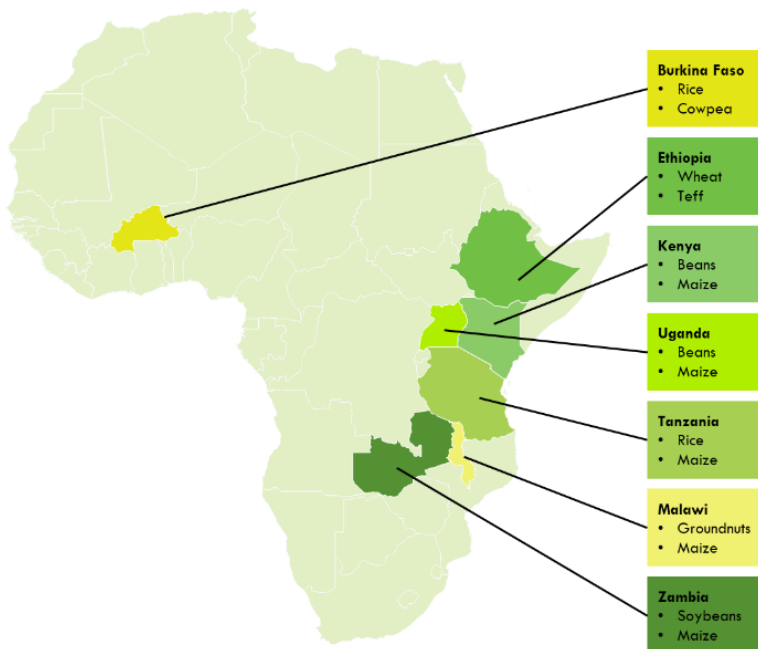


Figure 5. Selected crops for each of the seven target countries.

60. Within each country, a set of target regions have been identified, as outlined in Table 5. Within these sites, the specific beneficiaries will be selected following a set of eligibility criteria, as presented in Section 5.2, 5.6 and 6.4 of Annex 2. The selection of beneficiaries will involve social and environmental risk screening, including screening for the presence of indigenous communities¹⁵; in locations where indigenous communities are present, careful consideration and tailored approaches will be designed to address the unique circumstances and challenges faced by Indigenous communities in these areas with continuous engagement to proactively include their participation in the RE-GAIN programme.

Table 5. Priority regions within each target country.¹⁶

Country	Priority Regions		
Burkina Faso	Hauts Basins Centre Ouest	Cascades Sud Ouest	Boule du Mouhoun
Ethiopia	Arsi West Arsi	East Gojam Hadiya zone	North Shoa
Kenya	Makueni Kitui	Embu Tharaka Nithi	
Malawi	Kasungu Agricultural Development Division (ADD)	Lilongwe ADD Salima ADD	Mzuzu ADD
Tanzania	Morogoro Iringa Njombe Ruvuma	Mbeya Songwe Katavi Rukwa	Manyara Kigoma Tabora
Uganda	Bugisu Busoga	Sebei	

¹⁵ Specific guidance on stakeholder engagement processes for IPs and other vulnerable communities are outlined on Annex 7

¹⁶ See Annex 16 for maps of the target districts in which the programme will be implemented in each country.

Zambia	Southern Province Eastern Province	Central Province
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Component 1: Food Loss-Reduction Solutions (FL-RS) demand side development to increase the adoption of FL-RS by farmers

Outcome 1: Increased adoption of FL-RS by smallholder farmers

61. The first project component focuses on enhancing demand for, and adoption of, FL-RS in the target communities. It is this adoption of FL-RS that will ultimately reduce food losses and result in the core benefits of the programme for smallholder farmers – namely improved food security, increased resilience to climate shocks, and reduced GHG emissions. The approach for achieving this Outcome is grounded to two programme Outputs that will directly support the farmers through awareness, training, technology transfer and market linkages. However, given the integrated nature of the project approach, achieving the full impact of the Outcome will also rely on the supply-side interventions under Outcome 2, and the strengthening of the enabling environment under Outcome 3.

Output 1.1: Smallholder farmers supported to adopt FL-RS

62. At the heart of the demand-side development is the direct support provided to smallholder farmers to not only know what FL-RS options are available, but also to understand the need for adopting such options – particularly when considering the impacts of a changing climate – as well as how to optimise the use of FL-RS. Activities under this project will involve technical support targeting several core themes:
- Raising awareness among farmers of the impacts of climate change on food loss and the availability of improved, climate-resilient FL-RS options;
 - Training farmers on the use of improved FL-RS, as well as adapting existing post-harvest food loss-reduction technologies to climate change;
 - Training farmers on best practices for complementary crop management (including use of weather/climate information) to maximise the efficiency of FL-RS;
 - Actively demonstrating the effectiveness of FL-RS systems in local contexts; and
 - Supporting the transfer of technologies and boosting immediate availability of FL-RS solutions in target communities.
63. Training activities will cover three key stakeholder groups, namely: i) smallholder farmers; ii) extension workers and agro-dealers; and iii) agricultural traders and processors.
64. The activities under Output 1.1 will focus specifically on short-listed FL-RS interventions, tailoring the approach to each community's individual context. AGRA will train and engage VBAs (Village Based Advisors), who are community-based, and are usually young people who are lead farmers, unemployed youth and/or entrepreneurial farmers to provide extension services and agribusiness linkages. This extension model has been well developed and is operational in these target countries. They will play an important role in training smallholder farmers on FL-RS solutions. An evaluation of proposed physical FL-RS was conducted across all seven target RE-GAIN countries to identify those with the highest potential to reduce harvest and postharvest food losses and protect the harvests against growing impacts from climate hazards. The output of this assessment established a shortlist of FL-RS options tailored to each country and crop, as presented in Section 5 of Annex 2 – Feasibility Study. These solutions include: Communal storage structures; Warehouses; Harvesting machinery; Hermetic bags; Mechanical multi-crop threshers and shellers; Metal and plastic silos; Moisture meters; Storage protectants and control agents; Storage structures; and; Tarpaulins and plastic sheets. Further details on the nature of interventions per country are available in the Appendixes to Annex 2.

Activity 1.1.1: Conduct an awareness-raising activities targeting smallholder farming communities focussed on the impacts of climate change on post-harvest food losses and the availability of locally-appropriate and climate-resilient FL-RS.

65. The limited awareness on post-harvest food loss management at both the value chain and policy level, combined with inadequate communication on the importance of post-harvest food loss reduction has led to the limited adoption of available FL-RS as well as inadequate integration of food loss reduction strategies into relevant policies and strategies by governments in Africa. This has, in turn, resulted in

inadequate investment in solutions that reduce food losses¹⁷. Approximately 5,350 VBA will be in raising awareness, each targeting 200 to 250 farmers, resulting in a total reach of ~1,110,000 people trained.

66. To establish a foundation for increasing demand for FL-RS, awareness and capacity building focused actions and activities will be undertaken in each of the target communities across the seven countries, using local institutions, chiefdoms and community networks to increase the understanding of postharvest food losses among farmers, local stakeholders in the agriculture value chain, and the public. The awareness raising will target several key topics, as outlined below.
- RE-GAIN programme and its objectives to reduce food losses and for climate change adaptation.
 - Major drivers and causes of harvest and postharvest loss in the target locations, with emphasis on climate-food loss nexus
 - Impact of postharvest losses on food security, income, economy, and the environment (incl. climate change) and the importance to reduce FL.
 - Available best practices and improved technologies and methods (e.g., timing of harvesting, methods and technologies for harvesting, storage, etc.) to reduce harvest and postharvest losses and their benefits (food security, income environment).
 - Role of different actors (local government, extension services, farmer organisations, agro-dealers, financial institutions) to provide access for FL-RS across scales
 - Cross-cutting themes: climate change awareness, climate smart agriculture, farm management, marketing, product quality management, access to finance, gender and youths, etc.
67. The nature of awareness raising activities will be tailored to agricultural context of each country, accounting for the target crops, the availability of local information dissemination systems (including mass media such as radio, television, digital platforms and social media, as well as brochures, flyers and local extension services), cultural considerations and local languages. The campaign will collaborate with local governments and farmer organisations, build on existing outreach systems to maximise reach and participation. AGRA will closely monitor the awareness campaigns, assessing impacts and adapting outreach strategies as needed.

Activity 1.1.2: Train smallholder farmers on the use of improved climate-resilient FL-RS.

68. Training will be provided to develop the capacities of smallholder farmers on improved crop management and storage techniques to reduce food losses and to maintain quality of produce, increase farmers' income by reducing losses and improving marketability, and improve supply of financial services and FL-RS to smallholders and other value chain actors. AGRA will procure the services of suitable local entities (for example, NGOs, VBAs) to facilitate training in target communities. Training will then be provided through various channels — such as onsite demonstration of best practices, agricultural extension services, lead farmers and Village-Based Advisors (VBAs), farmer cooperatives, industry associations, and digital platforms — depending on the local context and the existing networks of the local service provider. Approximately 5,350 VBA will be engaged for training, each targeting 200 to 250 farmers, resulting in a total reach of ~1,100,000 people trained.
69. The nature and content of the training will be tailored to each group and demand as outlined below. Training will include baseline topics, including an introduction to the RE-GAIN programme and the core concepts of food security, post-harvest food losses and the impact of climate change – building on and leveraging material from the awareness campaign (Activity 1.1.1). The sequencing of the training will also, where feasible, follow Activity 1.1.5 to enable integration of the equipment demonstrations (Activity 1.1.6) into the training of farmers. In addition to this baseline, technical training will be provided on specific subject matter tailored to the local context of each community. A list of potential training topics has been identified, which will be refined for the local needs in each target community during the first year of implementation, once the communities are identified. This list includes:
- Major triggers and incidences of food loss and the impact they have on food security and health
 - Identification of the optimal timing of harvesting, thrashing and transport.
 - Methods to reduce contamination of produce by mycotoxins and other diseases at harvest and storage
 - Use of available weather forecast data.

¹⁷ Bellagio Statement 2018

- Key reasons of food losses during harvesting and postharvest management and storage.
- Major impacts of climate change on agriculture, with focus on food loss.
- Technical approaches on maintaining crop quality during harvesting, postharvest handling and storage.
- Approaches and solutions to prevent pest attacks, and proper storage methods.
Types of storage facilities and their advantages and disadvantages.
- An overview of FL-RS options and suppliers
- Approaches to measuring and keeping optimal moisture content in crops to prevent aflatoxin contamination.
- Best harvesting methods and tools, including mechanization to reduce food losses for various target crops.
- Proper use and maintenance of physical FL-RS, including operation and maintenance of machinery, and their environmental and safety aspects.
- Record-keeping, financial literacy and access to finance.
- Benefits of and processed for farmer aggregation.
- Packaging and marketing of crops.
- Methods and materials for proper on-farm storage, safe and proper use of pesticides and fungicides, pre-storage crop treatment and preparations, and monitoring storage losses and quality of crops during storage.

70. All training will be prepared with specific attention to gender responsiveness and opportunities for youth development, as well as environmental considerations for proposed solutions. The Gender and Inclusion Action Plan (Annex 8) stipulates how this will be achieved, with a strong focus on ensuring the VBA / Farmer Based Organisation (FBO) training content is gender-responsive and youth -oriented, and that these VBAs / FBOs are suitably trained on engaging specifically with women and youth. In addition to this targeted engagement, the Gender and Inclusion Action Plan also specifies that all aspects of the Programme's awareness raising and capacity building initiatives must include dedicated modules / components focussed on gender and youth inclusion and empowerment. These will seek to tackle issues of acceptance / sensitization of male counterparts by focussing on the benefits of working with / supporting women and youth -led initiatives. This approach extends to the training activities outlined below, which will ensure agricultural traders, agro-dealers and extension workers are equally exposed to this content.

Activity 1.1.3: Train agricultural traders and processors on complementary services for FL-RS

71. Building on the training of farmers, the programme will provide additional training to agricultural traders (agro-dealers, aggregators etc) and processors on complementary services that support and maximise the effectiveness of FL-RS. This training will include the same baseline concepts, including an introduction to the RE-GAIN programme and the core concepts of food security, post-harvest food losses and the impact of climate change. Beyond that, the training will include best-bet approaches to introduce FL-RS to smallholder farmers and the types of initial support required to enhance demand, tailored to the processing and trade of produce, covering topics such as:
- post-harvest food losses at the trade and processing stages, their causes and solutions;
 - improved quality management and adherence to quality standards;
 - improved storage, transport logistics and packaging (including food-safe packaging materials and methods);
 - the sustainable use of storage protectants;
 - farmer aggregation methods and systems and associated risks of food loss;
 - improved processing (including whole grain processing) and value addition;
 - supplier management;
 - effective marketing strategies; and
 - access to finance and credit.
72. Approximately 5,350 VBA will be engaged for training, each targeting four to five traders or processors, resulting in a total reach of ~22,000 people trained.

Activity 1.1.4: Implement a training-of-trainers programme targeting extension workers and agro-dealers.

73. In addition to the direct training of farmers under the programme, RE-GAIN will facilitate broader training opportunities through a training-of-trainers (ToT) approach, particularly for local leader extension agents, who will provide onward training to farmer. Using the same service providers procured for Activity 1.1.2, key stakeholders who engage regularly with farming communities — such as extension officers, VBAs and agro-dealers — will be trained as trainers, focussing on the subject matter outlined for Activity 1.1.2. In addition to ensuring that trainers have a context-specific extension content and firm understanding of the training content, they will be trained on the practical approaches needed to effectively reach and engage communities, including on the setup of trainings and the use of demonstrations and promotional materials, as well as concepts of gender-sensitivity and youth engagement. This training will enable these stakeholders to integrate the concepts of FL-RS promoted by the project into their day-to-day support provided to smallholder farmers, leading to further proliferation of knowledge to the broader farming community in the target districts. Training will target 5,350 VBAs or extension officers across the seven countries.

Activity 1.1.5: Conduct scoping missions to identify and engage manufacturers of FL-RS equipment.

74. Under this activity, AGRA will develop an inventory and characterization of FL-RS equipment manufacturers and agro-dealers that will be engaged in developing/strengthening local markets and enhancing the immediate availability of affordable FL-RS (establishing the basis for delivery models under Component 2). By mapping the market and building networks of local manufacturers and importers/agro-dealers, this activity will enable a better understanding of current suppliers of FL-RS, and their production scalability potential. To ensure the reach of FL-RS to remote and rural areas, the programme will facilitate engagement and increased collaboration between solution manufacturers and local agro-dealers in those target areas, thereby ensuring availability of solutions, and their accessibility by local farmers. Where feasible the programme will facilitate the development or strengthening of platforms that will bring together demand, supply and other stakeholders that enhance commerce and an exchange of goods and services.
75. The scoping missions will include assessments of the current market in each target region to better understand the market needs and opportunities. This will include identifying opportunities for exchanges between African manufacturers and companies in other developing countries on the continent and beyond to help introduce new technologies into the African market. Through these scoping missions, AGRA will facilitate knowledge and technology sharing and transfer between suppliers, farmers, and service providers.

Activity 1.1.6: Conduct demonstrations of locally-appropriate FL-RS

76. The programme will facilitate active demonstrations of short-listed FL-RS to target communities. This activity will be sequenced after and will build on the relationships developed under Activity 1.1.5, partnering with vendors for the implementation of product demonstrations, leveraging the increased sales of their solutions through the programme-supported outreach as an incentive for vendors to engage in demonstrations. Under this model, agro-dealers or suppliers set up and/or provide the inputs for an FL-RS demonstration that will be established either at the agro-dealers' property or on-farm in partnership with the smallholder farmers. These demonstration sites act as learning points, while also connecting farmers to the suppliers of the products. Where possible, demonstrations will be done in conjunction with the training under Activity 1.1.2 to improve efficiencies and ensure full connectivity between the theoretical training and practical demonstrations. Demonstrations, trainings and the awareness creation of the food loss–climate change nexus will be designed taking into account local context. They will take the form of a combination of on-farm and communal demonstrations, presented through one-on-one farmer trainings, innovation hubs, and by agro-dealers, targeting farmer and youth groups, aggregators and processors. The project will undertake detailed participatory assessment to design and implement the various modalities of demonstrations and trainings. At least 2 demonstrations will be done in each target district per year for five years, resulting in a total of 410 total demonstrations.

Activity 1.1.7: Provide technical assistance to food processors to facilitate a shift to whole-grain flour production.

77. In addition to the FL-RS approaches that will be promoted above, the programme will provide technical assistance to promote the transition from refining grains to wholegrain flour production. As depicted in

Error! Reference source not found., wholegrain production can produce almost 50% more bread from the same quantity of unprocessed wheat, helping reduce the “hidden food loss” associated with over processing. The intervention will target food producers – millers in particular – providing technical assistance along several themes, including: i) developing MSME capacity; ii) raising awareness of market opportunities for wholegrains; iii) support for technology transfers; and iv) facilitating access to finance. At least 10% of processors trained under 1.1.3 will be targeted for training on wholegrain production, targeting 2,200 people total.

78. To maximise demand for the increased production of whole grains, RE-GAIN will create linkages with major existing programmes and programmes in the areas of school meals and wholegrain. The activity will engage broader initiatives that are promoting wholegrain food for use in institutional markets such as schools that not only act a pull mechanism for markets but works at sustainably changing behaviours for future markets. For example, Rockefeller Foundation has large programs to promote the use of FWG in school meal programs. This will create the demand for FWG.

Output 1.2: Improved market linkages between agri-value chain actors

79. The proliferation of unstructured market systems is one of the major causes of post-harvest losses, with poor handling and low compliance with quality standards creating substantial risk for produce during the post-harvest phase of production. To address this challenge, RE-GAIN will provide technical assistance to enable the development of more structured relationships between producers, value-adding agents and off-takers. By structuring the linkages on the premise of increased food quality linked to improved post-harvest handling, these activities will create incentives for quality compliance and investments in climate-resilient FL-RS. Moreover, the improved market linkages will help de-risk potential external investments in FL-RS, enhancing opportunities for smallholder farmers and cooperatives to adopt FL-RS at scale.

Activity 1.2.1: Provide technical assistance to support the structuring of value chains and coordination between market actors

80. The technical assistance under this activity will facilitate market linkages between smallholders, MSMEs and potential buyers – such as retailers, processors and exporters – using the consortia model as tested and proven by AGRA. Other opportunities such as institutional markets will also be explored on this programme to optimise the role of markets in the reduction of food losses.
81. The process will start with an assessment of the existing market in each target community, drawing on and expanding the initial market assessment conducted during preparation of the RE-GAIN programme (see Annex 2). This extended market analysis will map out key actors in the agricultural value chains at the local level, including potential institutional markets (e.g. schools, hospitals, prisons) that are not yet fully accessible to smallholder farmers, but that have potential to act as reliable off-takers. Gaps in the value chains will be mapped, drawing connections between various actors to identify opportunities for filling gaps. Based on this stakeholder mapping, the programme will provide technical assistance to local actors to structure value chains for different crops and institutional markets. Actions to develop linkages will specifically consider existing constraints faced by women in the sector and identify and promote opportunities for women to actively engage in the market.
82. This activity will culminate in clear action plans for structuring value chains for target crops in each community, including an outline of business development opportunities (see Output 2.1) to fill gaps in the market. The implementation of value chain development will be further supported by establishing local coordination mechanisms and supporting the establishment and or strengthening of cooperatives where needed to facilitate improved coordination between actors.

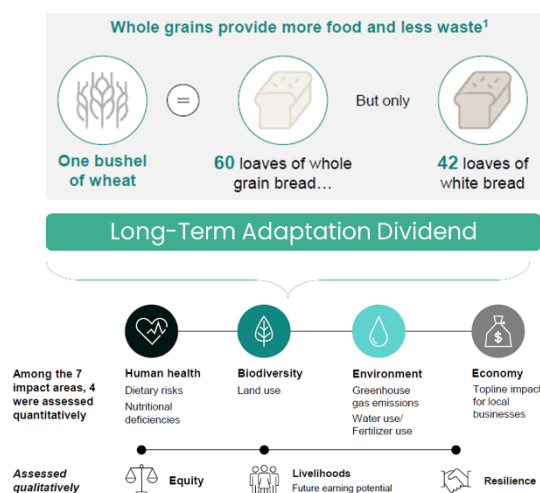


Figure 6. Benefits of wholegrain flour production as an approach to reducing food waste.

Activity 1.2.2: Facilitate linkages between farmers and institutional markets such as school feeding programs, food processors and other social safety programs.

83. Drawing on the structures identified under Activity 1.2.1, the RE-GAIN programme will engage key actors in the agri-value chain to link smallholder farmers to institutional markets such as school feeding programmes, food processors and other social safety programmes. AGRA will support local farmers and cooperatives in connecting them with other actors in the value chain and provide technical assistance to establish formal mechanisms that will secure offtake agreements for produce that meets the quality standards of the institutional market. This will include support for accessing institutional markets for wholegrain flour.

Component 2: FL-RS supply side development to increase availability and affordability of FL-RS

Outcome 2: Enhanced supply and affordability of FL-RS

84. The second component will complement the first by enhancing the supply and affordability of FL-RS. This will be achieved through two Outputs that address key gaps in the FL-RS value chain, introducing innovations that transform the market. The first Output will build the capacity of local MSMEs to provide access to FL-RS, either through improving local supply or offering services to farmers. This will be complemented by the second Output, which will implement innovative financial mechanisms that will catalyse investments in FL-RS through partnerships and conditional procurements¹⁸. Ultimately, when combined with the demand-side interventions under Component 1, the two Outputs under this component will unlock the local market, encouraging vendors of FL-RS equipment¹⁹ to invest in distribution channels, marketing, maintenance capacity and other needed infrastructure to make the equipment available and serviceable in a sustainable manner. This shift will lead to reduced prices and increased affordability, creating a self-sustaining system that will enable the phased out of programme support.

Output 2.1: Business development support provided to local MSMEs to facilitate the improved provision of FL-RS on local markets

85. Activities under this Output will develop the business case for investing in the FL-RS supply chain and support local MSMEs to develop business operations for the supply of FL-RS services. At the heart of this approach is the demonstration of solutions through pluralist extension services such as the VBA extension systems model and agro-dealers along with employing service models that enable shared access to larger, more expensive FL-RS equipment, such as harvesting machinery, mechanical multi-crop threshers and shellers (preferably solar-powered), moisture meters and communal storage structures. To ensure the sustainability of such models, the technical capacity of MSMEs will be enhanced, including for the development of sustainable business cases and improved business management practices. Activities under this Output will target at least 2,000 MSMEs.

Activity 2.1.1: Train new FL-RS service providers (MSMEs, cooperatives, incl. women- and youth-led initiatives) on the setup and operation of FL-RS service provision operations

86. The RE-GAIN programme will identify and train local MSMEs and cooperatives to develop FL-RS service provision operations in the target communities. This training will prioritise women- and youth-led initiatives to maximise the social benefits and build resilience among these vulnerable groups²⁰. The programme will adopt a rigorous process of scouting and in-depth assessment using standardized tools of the potential service providers and food processors complemented by intensive vendor support to make sure operations and maintenance skills for the equipment are transferred. The identification of suitable entities will be supported by government (decentralised and local) authorities as well as through digital platforms, social

¹⁸ Conditional procurements will act as a form of 'smart subsidy' through which AGRA will support the procurement of good in tandem with co-investments by the farmers or MSMEs.

¹⁹ African manufacturers might need additional technical skills to produce efficient and durable equipment at scale, while India- and China-based manufacturers might not see the market opportunity justifying their investment in the African market.

²⁰ This will include encouraging MSMEs to engage with informal youth groups for the provision of services on the ground, with the MSME playing a supervisory role over the youth groups, ensuring accountability and providing the youth group with an opportunity to build a track record of successful operations and governance.

media, industry players where potential service providers are connected or enrolled, and via the scoping exercises under Activity 1.2.1 (and will therefore be sequenced after Activity 1.2.1).

87. Once service providers are identified and enrolled on the programme, a rigorous assessment of their capacity and skills will be conducted, before organising intensive “boot camps” to train these entities on the FL-RS service provision. These training camps will cover several technical topics related to FL-RS service provision, including: i) service management and process optimization; ii) the safe, effective, efficient and sustainable operation of the equipment; iii) compliance with international and local quality standards for post-harvest services; and iv) the management of the fee-for-service model. In addition to the technical training on FL-RS service provision, entities will be supported on develop or strengthening their business delivery models into a sustainable entity. This will include training on basic business practices (such as financial literacy, cash flow management, record keeping, and risk management) and partnership development support, as well as training on value-add services such as certifications processes to enhance product value and market access. Guidance will also be provided on how to access to finance through innovative de-risking schemes under Output 2.2.
88. Vendors of the FL-RS equipment will be engaged in the implementation of this activity, incentivising their distribution channels to increase their outreach activities, including through VBAs and other last mile delivery channels. This model is not only sustainable in the long run but also creates an incentive for end users to scale up the technology in the very initial stages. Exchange visits will also be arranged to share experiences between service providers operating in nearby villages.

Activity 2.1.2: Provide business development support and market intelligence

89. To complement the development of MSME service providers, the RE-GAIN programme will support business development among manufacturers, importers and distributors of FL-RS to boost local markets. This will include providing market intelligence to increase marketing, distribution and after-sales services to develop the supply chain for FL-RS products.
 - Support for the youth groups to be able to run a “fee for service” business, including financial management, equipment management and marketing and business planning
 - Ecosystem support/market intelligence which includes mapping available FL-RS equipment, feasibility studies for FL-RS deployment, scoping of the potential demand for FL-RS services, mapping of financial services for FL-RS purchase, among others.
90. Through this training, RE-GAIN will showcase functional and validated components of FL-RS in specific country contexts and develop these as scenario-based business models that will be validated and used for expanding business opportunities. Entities will also be supported to develop partnerships with other actors in the supply chain, as well as to engage with innovative de-risking schemes piloted under Output 2.2. The business development support will focus on two main elements:

Activity 2.1.3: Capacity and market development for all supply-side market actors

91. In addition to the targeted support for FL-RS suppliers and service providers, RE-GAIN will provide general capacity development support to actors across the FL-RS supply chain to further develop the local supply market for FL-RS. The nature of this support will vary depending on the context of each target country and the actors involved but could include the convening of actors (including manufacturers, agro-dealers, service providers, MSMEs and financial institutions) and development of partnerships (this will be done by the CIU with support from the PIU), establishment of cooperatives to optimise the procurement and shared use of FL-RS among farmers (through other AGRA programming and other development partners’ work with smallholder farmers; including that of CNFA, Agriterra, ACDI VOCA, Technoserve, IFDC, SNV, and local NGO partners), market mapping and dissemination of market intelligence reports (using AGRA knowledge management and networks such as SAFIN and AMEA), and building network of local manufacturers and importers/agro-dealers (through support to private sector associations provided as part of AGRA and other partners’ other programming and interventions). Implementation of the supply-side market development support will include linkages with the agri-value chains market development initiatives under 1.2.1, sharing resources to optimise cost efficiency and co-develop both sides of the market.

Output 2.2: Financial mechanisms established to support smallholders and MSMEs to adopt FL-RS

92. One of the key transformative approaches in the RE-GAIN programme is the introduction of innovative financial models that will strategically use GCF grants as catalysts to develop the market for FL-RS. The approach taken in financial model design is focused on mitigating the current market dynamics and challenges faced by smallholder farmers in each of the programme's target countries, and to work with the additional activities in the programme to respond to some of the key challenges identified and the gaps in the current delivery of food-loss solutions. The goal of the financial model design is to specifically address these challenges and support the growth and development of the market, as a whole. In addition to the core financial mechanisms, the programme will support several other pathways to financing of FL-RS, including the inclusion of FL-RS in existing input packages provided by local financial institutions, and structuring prefinancing partnership arrangements that include FL-RS.

Activity 2.2.1. Facilitate the development and deployment of catalytic procurement models for FL-RS focussing on women and youth collectives, and enterprises as key beneficiaries.

93. Under this Activity, RE-GAIN will implement two financial models split into two distinct categories: interventions at a smallholder level; and interventions at a youth group/co-operative level. An overview of each of the two financial models is provided below, with further details – including descriptions of the key actors and eligibility criteria – provided in Annex 2: Feasibility Study. In all instances, the models draw on the outputs of the scoping exercises under Activities 1.1.5 and 1.2.1 to identify the potential actors and tailor the models to the local context. The models leverage partnerships between these actors to share risk and facilitate and incentivise the development of robust markets.
94. Interventions at a smallholder farmer level
95. Model 1 promotes the local supply of FL-RS interventions by using conditional procurements to effectively subsidise interventions at smallholder farmer level (referred to hereafter as 'smart-subsidies'). At its base, the smart-subsidy model enables agro-dealers to provide FL-RS to smallholder farmers at a discounted rate by using GCF funds to procure one item for every two items²¹ procured and sold by an agro-dealer. The subsidy is passed down to the smallholder farmers as a discount on the purchase price²².
96. The interventions are typically focused on smallholder farmers, with lower individual ticket sizes to the end-users — i.e. for FL-RS such as tarpaulins and plastic sheets, metal and plastic silos, and hermetic bags. The flow of funds will be facilitated through a trust/escrow account²³ at a local financial institution (FI) and released once proof of offtake by eligible smallholder farmers has been obtained. The instruction for release of the funds will be made by AGRA once the implementation unit is satisfied that the conditions have been met. On release, the funds will be transferred directly to the supplier. However, while funds will pass through an FI, it should be noted that no GCF funds will be transferred to a financial institution as the end beneficiary. The aim of the model is two-fold:
- to stimulate production and manufacturer capacity by placing pre-emptive orders of FL-RS, however, maintaining control over risk through a conditional release of funds to the manufacturer; and
 - to reduce the cost of interventions at smallholder farmer level, additional profitability will drive additional demand and facilitate knowledge sharing about the benefits of interventions.
97. An overview of Model 1 is presented in Figure 7, with more detailed descriptions of each step in the text that follows. Under Model 1, RE-GAIN will target ~540,000 households for adoption of FL-RS. Average ticket size is \$153 per household, of which approximately 30% on average will be subsidized through Model 1.

²¹ Note: the specific ratios will vary based on context of local markets in each country, with more developed markets receiving a smaller subsidy than less developed markets. A three-for-two model has been used as a base for demonstration purposes. The ratios for each target market will be finalised when the contract between suppliers FIs and AGRA are entered into. RE-GAIN staff will initiate negotiations on value for money in line with the state of the local market, which will be used to determine the ratios.

²² The maximum retail sales price and number of items per customer — as well as the eligibility criteria and geography where the sales at reduced cost can be made — will be included as part of the agreement between AGRA and the suppliers as a condition for participation in the scheme.

²³ The account will be held in AGRA's name with AGRA as the primary beneficiary. Depending on the agreements reached, the other parties (suppliers/financiers) will be included as co-signers as needed; however AGRA will maintain final say.

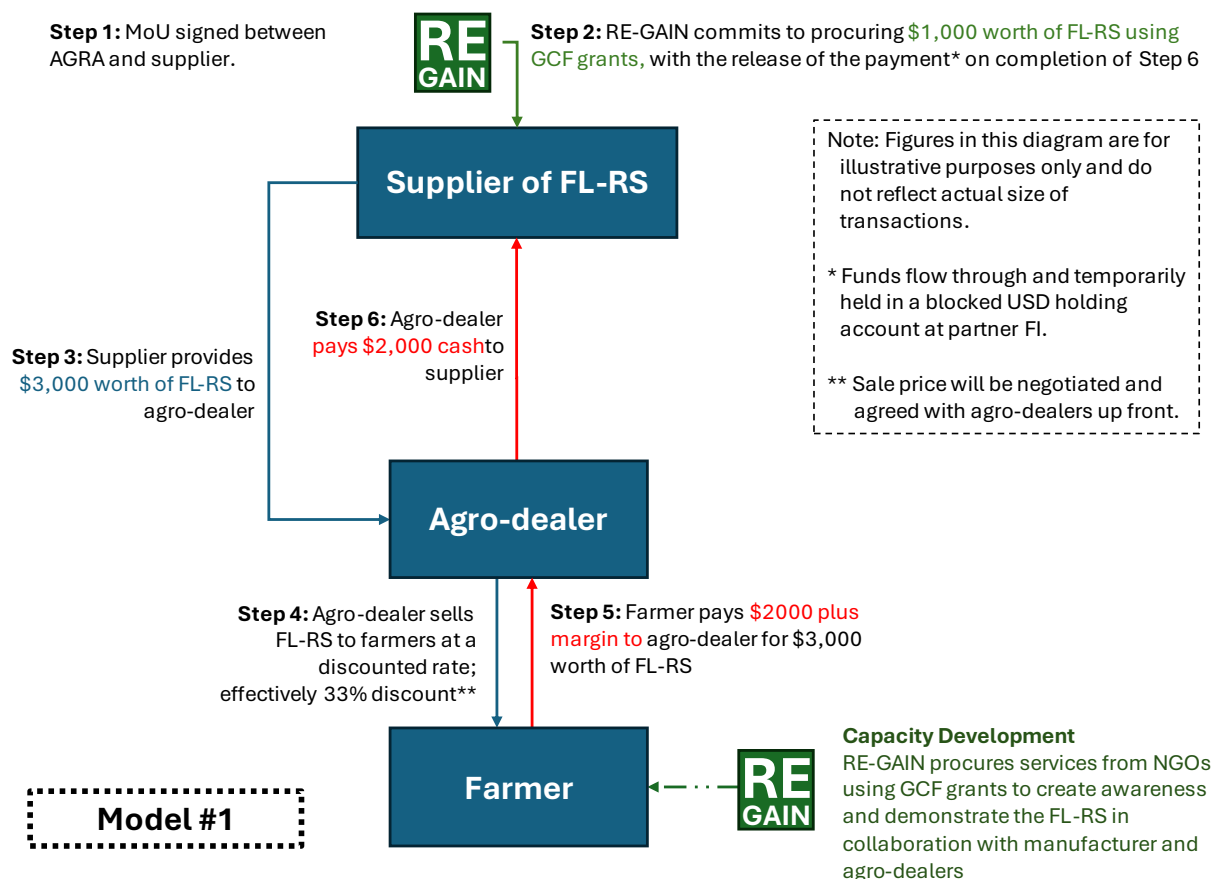


Figure 7. Schematic of Financial Model 1.

98. **Step 1:** RE-GAIN will facilitate the agreement and AGRA will enter into an agreement with a supplier. Each supplier will act through its agro-dealer network in the areas where eligible farmers are based. During this process, the details of the agreement will be negotiated between parties, including the level of smart subsidy provided and the conditions on final sale price that will be provided to the smallholder farmers. Through this process, AGRA will pass down eligibility criteria and the conditions of the agreement to the agro-dealers, ensuring that the benefits of the smart-subsidies are reaching the target groups.
99. **Step 2:** RE-GAIN places an order for FL-RS and deposits the value of the order into a holding account. Funds to be released on the completion of Steps 4-8.
100. **Step 3:** The supplier provides three units to the participating agro-dealers for every one unit procured by RE-GAIN. In some cases, the agro-dealers pay the supplier for the two non-subsidized units on delivery. In other cases, the suppliers give the units on credit (see Step 7).
101. **Step 4:** As per the agreement set out under Step 1, the agro-dealer offers the FL-RS to smallholder farmers at a discounted rate²⁴, transferring the full value of the smart-subsidy provided through GCF support to the farmers. Agro-dealers will take details of the buyers of the subsidized goods (there is a limit to how many bags per person can be bought to maintain the demonstration goal and prevent re-selling) and report on the sales made. This will help RE-GAIN to monitor the transfer of benefits to target groups and ensure impact is being achieved.
102. **Step 5:** Farmers buy the FL-RS at the discounted rate.²⁵
103. **Step 6:** The agro-dealer makes a payment to the manufacturer for two units for every one unit of the initial procurement from RE-GAIN (if not already paid on delivery; see Step 3). This payment — and the

²⁴ Note: Given that agro-dealers will need to apply a markup to the FL-RS to account for their income in the deal, the discount to the farmers might fall below 33% of the usual market price. However, agro-dealers bound by the nature of the agreement to transfer the full value of the subsidy to the farmers.

²⁵ If needed, the farmer may approach a micro-finance institution (MFI) to help secure a short-term bridging loan to purchase the FL-RS before or at harvest, when income might only be expected further down the line

submission of a corresponding report detailing the conclusion of the agreement — will trigger the release of the smart-subsidy payment through RE-GAIN to the supplier.

104. Should the suppliers, agro-dealers or farmers require additional finance for their role in the system, they will be able to approach local financial institutions that already exist in all the target countries. For example, if the supplier needs additional working capital or capital investment to satisfy the additional FL-RS demand²⁶, they can enter into a loan agreement with an FI to cover the additional liquidity needs for providing FL-RS. While AGRA can offer advice to suppliers or agro-dealers on this matter, the agreements themselves will be outside of the scope of the RE-GAIN Programme and will not include AGRA. The order placed by RE-GAIN will help de-risk the loan provision to the supplier, but no RE-GAIN Programme funds will be used to lend to the supplier or to make payments to an FI.
105. This model creates a win-win scenario for all actors, with the manufacturer ultimately receiving the full value of the FL-RS provided, the agro-dealer making income from their markup, and the farmers receiving FL-RS at a highly discounted rate. The market created will enable manufacturers to increase production with reduced risk, eventually bringing down the cost of FL-RS in the local market, enabling the smart-subsidies to be phased out.
106. **Interventions at a cooperative or SME level**
107. **Model 2** will use GCF grants to provide a unique approach to unlocking higher value items and food loss reducing equipment to farmer collectives or small to medium sized enterprises. The term ‘youth groups’ is used in this model description to highlight the priority given to this vulnerable demographic²⁷; however, the model will also apply to other cooperatives, including women’s groups and local MSMEs. The aim of Model 2 is to create multi-stakeholder agreements with suppliers, youth groups, and financial institutions, with the goal of:
 - Enhancing creditworthiness – through repurchase assurances from the suppliers that lower the loss given default.
 - Reducing borrowing costs – Through a combination of the lowered credit risk (as per above) and subsidies on the purchase price. The structure will ensure higher value FL-RS become more affordable and thus accessible to youth groups who provide services to smallholder farmers.
108. At the heart of Model 2 is the engagement of local youth groups that will be supported to act as service providers for FL-RS that require more expensive equipment that can service multiple farmers — such as mechanical multi-crop threshers and shellers (preferably solar-powered), moisture meters and communal storage structures. The establishment of the service operations will be supported through the business development under Output 2.1, ensuring that the youth groups have the necessary foundation to provide a reliable service²⁸. However, it is recognised that despite the proposed capacity development activities, some informal groups may still not fully meet the eligibility criteria for Model 2. In such instances, RE-GAIN will work with formalised MSMEs to partner with local youth groups; with the established MSMEs taking the lead in entering into the multistakeholder agreement (as per Step 1 below) and taking accountability for the management of the agreement. The MSMEs will recruit and work with informal youth groups from the community to operate the equipment and in the process develop their capacities²⁹. Through this MSME-supervised approach, the informal youth groups will have the opportunity to build a track record, demonstrating their governance structures and creditworthiness for future opportunities. This approach leverages several key concepts to unlock the targeted benefits:
 - Collectivism provides benefits to smallholder farmers in economies of scale through cost sharing and increased bargaining power with off-takers. These should promote further profitability and therefore additional demand for FL-RS.
 - Post-harvest handling will increase the quality and quantity of agricultural produce, allowing smallholder farmers to capture more value, hence increased incomes.

²⁶ For example, to finance working capital, additional processing lines or imports that are required to satisfy the additional demand for FL-RS

²⁷ Preferentially targeting youth groups will help build resilience among this vulnerable group of the population, as well as unlock the energy and innovation that youth can bring to the process.

²⁸ Including governance and leadership capacity, as well as financial literacy and business knowledge

²⁹ The income generated by the equipment is shared between the MSME and the youth group, whereby the MSME bears the credit risk and the informal group the operational risk.

- The inclusion of financiers will further unlock access to finance in a typically underserved market. The structure aims to reduce credit risk by providing a partial subsidy that will lead to lower borrowing costs (thanks to smaller loan size and interest payments).

109. While the concessionality under this model is first and foremost targeted at the youth groups themselves — as livelihood development support to these vulnerable members of the communities — when combined with the business development support, the RE-GAIN programme will enable youth groups to structure their fees to account for the actual (discounted) cost of the equipment and provide a service at a fair rate, which will in turn indirectly pass the benefits of the concessions down to the farmers using the service.
110. An overview of Model 2 is presented in Figure 8, with more detailed descriptions of each step in the text that follows. While RE-GAIN will facilitate the establishment of the full process, the active involvement of the RE-GAIN programme beyond Step 4 will focus on monitoring the delivery of services and providing continued support to youth groups. The ownership of Steps 5-9 lies directly with the three partners (youth groups, suppliers and FIs) who will enter into a separate loan agreement to which AGRA will not be a party. This model will target the 2,000 service providers supported under Output 2.1. Average ticket size is \$153 per household, of which approximately 30% on average will be subsidized through Model 2. The total market value of each ticket item is comprised of the collective inputs of between 100 and 200 people per solution.

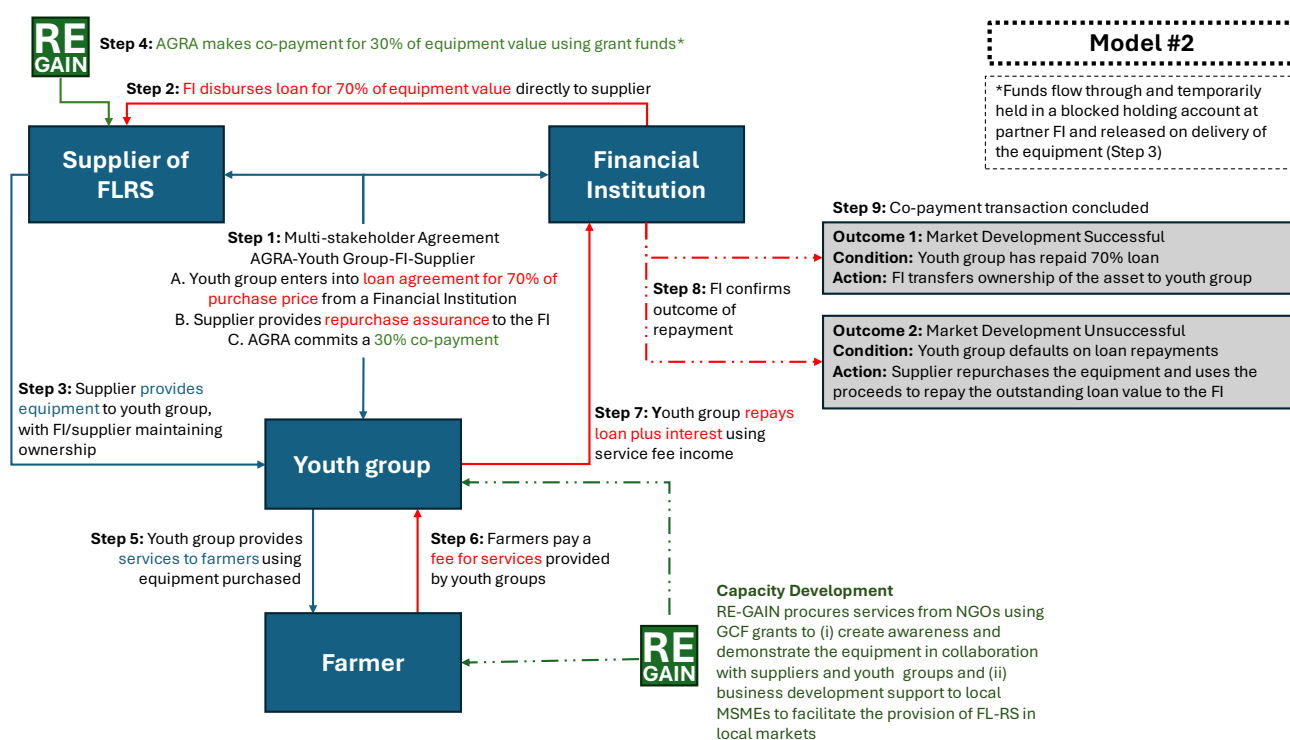


Figure 8. Schematic of Financial Model 2.

111. **Step 1:** Drawing on the scoping exercises under Activities 1.1.5 and 1.2.1, the RE-GAIN programme will match youth groups with suppliers and FIs, facilitating the signing of a multi-stakeholder agreement. Under this agreement, AGRA will commit to an upfront co-payment of 30%³⁰ of the purchase price for the targeted equipment, on condition that the youth group commits to the remaining 70%. The youth group payment will in turn be facilitated through a loan from the partner FI, with the supplier providing a repurchase assurance, thereby sharing the risk between the supplier and the FI. RE-GAIN will facilitate the negotiations, ensuring that all aspects of the agreement meet the eligibility criteria.
112. **Step 2:** Once the agreement is in place, the FI will deposit the 70% down-payment directly into the supplier's account on behalf of the youth group, triggering Step 3. At this stage, the remaining 30% co-

³⁰ Note: 30% is used as an average co-payment value for demonstration purposes; however, the specific co-payment will vary depending on the nature of the goods purchased and the state of development of the local market.

payment is deposited into a blocked USD holding account, pending delivery of the equipment to trigger its release (see Step 4).

113. **Step 3:** On receipt of the 70% payment from the FI, the supplier delivers the equipment to the youth group and reports on delivery to AGRA's RE-GAIN PIU.
114. **Step 4:** On receipt of the delivery report from the supplier, AGRA will release the 30% co-payment from the holding account to the supplier, completing the initial purchase agreement. At this stage, the youth group will have control over the use of the equipment, but ownership of the assets remains with the supplier or the FI (as agreed during Step 1).
115. **Step 5:** The youth group, now in possession of the equipment, provides an FL-RS service to local farmers. Support under Output 2.1 will ensure that the youth groups are adequately capacitated to operate the service enterprise.
116. **Step 6:** Smallholder farmers pay the youth group for the FL-RS service, with the youth group collecting income from multiple farmers; thereby distributing the cost of the equipment across multiple beneficiaries.
117. **Step 7:** The youth groups use the income from the services to make repayments to the FI on the loan, covering the cost of the loan and the agreed interest. The upfront co-payment through RE-GAIN reduces the repayment burden on youth groups as compared to a scenario where a 100% loan would have been required, thereby decreasing the loan loss given default.
118. **Step 8:** At the end of the agreed loan period, the FI concludes the transaction and reports on the outcome of the repayment, triggering Step 9.
119. **Step 9:** The conclusion of the transaction will lead to one of two possible outcomes, as outlined below.
 - a. Outcome 1: Market development was successful, indicated by the youth group operating an FL-RS service, enabling the full repayment of the loan. Under this scenario, the ownership of the asset will be formally transferred to the youth group, which will be able to continue to offer the service beyond the initial agreement, without the costs of servicing the loan.
 - b. Outcome 2: Market development was unsuccessful, indicated by the failure of the youth group to make the required repayments on the loan. Under this scenario, the supplier's repurchase assurance is triggered, through which the supplier buys back the asset (accounting for depreciation). The value of the repurchase will first go towards the repayment of any outstanding loan amount, and any associated transaction fees. Should the repurchase value exceed the outstanding loan amount, with any remaining value after transaction fees will be transferred to back to the youth group to compensate for any payments made before default.
120. Alternative variations to these models may be introduced depending on the local context and the nature of the FL-RS. For example, the agreements may be adjusted to incorporate a rent-to-own mechanism³¹ directly with the supplier, through which incentives would be created for employment of local youth to operate the machines, with these youth taking ownership of the machine once the original cost of the machines is paid back from income generated by the machine. In all cases, GCF grants will be used to make a co-payment on the equipment on behalf of the beneficiary (youth group or MSME), thereby reducing the financial burden of the transaction and de-risking the transaction for the suppliers or FIs involved in the agreement.

Activity 2.2.2. Support inclusion of FL-RS in climate-resilient input packages provided by local financial institutions
121. While access to credit for farmers is limited across all the target countries, there are increasing efforts to develop tailored agricultural input packages from various financiers and government input subsidy

³¹ The rent-to-own model is a variation of Model 2, where the supplier can provide the equipment to the youth group without the need to include an FI. Under this variation, the youth group will make monthly rental payments to the supplier, but the ownership will remain with the supplier until such time that the agreed purchase value has been reached. In such instances, the GCF contribution will be used as a downpayment to reduce the monthly rental fees, or to reduce the duration of the rent-to-buy agreement.

programs. Financing is provided mostly through rural financial institutions, cooperatives, digital and alternative lenders, that assist farmers in accessing agricultural inputs. However, these packages generally focus on initial production, often excluding FL-RS from the list of supported inputs. The RE-GAIN programme will, therefore, provide technical support through CIU and AGRA staff time to facilitate embedding the cost of FL-RS (e.g. hermetic technology) within the existing financing arrangements for input subsidies (in countries where such subsidy schemes exist like Ethiopia, Malawi and Zambia) to allow farmers to try out the technology and ask for inclusion in their input loan package in the following years. Under this activity, GCF funds will not be used to directly fund subsidies, but rather to promote the budling of FL-RS into existing subsidy programmes.

Activity 2.2.3. Structure prefinancing partnership arrangements that include FL-RS

122. To further unlock financing for accessing FL-RS, the RE-GAIN programme will provide technical assistance through its PIU, CIU and AGRA staff to key value chain players — for example off-takers, input suppliers and farmers — and financial institutions to support the structuring of prefinancing partnerships that will derisk the provision of finance. Assistance will include convening key actors (through workshops, meetings, or other collaborative platforms), and training/capacity development that will help structure financial products based on risk sharing and risk reducing arrangements. Some of these products are widely promoted by AGRA, after they have been successfully implemented in several of its focus countries, including Burkina Faso, Tanzania and Malawi.

Component 3: Enabling environment to ensure sustainability of the FL-RS market

Outcome 3: Strengthened enabling environment for the uptake of FL-RS

123. The impacts of climate change on food loss in Africa is aggravated by the lack of an inadequate policy environment to support the development of sustainable markets for FL-RS. These policy-level constraints include the limited: i) availability of pan-African platforms on climate change that would facilitate action to minimize post-harvest losses and that are supported by evidence and responding to African needs and priorities; ii) access to climate change adaptation incentives, particularly finance to catalyse action and support SME investments; iii) weak regulatory and enforcement mechanism that would have reduced food loss and iv) biosystems-specific climate policies and implementation strategies supporting the entrepreneurial skills of youth in minimizing food losses. In response to the challenges, Component 3 of the RE-GAIN programme will support the development of an enabling environment for the adoption of FL-RS at scale, working within the existing policy environment where needed, and supporting change where possible.

Output 3.1: Enhanced capacity of national institutions to enable investments in FL-RS

124. To increase the sustainability and scalability of the positive impact created by the improved demand and supply of FL-RS, and improved access to financing and established market linkages, RE-GAIN programme will support the revision and operationalization of policies in support of enabling FL-RS investments, including tax exemptions/incentives, certification and standards for FL-RS quality, and promote successful FL-RS business models for scaling-up and replication.
125. Active involvement and support from government organizations, both central and local, will be crucial. The programme will align with other projects and programmes to leverage synergies, utilize existing laws and policies on FL reduction, SME promotion, and smallholder support, and ensure effective and efficient programme management. This includes rigorous monitoring and incorporating lessons learned.
126. Effective stakeholder engagement is essential and will involve raising awareness, providing programme information, and ensuring inclusivity for women, youth, minority groups, and all value chain actors. A grievance mechanism will also be put in place, with programme funds being used for awareness creation and mobilization through stakeholder engagement meetings³². Additionally, ensuring the availability of quality FL-RS and access to finance is vital to support long-term continuation.

³² The cost of the grievance mechanism itself will be covered by AGRA using AE Fees.

Activity 3.1.1: Support the revision of policies in support of enabling FL-RS investments, including tax exemptions, reduced import duties, certification and standards for FL-RS quality

127. Under this Activity, AGRA will collaborate with government entities to identify critical policy gaps related to food loss, and revise and update key policies which are not responding to the current and future food losses and climate challenge scenario. This will include engaging relevant government ministries and agencies to review existing regulations and policies and proposing amendments to facilitate the development of local markets and support to farmers for the uptake of FL-RS. This will include introducing/strengthening national policies that provide financial incentives such as tax exemptions, subsidies, or reduced interest rates for FL-RS investments, making it more attractive for private and public sector investments³³. Evidence generated and lessons learned in FL-RS from target communities and counties will be used to develop capacity, influence policy and promote best practices at scale.
128. AGRA will also actively participate in policy advocacy initiatives, working with stakeholders to ensure that proposed changes support the long-term success of FL-RS initiatives as part of the broader food systems of the respective countries. In addition to support for policy revision, AGRA will support local entities in implementing existing policies and regulations that have potential to support the uptake of FL-RS but are currently not fully enforced or supported. These actions will result in a supportive policy environment that enables the successful implementation of the RE-GAIN programme and widespread adoption of FL-RS solutions.

Activity 3.1.2: Support institutional capacity building of key local entities to support the scaling of successful FL-RS models.

129. Through this activity, AGRA will provide technical assistance to local and national government entities engaged in crop value chains (directly or indirectly) to build their institutional capacity to support the scaling of successful FL-RS models and policies in their countries. This will include actively involving central and local government through the establishment of formal partnerships and facilitating regular meetings and consultations to ensure alignment of the RE-GAIN programme with national and regional development priorities. These partnerships will be complemented by technical assistance, including guidelines, targeting tools and technology delivery mechanisms, to develop the analytical capacity of partner institutions on the topics of climate change and the contribution of FL-RS to GHG reduction and food security — enabling these metrics to be tracked and reported on in relation to the NDCs — as well as for the scenario-based prediction of mycotoxins during post-harvest period under changing climate.
130. RE-GAIN will also work with governments and other development partners to promote private sector investment in post-harvest infrastructure, including storage facilities, processing units and cold chains, as well as supporting regulatory bodies to develop and implement quality standards and certification processes for FL-RS products and services, ensuring they meet required levels of reliability and efficiency. Support will also be provided to enhance the capacities of extension services, and climate advisory services to monitor food loss in relation to climate conditions and the impact of FL-RS in reducing loss, as well as identifying and reporting on opportunities for upscaling and replication.
131. Finally, the programme will leverage on other investments to create synergies with other players in building institutional capacity of governments and the private sector in FL-RS, which may include enhancing coordination among country actors and institutions dealing with post-harvest loss under changing climates.

Activity 3.1.3: Develop a business case for investments in FL-RS manufacturing, supply, service provision and financing to promote successful FL-RS business models for scaling-up and replication

132. A key step in unlocking investments and creating sustainable markets for FL-RS in Africa revolves around creating solid business cases that account for the needs of all actors – including manufacturers, suppliers, financiers, agro-dealers, service providers and farmers. RE-GAIN will, therefore, research and compile case studies of successful FL-RS initiatives implemented through the programme, as well as other best practices observed across the region and develop models and approaches for scaling and influence. This research will form the foundation of the business case for FL-RS, detailing best practice and benefits of

³³ Note: RE-GAIN will advocate at Government level to create a more enabling environment for FL-RS but will not directly cover the financing on tax incentives or subsidies.

various FL-RS related activities, including equipment selection, business model, profitability analysis, and financing models. The business case will be packaged into various communication materials, including reports, videos, and presentations, to share these models with potential investors, policy-makers, and other stakeholders, encouraging scaling-up and replication.

Activity 3.1.4: Implement a comprehensive monitoring, evaluation and learning system for FL-RS

133. The programme will develop a robust monitoring, evaluation and learning (MEL) framework that includes clear performance indicators, data collection methods, and regular reporting schedules. This system will build on the foundation of existing MEL frameworks within the AGRA system to track the progress of programme activities, assess the impact of interventions, and identify areas for improvement. The MEL framework will also establish mechanisms for continuous learning by regularly reviewing M&E data, conducting periodic evaluations, and gathering feedback from stakeholders. These insights will be used to make informed adjustments to the programme, ensuring that it remains relevant and effective.
134. Through the MEL framework, the RE-GAIN Programme will foster knowledge sharing, improve performance through a continuous feedback loop and collaboration among stakeholders. This will be supported through the activities under Components 1 and 2 to encourage the formation of farmer groups, cooperatives and networks where farmers, MSMEs and private sector actors can exchange experiences, share best practices and collectively address challenges related to post-harvest management. The knowledge generated will be also shared regionally through AGRA's networks to promote the scaling of the successful models across Africa and beyond. In the spirit of south-south cooperation, the programme will also coordinate with other projects to create synergies, identifying areas of overlap and collaboration. This will include developing joint action plans, sharing resources, and coordinating activities to maximize impact and avoid duplication of efforts.

B.4. Implementation arrangements

PSAA Applicant

135. AGRA — an African-led institution with a vision of contributing to a food system-inspired inclusive agricultural transformation across Africa, of reducing hunger, improving nutrition, and adapting to climate change, and has been scaling agricultural innovations that help smallholder farmers towards increased incomes, better livelihoods, and improved food security will serve as the PSAA Applicant for this project.
136. AGRA was founded in 2006, bringing 18 years of experience in a varied sectors related to food security, including seeds systems, sustainable farming, policy and state capability and inclusive markets and trade. Efforts across these main thematic areas are built on the foundation of several key cross-cutting themes, particularly climate change, nutrition and gender responsiveness.
137. The organisation operates in 12 countries, employing over 300 staff, and managing an average annual operating budget of USD 90 million. AGRA also hosts the Partnership for Inclusive Agriculture Transformation in Africa (PIATA); a partnership between AGRA and five other major donors (BMGF, Rockefeller, BMW, MCF, FCDO and USAID) with AGRA as the implementing partner. PIATA is focused on working with public and private partners to support 12 countries in Africa to create and sustain Inclusive Agriculture transformation with a budget of over USD 500M in the next five years.
138. As a continental institution in agriculture and food systems, AGRA is well placed to undertake the planned activities through its widespread infrastructure and partnerships on the continent, having strong AGRA country teams in the target countries. Using AGRA's continental forums, the Africa Food Systems Forum (AFSF) and Agribusiness Deal Room, where AGRA brings on board connections with governments, partners and other stakeholders to enhance ownership and coordinate key actors, knowledge and resources, which are fundamental to take food loss innovation to scale.
139. AGRA is known as a trusted broker by the governments it works with which will enable it to ensure the programme is aligned with national climate change policies, strategies, plans and instruments. Using AGRA's ongoing projects across the continent as leverage, AGRA will drive a paradigm shift for building the resilience of smallholder farmers and agricultural communities in the seven focus countries and across Africa. The lessons learnt, especially on financing food loss reduction and its relationship with climate change, will have the potential of generating new knowledge on ways of engaging MSMEs and smallholder

farmers in reducing their harvest and post-harvest losses and thus increasing their productivity and resilience to climate shocks.

140. The AGRA HQ senior leadership and technical leads will be responsible for the overall supervision and coordination of the project, including ensuring: i) funds are effectively managed to deliver results and achieve objectives; ii) the quality of project monitoring; and iii) liaison with the GCF. AGRA will also leverage expertise from its wider technical leadership and support by AGRA's Heads of Markets and Trade, Inclusive Finance, Sustainable Farming, Private-sector Partnerships, Strategy, Policy and State Capability, Monitoring and Evaluation and Knowledge Management. The AGRA HQ team will be the primarily liaison with the GCF.

Executing Entity (EE)

141. The project will be executed directly by AGRA through Programme Implementation Unit (PIU)³⁴. Through this unit, AGRA will provide key resources, including Finance, Grant Management and Procurement Officers who will provide financial and administrative management, overseeing financial, contractual, procurement and logistics aspects for the project from the Nairobi Headquarters. The unit will oversee planning and quality assurance; supervise programme monitoring, evaluation and reporting; ensure timely realization of all programme deliverables; provide leadership and technical support to implementing partners; and ensure smooth communication flow across all programme partners. This executing role will be fulfilled both through the Nairobi-based headquarters, and AGRA's country offices, and will report to the AGRA senior leadership.
142. The EE is responsible for:
- Execution of the project,
 - Procurement of services specifically (major procurement and contracting)
 - Facilitating partnerships,
 - Managing contracts, monitoring results,
 - Annual reporting by country offices to the PIU
143. AGRA deploys a diverse set of delivery models to deliver its country and institutional strategy. It offers services through its expert staff, placed at headquarters in Nairobi; at the East, Southern and West Africa regional offices; as well as at country offices. AGRA staff work with downstream partners and local organizations to implement specific components of a contracted programme area with the aim to improve local organizations' capacity, build institutional capacity and ensure long term ownership and sustainability of its interventions. AGRA provides Technical Assistance (TA) in the form of short- to medium-term expertise support (through consultants where needed) embedded within or seconded to mandated national, regional and continental institutions (e.g., government ministries, regional economic communities) to drive desired change, and in some instances consultants are hired to support specific assignments that require skilled expertise. AGRA is a convener (brings stakeholders together around a change agenda, e.g., the Africa Food Systems Summit) facilitating connections and interactions between different actors and stakeholders within the agriculture and food systems sector. AGRA utilizes advocacy and communication as key tools for change. The specific delivery models will be determined at the implementation stage and will depend on each country context.

Responsible Units

144. The EE team at the Nairobi HQ will be supported by AGRA country offices in each of the seven target countries who will serve as responsible units. These units will support on-the-ground coordination and implementation, as well as being mandated for specific outputs/activities.

Programme Governance

Programme Advisory Group:

³⁴ The PIU will fall under the same legal entity as the PSAA Applicant

145. AGRA will establish a Programme Advisory Group (PAG) made up of senior representatives from AGRA's Integrated Programme Management (IPM) unit³⁵ that will serve as the starting point to guide innovation, impact scale and adaptive thought leadership to shape the partnership at the continental level. AGRA envisions this Advisory Group will meet on quarterly as part of the IPM meetings.

Programme Implementation Unit

146. A central Programme Implementation Unit (PIU) will be established at AGRA's Nairobi headquarters to oversee implementation of the entire programme across all seven countries. This unit will report to the PAG and be comprised of two sub-groups; a Programme Management Unit (PMU) and a Technical Expert Group (TEG), as described below.

- *Programme Management Unit*
147. The Programme will establish a management unit that will be functional for the entire duration and be responsible for day-to-day implementation of the project. The PMU will offer overall management, implementation and general technical direction of the entire programme, ensuring an integrated vision among different components. The PMU will consist of five full time positions: i) PMU Lead; ii) Senior Finance Officer; iii) Procurement Officer; iv) Project Analyst; and v) M&E Officer. The PMU will be based in AGRA Nairobi Headquarters, with in-country support from responsible units in the country offices.
- *Technical Expert Group*
The TEG, also situated within the Nairobi Headquarters, will provide expertise to assist the PMU in the technical implementation of the RE-GAIN programme. The TEG will include several full-time positions, including: i) Program Officer — Gender, Youth and Inclusion; ii) Technical Advisor — Inclusive Finance and BDS; iii) Technical Advisor — Extension and Value Chain Development. These full-time roles will be supported by several part-time technical team members, including: i) Technical Advisor — Inclusive Markets and Finance; ii) Lead — Sustainable Farming, Distribution and Youth in Extension; iii) Technical Advisor — Livelihood Resilience and Climate Adaption; iv) Head: M&E; and v) Technical Advisor — Food Loss Reduction Analytics.

Country-level Implementation Units

148. The PIU will be assisted in project implementation within each target country by a country-level implementation unit (CIU) which will be established in each of the AGRA country offices³⁶ and will be comprised of country-office staff. The CIUs will be responsible for managing day-to-day operations in each country, reporting directly to the PIU, as well as providing regular reports to the relevant Project Steering Committee (see below). Specific responsibilities of the CIU for each project Output are outlined in the first row of Table 6 below.

Programme Steering Committee

149. At the country level, the programme will be implemented under the overall guidance of a Programme Steering Committee (PSC) co-chaired by a representative of the NDA, and AGRA country managers³⁷. The PSC will include representatives of other key government departments and agencies, the private sector and civil society organizations. These partners will likely include Ministries of Agriculture and their Departments for Land Resources Conservation, Crop Development, Agriculture Extension Services and Agriculture Planning Services. The role of the PSC will be to: i) provide overall guidance and direction to the project in country; ii) address project issues as raised by the advisory group; iii) review the project progress and provide direction and recommendations to ensure that the agreed deliverables are produced satisfactorily and within the approved project framework; iv) review and approve annual work plan and budget (AWPB) and provide necessary strategic guidance for its implementation; v) appraise the annual project implementation report, including the quality assessment rating report; vi) make recommendations for subsequent work plans to build on achievements and address any shortcomings; and vi) provide ad

³⁵ Vice presidents, relevant business line or programme directors/heads, Lead of PMU , Head of MEL

³⁶ Which fall under the same legal entity as the PSAA Applicant

³⁷ The final composition and chairing of the PSC will be finalized at inception stage in collaboration with the NDA in each country.

hoc direction and advice for exceptional situations or when requested by the GCF, strategic advisory group or PSC members.

150. Each national PSC will include representatives of private sector actors in addition to key government institutions. A list of potential private partners is presented in Appendix 9 of Annex 2. The selection of specific partners for each country will be led by AGRA and will be dependent on specific criteria as outlined in Annex 2. At country level there will annual forums for feedback and policy dialogues that will be organized by each county office. The lessons learned through the project monitoring, evaluation and learning systems in each participating country will be shared to all other participating countries through two approaches: i) Cross-country presentations at AGRA's internal Quarterly Performance Review Meeting, where all country directors and program officers participate; and ii) an annual planning and review session organized by the PMU in which all countries and partners participate to promote cross country learnings, exposure and innovation. In addition, at continental level, the AFSF will organization special sessions for cross country learning and feedback.
151. Each National PSC will convene in an interval of 3 months (quarterly) with a provision for additional extraordinary meetings when required and to be called by the chair and co-chair or if requested by members. The PSC will report to the NDA who oversees all GCF project in the individual countries.

Stakeholder Engagement

152. Across the different countries, AGRA will liaise with different governmental agencies during the implementation of the different outputs to ensure that the RE-GAIN programme is aligned with country-specific policies. A non-exhaustive list of these stakeholders is provided below and will be further updated through engagement with the NDA's selected representative in each country. Further information of the key players working across the FL-RS adoption in the key markets is available in Annex 2, with a full list of stakeholders already engaged during the design of the RE-GAIN programme in Annex 7's Appendices.
153. The table below also outlines the key roles and responsibilities that AGRA will have in each countries in the implementation across the different outputs.

Table 6. List of National Stakeholders for Engagement Across RE-GAIN's Outcomes

Output 1.1	Output 1.2	Output 2.1	Output 2.2	Output 3.1
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AGRA CO role across all counties		- Strengthen capacities of extension systems to deliver FL-RS at farm level (Policy and Institutional capacities) - Assess level of public sector integration in planning and implementation of food loss reduction programmes - Capacity building for climate advisory services and related agronomic practices - Institutional level engagement on capacity development for extension systems on creating awareness for FL-RS	- Scoping and assessment of available market stakeholder in FL-R technologies and support services - Capacity assessment these stakeholders to respond to market demands - Facilitate the creation of coordination systems or consortia platforms for evidence generation and knowledge sharing on value chain development and linkages for market opportunities	- Scoping of available MSMEs and other supporting institutions in supporting the deployment of FL-RS and technologies - Facilitate market linkages between market actors on both the demand, supply and after services sides of FL-RS - Technical support to market actors on farmer facing use and promotion of FL-RS - Training MSMEs on FL-RS - Formation of or strengthening of existing youth groups to FL-RS, market opportunities and available financing products - Capacity development of local actors of business development for FL-RS	- Market assessment of existing FIs to engagement on the financial model development of the FL-RS deployment - Technical assistance to FIs on the financial services models - TA on contracting and contract performance management - Record keeping and BDS capacities for FL-RS marking and support services - Development and promotion of financial products and creation of awareness on access to finance for FL-RS	- Capacity assessment of institutions on food loss reduction - Policy assessments and implementation support for food loss reduction - Policy and institution coordination
Burkina Faso	AGRA	Managed by HQ with CO logistic support	Managed by CO with close HQ Oversight	Managed by CO with close HQ Oversight	Managed by HQ with CO logistic support	Managed by HQ with CO support
	National Partners	The Ministry of Agriculture (MoA) of Burkina Faso	- MoA of Burkina Faso - Fédération des Professionnels Agricoles du Burkina (FEPAB) - Union Nationale des Producteurs Agricoles du Faso (UNPA-BF)	- MoA of Burkina Faso - UNPA-BF - FEPAB	- The Fonds de Développement Agricole (FDA) Banque - Agricole du Faso (BAF)	MoA of Burkina Faso
Ethiopia	AGRA	Managed by HQ with CO logistic support	Managed by CO with close HQ Oversight	Managed by CO with close HQ Oversight	Managed by HQ with CO logistic support	Managed by HQ with CO support
	National Partners	- MoA of Ethiopia - Agricultural Transformation Agency (ATA)	- MoA of Ethiopia - Ethiopian Farmers' Federation (EFF) - ATA	- MoA of Ethiopia - ATA - Federation of Ethiopian Farmers' Cooperative Unions (FEFCU)	- Commercial Bank of Ethiopia (CBE) - Development Bank of Ethiopia (DBE)	MoA of Ethiopia
Kenya	AGRA	Managed by HQ with CO logistic support	Managed by CO with close HQ Oversight	Managed by CO with close HQ Oversight	Managed by CO with close HQ Oversight	Managed by HQ with CO support
	National Partners	MoA of Kenya	- MoA of Kenya - Kenya National Farmers Federation (KENAFF)	- MoA of Kenya - Agriculture Sector Network (ASNET)	Agricultural Finance Corporation (AFC)	MoA of Kenya



Malawi	AGRA CO	Managed by HQ with CO logistic support	Managed by CO with close HQ Oversight	Managed by CO with close HQ Oversight	Managed by CO with close HQ Oversight	Managed by HQ with CO support
	National Partners	MoA of Malawi	MoA of Malawi	MoA of Malawi	- National Bank of Malawi - Malawi Agriculture and Industrial Investment Corporation (MAIIC)	MoA of Malawi
Tanzania	AGRA	Managed by HQ with CO logistic support	Managed by CO with close HQ Oversight	Managed by CO with close HQ Oversight	Managed by CO with close HQ Oversight	Managed by HQ with CO support
	National Partners	MoA of Tanzania	- MoA of Tanzania - Tanzania Farmers Association (TFA)	- MoA of Tanzania - Tanzania Federation of Cooperatives (TFC)	Tanzania Agricultural Development Bank (TADB)	MoA of Tanzania
Uganda	AGRA	Managed by HQ with CO logistic support	Managed by CO with close HQ Oversight	Managed by CO with close HQ Oversight	Managed by CO with close HQ Oversight	Managed by HQ with CO support
	National Partners	- Ministry of Agriculture, Animal Husbandry and Fisheries (MAAIF) - National Agricultural Research Organisation (NARO)	- MAAIF - NARO	- MAAIF - Uganda National Farmers Federation (UNFFE) - Young Farmers' Federation of Uganda (UNYFA)	- Agricultural Credit Facility (ACF) - Uganda Development Bank (UDB)	MAAIF
Zambia	AGRA	Managed by HQ with CO logistic support	Managed by CO with close HQ Oversight	Managed by CO with close HQ Oversight	Managed by HQ with CO logistic support	Managed by HQ with CO support
	National Partners	MoA of Zambia	- MoA of Zambia - National Association for Small Scale Farmers (NUZFAZ)	- MoA of Zambia - National Association for Small Scale Farmers (NUZFAZ) - Zambia National Farmers' Union	- Agricultural Finance Company (AFC), - Zambia Agricultural Development Bank (ZADB) - Sustainable Agriculture Financing Facility (SAFF)	MoA of Zambia

Flow of Funds

154. To enable transparent financial management, AGRA will establish a separate interest-bearing USD bank account for the RE-GAIN programme, which the PSAA Applicant will manage with oversight from the AGRA senior leadership and technical leads. Funds disbursed from GCF to the AE will be deposited in this account, where they will be monitored and reported on by the Finance Officer. AGRA co-finance that is not used internally for staffing costs, office operating costs and travel will flow to service providers under professional services contracts.

155. The Procurement officer will be responsible for overseeing all procurement under the programme. Procurement of small ticket items (under USD10,000) will be managed directly by the AGRA country offices, with funds disbursed from the central account to the country offices for pre-determined items through country-specific ledgers within the separate bank account open for the RE-GAIN programme. Any larger procurement items (greater than USD10,000) will be procured directly by the PMU procurement office following approval by AGRA HQ senior management. See Figure 15 for an overview of the overarching programme implementation arrangements, flow of funds, and reporting lines. A focused example of the implementation arrangements as they will apply to each individual country is presented in Figure 10.

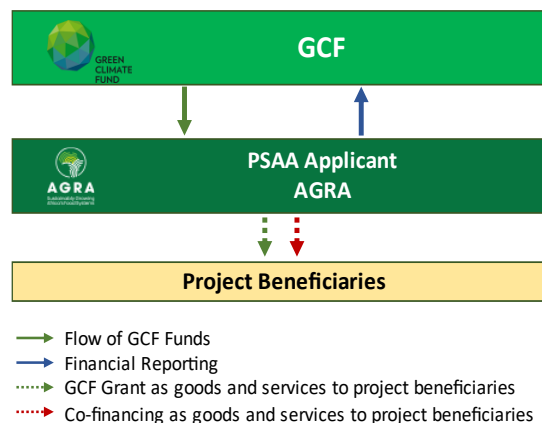


Figure 9. RE-GAIN Programme flow of funds.

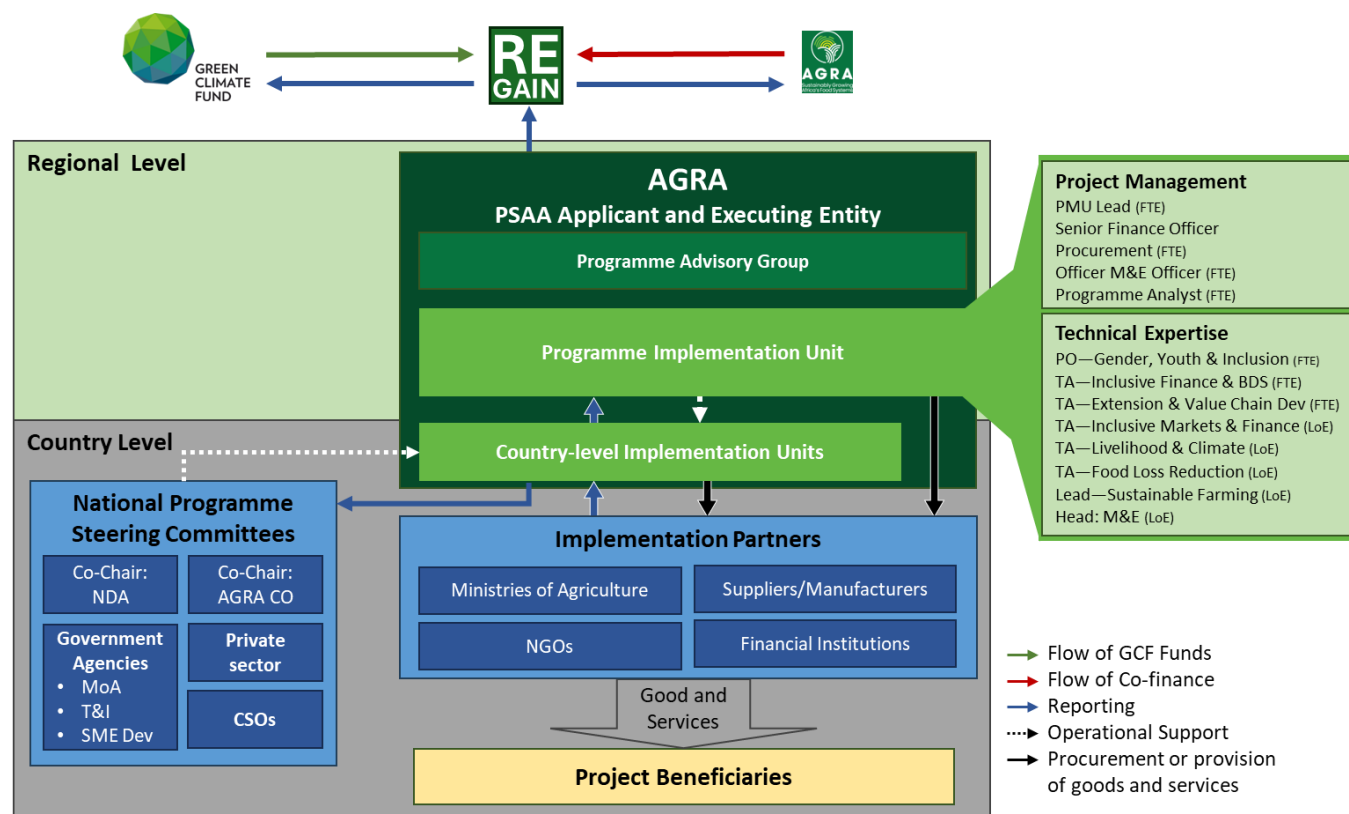


Figure 10. Overarching programme implementation arrangements. This diagram depicts the hierarchy of roles and responsibilities at the country level. The same structure will apply to each country individually.

B.5. Justification for GCF funding request

156. Africa's food insecurity challenge has been exacerbated by climate change. Sub-Saharan Africa stands at a crossroads with an unprecedented opportunity for agricultural transformation, driven by the demands of a rapidly growing population of 1.2 billion and the pressures of a changing climate³⁸. The continent faces significant development challenges including food insecurity, resource degradation, poverty, gender inequality, and social exclusion. The vicious cycle of poverty and environmental degradation in Africa is

³⁸ World Bank, 2023

evident in low crop productivity, deforestation, land degradation, conflict, migration, and vulnerability to climate shocks, which perpetuate persistent food insecurity and poverty. These challenges are being exacerbated by climate change, which is impacting the full spectrum of food production system. Most relevant of these to RE-GAIN is the expected increase in post-harvest losses in African food systems resulting from increases in pests and diseases, unpredictable rainfall pattern, increased humidity, and changes in precipitation patterns — all of which will affect the harvest and post-harvest food handling. The effects of climate change are expected to be particularly severe in Africa, where the capacity to adapt and respond to a changing climate is weak. Climate finance support from the global community is therefore essential to shift development pathways in the post-harvest food systems in the seven targeted African countries.

157. Low productivity, aggravated by harvest and postharvest loss, has been the major driver of climate vulnerability as farming is expanding to fragile ecosystems, wetlands and mountains, causing deforestation, land degradation and drying of water resources, exposing farmers to extreme events.
158. In response to these climate change threats on food security, the RE-GAIN programme aims to enhance the climate resilience and adaptive capacity of smallholders by promoting the widespread adoption of FL-RS in seven African countries. AGRA, as the lead agency, is requesting GCF grant funds to support the roll out of a suite of physical and management interventions alongside capacity building and enhanced financial and market access. The rationale for GCF concessional support – beyond the basic climate change attribution described above – is presented below.

Barriers to public or private finance alternatives

159. Experiences to date show that countries rarely invest in climate change adaptation, less so in reducing food loss. This is largely due to the constrained fiscal space in the public sector, which limits opportunity to invest in catalytic programme such as RE-GAIN using public funds. Constraints on public finance are further compounded by the already high levels of external debt, which prevents governments access to loan finance without putting considerable strain on the national economy. Given these constraints and the resulting high levels of external dependency, the target countries are reliant on external funding and support for climate change adaptation initiatives. However, international aid and climate finance flows may be insufficient or unpredictable, making it difficult for these countries to plan and implement long-term adaptation and mitigation strategies.
160. Looking to the private sector as an alternative, it is highly challenging to secure finance from conventional sources in the market for such resilience-building interventions unless market demand is created and appropriate market linkages are put in place. The return on investment for such programmes is often seen as unattractive for private sector investors, and the element of innovation that characterizes the necessary climate adaptation initiatives is viewed by the market as untested, with the absence of a track record of financial soundness translating to a high estimation of risk among potential investors. This challenge is compounded by the fact that resource-poor smallholder farmers may not be willing to invest in the FL-RS without a clear demonstration of the effectiveness of the solutions – particularly given that some of the technologies are unaffordable and farmers need a quasi-guarantee about the viability of these solutions before investing.
161. Based on the stakeholder mapping AGRA conducted during the country strategy development, existing interventions and support focus mainly on last mile delivery, with few actors willing or able to invest in unlocking the systemic challenges that would facilitate agriculture and food systems transformation. AGRA sees this as a gap and a potential opportunity for creating significant impact if interventions incentivize the markets inclusively, the right level of awareness and adoption is created and access to finance is de-risked.
162. Building the necessary capacity to develop and implement effective adaptation plans and investments requires substantial investments in education, training, and institutional strengthening, which, although catalytic, are generally considered public goods with limited direct return. International aid is, therefore, necessary to initiate a paradigm shift in the sector, supporting Africa-led solutions to tackle national priorities and investment plans.

Consideration of alternative funding models.

163. In the process of designing the RE-GAIN programme, numerous other alternative finance models were considered. While each option has some merit, each faces considerable barriers that prevent these models from working on their own. These alternative models are summarised below and have been factored into the design of innovative finance models that the proposed programme will initiate through Component 2.

- Farmers pay cash for the hermetic technologies and for using the FL-RS services if they have savings and or alternative sources of income. This is often not the case in most contexts and countries.
- Farmers can get the FL-RS on credit from the supplier of the technology (agro-dealers) and/or service providers if these MSMEs have the liquidity to give credit to their customers. This is often not the case in most countries and contexts. A mix of finance from vendors and suppliers, own contribution from farmers and additional funding from GCF can solve for the access to finance.
- Service providers offering FL-RS could get equipment financing through wealthy individuals, family and friends or through financial institutions or equipment vendors. However, there is inherent risk, both for the lender and borrowers, with several barriers currently preventing access to finance. Innovative finance models are needed to overcome this barrier.
- While credit options are possible, borrowers would need to offset the risk for lenders through co-contributions as well as the provision of some form of guarantee or collateral.
- This option is further hindered by the high interest rates in some countries (particularly those with high inflation) and transaction costs for accessing financing are high in rural contexts (documentation, transport).
- There are risks for lenders (first time borrowers and their lack of collateral and transaction history) as the financing of FL-RS in food crops have been minimal till date.
- The costs for lenders (loan origination, assessment and monitoring) are higher than for other sectors and loan products
- There is also risk to potential borrowers (market risk, profitability risk), as the business case for many of the FL-RS have not yet been proven in rural contexts for staple crops produced by small scale producers.
- Extension services (for example the provision of information on weather and harvesting practices) could be paid for with Government budget. However, most Governments are constrained in their budgets for extension services (they are over-indebted and have many competing priorities), hence the need for donor funds to complement the government budget.
- The proposed awareness activities could be financed by the suppliers of the FL-RS; however their reach is limited, and they might not be able to pay 100% of the investment costs for creating awareness in rural areas as the profit margins do not make up for that investment. Private sector does invest in the activities that give the highest return on investment and that might not be on products sold in rural areas to low-income small-scale farmers.

Rationale for concessional funding

164. AGRA is requesting concessional finance in the form of grants for the RE-GAIN programme. The need for full concessional finance is based on the nature of the investments, which require targeted capacity development support combined with strategic conditional procurements to catalyse a paradigm shift in the post-harvest value chain using innovative FL-RS in the seven target countries. The specific details on the nature of intervention, their suitability for concessional finance, and how the programme will transfer the benefits of concessional finance directly to farmers are described in more detail below.

- De-risk first time users to invest in new technology that remains unproven in their context. The “pay for two, get one free” type of investment will make sure the concessional funding end up in farmers pockets.
- Reduce the cost of distributing FL-RS to rural areas where demand is not yet confirmed (the concessional funding will de-risk agro-dealer and distributors from remaining with unsold stock).
- Reduce the risk for FL-RS service providers to invest in equipment for which the business case is not yet proven. The manufacturers will be paid, and the service providers will not be left with unpaid debts in case the equipment is not able to be operated at profitable scale.
- De-risk investments in awareness creation by sellers and distributors of FL-RS (extension, demonstrations, etc.) because awareness creation does not necessarily translate in adoption of FL-RS and profit for the sellers of FL-RS.

- Incentivize farmers to adopt FL-RS by reducing the cost of the FL-RS and enhance the availability of FL-RS at the right time.

165. The level of concessionality transferred to the end user through the 'smart-subsidies' will depend on a number of factors which differ per country and per FL-RS. The less developed the local context is on any of the following factors, the greater the subsidy relative to the co-contribution from the recipient. These factors include:

- Is the FL-RS technology proven to be profitable in financial terms for all actors: farmers, agro-dealers, service providers?
- Is the distribution network of FL-RS already in place and reaching all areas where the FL-RS is needed?
- Are farmers already aware of the FL-RS and ready to adopt? Is there a ready market?
- Are manufacturers capable of forecasting demand and prefinance stock of FL-RS where the demand is?
- Does the Government have the knowledge and tools to provide the FL-RS information in time and with sufficient accuracy to help farmers harvest in the right time using the right practices?

B.6. Exit strategy

166. Long-term sustainability is at the core of the programme design, with GCF investments seen as catalysers of change within local systems. A key aspect of this GCF intervention is having an exit strategy through the creation of an enabling environment to shift away from the smart-subsidies, catalysing further investments in FL-RS under Component 2. This approach to sustainability is centred on market development and stimulating the local supply chains to bring FL-RS to smallholder farmers at affordable rates — working through the private sector to foster innovation and commercial opportunities. To establish sustainable supply chains, the programme will forge market-led relationships with value chain actors such as off-takers, processors, aggregators, business development service providers and financial institutions, deepening and strengthening the networks that connect farmers to market. When combined with the demand-oriented actions of awareness, demonstration and market development under Outcome 1, the programme will result in a balance of supply, demand, and finance opportunity that will enable these relationships and commercial opportunities to persist within the target communities and expand to spill over areas beyond initial GCF investment period. Strengthening a market-led system is particularly important for the economic value that is being created through the adaptation components of RE-GAIN; these will ensure that the farmers and the supporting value chain actors continue to see value in the deployment of FL-RS even after the GCF project comes to an end. Further, the strengthening of an enabling public policy and institutions will ensure continued promotion of FL-RS through extension systems beyond the life of the project. Support for these systems among communities will be further strengthened by the long-term adaptation support provided through the environmental and social co-benefits. Such benefits, when combined with the strengthened awareness of climate risk, will further incentivise farmers to continue with FL-RS beyond the project lifespan.

167. At the community level, the programme is expected to actively strengthen and grow agricultural MSMEs (including farmer cooperatives, women's and youth groups) through access to finance and business development services to enhance development of FL-RS technologies, helping them attain a higher level of self-sufficiency that is expected to increase the sustainability of the full agriculture value chain, especially at post-harvest. It is also expected that by demonstrating a positive track record of successful direct investments in rural MSMEs and farmer organization, as well as by funding financial intermediaries for the creation or expansion of FL-RS investment portfolios, the GCF fund will incentivize the wider financial sector to focus on the agriculture sector through the design and development of appropriate products, thereby contributing to its increased sustainability of the programme. This will further enhance opportunities for continued investment in FL-RS beyond the initial subsidised finance models.

168. To ensure balance in interventions and sustainability of the programme, an end-to-end process approach will be used to for inclusion of stakeholders, from food system input processes, productivity, output processes, post-harvest handling and marketing. This will be context-specific, and the depth of investments will depend on existing AGRA or partners' investments. The programme will take a market-led approach to ensure sector sustainability, innovation and scale. Through the various capacity development activities, the programme will invest in strengthening the capacity of VBA extension workers

and farmers on the topic of climate-resilient agriculture and FL-RS. The programme will build and strengthen a network of village-based advisors and agro-dealers who will build the needed extension capacity within the farmer communities. They will be able to maintain and support other similar extension interventions inside and outside the targeted areas within focus countries, beyond the programme lifespan.

169. The proposed programme is designed through the participation of a wide array of public and private institutions, and civil society (including national and international and development actors) to promote ownership of the programme during implementation, the activities will be executed in close collaboration with relevant government departments in the focus countries, engaging with key stakeholders to enhance their capacity and create the enabling environment for continued investment. The programmatic monitoring, evaluation and learning activities will ensure that the lessons learned, documents produced, and the activities implemented during the programs are synthesized, stored and disseminated to the benefit of other programs and enable adaptive management of FL-RS initiatives.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	75.62			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	75,623,754 USD					
(ii) Subordinated loans						
(iii) Equity						
(iv) Guarantees						
(v) Reimbursable grants						
(vi) Grants						
(vii) Results-based payments						
(b) Co-financing information	Total amount			Currency		
	29.38			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor and grace	Pricing	Seniority
AGRA	Grant	29.38	million USD (\$)			
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Click here to enter text.						
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	105			million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	Please explain if any of the financing parties including the AE would benefit from any type of guarantee (e.g. sovereign guarantee, MIGA guarantee). Please also explain other contributions such as in-kind contributions including tax exemptions and contributions of assets. Please also include parallel financing associated with this project or programme (refer to the co-financing policy).					



170. The budget presented below presents the total RE-GAIN Programme budget, totalled for all seven countries. Budget will be distributed equally between countries, as indicated in Annex 17.

Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
Food Loss- Reduction Solutions (FL-RS) demand side development to increase the adoption of FL-RS by farmers	1.1: Smallholder farmers supported to adopt FL-RS	28.28	19.6	Grants	8.68	Grants	AGRA
	1.2: Improved market linkages between agri- value chain actors	14.46	3.17	Grants	11.29	Grants	AGRA
FL-RS supply side development to increase availability and affordability of FL-RS	2.1: Business development support for the improved provision of FL-RS on local markets	28.11	26.38	Grants	1.73	Grants	AGRA
	2.2: Financial mechanisms for smallholders and MSMEs to support the adoption of FL-RS	15.28	14.50	Grants	0.79	Grants	AGRA
Enabling environment to ensure sustainability of the FL-RS market	3.1: Enhanced capacity of national institutions to enable investments in FL- RS	11.34	6.34	Grants	5.00	Grants	AGRA
M&E		2.5	2.05	Grants	0.45	Grants	AGRA
PMC		5.02	3.59	Grants	1.44	Grants	AGRA
Indicative total cost (USD)		105	75.62		29.38		

*This table should match the one presented in the term sheet and be consistent with information presented in other annexes including the detailed budget plan and implementation timetable.
In case of a multi-country/region programme, specify indicative requested GCF funding amount for each country in annex 17, if available.*

C.3. Capacity building and technology development/transfer

C.3.1 Does GEF funding finance capacity building activities	Yes ☒ No ☐
C.3.2 Does GCF funding finance technology development	Yes ☐ No ☒

171. Capacity development is a key aspect of the programme design, cutting across all three project components. Under Component 1, capacity development will focus on training smallholder farmers on the use of improved climate-resilient FL-RS (USD 6M) and agricultural traders and processors on complementary services for FL-RS (USD 1.4M). The capacity of extension workers and agro-dealers will also be enhanced to further scale the provision of training support to farmers, traders and processors, with specific training of trainers programme (USD 2M).
172. Under Component 2, similar capacity development support will be provided to train new FL-RS service providers such as MSMEs, cooperatives, incl. women- and youth-led initiatives (USD 3.2M), including business development support and market intelligence (USD 1.8M), as well as capacity and market development for all supply-side market actors (USD 1.8M).
- Under Component 3, the programme will provide institutional capacity development support to key local entities to support the scaling of successful FL-RS models (USD 2.6M).
173. In terms of technology, the project will not develop new technologies but will invest in technology transfers for existing FL-RS. This will be approached primarily through the development of innovative finance mechanisms under Component 2, using conditional procurements to leverage private finance through partnerships with financial service providers, agro-dealers, MSMEs, youth groups, and smallholder farmers (USD 32.7M).

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

D.1. Impact potential

Adaptation Impact

174. The proposed programme will yield significant adaptation benefits for vulnerable smallholder farmers, their families, and communities across the seven target countries. At the core of these benefits is the expected reduction in climate change-induced food losses that are increasingly being experienced across Africa. Such food loss reductions will have multiple positive impacts on the lives and livelihoods of these highly vulnerable communities, mainly by enhancing the absorption capacity of smallholder farmers to seasonal climate and other shocks, building their resilience in the face of climate change. At the most fundamental level, reduced food loss will directly contribute to food security in the target communities — thereby contributing to GCF Adaptation Result Area (ARA) 2: Health and well-being, and food and water security. While climate change is known to impact the full spectrum of food systems — from production through to consumption — the proposed programme's focus on post-harvest food loss addresses a critical, but somewhat under-served, component of food security. Reduced food loss through improved management at harvest and postharvest phases will not only improve food security, but also increase farmers income and farm-level profitability. Additionally, reduction of food loss also leads to preserving the integrity of natural ecosystems that further has a positive impact on the resilience of farmers. The impacts of this programme are, therefore, considered as complementary to the broader adaptation strategies for agricultural resilience within each target country, filling a key gap in the broader landscape of food security. However, despite the complementarity of this approach with others focused more on primary production, the expected impacts on food security would remain even when viewed in isolation.
175. The nature of the approach – targeting smallholder farmers, with particular focus on women, youth groups, and MSMEs — means that the resilience benefits will apply to the most vulnerable people and communities – thereby contributing to GCF ARA 1. In addition to the increased food security, these benefits will include the increased resilience of rural livelihoods, with significant investments being made in developing local markets and building up farmer cooperatives, youth and women's groups, and MSMEs. The resulting investments extend beyond just the initial GCF contribution, with catalytic grants creating sustainable systems that will leverage partnerships with financial institutions and private sector actors to mobilise significant investment beyond the proposed project.
176. Through these results, the project will directly benefit a total of 2.62 million people across the seven countries, distributed across the even target countries as shown in Table 7. Further details on the methodology for determining beneficiary numbers are presented in Supplementary Annex 1: Adaptation Impact Estimation.

Table 7: Direct and Indirect beneficiaries disaggregated by country and sex

Country	Direct Beneficiaries			Indirect Beneficiaries		
	Total	Female	Male	Total	Female	Male
Burkina Faso	451,524	226,665	224,859	466,248	234,056	232,192
Ethiopia	359,844	179,202	180,642	374,568	186,535	188,033
Kenya	358,889	180,880	178,009	373,613	188,301	185,312
Malawi	329,284	160,361	168,923	344,008	167,532	176,476
Tanzania	375,124	189,813	185,311	389,848	197,263	192,585
Uganda	375,124	189,438	185,686	389,848	196,873	192,975
Zambia	382,764	194,061	188,703	397,488	201,526	195,962

D.2. Paradigm shift potential

177. The proposed programme has been designed to elicit a paradigm shift in the harvest and post-harvest development pathways for the seven target countries and continentally in Africa. At the core of this shift is the use of GCF grants as a catalyst for change, stimulating markets and mobilising investment at scale. This approach will lean on AGRA's convening power in Africa, drawing together key actors across the

post-harvest value chain — including farmers, agro-dealers, suppliers, off-takers and decision-makers — to establish sustainable programmes that enable farmers to adopt improved FL-RS in a manner that is financially viable and climate resilient. A clear focus on achieving scale and sustainability is embedded in the programme design, both through the program’s continental reach and its ambition to work on the ground within seven countries through employing tested and functional last mile delivery systems. This commitment to achieving scale and sustainability is underpinned by the programme’s emphasis on achieving transformational systems change, as described below.

Potential for scaling up and replication

178. The nature of the programme approach is such that it will stimulate local supply of affordable FL-RS options that will enable the scaling of interventions beyond the scope of the GCF investment. Specifically, the innovative finance mechanisms under Output 2.2, along with the MSME business development and market support under Output 2.1, will catalyse the establishment of and/or strengthen local supply chains that will bring FL-RS to smallholder farmers at affordable rates. A key feature of this approach is the notion of working through the private sector to foster innovation and commercial opportunity. AGRA’s delivery model will demonstrate and scale innovations that enable smallholder farmers to participate in both input and output markets, and in doing so to forge commercial relationships with off-takers, processors, exporters, commercial aggregators, business development service providers and financial institutions. This ecosystem focus builds sustainability and strengthens resilience by deepening and strengthening the networks that connect farmers to market. The FL-RS solutions will be disseminated at scale by integrating best practices into major national and regional flagship programs and anchor firms. When combined with the demand-oriented actions of awareness, demonstration and market development under Outcome 1, the programme will result in a balance of supply, demand, and finance opportunity that will enable the scaling of interventions in the target communities beyond initial GCF investment.
179. In addition to the scaling potential noted above, the programme is structured such that the approaches to stimulating the FL-RS market can be replicated; not only within other communities in the target countries, but also regionally across Africa. This replication potential is initially rooted in the demonstration effect the project will have, which when combined with the convening power of AGRA across Africa will attract additional market actors to engage in similar investment schemes. To further facilitate replication, Component 3 of the project will establish an enabling environment for sustainable FL-RS markets (see below). While these enabling activities will focus on replication at the national level, AGRA will facilitate collaboration with continental and regional bodies and national governments, including African regional climate institutions and Regional Economic Communities, to initiate the replication of the programme’s innovative models for scaling FL-RS across the region.

Potential for knowledge sharing and learning

180. The innovations, especially on innovative financing and business models, within the programme have the potential to generate new knowledge on ways of engaging MSMEs and smallholder farmers in reducing their food losses and thus increasing their productivity and their resilience to climate shocks. Activities under Component 3 will capture these lessons and develop knowledge products to share best practices at the regional level. This knowledge sharing component will be fundamental to the replication potential described above.

Contribution to the regulatory framework and creation of an enabling environment

181. Through the partnerships that will be established under the innovative finance and business models, the programme will address key barriers that limit the scaling up of FL-RS. This will be complemented by the policy focus with national governments in partner countries to help them realize their commitments to tackling climate change and improving food security in their countries. This will include supporting national governments to enact forward-thinking policies and regulations that seek to integrate climate-resilient FL-RS into the agriculture sector, as well as advocating for the establishment of tax incentives, and the uptake of certifications and standards for FL-RS.

D.3. Sustainable development

182. In addition to the core impacts described in D.1 above, the proposed programme will elicit multiple co-benefits that will contribute to the sustainable development goals (SDGs) of the target countries. Key contributions to the SDGs are listed in Table 8.

Table 8. Key Sustainable Development Goal Contributions

SDG	How the project contributes
SDG 2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture.	The programme will introduce FL-RS that will reduce food loss, thereby improving the net productivity of farms, contributing to food security.
SDG 5: Achieve gender equality and empower all women and girls.	The proposed project has a strong focus on gender-responsive actions that will empower women within the post-harvest agriculture value chain.
SDG 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.	The business development and market strengthening approaches under the proposed programme will create inclusive employment opportunities, with a particular focus on opportunities for vulnerable groups such as women and the youth.
SDG 13: Take urgent action to combat climate change and its impacts.	The programme is directly focused on building resilience among smallholder farming communities by addressing the impacts of climate change on post-harvest food loss.

Environmental co-benefits

183. Programme interventions to reduce post-harvest food loss will have several environmental co-benefits.
- *Reduced land degradation and improved water use efficiency:* Reducing post-harvest food loss will increase the net production of agricultural land, thereby reducing area of land and amount of water needed to produce sufficient food to meet demand. FL-RS will, therefore, improve the efficiency of land and water use in the agriculture sector, thereby reducing the impact of agriculture on land and water resources. This includes curtailing some of the drivers of agricultural expansion associated with the need to compensate for food loss. The resultant reduction in the rate of land use change will help preserve natural ecosystems and the important services they provide.
 - *Decreased in net use of fertilisers and pesticides:* The improved net-productivity of agriculture will reduce the environmental impact of fertilisers and pesticides, as well as the related energy demands, particularly as related to regrowing of lost food.
 - *Reduction in air pollutants from waste food disposal:* Reduced food waste will limit the potential environmental impacts of improper disposal of spoilt food – for example the release of air pollutants when waste is burned, or the production of methane during decomposition (see mitigation benefits in Annex 22).
 - *Increased greening and improved land use:* Reduced expansion of farming will release more land to reforestation, land rehabilitation and other services.
 - *Reduced loss of biodiversity:* Land expansion into forests and fragile areas is the major cause of biodiversity loss on the continent, which is threatening the livelihoods of current and future generations.

Mitigation Impact

Through the process of reducing food losses and creating capacity of key players in sustainable FL-RS solutions, the proposed programme will reduce greenhouse gas emissions; contributing to each country's mitigation targets. Emission reductions will be achieved through two primary pathways. The first, and most direct pathway is the reduction of emissions arising from the production of compensatory food (that is, food produced to compensate for food lost in harvest and post-harvest without considering land expansion). The second emission reduction pathway is through the reduction of food loss as a driver of land use change linked to compensatory agricultural expansion. Combined, these pathways will result in an estimated total emission reduction of 111,275 tCO₂e

over 10 years. Details on the emission reduction methodology and justifications are presented in Annex 22, with a country-specific breakdown provided in Table 10.

Table 9: Emission reductions disaggregated by country

Country	Emission Reduction (tCO ₂ e)
Burkina Faso	16,846
Ethiopia	5,083
Kenya	6,374
Malawi	34,632
Tanzania	16,161
Uganda	4,478
Zambia	27,700

Socioeconomic and health co-benefits

184. In targeting the livelihoods of highly vulnerable smallholder farming communities, the proposed programme will yield multiple socioeconomic co-benefits. Some of these co-benefits arise as a direct result of the core impact of improved food security, while others arise as a result of the actions to develop the FL-RS market in target communities.

- *Increased food quality and safety:* Improved post-harvest processes that reduce food waste will also improve the quality of food reaching consumers, which will enhance the nutritional value and reduce the risk of food-related illnesses resulting from contamination by pests, molds, aflatoxins and other pathogens. This will not only benefit individuals, but a more stable and safe food supply will enhance public health systems through reduced incidence of nutrition-related illness.
- *Job creation:* By working with local suppliers and agro-dealers, while also supporting the business development of MSMEs as service providers in the FL-RS value chain, the proposed programme will generate employment opportunities that will extend beyond the initial investment. This will particularly benefit vulnerable groups such as women and youth that will be actively targeted for support in business development initiatives.
- *Market stability:* Reducing food loss can help stabilize food prices by ensuring a more consistent supply. This can protect both producers and consumers from price volatility.

Gender-sensitive development benefit

185. The proposed programme has been designed with gender sensitivity at its core; with each component carrying key gender-responsive actions to not only ensure equal participation of women, but also actively empower women within the FL-RS value chain. This includes business development opportunities directly targeted at women's groups and women-led enterprises, recognising them as key actors in the market and supporting these groups to establish robust, sustainable services in the FL-RS value chain. The delivery mechanism of the project which primarily focuses on targeting farmer groups provide a platform that are likely to promote gender and youth integration and therefore leads to development outcomes that are gender sensitive.

D.4. Needs of recipient

186. RE-GAIN targets seven African countries that are all recognised as being highly vulnerable to climate change, which impacts all sectors of society. Of particular concern is the highly vulnerable agriculture

sector, which employs between 40 and 80% of total population and contributes between 20 and 40% of GDP across the target countries³⁹. Given the high rates of employment and economic contribution of the sector, climate change impacts on agriculture extend beyond the farmers themselves to the national economies as a whole. In addition to their sensitivity to climate impacts, all seven of these target countries are recognized as having low adaptive capacity⁴⁰ and low readiness⁴¹ to adapt; highlighting their need for support.

Table 10. ND-GAIN index for 2021

Country	Adaptive Capacity Rank (out of 192)	Readiness Rank (out of 192)
Burkina Faso	154 (low adaptive capacity)	158 (low readiness)
Ethiopia	159 (low adaptive capacity)	156 (low readiness)
Kenya	133 (low adaptive capacity)	152 (low readiness)
Malawi	150 (low adaptive capacity)	157 (low readiness)
Tanzania	141 (low adaptive capacity)	151 (low readiness)
Uganda	147 (low adaptive capacity)	163 (low readiness)
Zambia	103 (low adaptive capacity)	141 (low readiness)

187. As noted in Section B.1, climate change impacts the full spectrum of food production systems in Africa, from preparation of the land through the production and harvest, to the processing, storage and marketing of the produce. While the impacts across the whole system are well documented, the lion's share of research and resources for resilience interventions in the agricultural sector have been focused thus far on production, leaving a considerable gap in investments in the post-harvest stages of the process. In the absence of innovative initiatives to develop post-harvest food loss reduction systems, the target populations are left vulnerable to the impacts of post-harvest losses on their food security, which are being exacerbated by climate change. Given the high levels of poverty, weak markets, and low adaptive capacity experienced in smallholder farming communities, farmers are left unable to adopt climate-resilient FL-RS without initial external support. Moreover, the national governments of each country, while prioritising climate-resilient agriculture development, lack the financial, institutional and technical capacity to establish sustainable markets for FL-RS, with the majority of limited resources being directed towards production.
188. The RE-GAIN programme targets this vulnerable stage of the value chain that, despite the urgent need, has remained relatively underdeveloped at the global scale – particularly when considering smallholder farmers. By targeting local suppliers and agro-dealers, while also supporting the business development of MSMEs as service providers in the FL-RS value chain, RE-GAIN will build resilience directly within the target populations, integrating the solutions into local systems that meet the individual needs of each community.
189. More details on the needs of each country and the tailoring of interventions to each local context are provided in the country-specific Feasibility Studies appended to Annex 2.

D.5. Country ownership

190. The design of RE-GAIN directly emerged from a national-driven process that reflects country ownership by seven target countries — with significant effort applied throughout development to ensure country ownership of this programme. AGRA co-developed and submitted project idea notes and summary concept notes to NDAs and the various ministries in the respective country for demand creation, commenting on relevance and content and have jointly developed the potential, context-specific solutions together with the respective ministries and key stakeholders of the respective countries. This including having established demand by conducting initial consultation with key government institutions, followed by multiple local and national stakeholder workshops. Hence, the focus on post-harvest loss reduction is aligned with key agriculture and adaption strategies, as outlined in various policy documents and

³⁹ The percentage employment in the sector increases considerably when directly considering the rural population, as does the relative contribution to the local economy.

⁴⁰ Under the ND GAIN Index, 'Adaptive Capacity' is the ability of society and its supporting sectors to adjust to reduce potential damage and to respond to the negative consequences of climate events. This captures a collection of means, readily deployable to deal with sector-specific climate change impacts.

⁴¹ Under the ND GAIN Index, 'Readiness to make effective use of investments for adaptation actions thanks to a safe and efficient business environment. Readiness has three components: economic readiness, governance readiness, and social readiness.

development strategies in each of the seven focus countries (Table 11). Further, the programme is fully aligned with national policies, programmes in the various thematic areas of climate change, food and agriculture development, SME development, financial system and the national food security plans. the project will further leverage on existing national and partner programmes to ensure complementarity of programmes. These factors will enhance national and local ownership of the programme.

Table 11. Key alignment with national policies and priorities for climate change and agriculture in each of the target countries.

CC Strategies and Policies	
Burkina Faso	
Burkina Faso's National Sustainable Development Policy (PNDD) includes a key focus on reducing post-harvest losses through improved storage, processing, and distribution techniques to enhance food security and promote sustainable agricultural practices. This is complemented by the National Economic and Social Development Plan II 2021-2025 which highlights improving the handling, storage, and processing of agricultural products to reduce post-harvest losses as part of the strategy to restore security and peace, strengthen national resilience, and transform the economy for inclusive and sustainable growth. The National Climate Change Adaptation Plan (PNA) also addresses the need for resilience in agricultural systems amid climate variability and change, underscoring adaptive agricultural practices and sustainable methods to secure food and nutritional needs. The National Agricultural Investment Program further reinforces these efforts, providing strategic guidance to the agricultural sector to reduce post-harvest losses and improve food security by addressing aflatoxin-triggered losses, among other issues. Similarly, the National Strategy for the Development of Agricultural Entrepreneurship by 2025 includes focus on reducing post-harvest losses through improved storage facilities, processing techniques, and investments in modern infrastructure and training for farmers, while the Agricultural Value Chain Development Strategy 2019-2023 focuses on increasing productivity and reducing harvest and post-harvest losses through disseminating conservation technologies, building capacity on good practices, and constructing storage facilities. This approach is also promoted in the Agro-sylvo-pastoral Production Sectoral Policy 2018-2027. Finally, Burkina Faso's revised its Nationally Determined Contribution for the period 2021–2025 sets ambitious targets for reducing greenhouse gas emissions and adapting to climate change—including through the promotion of regenerative agriculture and food systems.	
Ethiopia	
Ethiopia's Agricultural Transformation Agenda focuses on improving productivity and reducing post-harvest losses through infrastructure development and training. The Postharvest Management Strategy in Grains (2018) and the Working Strategy for Strengthening Ethiopia's Teff Value Chain (2015) also both focus on reducing post-harvest losses and enhancing value chains. In addition to these policies aimed specifically at post-harvest, numerous national policies and strategies identify agriculture as a key sector for sustainable and climate resilient development. This includes the Ten-Year Development Plan (2021-2030), the Homegrown Economic Reform Agenda, and Climate Resilient Green Economy Strategy which focus on agriculture and private sector involvement as a driver to reach middle-income status by 2025. The programme also aligns with investment roadmaps for prioritizing agricultural investments laid out in the National Agriculture Investment Plan (2022) and the Agricultural Sector Policy and Investment Framework (2010-2020). Ethiopia's updated NDC highlights the mitigation potential in land use and forestry, targeting significant GHG reductions by 2030. The National Adaptation Plan (2017) focuses on agriculture and industry, aiming to reduce food losses and increase resilience. Additionally, the Agricultural Extension Strategy (2017-2030) integrates best practices for effective extension services, and the Nutrition Sensitive Agriculture Strategy (2017-2021) aims to mainstream nutrition in agricultural programs.	
Kenya	
Kenya's Vision 2030 aims to transform the country into a newly industrializing middle-income nation by 2030, focusing on modernizing agriculture and achieving a 10% annual economic growth rate. The National Climate Change Action Plan (2018-2022) strengthens the path towards sustainable, climate-resilient development, addressing agricultural emissions and enhancing productivity and resilience. Kenya's Agricultural Sector Development Strategy (2010-2020) and the National Agriculture Investment Plan (2019-2024) promote market-oriented production and post-harvest loss reduction. Kenya's Nationally Determined Contribution (2020-2030) seeks to abate GHG emissions by 30% by 2030, with the agriculture sector playing a key role. The National Adaptation Plan (2015-2030) enhances climate resilience towards Vision 2030, promoting climate-smart agriculture and efficient food systems. The Agricultural Sector Transformation and Growth Strategy (2019-2029) emphasizes increasing incomes, output, and food resilience, with specific targets for SME development and market access. The Climate Risk Management Framework (2016) harmonizes climate change and disaster risk policies. The Kenya Climate Smart Agriculture Implementation Framework (2018-2027) provides guidance for innovative and transformative initiatives, focusing on productivity, resilience, and emissions reduction.	

Malawi
The programme approach aligns with the priorities of the two core development agendas for Malawi, namely Malawi 2063 and the Malawi Growth Development Strategy, which both include focus on long-term agricultural development and economic transformation. The need for addressing climate change impacts and post-harvest food losses are recognised in the National Agriculture Policy (NAP 2016-2020), which propose measures that include the use of CIEWS and community-based storage. This was further highlighted in the Vision 2020 - The National Long Term Development Perspective, which highlighted the need for clear policies and technologies to reduce post-harvest losses. Malawi's updated NDC (2021) also identifies agriculture as the largest contributor to GHG emissions, emphasizing mitigation and adaptation measures to enhance food security and resilience.
Tanzania
Tanzania's comprehensive policy framework addresses agricultural productivity, climate resilience, and post-harvest loss management. The Tanzania Development Vision 2025 and the Long-Term Perspective Plan (2011-2021) set long-term development goals, emphasizing food security and sustainable development. The National Strategy for Growth and Reduction of Poverty II and the Third National Five-Year Development Plan focus on post-harvest loss management and agricultural productivity. The Agricultural Sector Development Programme Phase II (2017/18-2027/28) and the Agricultural Sector Development Strategy (2015/2016-2024/2025) aim to modernize agriculture, improve infrastructure, and enhance market access. The National Climate Change Strategy (2012-2018) and the updated Nationally Determined Contribution (2021) target GHG emission reductions and climate resilience in agriculture. The National Postharvest Management Strategy (2019-2029) and its Implementation Plan (2019-2024) focus on reducing crop losses through improved storage, technology adoption, and market systems. Tanzania's policies also include the Agriculture Climate Resilience Plan (2014-2019) and the National Agriculture Policy (2013) to promote sustainable agricultural practices and resilience. The ASDP II Communication Strategy (2020-2028) identifies key information needs for smallholders and strategic interventions to reduce post-harvest losses. The National Environment Policy (NEP 2021) and the National Climate Change Response Strategy (2021-2026) outline environmental and climate change goals.
Uganda
Uganda's policy landscape includes long- and mid-term strategies focused on agricultural development, climate resilience, and post-harvest loss reduction. Uganda Vision 2040 aims to transform Uganda into a competitive upper-middle-income country by 2040, emphasizing agricultural development and reducing post-harvest losses through expanded market infrastructure. The Green Growth Development Strategy (2017/18-2030/31) ensures sustainable development in agriculture, natural capital management, and climate resilience, focusing on income enhancement, job creation, climate change adaptation, and resource use efficiency. The Third National Development Plan (2020-2025) prioritizes reducing post-harvest losses to enhance food security and increase agricultural productivity, with a focus on agro-industrialization. The National Agriculture Policy (2013) aims to promote food security and household incomes through sustainable agricultural productivity and value addition. It emphasizes reducing post-harvest losses through improved storage, processing, and value addition. The National Cooperative Policy (2011) seeks to improve governance, production, value addition, and marketing capacities of cooperatives, addressing challenges like poor storage facilities and high post-harvest losses. The National Agricultural Extension Strategy (NAES 2016/17-2020/21) aims to establish a sustainable farmer-centred agricultural extension system, enhancing capacity in production, agribusiness, value addition, and post-harvest management. The National Grain Trade Policy (2015) targets improved grain handling and storage to reduce post-harvest losses, which range between 26% to 37%. It focuses on enhancing pre- and post-harvest practices, standardizing warehouses, and improving market access. The National Climate Change Policy (2015) promotes climate-resilient and low-carbon development, with adaptation strategies to enhance agricultural systems' resilience and improve food storage and management. Uganda's updated Nationally Determined Contribution (NDC 2022) prioritizes adaptation in agriculture, aiming to reduce post-harvest losses from 37% to 12% by 2025 and to 3% by 2030. The National Organic Agriculture Policy (2020) emphasizes enhancing post-harvest handling and value addition to improve market opportunities and job creation. It promotes access to appropriate agricultural technologies for post-harvest management.
Zambia
Zambia has a comprehensive policy framework aimed at enhancing agricultural productivity, reducing post-harvest losses, and promoting climate resilience. The 8th National Development Plan (2022-2026) focuses on improving living standards, reducing poverty and inequality, and promoting economic transformation through agricultural development. It emphasizes increasing production, productivity, and value addition, along with enhancing storage and logistics for small-scale farmers. The Second National Agricultural Policy (2016) supports sustainable agricultural diversification, commercialization, and private sector participation, targeting high post-harvest losses by constructing and rehabilitating storage facilities and providing fumigation services. The revised National Agriculture Policy (2012-2030) guides sustainable agricultural growth, focusing on reducing post-harvest losses through better storage and handling practices. The National Food and Nutrition Policy includes

provisions for reducing post-harvest losses to ensure food security. Zambia's Climate-Smart Agriculture Investment Plan promotes practices that increase productivity, enhance resilience, and reduce GHG emissions, emphasizing strategies for reducing post-harvest losses and the need for market infrastructure and training on improved post-harvest management.

The National Policy on Climate Change (2016) provides a framework for coordinating climate change programs to ensure climate-resilient and low-carbon development towards Zambia's Vision 2030. It includes measures to strengthen early warning systems, control climate-related pests and diseases, and promote climate-smart agricultural technologies. Zambia's Nationally Determined Contribution (NDC 2015-2030) pledges to reduce GHG emissions by 25% (20,000 Gg CO₂eq) by 2030 under a Business As Usual (BAU) scenario, with a potential reduction of 47% (38,000 Gg CO₂eq) with substantial international support. The NDC emphasizes sustainable agriculture, forest conservation, and waste management, including food loss reduction as key areas for mitigation. Zambia's revised NDC (2021) broadens the scope of mitigation and adaptation actions, focusing on sustainable agriculture, productive infrastructure, capacity building, and research.

191. The design of the proposal has been built on a strong foundation of consultations with communities as well as local and national governmental and non-governmental entities. These consultations, together with substantial desktop research and the extensive local experience of AGRA in the target countries, were used to establish a strong baseline for the project, identifying key technical, institutional and market gaps that need to be filled to build resilience of the target smallholder farmers. This process will continue through the project implementation, ensuring that the solutions are tailored to the individual needs of each target community. In particular, the innovative finance and catalytic grant programmes under Component 2 will be specifically tailored to the context and needs of local stakeholders.
192. While the project will be implemented by AGRA, the co-development process with the target countries, underpinned by the consultative process noted above, ensures that the project retains a strong sense of country ownership. In addition to the consultation held specifically for the development of this proposal, the foundation of the approach arose from extensive engagements AGRA has conducted in each of the target countries during the development of its five-year country strategies (2023-2027). These engagements resulted in a gap analysis across the agriculture and food system sector of each country, and the region more generally, identifying areas for collaboration and complementarity to enhance a food system transformation. AGRA prioritizes collaboration with government – through its policy and state capacity strategic support that involves alignment with national climate change strategies, policies, plans, and instruments as mentioned earlier in this concept note. Interventions in this program support national priorities, visions and plans, fostering a natural ownership by the government. For more details on country stakeholder analysis, please refer to Annex 7.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

Financial Structure – adequacy, reasonableness and return on investment

193. Affordability, access to finance, market access, financial literacy, and consumer awareness are key challenges for smallholder farmers seeking to adopt practices that reduce food loss in the agricultural value chain. Key to viably addressing these challenges is RE-GAIN's programmatic approach that combines demand-side, supply-side, and enabling environment components.
194. The combination of both financial and non-financial approaches effectively ensures market failures resulting from both the supplier and consumers side of the agricultural value chain are addressed.
195. The two key financial models—1) catalytic grants for smallholder farmers and 2) catalytic grants for agricultural MSMEs—use concessional funding to incentivize private sector manufacturers and financial institutions (FIs) to
 - Provide physical solutions that are within the affordability envelope of farmers while remaining commercially viable to private sector suppliers.
 - Provide tailored financial products that are affordable to farmers, while mitigating credit risk to financial institutions.

196. By focusing on a 5-year period, the program aims to demonstrate long-term market viability to suppliers (both physical and financial), build the necessary credit history required to access financial products, and encourage adoption of physical solutions by farmers.
197. Non-financial interventions drive adoption of these solutions from farmers by focusing on building capacity and awareness through training, educational materials, and village-based agent (VBA) demonstrations, addressing information gaps and promoting financial literacy. These interventions are critical to creating linkages between private sector manufacturers, MSMEs and Small hold farmers.
198. This combined approach of addressing both supply and demand side constraints is expected to mobilize US\$ 144.95 million worth of physical solutions and reach 553,900 smallholder farmers households over the next 5 years. Given the programme cost of US\$ 105 million this represents a catalytic effect or return of 1.38 times the original investment from the GCF. The financial return on investment exceeds 1 time the original investment across each of the seven countries. This return excludes the long-term impact of the programme's ability to demonstrate the market viability of these physical solutions to both users (farmers) and suppliers (manufactures and FIs), which will catalyse new private sector markets while addressing food-loss and associated climate related challenges.

Economic Efficiency and effectiveness of the proposed project/programme and return

199. The proposed programme is efficient and effective, with a total investment of US\$ 105 million expected to generate significant climate adaptation outcomes. By reducing post-harvest food losses, the programme contributes directly to increased food availability, with co-benefits of reduced greenhouse gas (GHG) emissions; projected reduction of around 0.1 gigatonne of CO₂-equivalent emissions. This mitigation co-benefit is coupled with economic and social benefits such as enhanced food security, improved household incomes, and better health outcomes through reduced aflatoxin exposure, improved food safety and availability.
200. An economic analysis for the programme was undertaken using a cost-benefit analysis approach. Programme costs and benefits were assessed over a 10-year period and using country-specific social discount rates as presented in Annex 3. The analysis weighed up the total programme costs against the benefits associated with the uptake of equipment by smallholder farming households. These benefits included the income gains associated with the value of food that would otherwise have been lost in the absence of the programme interventions, the health benefits from reductions in aflatoxin exposure, the improved nutritional status (particularly of infants) from avoided food losses, and the climate benefits associated with reduced greenhouse gas emissions from food production. The results of the economic analysis suggest that, under the current set of assumptions, the total package of interventions is cost-beneficial, with the total programme investment generating an economic Net Present Value (ENPV) of US\$ 69 million, with a benefit to cost ratio (BCR) of 8:1 and an Economic Internal Rate of Return (EIRR) of around 56%. On this basis, there is a clear economic case for implementing the proposed programme. Additional analyses were conducted to test the sensitivity of the findings to changes in the underlying assumptions, including around the choice of discount rate, the shadow price of carbon, and the proportion of existing food production for which the quality (and safety) improves as a result of programme interventions. The results of these analyses are presented in Annex 3 and show that, at a programme level, the economic summary statistics are robust to changes in the input values.
201. At a country level, both the ENPV and BCR is positive across all seven countries. The highest net benefits are experienced in Burkina Faso and Ethiopia and can be explained, at least in part, by the choice of crops used to illustrate the impacts of interventions, and which have comparatively higher prices; the relatively high prevalence of aflatoxin-related illnesses in Burkina Faso

Application of best practices

202. An evaluation of proposed physical food loss reduction solutions (FL-RS) was conducted across all seven target RE-GAIN countries to identify those with the highest potential to reduce postharvest food losses and protect the harvests against growing impacts from climate hazards. Stakeholder engagements in all seven countries provided critical nuances including advantages, disadvantages, and barriers for use,

particularly for smallholder farmers, while a detailed assessment of best practices was used to underpin the assessment.

203. In addition to the above-mentioned prioritizations following the climate rationale, the final selection of solutions considered additional prioritization factors to ensure the success of the RE-GAIN programme and achieve lasting systemic changes in all target countries. Those factors included solution's estimated potential in reducing food losses, estimated level of environmental pollution/ GHG emissions during the use of the solutions, current level of awareness of the farmers about the solution's proper use and maintenance, availability of selected FL-RS in the country, and potential for their supply scalability and job creation through locally produced or assembled solutions and improving market linkages. The results of the evaluation, including major climate hazards for each country, a package of proposed solutions, together with their food loss reduction potential and existing barriers for their increased adoption is presented in Annex 2.
204. Based off the above-described analysis, basket of options approach was selected for the RE-GAIN programme FL-RS adaptation strategy – adopting bespoke combinations of FL-RS tailored to the context of each country. The key best practice innovations for FL-RS identified include mechanical multi-crop threshers and shellers (preferably solar-powered) combined with moisture meters for monitoring the level of moisture in the target crops, and communal storage structures, with the FL-RS uses on the individual farm level, such as tarpaulins and plastic sheets for drying crops, hermetic storage technologies (hermetic bags, silos) used for storage of the crops, and storage protectants and control agents, preferably biological origin.
205. In addition to the technical FL-RS designs described above, the overall project design considers AGRA's experience and lessons learnt from previous projects. These lessons and best practice are described below.
 - AGRA's last mile delivery model through training and supporting VBAs (village-based advisors) and agro-dealers in the target countries, who are largely in place to take the FL-RS solutions to farmers and their institutions.
 - AGRA has multiple and largescale projects working on food security and resilience in the target countries, which will allow this project to leverage at a scale and have a jump start.
 - AGRA has been working on multiple financial models, which could be used to move this project forward.
 - AGRA has a well fledged infrastructure and country teams in the target countries that will ensure a jump start and successfully delivery.
 - It was working with the governments of these target countries on major food security, climate and market issues, which this project will be leveraging both the political good will but also the infrastructure.
 - Farmer Education is a requisite. Partnerships with input providers and loss reducing technology manufacturers can help in training smallholder farmers to promote the sustainability of loss reducing efforts in the value chain.
 - Financial products should consider the production cycle and set requirements for repayment that are informed by the crop season.
 - Financing models should include risk sharing strategies for different market actors in the value chain
 - Partnerships and brokering agreements between financial institutions and strategically positioned market actors with the aim of unlocking barriers to accessing finance and encourage the development of tailor-made financial products that enable actors to innovate, invest for the expansion of operations and increase equipment acquisition for light to medium processing.
 - Create incentives for SMEs to invest in food loss reduction technologies.
 - Technical assistance to assist value chain actors to develop loan proposals and business plans for quick processing of loans.
 - Smallholder farmers must be convinced of the return on their investment requiring linkages to a reliable market that will provide a fair price for the product's quality.
 - Promoting technologies for multiple crops to increase uptake of post-harvest technologies that are introduced through a program.
 - The involvement of technology distribution companies leads to faster adoption by farmers.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

- ☐ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	The uptake of food loss-reducing solutions in the target countries is low, particularly among smallholder farmers. While solutions are generally available, local supply markets are underdeveloped, restricting access and inflating prices. The combination of limited awareness, limited access, high prices, and low access to finance results in a very low adoption rate for FL-RS. As a result, farmers experience considerable levels of food loss which threatens food security. In the absence of FL-RS, there are few processes in place to mitigate the GHG emissions associated with food loss and the related land use change. Moreover, the limited demand constrains interest among manufacturers and suppliers to develop the local market, preventing the development of self-sustaining systems. This problem is exacerbated by a lack of innovative financial mechanisms that can catalyse investment in local market development.	Low	Smallholder farmers understand the need for adopting FL-RS and have access to well-developed local supply markets that increase access and affordability of FL-RS. Strong partnerships are in place between stakeholders that facilitate the development of such markets, with manufacturers, agro-dealers, finance providers and services providers working together to enable smallholder farmers to access post-harvest technologies and services at an affordable rate. The increased demand and demonstration of effectiveness of the solutions drives further market development, enabling the scale of impact to extend within the target communities, including to other farmers as well as creating the pathway for use of FL-RS on other crops.	Under the RE-GAIN programme, AGRA will play a facilitation role in establishing the partnerships for the innovative financial models that underpin the transformation of FL-RS solutions in the target countries. Moreover, by providing smart subsidies and investing in MSME business development and market support, will catalyse the establishment of local supply chains that will bring FL-RS to smallholder farmers at affordable rates. AGRA's delivery model will demonstrate and scale innovations that enable smallholder farmers to participate in both input and output markets, and in doing so to forge commercial relationships with off-takers, processors, exporters, commercial aggregators, business development service providers and financial institutions. When combined with the demand-oriented actions of awareness, demonstration and market development under Outcome 1, the programme will result in a balance of supply, demand, and finance opportunity that will enable the scaling of interventions in the target communities beyond initial GCF investment.
Replicability	There are few robust investment models targeting food security through resilient post-harvest systems that reduce food losses. Some models have been piloted in individual countries or communities, but there are few systematic initiatives that demonstrate replicable models that are not reliant on continued subsidies	Low	Mechanisms are in place that demonstrate the business case for and viability of FL-RS market development among smallholder farming communities in Africa. Robust financial models and clear demonstration of impact facilitate scaling into other communities, regions and countries across Africa.	The programme is structured such that the approaches to stimulating the FL-RS market can be replicated; not only within other communities in the target countries, but also regionally across Africa. This replication potential is initially rooted in the demonstration effect the project will have, which when combined with the convening power of AGRA across Africa will attract additional market actors to engage in similar investment schemes. To further facilitate replication, Component 3 of the project will establish an enabling environment for sustainable FL-RS markets (see below). While these enabling activities will focus on replication at the national level, AGRA will facilitate collaboration with continental and regional bodies and national governments, including African regional climate

				institutions and Regional Economic Communities, to initiate the replication of the programme's innovative models for scaling FL-RS across the region.
Sustainability	There is a general lack of partnership between manufacturers, suppliers, agro-dealers, financiers, service providers and farmers to sustain local markets for FL-RS. The policy and institutional environment in the target countries does not adequately incentivise such partnerships, with few instruments (such as tax breaks or quality standards) in place or enforced to incentivise investment.	Low	Long-term sustainability is at the core of the programme design, with GCF investments seen as catalysers of change within local systems. A key aspect of the sustainability strategy is the enabling environment to shift away from the smart subsidies that will catalyse investments in FL-RS under Component 2. This approach to sustainability is centred on market development, stimulating the local supply chains to bring FL-RS to smallholder farmers at affordable rates — working through the private sector to foster innovation and commercial opportunities.	To establish sustainable supply chains, the programme will forge commercial relationships with off-takers, processors, exporters, commercial aggregators, business development service providers and financial institutions, deepening and strengthening the networks that connect farmers to market. When combined with the demand-oriented actions of awareness, demonstration and market development under Outcome 1, the programme will result in a balance of supply, demand, and finance opportunity that will enable these relationships and commercial opportunities to persist within the target communities beyond initial GCF investment period.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁴²	
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	Farmer surveys AGRA implementation partner reports	0	Direct Women: 660,210 Men: 656,066 Total: 1,316,276 Indirect Women: 686,043 Men: 81,767 Total: 1,367,810	Direct Women: 1,320,420 Men: 1,312,133 Total: 2,632,553 Indirect Women: 1,372,087 Men: 1,363,534 Total: 2,735,621	
	Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options	Agreements under Model 2 AGRA implementation partner reports	0	Men: 11,668 Women: 7,778 Total: 19,446	Men: 23,335 Women: 15,557 Total: 38,892	Agro-dealers are involved in the supply to communities of climate-resilient FL-RS Youth groups and MSMEs will take up opportunities to develop

⁴² The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

						service-based livelihoods using FL-RS. The establishment of an F-LRS service will benefit the livelihood of all members of the group offering the service.
	Supplementary 2.5: Beneficiaries (female/male) adopting innovations that strengthen climate change resilience	Farmer surveys AGRA implementation partner reports	0	Women: 660,210 Men: 656,066 Total: 1,316,276	Women: 1,320,420 Men: 1,312,133 Total: 2,632,553	Capacity development and financial models will enable farmers and MSMEs to adopt innovative technologies for FL-RS, with 50% of the smallholder farmers engaging in the activities are successfully implementing those .
ARA2 Health, well-being, food and water security	Core 2: Direct and indirect beneficiaries reached	Farmer surveys using FIES methodology AGRA implementation partner reports	0	Direct Women: 45,284 Men: 45,107 Total: 90,391 Indirect Women: 696 Men: 693 Total: 1,388	Direct Women: 90,568 Men: 90,214 Total: 180,782 Indirect Women: 1,391 Men: 1,385 Total: 2,776	
	Supplementary 2.2: Beneficiaries (female/male) with improved food security	Farmer surveys using FIES methodology AGRA implementation partner reports	0	Women: 45,284 Men: 45,107 Total: 90,391	Women: 90,568 Men: 90,214 Total: 180,782	Reduction in post-harvest food losses will increase food security to 50% of the likely food insecure smallholder farmers engaged in the RE-GAIN programme and actively implementing the initiatives.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
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Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	Development of national policies, strategies, frameworks and implementation plans aiming to address agricultural sector and climate change challenges has been a priority for many of the African governments in the recent years. Despite this, there are still considerable gaps in the policy and regulatory environment across the target countries that impede the mainstreaming of sustainable FL-RS in the agricultural sector. This gap is underpinned by several factors. First, there is limited or no use of the scientific evidence base in the formulation of policies and strategies on post-harvest loss reduction. Second, policies generally lack coordination and harmonisation across sectors, limiting opportunities for integrated approaches to building resilience in the agriculture sector. Third, where policies or regulatory frameworks are in place, they are often constrained to agricultural input subsidies — mainly for fertilizer and seeds — with no subsidy on post-harvest technologies. This further extends to the focus of information and extension services, which are generally geared towards primary production, and less so to post-harvest processes. Similarly, existing taxation regimes do not favour or incentivise investments in storage and processing technologies for food in the smallholder system.	low	Policies and incentives are in place to facilitate the development of local markets and support to smallholder farmers for the uptake of FL-RS in each of the seven target countries. Manufacturers, importers and suppliers are incentivised to establish local markets through tax exemptions, subsidies and affordable access to finance.	The RE-GAIN programme will support the revision and operationalization of policies in support of enabling FL-RS investments, including tax exemptions/incentives, certification and standards for FL-RS quality, and promote successful FL-RS business models for scaling-up and replication. This will include collaborating with government entities to revise and update key policies which are not responding to the current and future food losses and climate challenge scenario. RE-GAIN will engage relevant government ministries and agencies to review existing regulations and policies and proposing amendments to facilitate the development of local markets and support to farmers for the uptake of FL-RS. This will include introducing/strengthening national policies that provide financial incentives such as tax exemptions, subsidies, or reduced interest rates for FL-RS investments, making it more attractive for private and public sector investments.	Multi-countries
Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation	The deployment of FL-RS in the target countries is extremely limited, particularly among smallholder farmers. This includes extremely limited penetration of suppliers into rural markets. There are also few innovative financial mechanisms that can unlock local markets for FL-RS.	low	FL-RS technologies are disseminated in the local communities, both directly to farmers (for smaller, low-cost solutions) and through MSME service providers for larger solutions that can be shared among farmers. Innovative finance mechanisms are in place in the target communities linking manufacturers, suppliers, agro-dealers, financiers, service providers and farmers.	FL-RS technologies will be transferred to smallholder farmers, MSMEs, farmer cooperatives and youth groups through innovative finance models and smart subsidies. The RE-GAIN programme will facilitate partnerships for technology transfer through the innovative finance models, using the smart subsidies to derisk investments for suppliers and financiers. Beneficiaries will also be trained on the effective use of FL-RS, including the operation of service provision businesses.	Multi-countries

Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level	Local markets for FL-RS are underdeveloped, particularly among smallholder farming communities. Limited awareness and financial capacity of farmers constrains demand for FL-RS, which in turn constrains interest among manufacturers and suppliers to develop the local market. This problem is exacerbated by a lack of innovative financial mechanisms that can catalyse investment in local market development.	low	The local market is unlocked, encouraging vendors of FL-RS equipment to invest in distribution channels, marketing, maintenance capacity and other needed infrastructure to make the equipment available and serviceable in a sustainable manner. This shift will lead to reduced prices and increased affordability, creating a self-sustaining system that will enable the phased out of programme subsidies.	RE-GAIN will build the capacity of local MSMEs to provide access to FL-RS, either through improving local supply or offering services to farmers. MSMEs and youth groups will be trained on technical topics related to FL-RS service provision, including: i) proper service management and process optimization; ii) the safe, effective, efficient and sustainable operation of the equipment; iii) compliance with international and local quality standards for post-harvest services; and iv) the management of the fee-for-service model. This will be complemented by the development of innovative financial mechanisms that will catalyse investments in FL-RS through partnerships and smart subsidies.	Multi-countries
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	While farmers are distinctly aware of the impacts of food loss on food security, many do not fully understand the specific causes of food loss, or the potential solutions available to reduce food loss. This includes limited understanding of basic harvesting practices that can reduce risk of food loss, or the related post-harvest practices and tool available. Moreover, many farmers are not fully aware of the impacts that CC has on food loss and the consequent need to change post-harvest practices.	low	Farmers are aware of the need and options available for FL-RS, thereby enhancing demand for, and adoption of, FL-RS in the target communities. This adoption of FL-RS will ultimately reduce food losses and result in the core benefits of the programme for smallholder farmers – namely improved food security, increased resilience to climate shocks, and reduced GHG emissions.	RE-GAIN will provide direct support to smallholder farmers to not only know what FL-RS options are available, but also to understand the need for adopting such options – particularly when considering the impacts of a changing climate – as well as how to optimise the use of FL-RS. Moreover, RE-GAIN programme will identify and train local MSMEs and cooperatives to develop FL-RS service provision operations in the target communities.	Multi-countries

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (Outcomes/Outputs)	Project/programme-specific Indicator	Means of Verification (MoV)	Baseline	Target ⁴³		Assumptions / Note
				Mid-term	Final	
Outcome 1: Increased adoption of FL-RS by smallholder farmers	Outcome indicator 1: Number of targeted smallholder farmers adopting one or more FL-RS	Household outcome survey report	<i>To be confirmed from baseline survey</i>	Year 3: 1,024,371 of targeted smallholder farmers adopt one or more FL-RS	Year 5: 1,536,557 of targeted smallholder farmers adopt one or more FL-RS	Given the integrated nature of the project approach, achieving the full impact of the Outcome will also rely on the supply-side interventions under Component 2, and the

⁴³ These are to be confirmed following baseline information and updated during Project Inception

						strengthening of the enabling environment under Component 3
Output 1.1: Smallholder farmers supported to adopt FL-RS	Output indicator 1:1 Number of targeted smallholder farmers trained and received support on the use of climate-resilient FL-RS interventions	AGRA implementation partner reports	No training content has been developed and delivered	Year 3: 1,536,557 of targeted smallholder farmers trained, of which 384,139 are women 2,674 VBAs trained	Year 5: 2,560,928 of targeted smallholder farmers trained with 1,024,371 female participation ⁴⁴ 5,348 VBAs trained	Suitable local entities exist in target communities to facilitate training. Where relevant, agricultural extension services, lead farmers and Village-Based Advisors (VBAs), farmer cooperatives, and industry associations are available and willing to conduct the training Smallholder farmers attend trainings that are made available to them through various channels. There is a gender disparity in productivity between male and female-led farming practices. Assumes women will be slower to participate in these sessions
Output 1.2: Improved market linkages between agri-value chain actors	Output Indicator 1.2 Percentage of targeted smallholder farmers' harvest sold into structured market linkages	AGRA implementation partner reports	<i>To be confirmed through baseline survey</i>	Y3: Between 20%-30% of harvests sold into structured market linkages	Y5: >50% of harvests sold into structured market linkages	Local farmers, cooperatives and actors in the value chain are willing to connect with other
Outcome 2: Enhanced supply and affordability of FL-RS	Outcome Indicator 2: Increased affordability and accessibility of FL-RS to targeted smallholder farmers	Affordability Index	<i>To be confirmed through baseline survey</i>	Y3: 5% Reduction to Affordability Index	Y5: 15% Reduction to Affordability Index	Wages to purchase FL-RS don't decrease There are no major economic downturns or adverse conditions that would negatively impact purchasing power. Affordability Index located in the household surveys leveraging data collected from questions about income and purchasing power
Output 2.1: Business development support provided to local	Output Indicator 2.1 Number of smallholder farmers through MSMEs engaging with service	AGRA implementation partner reports	No business development support provided	Y3: 13,370 of targeted smallholder farmers	Y5: 26,740 of targeted smallholder	Manufacturers, importers and distributors of FL-RS are accepting of business development support

⁴⁴ This aligns with the Gender & Inclusion Action Plan that outlines both qualitative and quantitative targets, aiming to ensure 40% of all smallholders benefitting from the programme's FL-RS are female.

MSMEs to facilitate the improved provision of FL-RS on local markets	models that enable shared access to larger, more expensive FL-RS equipment			engage with access to service model, with 5,348 being women and youth smallholder farmers	farmers engage with access to service model, with 10,696 being women and youth smallholder farmers	
Output 2.2: Financial mechanisms established to support adoption of FL-RS	Output Indicator 2.2: Number of smallholder farmers, MSMEs and other relevant players who engage with Model 1 and Model 2 financial mechanisms introduced to develop the market for FL-RS	AGRA implementation partner reports AGRA audits	No financial mechanisms have been introduced	Y3: 800,000 smallholder farmers engage with Model 1 7,500 agro-dealers engage with Models 1 and 2 3,750 suppliers engage with Model 2 7 financial institutions engage with Model 2 300 MSMEs engage with Model 2	Y5: 1,200,000 smallholder farmers engage with Model 1 15,000 agro-dealers engage with Model 1 and Model 2 7,500 suppliers engage with Model 2 14 financial institutions engage with Model 2 600 MSMEs engage with Model 2	Financial mechanisms exist to develop the market for FL-RS Smallholders and MSMEs use financial mechanisms established Country-level disaggregation will be provided during reporting
Outcome 3: Enabling environment for the uptake of FL-RS	Outcome Indicator 3 Number of FL-RS and gender and youth responsive enabling policies, reforms, regulations, certifications, standards supported in their development	AGRA baseline surveys	Limited FL-RS enabling policies, reforms, regulations, certifications, and/or standards have been supported	Y3: development of 7 FL-RS supported policies (1 per country)	Y5: development of 14 FL-RS supported policies (2 per country)	National and regional bodies/institutions are supportive of the proposed policies etc. AGRA plays a significant role in influencing, supporting, and facilitating the policy process through stakeholder engagement and advocacy.
Output 3.1: Enhanced capacity of national institutions to enable investments in FL-RS	Output Indicator 3.1 Number of national and regional bodies/institutions provided with technical support, secondment, and/or policy analysis and inter-	AGRA implementation partner reports	No national and regional bodies/institutions have been	Y3: 50% of target bodies/institutions are provided with technical support,	Y5: 100% of target bodies/institutions are provided with technical support, secondment	National/regional bodies/institutions are in demand of support provided.

	intra-sector coordination tools and frameworks, and joint sector reviews		provided with support	secondment and/or policy analysis	and/or policy analysis	Assumes no other policy or regulation is passed to undermine the support provided. The exact number of target bodies/institutions will be defined with the National Programme Steering Committee in each of the target geographies
Project/programme co-benefit indicators						
Co-benefit 1 Increased food safety	Co-benefit indicator 1: Percentage of produce meeting quality standards	Farmer surveys AGRA implementing partner reports	<i>To be confirmed following initial baseline survey</i>	Y3: 25% Reduction in the volume of grain rejected due to quality concerns associated with aflatoxins/mycotoxins	Y5: 50% reduction in the volume of grain rejected due to quality concerns associated with aflatoxins/mycotoxins	Assumes no other impacts to food safety (positive or negative) outside of uptake of FL-RS <i>Please note that the final targets will be confirmed through an initial baseline survey</i>
Co-benefit 2 Increased support for women/youth entrepreneurs	Co-benefit indicator 2: Percentage of employment opportunities made available particularly for women and youth through the RE-GAIN programme	AGRA implementing partner reports	No dedicated employment opportunities for women and youth	Y3: 20% of employment opportunities created by the RE-GAIN are designated for women and youth	Y5: 40% of employment opportunities created by the RE-GAIN are designated for women and youth	Women and youth are able and willing to undertake employment in FL-RS in communities
Co-benefit 3 Reduced GHG emissions	Co-benefit indicator 1: GHG emissions (tCO ₂ e) reduced, avoided, or sequestered through improved land use practices	AGRA implementation partner reports FAO ExACT Tool Agro-Chain Greenhouse gas Emissions (ACE) tool	0	Y3: 48,615 tCO ₂ e	Y5: 88,438 tCO ₂ e Y10: 111,275 tCO ₂ e	The reduction in post-harvest losses will be sufficiently impactful to prevent further compensatory agricultural expansion and the associated emissions

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
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Activity 1.1.1: Conduct an awareness-raising campaign targeting smallholder farming communities focused on the impacts of climate change on post-harvest food losses and the availability of locally-appropriate and climate-resilient FL-RS.	An awareness campaign will be conducted in each of the target communities across the seven countries, using local institutions, chiefdoms and community networks to increase the understanding of postharvest food losses among farmers, local stakeholders in the agriculture value chain, and the public, thereby establishing a foundation for increasing demand for FL-RS. The nature of the campaign will be tailored to agricultural context each country, accounting for the target crops, the availability of local information dissemination systems (including mass media such as radio, television, digital platforms and social media, as well as local extension services), cultural considerations and local languages. The campaign will collaborate with local governments and farmer organisations, build on existing outreach systems to maximise reach and participation. AGRA will closely monitor the awareness campaigns, assessing impacts and adapting outreach strategies as needed.	- Develop an awareness-raising campaign for each target country. - Deliver awareness campaigns in target communities in each of the seven countries.	7 Awareness raising campaigns developed and delivered.
Activity 1.1.2: Train smallholder farmers on the use of improved climate-resilient FL-RS.	Training will be provided to educate smallholder farmers on improved crop management and storage techniques to reduce food losses and to maintain quality of produce, increase farmers' income by reducing losses and improving marketability, and improve supply of financial services and FL-RS to smallholders and other value chain actors. AGRA will procure the services of suitable local entities (for example, NGOs) to facilitate training in target communities. Training will then be provided through various channels — such as onsite demonstration of best practices, agricultural extension services, lead farmers and Village-Based Advisors (VBAs), farmer cooperatives, industry associations, and digital platforms — depending on the local context and the existing networks of the local service provider. The nature and content of the training will be tailored to each group. Training will include baseline topics, including an introduction to the RE-GAIN programme and the core concepts of food security, post-harvest food losses and the impact of climate change – building on and leveraging material from the awareness campaign (Activity 1.1.1). In addition to this baseline, technical training will be provided on specific subject matter tailored to the local context of each community.	- Develop training material for target crops in each country. - Engage local partners (including suppliers and agro-dealers) to identify and agree on training delivery models for each community. - Deliver training to target communities. - Integrate the use of FL-RS into extension services curricula at the local level - Establish demonstration/learning sites in each of the project target areas	14 Training methodologies (two crops in each of the seven countries). - Training delivered in 7 countries. - Two demos per year across each of the priority regions per country
Activity 1.1.3: Train agricultural traders and processors on complementary services for FL-RS	Building on the training of farmers, the programme will provide additional training to agricultural traders and processors on complementary services that support and maximise the effectiveness of FL-RS. This training will include the same baseline concepts, including an introduction to the RE-GAIN programme and the core concepts of food security, post-harvest food losses and the impact of climate change. Beyond that, the training will include on best-bet approaches to introduce FL-RS to smallholder farmers and the types of initial support required to enhance demand, tailored to the processing and trade of produce.	- Develop training material for complementary services in each target country. - Deliver training to agricultural traders and processors	Training delivered in 7 countries.
Activity 1.1.4: Implement a training of trainers programme targeting extension workers and agro-dealers.	In addition to the direct training of farmers under the programme, RE-GAIN will facilitate broader training opportunities through a training of trainers (ToT) approach, particularly for VBAs. Using the same service providers procured for Activity 1.1.2, key stakeholders who engage regularly with farming communities — such as extension officers, VBAs and agro-dealers — will be trained as trainers, focusing on the subject matter outlined for Activity 1.1.2. In addition to ensuring that trainers have a firm understanding of the training content, they will be trained on the	- Develop training material. - Identify potential trainers in local organisations and VBAs. - Host ToT workshops in each target country.	7 ToT workshops.

	practical approaches needed to effectively reach and engage communities, including on the setup of trainings and the use of demonstrations and promotional materials, as well as concepts of gender-sensitivity and youth engagement.		
Activity 1.1.5: Conduct scoping missions to identify and engage manufacturers of FL-RS equipment.	Develop an inventory of FL-RS equipment manufacturers and agro-dealers that can be engaged in developing local markets and enhancing the immediate availability of affordable FL-RS (establishing the basis for delivery models under Component 2). By mapping the market and building network of local manufacturers and importers/agro-dealers, this activity will enable a better understanding of current suppliers of FL-RS, and their production scalability potential. To ensure the reach of FL-RS to remote and rural areas, the programme will facilitate engagement and increased collaboration between solution manufacturers and local agro-dealers in those target areas, there ensuring availability of solutions, and their accessibility by local farmers. The scoping missions will include assessments of the current market in each target region to better understand the market needs and opportunities. This will include identifying opportunities for exchanges between African manufacturers and companies based in other developing nations to help introduce new technologies into the African market. Through these scoping missions, AGRA will facilitate knowledge and technology sharing and transfer between suppliers, farmers, and service providers.	- Host scoping mission in each target community and surrounds areas to identify equipment manufacturers and agro-dealers. - Develop a stakeholder map, linking stakeholders across various stages of the value chain.	7 Stakeholder maps, one for each target country.
Activity 1.1.6: Conduct demonstrations of locally-appropriate FL-RS	The programme will facilitate active demonstrations of short-listed FL-RS to target communities. This activity will build on the relationships developed under Activity 1.1.5, partnering with vendors for the implementation of product demonstrations, leveraging the incentives for vendors to increase the sales of their solutions through the programme-supported outreach. Where possible, demonstrations will be done in conjunction with the training under Activity 1.1.2 to improve efficiencies and ensure full connectivity between the theoretical training and practical demonstrations.	- Engage project partners in target countries to develop demonstration strategies. - Conduct demonstrations of non-physical solutions - Conduct demonstrations of physical solutions	Demonstrations of at least one physical and one non-physical intervention in each target community per year. 410 total demonstrations over 5 years
Activity 1.1.7: Provide technical assistance to food processors to facilitate a shift to whole-grain flour production	In addition to the FL-RS approaches that will be promoted above, the programme will provide technical assistance to promote the transition from refining grains to wholegrain flour production. As depicted in Figure 4, wholegrain production can produce almost 50% more bread from the same quantity of unprocessed wheat, helping reduce the “hidden food loss” associated with over processing. The intervention will target food producers – millers in particular – providing technical assistance along several themes, including: i) developing MSME capacity; ii) raising awareness of market opportunities for wholegrains; iii) support for technology transfers; and iv) facilitating access to finance. The activity will form part of a broader initiative that is promoting wholegrain food for use in school meal programmes, which the RE-GAIN programme will leverage and complement.	- Develop awareness raising material on wholegrain production - Host capacity development workshops in target communities. - Connect processors with technology providers for improved processing equipment suited to wholegrain production	At least one capacity development workshop in each target community.
Activity 1.2.1: Provide technical assistance to support the structuring of value chains and coordination between market actors	The technical assistance under this activity will facilitate market linkages between smallholders, MSMEs and potential buyers – such as retailers, processors and exporters – using the consortia model as tested and proven by AGRA. The process will start with an assessment of the existing market in each target community, drawing on and expanding the initial market assessment conducted	- Conduct a market assessment (including identifying gaps) in each target community - Host technical workshops in target communities to structure	- Value chain map and gap assessment for each community. - At least one technical workshop per community

	during preparation of the RE-GAIN programme (see Annex 2). This extended market analysis will map out key actors in the agricultural value chains at the local level, including potential institutional markets that are not yet fully accessible to smallholder farmers, but that have potential to act as reliable off-takers. Gaps in the value chain will be mapped, drawing connections between various actors to identify opportunities for filling gaps. Based on this stakeholder mapping, the programme will provide technical assistance to local actors to structure value chains for different crops and institutional markets. Actions to develop linkages will specifically consider existing constraints faced by women in the sector and identify and promote opportunities for women to actively engage in the market. This activity will culminate in clear action plans for structuring value chains for target crops in each community, including an outline of business development opportunities (see Output 2.1) to fill gaps in the market. The implementation of value chain development will be further supported by establishing local coordination mechanisms and supporting the establishment of cooperatives where needed to facilitate improved coordination between actors.	- improved value chains and train local stakeholders on their implementation. - Develop action plans for newly identified value chain development opportunities.	At least one action plan per community.
Activity 1.2.2: Facilitate linkages between farmers and institutional markets such as school feeding programs, food processors and other social safety programs.	Drawing on the structures identified under Activity 1.2.2, the RE-GAIN programme will engage key actors in the agri-value chain to link smallholder farmers to institutional markets such as school feeding programmes, food processors and other social safety programmes. AGRA will support local farmers and cooperatives in connecting with other actors in the value chain and provide technical assistance to establish formal arrangements that will secure off take agreements for produce that meets the quality standards of the institutional market. This will include support for accessing institutional markets for wholegrain flour.	- Host roadshows to promote value chain development and bring together agri-value chain actors. - Facilitate meetings between agri-value chain actors.	- Agri-value chain roadshows in each target community. - Offtake agreements signed in each target community.
Activity 2.1.1: Train new FL-RS service providers (MSMEs, cooperatives, incl. women- and youth-led initiatives)	The RE-GAIN programme will identify and train local MSMEs and cooperatives to develop FL-RS service provision operations in the target communities. This training will prioritise women- and youth-led initiatives to maximise the social benefits and build resilience among these vulnerable groups. The programme will adopt a rigorous process of scouting and in-depth assessment using standardized tools of the potential service providers and food processors complemented by intensive vendor support to make sure operations and maintenance skills for the equipment are transferred. The identification of suitable entities will be supported by government (decentralised and local) authorities as well as through digital platforms, social media, industry players where potential service providers are connected or enrolled, and via the scoping exercises under Activity 1.2.1. Once service providers are identified and enrolled in the programme, a rigorous assessment of their capacity and skills will be conducted, before organising intensive “boot camps” to train these entities on the FL-RS service provision. These training camps will cover several technical topics related to FL-RS service provision, including: i) proper service management and process optimization; ii) the safe, effective, efficient and sustainable operation of the equipment; iii) compliance with international and local quality standards for post-harvest services; and iv) the management of the fee-for-service model. In addition to the technical training on FL-RS service provision, entities will be supported to develop their business into a sustainable entity. This will include training on basic business practices (such as financial literacy, cash flow management, record keeping, and risk management) and	- Host scouting missions to identify local MSMEs and cooperatives in each target community - Develop capacity development training material - Engage local vendors to support in training initiatives - Host training bootcamps in target community. - Host exchange visits between nearby communities	- Shortlist of suitable MSMEs and Cooperatives - Agreements with local vendors signed to facilitate training - At least one training bootcamp in each target community - At least one exchange visit per community.

	partnership development support, as well as training on value-add services such as certifications processes to enhance product value and market access. Guidance will also be provided on how to access to finance through innovative de-risking schemes under Output 2.2. Vendors of the FL-RS equipment will be engaged in the implementation of this activity, incentivising the recruitment of staff and the increase of outreach activities through VBAs and other last mile delivery channels. This model is not only sustainable in the long run but also creates an incentive for end users to scale up the technology in the very initial stages. Exchange visits will also be arranged to share experiences between service providers operating in nearby villages.		
Activity 2.1.2: Provide business development support and market intelligence	To complement the development of MSME service providers, the RE-GAIN programme will support business development among manufacturers, importers and distributors of FL-RS to boost local markets. This will include providing market intelligence to increase marketing, distribution and after-sales services to develop the supply chain for FL-RS products. Entities will also be supported to develop partnerships with other actors in the supply chain, as well as to engage with innovative de-risking schemes piloted under Output 2.2.	▪ Prepare business development support material, including compiling market intelligence reports ▪ Host training workshops targeting MSME and cooperative service providers.	▪ At least one market intelligence report per country. ▪ At least one training workshop per community.
Activity 2.1.3: Capacity and market development for all supply-side market actors	In addition to the targeted support for FL-RS suppliers and service providers, RE-GAIN will provide general capacity development support to actors across the FL-RS supply chain to further develop the local supply market for FL-RS. The nature of this support will vary depending on the context of each target country and the actors involved but could include the convening of actors and development of partnerships, establishment of cooperatives to optimise the procurement and shared use of FL-RS among farmers, market mapping and dissemination of market intelligence reports, and building network of local manufacturers and importers/agro-dealers. Implementation of the supply-side market development support will include linkages with the agri-value chains market development initiatives under 1.2.1, sharing resources to optimise cost efficiency and co-develop both sides of the market.	▪ Convene actors and facilitate the development of partnerships. ▪ Support the establishment of new cooperatives ▪ Develop and disseminate market intelligence reports ▪ Build a network of local manufacturers and imports/agro-dealers.	▪ Deliverable will differ per country based on local needs, but will include at least one market development tool per target community.
Activity 2.2.1. Facilitate the development and deployment of catalytic procurement models for FL-RS focussing on women and youth collectives and enterprises as key beneficiaries.	Implement three financial models, with varying levels of complexity and potential impact. Models are split into two distinct categories: interventions at a smallholder level; and interventions at a youth group/co-operative level. An overview of each of the three financial models is provided in Annex 2: Feasibility Study. In all instances, the models draw on the outputs of the scoping exercises under Activities 1.1.5 and 1.2.1 to identify the potential actors and tailor the models to the local context. The models leverage partnerships between these actors to share risk and facilitate and incentivise the development of robust markets. RE-GAIN will play a facilitation role in the establishment of the financial models, using smart-subsidies for farmers to stimulate demand in the market while helping derisk the investment models.	▪ Draft an operations manual for each financial model. ▪ Identify suitable partners in each community ▪ Facilitate engagements with partners to agree on which financial model is most suitable to the local context and needs. ▪ Establish smart-subsidy programmes based on agreed modalities by signing of MoUs with partners ▪ Disburse subsidies based on agreed triggers.	▪ At two financial mechanisms per community – one based on models 1 or 2, and one based on model 3.
Activity 2.2.2. Support inclusion of FL-RS in climate-resilient input packages provided by local financial institutions	The RE-GAIN programme will provide technical support to facilitate embedding the cost of FL-RS (e.g. hermetic technology) within the financing arrangement for input subsidies to allow farmers to try out the technology and ask for inclusion in their input loan package in the following years.	▪ Prepare business cases for directing agricultural subsidies to FL-RS	▪ Briefing papers on the integration of FL-RS into

		- Facilitate meetings with local institutions to advocate for inclusion of FL-RS in agriculture subsidy programmes - Provide technical training to decision makers on how to structure FL-RS targeted subsidies.	subsidy programmes for each target country.
Activity 2.2.3. Structure prefinancing partnership arrangements that include FL-RS	To further unlock credit for accessing FL-RS, the RE-GAIN programme will provide technical assistance to key value chain players — for example off-takers, input suppliers and farmers — and financial institutions to support the structuring of prefinancing partnerships that will derisk the provision of finance. Assistance will include convening key actors (through workshops, meetings, or other collaborative platforms), and training/capacity development that will help restructure banks' perceptions of risk as well as directly reducing actual risks. This will, in turn, release additional credit for financing access to FL-RS. This model, which is widely promoted by AGRA, has been successfully implemented in several of its focus countries, including Burkina Faso, and can be replicated in other countries.	- Facilitate meetings between key value chain players - Host training workshops for finance providers	At least one pre-financing arrangement developed per target country.
Activity 3.1.1: Support the revision of policies in support of enabling FL-RS investments, including tax exemptions, reduced import duties, certification and standards for FL-RS quality	AGRA will collaborate with government entities to revise and update key policies which are not responding to the current and future food losses and climate challenge scenario. This will include engaging relevant government ministries and agencies to review existing regulations and policies and proposing amendments to facilitate the development of local markets and support to farmers for the uptake of FL-RS. This will include introducing/strengthening national policies that provide financial incentives such as tax exemptions, subsidies, or reduced interest rates for FL-RS investments, making it more attractive for private and public sector investments. Evidence generated and lessons learned in FL-RS from target communities and countries will be to develop capacity, influence policy and promote best practices at a scale. AGRA will also actively participate in policy advocacy initiatives, working with stakeholders to ensure that proposed changes support the long-term success of FL-RS initiatives. In addition to support for policy revision, AGRA will support local entities in implementing existing policies and regulations that have potential to support the uptake of FL-RS, but are currently not fully enforced or supported. These actions will result in a supportive policy environment that enables the successful implementation of the RE-GAIN programme and widespread adoption of FL-RS solutions.	- Review existing regulations and policies - Draft policy briefs for amendments to facilitate the development of local markets and support to farmers for the uptake of FL-RS. - Engage relevant government ministries and agencies to sensitise them to the need for policy revision as identified in the briefs. - Participate in policy advocacy initiatives in target countries. - Train local entities on the implementation of new and existing policies.	- At least one policy brief per country. - At least one training workshop per target district in each country.
Activity 3.1.2: Support institutional capacity building of key local entities to support the scaling of successful FL-RS models.	AGRA will provide technical assistance to local and national government entities engaged in crop value chains (directly or indirectly) to build their institutional capacity to support the scaling of successful FL-RS models and policies in their countries. This will include actively involving central and local government through the establishment of formal partnerships and facilitating regular meetings and consultations to ensure alignment of the RE-GAIN programme with national and regional development priorities. These partnerships will be complemented by technical assistance, including guidelines, targeting tools and technology delivery mechanisms, to develop the analytical capacity of partner institutions on the topics of climate change and the	- Host training workshops at the national level to build their institutional capacity to support the scaling of successful FL-RS models and policies in their countries - Host training workshops at district level to build institutional capacity for FL-RS support, including	- At least one national workshop per country - At least one local-level workshop per district in each country - At least one decision support tool developed per country

	contribution of FL-RS to GHG reduction and food security — enabling these metrics to be tracked and reported on in relation to the NDCs — as well as for the scenario-based prediction of mycotoxins during post-harvest period under changing climate. RE-GAIN will also work with governments and other development partners to promote private sector investment in post-harvest infrastructure, including storage facilities, processing units and cold chains, as well as supporting regulatory bodies to develop and implement quality standards and certification processes for FL-RS products and services, ensuring they meet required levels of reliability and efficiency. Support will also be provided to enhance the capacities of extension services, and climate advisory services to monitor food loss in relation to climate conditions and the impact of FL-RS in reducing loss, as well as identifying and reporting on opportunities for upscaling and replication. Finally, the programme will leverage on other investments to create synergies with other players in building institutional capacity of governments and the private sector in FL-RS, which may include enhancing coordination among country actors and institutions dealing with post-harvest loss under changing climates.	- monitoring and evaluation and ongoing extension support. - Develop guidelines and tools to support local institutions in scaling and monitoring FL-RS - Facilitate regular meetings and consultations to ensure alignment of the RE-GAIN programme with national and regional development priorities - Train key stakeholders on analytical approaches to climate-responsive post-harvest management.	At least one meeting per year convened in each country.
Activity 3.1.3: Develop a business case for investments in FL-RS manufacturing, supply, service provision and financing to promote successful FL-RS business models for scaling-up and replication	A key step in unlocking investments and creating sustainable markets for FL-RS in Africa revolves around creating solid business cases that account for the needs of all actors – including manufacturers, suppliers, financiers, agro-dealers, service providers and farmers. RE-GAIN will, therefore, research and compile case studies of successful FL-RS initiatives implemented through the programme, as well as other best practices observed across the region and develop models and approaches for scaling and influence. This research will form the foundation of the business case for FL-RS, detailing best practice and benefits of various FL-RS related activities, including equipment selection, business model, profitability analysis, and financing models. The business case will be packaged into various communication materials, including reports, videos, and presentations, to share these models with potential investors, policy-makers, and other stakeholders, encouraging scaling-up and replication.	- Conduct an assessment of pilot initiatives and programme outputs to identify best practices - Compile case studies of successful FL-RS initiatives - Develop business models - Package business models into various communication materials.	At least two business models per country packaged into locally relevant media for distribution.
Activity 3.1.4: Implement a comprehensive monitoring, evaluation and learning system for FL-RS	Develop a robust monitoring, evaluation and learning (MEL) framework that includes clear performance indicators, data collection methods, and regular reporting schedules. This system will build on the foundation of existing MEL frameworks within the AGRA system to track the progress of programme activities, assess the impact of interventions, and identify areas for improvement. The MEL framework will also establish mechanisms for continuous learning by regularly reviewing M&E data, conducting periodic evaluations, and gathering feedback from stakeholders. These insights will be used to make informed adjustments to the programme, ensuring that it remains relevant and effective. The knowledge generated will be shared regionally through AGRA's networks to promote the scaling of the successful models across Africa and beyond. In the spirit of south-south cooperation, the programme will also coordinate with other projects to create synergies, identifying areas of overlap and collaboration. This will include developing joint action plans, sharing resources, and coordinating activities to maximize impact and avoid duplication of efforts.	- Develop M&E tools - Collate lessons learned - Generate knowledge products for sharing lessons learned.	- M&E tool tailored to each country. - Knowledge products for each country.

E.7. Monitoring, reporting and evaluation arrangements

206. The M&E structure for RE-GAIN has been developed to be in-line with AGRA's existing M&E structure, to ensure a cohesive approach, as well as with GCF's policy requirements such as Monitoring and Accountability Framework and Integrated Results Management Framework. It has also been designed to target the GCF-supported project Outcome and Output key performance indicators (KPIs; listed above) and the broader programme performance, respectively. A baseline survey will be conducted during the inception phase of the programme, and AGRA will update the performance indicators where applicable based on the results alongside the baseline results as part of the inception report. The performance monitoring will be conducted on an ongoing basis by a dedicated M&E officer with close oversight from the AGRA M&E Team and Board. The GCF-supported project KPIs will be monitored during the implementation period as a subset of the broader performance monitoring of the programme over the long-term. The responsibilities of the M&E officer will include:
- Data collection and performance measurement.
 - The M&E officer will work closely with each project intermediary, to aggregate data on the non-financial fund metrics obtained directly from the intermediaries. This will include data to support the development of site-specific baselines; Economic activity; Market share; Production; Sales volume and value. Additional data will be periodically and systematically aggregated by intermediaries during implementation relating to production, sales volumes and value, human resources, among others.
 - Periodic site visits will be conducted to assess and verify the results presented by the intermediaries.
 - When appropriate and deemed necessary, external consultants may support the data collection and measurement process.
 - Performance assessment and recommending corrective actions: Based on the predefined output and outcome targets and metrics, corrective actions may be defined by engaging with the intermediary.
 - Reporting: Performance against the Outcome and Output KPIs are reported to the GCF on an annual basis.
207. The monitoring and reporting on the Outcome and Output KPIs will be integrated into the broader performance monitoring system for AGRA. AGRA has a comprehensive MRV system that extends from grantees, to country, region, corporate level review and verification processes supported by an online Management Information System (MIS). AGRA also has standard Data Quality Assurance (DQA) mechanism to ensure quality performance data are collected and reported. AGRA will integrate RE-GAIN into the existing MRV system. The M&E system for RE-GAIN will be validated and finalised by the AGRA Board, in collaboration with the AGRA M&E Team, during the programme's inception and will continue to be updated as the programme progresses. The AGRA Board and M&E Team will work with the M&E officer to evaluate the interplay between project indicators and results to assess the effectiveness of the M&E system and amend it where appropriate.

Evaluation

208. The performance will be independently evaluated against the Outcome and Output KPIs at the mid-term and final stages of the project implementation period, specifically in years 3 and 5, as per GCF evaluation policy. These evaluations will be conducted by an independent evaluation specialist, under the oversight of AGRA.

Financial Reporting

209. AGA will track the program's finances and will produce semi-annual reports that summarizes material developments of these. AGRA will develop valuation processes and procedures for the underlying portfolio investments and ensure these are done in accordance with Accounting Standards Codification (ACS) 820-10 on a semi-annual basis on June 30th and December 31st.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigation measures

Selected Risk Factor 1		
Category	Probability	Impact
Technical and operational	Low	Medium
Description		
Lack of programme ownership by beneficiaries and target governments leads to weak uptake and/or sustainability of FL-RS		
Mitigation Measure(s)		
Detailed stakeholder engagement has been conducted throughout the programme design stage, and will continue throughout implementation to ensure that interventions are fully owned by and meet the individual needs of the target countries. This consultative process will ensure that the probability of this risk remains low.		
Selected Risk Factor 2		
Category	Probability	Impact
Technical and operational	Medium	High
Description		
Financial crisis caused by unstable macro-economic variables impacts the market development and increases the risk profile of financial mechanisms.		
Mitigation Measure(s)		
The approach to designing the financial models was built on a foundation of a detailed analysis of local markets and economic fundamentals, ensuring that the mechanisms are robust to withstand fluctuations in the macro-economic situation. Multiple derisking approaches have been integrated into the market development models, which will provide a buffer to reduce the potential impact of economic fluctuations on the effectiveness of the programme.		
Selected Risk Factor 3		
Category	Probability	Impact
Governance	Medium	Medium
Description		
Political instability in some target countries leads to disruptions to the implementation of project interventions, or ineffectiveness of enabling environment actions.		
Mitigation Measure(s)		
During times of political instability, AGRA will increase engagement with national stakeholders to ensure investments and programmes are not disrupted, reducing the potential impact of political instability on programme outcomes.		
Selected Risk Factor 4		
Category	Probability	Impact
Technical and operational	Medium	High
Description		
Lack of willingness from end users to adopt the FL-RS reduces market transformation potential and long-term impacts of the project.		
Mitigation Measure(s)		
The RE-GAIN programme will engage in extensive awareness campaigns and capacity development to ensure that farmers fully understand the threats posed by climate change on post-harvest loss, the benefits of FL-RS for reducing losses, and the financial viability of FL-RS solutions. Combined with supply-side investments that increase availability and affordability of FL-RS, these actions will reduce the probability of this risk to low.		
Selected Risk Factor 5		
Category	Probability	Impact
Credit	Medium	Medium
Description		
Credit default and end beneficiaries' failure to repay the loans and meet their contractual obligations may reduce the effectiveness of financial models and erode trust in the system.		
Mitigation Measure(s)		
The credits are administrated by financial institutions with excellent knowledge of their customers and the market within which they operate. The financial models have also been designed with multiple levels of derisking, including smart subsidies to reduce burden on end beneficiaries (which will increase likelihood of being able to pay), as well as broad market development in agriculture value chains and capacity development of farmers and MSMES. Assessments will also be conducted by AGRA to ensure alignment of credit providers with policies and procedures on compliance and AML		
Selected Risk Factor 6		
Category	Probability	Impact
Governance	Low	Medium
Description		
Regulatory constraints impede the implementation of the project interventions in individual countries.		
Mitigation Measure(s)		
The interventions have been designed to account for legal frameworks and internal policies to ensure that the activities align with regulations in each country, ensuring the project can be implemented within the programme timeline.		

G. GCF POLICIES AND STANDARDS

G.1. GCF Policies and social risk assessment

Risk Categorisation

210. The programme is assigned a Category B risk rating as per the GCF Revised Environmental and Social (E&S) Policy definition. The rationale for the risk rating is as follows:
- The Programme will target established smallholder farms and MSMEs and as such will not be associated with any greenfield development or land acquisition;
 - The Programme is designed to facilitate the adoption of FL-RS solutions by smallholder farmers and MSMEs through existing supply chains in the countries. As such, the Programme will not directly be involved in the manufacturing, transportation and/or installation of the FL-RS solutions;
 - The E&S risks associated with the Programme are mainly associated with the regions targeted by the Programme in each in-scope country, the contextual circumstances of the existing smallholder farms targeted to participate in the interventions and the in-country partner(s) chosen to facilitate the interventions; and
 - The risk assessment conducted on the Programme highlighted limited adverse E&S risks and impacts that are few (see *Programme E&S Risk Assessment*) and can be readily addressed through actionable mitigation measures (see *Programme EandS Risk Management Framework*).

Programme E&S Risk Assessment

211. In order to meet the GCF EandS due diligence requirements expected on a Category B type project, AGRA conducted two desktop assessments, namely a country and region contextual EandS risk analysis for each in-scope country and focus crops and an intervention level EandS risk assessment taking into account the envisaged Programme activities. The key EandS risks stemming from these assessments are summarised as follows:

Overall Programme E&S Risks

212. i) AGRA procures the services of local NGOs and/or private sector actors to support the building of the intervention ecosystem (i.e., to build the connections between manufacturers, agro-dealers, and the end recipients such as smallholder farmers and MSMEs). Reputational risks may exist associated with the Programme partners chosen due to poor E&S performance and/or reputation.
213. ii) AGRA will preferentially identify and partner with locally available manufactures of FL-RS technologies for inclusion into the Programme. Reputational risks associated with manufactures due to EandS related risk factors such as:
- Sub-standard labour and working conditions including potential for poor occupational health and safety practices;
 - Limited pollution control measures and waste management; and/or
 - Road transport health and safety risks associated with transporters (internal/external) of FL-RS technologies to agro-dealers.
214. iii) AGRA will partner with locally available financial institutions to provide financing at preferential rates to either manufacturers or smallholder farmers / MSMEs. Reputational risks associated with financial institutions due to poor and/or unethical lending processes (especially those related to microfinance institutions) which may lead to smallholder farmer indebtedness.

Contextual E&S Risks

215. i) Conflict areas were identified in certain regions in Burkina Faso and Ethiopia where the Programme is intended to be implemented may lead to challenges in meeting Programme objectives including potential health, safety and security risks to Programme personnel.

216. ii) Depending on the country, maturity and oversight of labour regulations, reputational risks associated with labour and working conditions may exist on the target smallholder farms and/or MSMEs. These may include the presence of child and/or forced labour practices on these farms, poor health and safety practices, non-compliance with labour laws and non-alignment with the requirements of international standards such as those set out in the ILO and IFC PS 2.
217. iii) AGRA conducted a desktop Indigenous Peoples (IPs) screening and vulnerability assessment of the Programme. Of the seven countries assessed, no recognized IPs were identified in Zambia and Malawi. However, potential IPs were identified in specific regions within Burkina Faso, Ethiopia, Kenya, Tanzania, and Uganda. There are no circumstances in which the Programme will result in any form of displacement of IPs, affect access to land or natural resources, nor will the Programme interventions be likely to include activities that significantly alter the physical or social context of the local areas in which the interventions will be implemented. However, the identification of the IPs underscores the need for careful and proactive consideration of any risk that might arise during program implementation and requires tailored approaches to address the unique circumstances and challenges of indigenous communities in the identified areas. In this regard, the Programme will proactively work with IPs and their representatives to ensure that the programme (i) does not cause any significant harm to their way of life and livelihoods and (ii) enhances or complements their existing livelihoods. Further, using appropriately translated ESS and programme information to local languages, IPs will also be continuously engaged before and during programme implementation, with feedback and suggestions from leaders and representatives used as proactive measures to ensure a successful implementation of the programme.

Intervention Level E&S Risks

218. i) The Programme includes the introduction and promotion of novel FL-RS technologies. The innovative technology introduced to small holder farmers and/or MSMEs may lead to unintended health, safety and/or environmental risks such as injury or increased pollution generation as a consequence of recipients not being adequately trained on their correct use and/or waste disposal.
219. ii) AGRA procures NGOs to create awareness and demonstrate the FL-RS in collaboration with manufacturer and agro-dealers. These trainings are conducted at selected smallholder farms. NGO staff may not have adequate technical competencies to provide training on unintended health, safety and/or environmental risks associated with the FL-RS technologies. NGOs may deliver training that is not culturally suited to the target audience and may be biased from an inclusion and gender perspective.

Programme E&S Risk Management Framework

220. AGRA has in place a centralised Environmental and Social Management System (ESMS) that will be applicable to all interventions initiated in AGRA's focus countries. The ESMS is designed to ensure that potential negative E&S impacts of AGRA interventions are minimized, while positive effects on communities and the environment are enhanced. It follows the guidance of IFC Performance Standard 1 and includes an EandS policy and procedures, identification of risks and impacts, management programs, organisational capacity and competency, emergency preparedness and response, stakeholder engagement, grievance management, monitoring, review and reporting.
221. In addition, AGRA has developed a Programme E&S Action Plan (ESAP), which provides a framework for E&S action across the entire RE-GAIN Programme. Once the specific communities and beneficiaries have been selected, the Programme ESAP will be adapted where appropriate to be specific to the E&S risks and impacts associated with the individual country, region, crop and combination of FL-RS applicable to the specific intervention envisaged. The Programme ESAP sets out mitigation measures for each of the risks identified in the E&S risk assessments conducted.
222. Full details on Programme E&S risk identification and mitigation are provided in Annex 6 of the Funding Proposal.

G.2. Gender assessment and action plan

223. To ensure that the RE-GAIN Programme is designed, financed, implemented and monitored in a gender-responsive and socially inclusive manner, a comprehensive Gender Assessment and Gender and Inclusion Action Plan has been developed (Annex 8). These outputs comply with the GCF's Gender Policy, as well as AGRA's own Gender and Inclusion Strategy, ESMS Manual and Safeguarding of Vulnerable Persons and Youth Policy.
224. RE-GAIN's Gender Assessment considers the various factors (including those exacerbated by climate change) that influence gender inequality in each of the seven target countries' agriculture sectors and identifies opportunities for women, men and youth to contribute to inclusive resilience building within post-harvest value chains. These opportunities are captured in a Programme-wide Gender and Inclusion Action Plan, which defines activities, indicators and targets specific to women and youth in their capacities as smallholders, cooperative networks, MSMEs and FL-RS vendors.
225. The Gender Assessment combines extensive literature review, carried out on a country-by-country basis on the key contextual factors that enable and / or impede gender equality progress at a national -level, with primary findings drawn from 28 national and local stakeholder engagements spread evenly across the seven countries.
226. Using these information sources to examine the disparities (and underlying drivers) between women, men and youth within the context of agriculture and climate change in each of RE-GAIN's countries, several clear, common and consistent themes on gender and inclusion emerge: Improved, inclusive access to finance for women and youth is a fundamental enabler for increased and more equitable participation in FL-RS. The upfront cost of many solutions is a primary barrier to adoption by women and youth. It is therefore critical that inclusive and diverse finance options and advisory services are made available to these groups. Inclusive and accessible training and information on FL-RS for women and youth is necessary to empower and capacitate them as farmers and business owners. This includes both technical and financial advisory support on: FL-RS tools, equipment and structures; the benefits of changing current practices or preferences; and on financial management and marketing – including options available, budgeting, saving, repayment processes, pricing, market quality standards and control etc. Such training must be designed in a gender sensitive and highly context specific manner. This must be coupled with sensitization of male community members, to overcome persistent disparities between women and men's access to financial, technical, infrastructural and capacity building resources. The deliberate and targeted appointment of extension service delivery partners is fundamental to achieving many of these recommendations, meaning the selection of RE-GAIN's implementation partners must incorporate a specific technical competency on demonstrated experience of working with women, youth and other marginalised groups. Adopting a dual top-down bottom-up approach to amplify women and youth's voices on FL-RS is needed to create sustainable shifts in the way women and youth engage in, and benefit from, RE-GAIN's support. This means that on-the-ground, community centred efforts must be coupled with more inclusive, enforceable FL-RS policies. These recommendations, together with RE-GAIN's overarching goal, outcomes and outputs, were used to develop the Gender and Inclusion Action Plan which includes both qualitative and quantitative targets over the five-year period. Implementing the measures set out in this Plan will result in 1,284,321 women (~50% of the total beneficiaries) benefitting from RE-GAIN's support. A dedicated RE-GAIN Gender Expert will oversee the delivery and M&E of the Gender and Inclusion Action Plan, and ensure continued compliance with related processes, including application of the grievance mechanism, safeguarding protocols and ESMS. The cost of this expert has been budgeted for the full five years, along with related expenses required to deliver all activities.

G.3. Financial management and procurement

227. AGRA has a comprehensive financial management system designed to ensure accountability, transparency, and efficient use of resources. Below is an overview of AGRA's financial management structure, including financial monitoring systems, financial accounting, auditing, and disbursement

methods. All the structures listed below will be applied throughout the implementation of the RE-GAIN programme.

1. Financial Monitoring Systems

228. AGRA transitioned from Navision to an integrated system - Oracle system. It employs robust financial monitoring systems to oversee the management and utilization of funds. These systems are designed to ensure that funds are used effectively and in line with AGRA's strategic goals. Key components include:

- **Budget Monitoring:** AGRA uses a budgetary control system that monitors expenditures against approved budgets. This allows for real-time tracking of financial performance and ensures that any variances are identified and addressed promptly.
- **Financial Reporting:** AGRA's financial monitoring includes regular financial reporting. These reports provide detailed insights into the financial health of the organization and are used by management to make informed decisions. The reports are typically produced on a quarterly basis and include budget variance analysis, fund utilization reports, and financial forecasts.
- **Grant Management System (AMIS):** For managing funds disbursed through grants, AGRA uses a dedicated grant management system-AMIS. This system tracks grant disbursements, monitors compliance with grant terms, and ensures that grantees are using the funds as intended.

2. Financial Accounting

229. AGRA's financial accounting practices are designed to ensure accuracy, consistency, and compliance with international accounting standards. The key elements include:

- **Accrual and Modified Cash Accounting:** AGRA uses both accrual and modified cash accounting system, which records financial transactions when they are incurred and in some cases its only when cash is exchanged as this is a requirement for some of our donors. This method provides a more accurate picture of the organization's financial position.
- **Internal Controls:** AGRA has a set of internal controls in place to prevent and detect errors or fraud. These controls include segregation of duties, authorization of transactions, and reconciliation of accounts.
- **Financial Software:** AGRA uses Oracle financial management software to record and manage its accounting activities. This software integrates various aspects of financial management, including accounts payable, accounts receivable, general ledger, and fixed assets management.

3. Auditing

230. AGRA places significant emphasis on auditing to ensure the integrity of its financial management. The auditing process includes:

- **Internal Audits:** AGRA conducts regular internal audits to assess the effectiveness of its internal controls, compliance with policies and procedures, and the accuracy of financial records. The internal audit function reports directly to the Audit Committee of the Board to maintain independence.
- **External Audits:** In addition to internal audits, AGRA is subject to annual external audits conducted by independent audit firms. These audits assess the accuracy of AGRA's financial statements, compliance with relevant regulations, and the effectiveness of its financial management systems. The external audit reports are shared with stakeholders, including donors and board members, to ensure transparency.
- **Audit Committee:** AGRA has an Audit Committee that oversees both internal and external audit processes. The committee is responsible for reviewing audit findings, ensuring that management takes corrective actions, and making recommendations to the Board on matters related to financial governance.

4. Disbursement Structure and Methods

231. AGRA's disbursement structure is designed to ensure that funds are distributed efficiently and in accordance with donor agreements and organizational objectives. The key elements include:

- **Procurement:** AGRA provides support through the procurement of various partner organizations/implementing agencies, including governments, research institutions, and NGOs. AGRA follows stringent procurement processes to ensure value for money. All procurement activities are conducted in accordance with AGRA's procurement policy, which includes competitive bidding, vendor evaluation, and contract management. Project payments are typically made in tranches, contingent upon the meeting of specific deliverables and reporting requirements. This ensures that funds are used effectively and for their intended purposes.
- **Direct Payments:** For certain projects and operational costs, AGRA makes direct payments to vendors or service providers. These payments are processed through AGRA's financial management system, ensuring proper documentation and authorization.
- **Banking and Treasury:** AGRA's treasury management includes maintaining relationships with banking partners, managing cash flows, and ensuring that funds are available to meet the organization's financial obligations. AGRA uses secure and reliable banking systems to process disbursements and manage its funds.

AGRA Procurement Process Overview.

232. AGRA has a fully resourced procurement unit with clear policies and procedures that meet best practices. The procurement policy and procedures manual are approved by the board and clearly outlines the detailed processes to be followed and the approval matrix.

233. As a policy requirement, we follow competitive processes in sourcing goods and services for most of our needs. The procurement process is fully automated starting from budget approval all the way through to contracting.

Key steps in the procurement process

- **Planning and Budgeting** - Initiated by user units through annual budgeting and procurement planning.
- **Vendor Management** - Handled by the procurement unit and includes vendor due diligence, prequalification, registration and performance management.
- **Requisitioning** - User units initiate requests through AGRA's automated SCM management system.
- **Sourcing** – Procurement initiates sourcing through AGRA's automated SCM management system. This includes evaluation of proposals/quotations by the Technical Evaluation Panels, negotiations and review of proposed awards by the Procurement Committee, Executive Committee and the Board as per the approval matrix.
- **Contracting** – this involves drafting of contracts, sharing with proposed consultants/vendors and signing of the agreements. This is also done through AGRA's automated SCM management system.
- **Contract management** – this includes management of all contracts by the requesting units, vendor payments, and performance management.

Escrow Accounts

234. During the implementation of the financial models under Output 2.2, escrow accounts will be used to hold funds until specific conditions have been met. In these instances, AGRA will open an account either at the bank of the supplier, at the partner financial institution, or at its own bank (depending on the specific agreement reached with the supplier and FI). This will be a USD denominated deposit account to which AGRA is a signatory. Depending on the agreements reached, the other parties (suppliers, financiers) will be included as co-signers as needed. Funds can only be released if the trigger in the agreement has materialized.

Governance and Oversight

235. The roles and responsibilities are clearly outlined in the procurement policy and procedures manual. The Board is responsible for reviewing and approving the policies. The President appoints the Procurement Committee comprising of nine members including the Chair, Vice chair, Secretary and Members who are responsible for providing oversight and ensuring that the procurement system is fair, transparent, competitive, and cost-effective. Internal Audit Function provides independent review and assurance on risk management, controls, and governance processes.

Procurement Risk Assessment and management

236. AGRA has an elaborate Risk Management Framework that is overseen by the Enterprise Risk Management Unit. AGRA defines procurement risk as the possibility that an event may impact its ability to achieve its objectives, either adversely or favorably. To address this, AGRA's risk management process involves identifying, assessing, measuring, and managing procurement risks to ensure they remain within acceptable tolerance levels. This proactive approach aims to enhance transparency, accountability, and stakeholder confidence in the procurement process. The procurement unit and other units have risk registers that are updated on quarterly basis.

Other policies

237. AGRA has several other policies that complement the procurement policy. These include Human Resources Policy, Code of Conduct Policy, The Whistle-blower Policy, Anti-Fraud Policy, and Anti-Money Laundering and Counter Terrorism Financing policy. To gain further insight into our procurement procedures, please refer to our procurement policy document embedded below.

G.4. Disclosure of funding proposal

☒ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 EandS document corresponding to the EandS category (A, B or C; or I1, I2 or I3):
 - [\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☒ Environmental and Social Management System (ESMS)
 - ☐ Others (Please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People’s Plan, Land Acquisition Plan, Etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 LEGAL DUE DILIGENCE (REGULATION, TAXATION AND INSURANCE)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☒ Annex 16 Map(s) indicating the location of proposed interventions
- ☒ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☒ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)⁴⁵
- ☒ Annex X Other references

* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents

⁴⁵ Annex 22 is mandatory for mitigation and cross-cutting projects.

PRIMATURE

SECRETARIAT GENERAL

SECRETARIAT EXECUTIF
DU FONDS VERT POUR LE CLIMAT
AU BURKINA FASO



**GREEN
CLIMATE
FUND**

Ouagadougou, September 13, 2024

N° 2024-041 /PM/SG/SE-FVC/BF

National Designated Authority

To

Executive Director

Secretariat of the Green Climate Fund

***175 Art Center-daero
Yeonsu-gu, Incheon 406-840***

Republic of Korea

Subject: Expression of nomination and no-objection for the application for the Green Climate Fund Project-specific Assessment Approach (PSAA)

Dear Madam,

We refer to the programme titled RE-GAIN- Scaling solutions for food loss in Africa under the Project-Specific Assessment Approach ("PSAA") submitted by AGRA to us on September 13, 2024

The undersigned is the duly authorized representative of the National Designated Authority of Burkina Faso.

Pursuant to the GCF Board decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

Additionally, pursuant to paragraph 47 of the Governing Instrument for the Green Climate Fund and the GCF Board decision B.31/06 on the implementation of the PSAA in my capacity as representative of the National Designated Authority for Burkina Faso, I hereby nominate the entity below in connection with the funding proposal:

Legal name of the **applicant entity**- AGRA
Contact person: name- DR. Agnes Kalibata
Contact details: telephone, address, email address
+254(20) 3675 000
P.O Box 66773, westlands 00800, Nairobi, Kenya
AKalibata@agra.org

The NDA may wish to include information on how the entity and the proposal can contribute to the country's climate priorities.

By communicating our no-objection, it is implied that:

- (a) The government of Burkina Faso has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Burkina Faso;
- (c) In accordance with the GCF's environmental and social safeguards, the **programme** as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

In the case that the funding proposal is for a programme, we also confirm that our no-objection applies to all projects or activities to be implemented within the approved framework.

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely,



Nebyida Lamech KABORE



FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA
MINISTRY PLANNING AND DEVELOPMENT

To: The Green Climate Fund (“GCF”)

Addis Ababa, Sept 12, 2024

Re: Expression of nomination and no-objection for the application for the Green Climate Fund Project-specific Assessment Approach (PSAA)

Dear Madam, Sir,

We refer to the Program titled RE-GAIN- Scaling solutions for food loss in Africa under the Project-Specific Assessment Approach (“**PSAA**”) submitted by AGRA to us on 11 Sept 2024.

The undersigned is the duly authorized representative of the Ministry of Planning and Development, Abas Mohammed Ali, the National Designated Authority of Ethiopia.

Pursuant to the GCF Board decision B.08/10, the content of which we acknowledge to have reviewed, [in my capacity as representative of the National Designated Authority] OR [in my capacity as focal point], we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Ethiopia has no-objection to the programme as included in the funding proposal.
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Ethiopia.
- (c) In accordance with the GCF’s environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

Additionally, pursuant to paragraph 47 of the Governing Instrument for the Green Climate Fund and the GCF Board decision B.31/06 on the implementation of the PSAA, [in my capacity as representative of the National Designated Authority for Ethiopia, I hereby nominate the entity below in connection with the funding proposal:

Legal name of the applicant entity- AGRA

Contact person: name- DR. Agnes Kalibata

Contact details: telephone, address, email address.

+254(20) 3675 000

P.O Box 66773, westlands 00800, Nairobi, Kenya

AKalibata@agra.org



FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA
MINISTRY PLANNING AND DEVELOPMENT

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

In the case that the funding proposal is for a programme, we also confirm that our no-objection applies to all projects or activities to be implemented within the approved framework.

We acknowledge that this letter will be made publicly available on the GCF website.

Abas Mohammed Ali, CEO Environment and Climate change Policy, Strategy and Plan

NDA for GCF projects in Ethiopia

Ministry of Planning and Development

Addis Ababa, Ethiopia

Cell: [+251 91 154 45 26](tel:+251911544526)

P. O. Box 4472 (Office)



**REPUBLIC OF KENYA
THE NATIONAL TREASURY AND ECONOMIC PLANNING**

Telegraphic Address: 22921
FAX NO. 310833
Telephone: 2252299

**THE NATIONAL TREASURY
P O BOX 30007 - 00100
NAIROBI**

When Replying Please Quote

Ref: TNT/CONF/36/021/C/ PART H (10)

Date: September 18, 2024

**Ms. Mafalda Duarte
Executive Director
Green Climate Fund
175 Art Centre-daero, Yeonsu-ngu
Incheon City, Republic of Korea**

Dear



Subject: Expression of no-objection for the application for the Green Climate Fund Project-specific Assessment Approach (PSAA)

We refer to the programme titled *Re-Gain: Scaling Solutions for Food Loss in Africa* under the Project-Specific Assessment Approach ("PSAA") submitted by Alliance for a Green Revolution in Africa (AGRA) to us on 9 September 2024.

The undersigned is the duly authorized representative of the National Treasury, the National Designated Authority of the Republic of Kenya.

Pursuant to the GCF Board decision B.08/10, the content of which we acknowledge to have reviewed, in my capacity as representative of the National Designated Authority, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of the Republic of Kenya has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of the Republic of Kenya;

(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Yours 

DR. CHRIS K. KIPTOO, CBS
PRINCIPAL SECRETARY/THE NATIONAL TREASURY

Copy to: **Dr. Agnes Kalibata**
President
Alliance for a Green Revolution in Africa (AGRA)
NAIROBI

Telephone: 265 1 771111
Telefax No: 265 1 773379

Correspondence should be addressed to:
The Director of Environmental Affairs



ENVIRONMENTAL AFFAIRS DEPARTMENT
LINGADZI HOUSE
CITY CENTRE
PRIVATE BAG 394
LILONGWE 3
MALAWI

Mafalda Duarte
Executive Director
Secretariat of the Green Climate Fund
175 Art Center-daero
Yeonsu-gu, Incheon 406-840
Republic of Korea

Date: 09 September, 2024
Reference: EAD/99/06/4L

Subject: Expression of nomination and no-objection for the application for the Green Climate Fund Project-specific Assessment Approach (PSAA)

Dear Madam,

We refer to the programme titled **RE-GAIN- Scaling solutions for food loss in Africa** under the Project-Specific Assessment Approach ("PSAA") submitted by **AGRA** to us on 29 August, 2024.

The undersigned is the duly authorized representative of Environmental Affairs Department, the National Designated Authority of Malawi.

Pursuant to the GCF Board decision B.08/10, the content of which we acknowledge to have reviewed, in my capacity as the National Designated Authority and focal point, we hereby communicate our no-objection to the programme as included in the funding proposal.

Additionally, pursuant to paragraph 47 of the Governing Instrument for the Green Climate Fund and the GCF Board decision B.31/06 on the implementation of the PSAA, in my capacity as the National Designated Authority and focal point for Malawi, I hereby nominate the entity below in connection with the funding proposal:

Legal name of the applicant entity: AGRA
Contact person: DR. Agnes Kalibata
Contact details: AKalibata@agra.org, +254(20) 3675 000
P.O Box 66773, westlands 00800, Nairobi, Kenya

By communicating our no-objection, it is implied that:

- (a) The government of Malawi has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Malawi;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

In the case that the funding proposal is for a programme, we also confirm that our no-objection applies to all projects or activities to be implemented within the approved framework.

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely,



Tawonga Mbale-Luka
GCF NDA for Malawi

Director of Environmental Affairs
Ministry of Natural Resources and Climate Change

**UNITED REPUBLIC OF TANZANIA
VICE PRESIDENT'S OFFICE**

Telegram: "MAKAMU",
Phone No: +255 026 2329006
Fax No: +255 026 2329007
Email: ps@vpo.go.tz



Government City,
Mtumba Area,
P.O. Box 2502,
40406 **DODOMA,**
TANZANIA.

Ref. No. BD. 38/90/01A

11th September, 2024

Executive Director,
Secretariat of the Green Climate Fund (GCF),
175 Art Center-daero,
Yeonsu-gu, Incheon 406-840,
Republic of Korea.

**REF: EXPRESSION OF NOMINATION AND NO-OBJECTION FOR THE
APPLICATION FOR THE GREEN CLIMATE FUND PROJECT-SPECIFIC
ASSESSMENT APPROACH (PSAA)**

Reference is made to the heading above.

2. We refer to the programme titled RE-GAIN- Scaling solutions for food loss in Africa under the Project-Specific Assessment Approach ("PSAA") submitted by AGRA to us dated August 28, 2024. The undersigned is the duly authorized representative of The Permanent Secretary, Vice President's Office, the National Designated Authority /Focal Point of the United Republic of Tanzania.
3. Pursuant to the GCF Board decision B.08/10, the content of which we acknowledge to have reviewed, [in my capacity as representative of the National Designated Authority] OR [in my capacity as focal point], we hereby communicate our no-objection to the programme as included in the funding proposal.
4. Additionally, pursuant to paragraph 47 of the Governing Instrument for the Green Climate Fund and the GCF Board decision B.31/06 on the implementation of the PSAA, [in my capacity as representative of the National Designated Authority] OR [in my capacity as focal point] for [country name], I hereby nominate the entity below in connection with the funding proposal:

Legal name of the applicant entity- AGRA
Contact person: name- DR. Agnes Kalibata
Contact details: telephone, address, email address
+254(20) 3675 000
P.O Box 66773, westlands 00800, Nairobi, Kenya
AKalibata@agra.org

5. The NDA/focal point may wish to include information on how the entity and the proposal can contribute to the country's climate priorities. By communicating our no-objection, it is implied that:
 - (a) The government of Tanzania has no-objection to the programme as included in the funding proposal;
 - (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Tanzania; and
 - (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.
6. We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.
7. In the case that the funding proposal is for a programme, we also confirm that our no-objection applies to all projects or activities to be implemented within the approved framework.
8. We acknowledge that this letter will be made publicly available on the GCF website.
9. I thank you for your cooperation.



Abdallah H. Mitawi

ACTING PERMANENT SECRETARY

Telephone: 256 41 4341305/230487
Fax : 256 41 4233524
Email : finance@finance.go.ug
Website : www.finance.go.ug
Plot No. 2-8 Apollo Kaggwa Road
In any correspondence on
This subject please quote No. ALD 79/251/03



Ministry of Finance, Planning &
Economic Development,
P.O Box 8147
Kampala, Uganda

17th September 2024

Ms. Mafalda Duarte,
Executive Director,
Secretariat of the Green Climate Fund,
175 Art Center-daero,
Yeonsu-gu, Incheon 22004,
REPUBLIC OF KOREA.

**LETTER OF NO OBJECTION TO THE GREEN CLIMATE FUND FOR
A PROJECT TITLED "RE-GAIN: SCALING SOLUTIONS FOR FOOD
LOSS IN AFRICA"**

We refer to the above Programme as included in the funding proposal submitted to you by the Alliance for a Green Revolution in Africa (AGRA) for funding by the Green Climate Fund.

The undersigned is the duly authorized representative of the Ministry of Finance, Planning and Economic Development, the National Designated Authority/focal point of Uganda

Pursuant to the GCF decision B.08/10 and B.13/21, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal worth USD 75 Million with Uganda's component amounting to USD 10 Million.

By communicating our no objection, it is implied that:

- (a) The Government of the Republic of Uganda has no objection to the Programme as included in the Funding Proposal;
- (b) The Programme as included in the Funding Proposal is in conformity with Uganda's National Priorities, Strategies and Plans of Uganda.

Mission

"To formulate sound economic policies, maximize revenue mobilization, ensure efficient allocation and accountability for public resources so as to achieve the most rapid and sustainable economic growth and development"

(c) In accordance with the GCF's Environmental and Social Safeguards, the programme as included in the Funding Proposal is in conformity with relevant laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Patrick Ocailap



**FOR: PERMANENT SECRETARY/ SECRETARY TO THE
TREASURY / NATIONAL DESIGNATED AUTHORITY FOCAL POINT
FOR THE GREEN CLIMATE FUND.**

Copy to: - Hon. Minister of Finance, Planning and Economic Development
- Permanent Secretary, Ministry of Agriculture, Animal Industry and Fisheries.

Mission

"To formulate sound economic policies, maximize revenue mobilization, ensure efficient allocation and accountability for public resources so as to achieve the most rapid and sustainable economic growth and development"

All communication should be addressed to the
Permanent Secretary
Telephone: 0211-252395
0211-252394
0211-252391



In reply please quote

No.:.....

REPUBLIC OF ZAMBIA
MINISTRY OF GREEN ECONOMY AND ENVIRONMENT
OFFICE OF THE PERMANENT SECRETARY

Corner of John Mbita & Nationalist Roads
P.O. BOX 30147
Lusaka-Zambia

Mafalda Duarte
Executive Director
Secretariat of the Green Climate Fund
175 Art Center-daero
Yeonsu-gu, Incheon 406-840
Republic of Korea

Date: 16th September 2024
Reference: NDA/71/21/1
Page:

**Subject: Expression of no-objection for the application for the Green Climate Fund
Project-specific Assessment Approach (PSAA)**

Dear Madam, Sir,

We refer to the programme titled RE-GAIN: Scaling Solutions for Food Loss in Africa under the Project-Specific Assessment Approach ("PSAA") submitted by AGRA to us on 3rd September 2024.

The undersigned is the duly authorized representative of the Ministry of Green Economy and Environment (MGEE), the National Designated Authority for Zambia.

Pursuant to the GCF Board decision B.08/10, the content of which we acknowledge to have reviewed, in my capacity as representative of the National Designated Authority, we hereby communicate our no-objection to the programme as included in the funding proposal.

The RE-GAIN Food Loss Reduction Programme offers a comprehensive solution that aligns with Zambia's climate priorities. By minimizing food waste, the programme contributes to mitigating climate change through reduced greenhouse gas emissions. Additionally, it supports adapting to climate change by ensuring food security and promoting resilient agricultural practices.

This approach is in line with Zambia's national climate action plan, which prioritizes sustainable agriculture, food security, and climate resilience. The RE-GAIN Programme's focus on reducing food loss enhances agricultural efficiency, reduces reliance on external inputs, and strengthens local food systems, thereby contributing to these key objectives.

By communicating our no-objection, it is implied that:

- (a) The government of the Republic of Zambia has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Zambia;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

In the case that the funding proposal is for a programme, we also confirm that our no-objection applies to all projects or activities to be implemented within the approved framework.

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely,



Billy Katontoka

National Coordinator

National Designated Authority

MINISTRY OF GREEN ECONOMY AND ENVIRONMENT

**ESS disclosure report form for projects or programmes submitted under the
Project-specific Assessment Approach (PSAA) Pilot**

Environmental and social report(s) disclosure

Basic project or programme information	
Project or programme title	RE-GAIN: Scaling solutions for food loss in Africa
Existence of subproject(s) to be identified after GCF Board approval	No
Sector (public or private)	Public
Name of entity	AGRA
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Target countries include: • Burkina Faso; • Ethiopia; • Kenya; • Malawi; • Tanzania; • Uganda; and • Zambia
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on entity's website	English: Tuesday, January 1, 2019 French: Friday, January 10, 2025 Amharic: Friday, January 10, 2025 Swahili: Friday, January 10, 2025
Language(s) of disclosure	English, French, Amharic, and Swahili
Explanation on language	These are the official languages of target countries.
Link to disclosure	English: https://agra.org/wp-content/uploads/2025/01/RE-GAIN_EN.pdf French: https://agra.org/wp-content/uploads/2025/01/RE-GAIN-FRENCH.pdf Amharic: https://agra.org/wp-content/uploads/2025/01/Appendix1 Annex 6 RE-GAIN ES Risks Assessment and Mitigation-Final.pdf Swahili: https://agra.org/wp-content/uploads/2025/01/REGAIN-SWAHILI.pdf
Other link(s)	N/A
Remarks	An ESIA consistent with the requirements for a Category B programme is contained in the 'Environmental and Social Baseline and Risk Assessment'.
Environmental and Social Management Plan (ESMP) (if applicable)	

Date of disclosure on entity's website	English: Tuesday, January 1, 2019 French: Friday, January 10, 2025 Amharic: Friday, January 10, 2025 Swahili: Friday, January 10, 2025
Language(s) of disclosure	English, French, Amharic, and Swahili
Explanation on language	These are the official languages of target countries.
Link to disclosure	English: https://agra.org/wp-content/uploads/2025/01/RE-GAIN_EN.pdf French: https://agra.org/wp-content/uploads/2025/01/RE-GAIN-FRENCH.pdf Amharic: https://agra.org/wp-content/uploads/2025/01/Appendix1 Annex 6 RE-GAIN ES Risks Assessment and Mitigation-Final.pdf Swahili: https://agra.org/wp-content/uploads/2025/01/REGAIN-SWAHILI.pdf
Other link(s)	N/A
Remarks	An ESMP consistent with the requirements for a Category B programme is contained in the 'Environmental and Social Baseline and Risk Assessment'.
Environmental and Social Management System (ESMS)	
Date of disclosure on entity's website	English: Tuesday, January 1, 2019 French: Friday, January 10, 2025 Amharic: Friday, January 10, 2025 Swahili: Friday, January 10, 2025
Language(s) of disclosure	English, French, Amharic, and Swahili
Explanation on language	These are the official languages of the target countries.
Link to disclosure	English: https://agra.org/wp-content/uploads/2025/01/RE-GAIN_EN.pdf French: https://agra.org/wp-content/uploads/2025/01/RE-GAIN-FRENCH.pdf Amharic: https://agra.org/wp-content/uploads/2025/01/Appendix1 Annex 6 RE-GAIN ES Risks Assessment and Mitigation-Final.pdf Swahili: https://agra.org/wp-content/uploads/2025/01/REGAIN-SWAHILI.pdf
Other link(s)	N/A
Remarks	The AGRA ESMS will be applied to the Programme in each target country without exception. The ESMS is also available on the AGRA website here .

Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report	Environmental and Social Baseline and Risk Assessment
Date of disclosure on entity's website	English: Tuesday, January 1, 2019 French: Friday, January 10, 2025 Amharic: Friday, January 10, 2025 Swahili: Friday, January 10, 2025
Language(s) of disclosure	English, French, Amharic, and Swahili
Explanation on language	These are the official languages of the target countries.
Link to disclosure	English: https://agra.org/wp-content/uploads/2025/01/RE-GAIN_EN.pdf French: https://agra.org/wp-content/uploads/2025/01/RE-GAIN-FRENCH.pdf Amharic: https://agra.org/wp-content/uploads/2025/01/Appendix1 Annex 6 RE-GAIN ES Risks Assessment and Mitigation-Final.pdf Swahili: https://agra.org/wp-content/uploads/2025/01/REGAIN-SWAHILI.pdf
Other link(s)	N/A
Remarks	AGRA has conducted the following studies in respect of the Programme (included in Environmental and Social Baseline and Risk Assessment): - Country and Region Contextual E&S Risk Analysis (7 countries) - Indigenous Peoples Planning Framework - Programme Intervention Level E&S Risk Assessment - Programme Environmental and Social Action Plan
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Monday, September 30, 2024
Place	AGRA, Headquarters, 4th Floor, West End Towers, Wayaki Way, Nairobi, Kenya Focal point: Assan Ng'ombe, Environment and Social Management Officer (email- angombe@agra.org). Burkina Faso: AGRA Country Office, WASCAL Competence Centre, 06 BP9507 Ouagadougou Focal point: Raoul Ouedraogo, Environment and Social Management focal point (email- rouedraogo@agra.org). Ethiopia: AGRA Country Office, ILRI Campus, Room 08, Addis Ababa Focal point: Nicolas Obare, Country Programme Lead (email- nobare@agra.org).

	Kenya: AGRA Country Office, 4th Floor Westend Towers, Wayaki Way, Nairobi Focal point: Edward Agaba, Country Programme Lead (email- eagaba@agra.org). Malawi: AGRA Country Office C/O CIAT, Golden Peacock Complex, Presidential Way, City Centre, Lilongwe Focal point: Jannike Kamfose, Safeguarding Officer (email- jkamfose@agra.org). Tanzania: C/O IITA – Tanzania, Plot 25, Mwenge Cocacola Road Mikocheni, Dar es Salaam Focal point: Aikade Nkini, Safeguarding Officer (email- ankini@agra.org). Uganda: 42 Princess Anne Drive, Bugoloobi, Kampala Focal point: Liston Njoroge, Country Programme Lead (email- lnjoroge@agra.org). Zambia: C/O COMESA, COMESA HQ, Lusaka Focal point: Maziko Phiri, Country Manager (email- mphiri@agra.org).
Date of Board meeting in which the FP is intended to be considered	
Date of entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, February 17, 2025

Note: This form was prepared by the entity stated above.

Secretariat's assessment of the project-specific assessment approach applicant and FP257

The Secretariat's assessment is presented in two sections:

1. Secretariat's assessment of the applicant under the project-specific assessment approach
2. Secretariat's assessment of the funding proposal (FP257)

Section 1. Secretariat's assessment of the applicant under the project-specific assessment approach

1. The document “**PSAA accreditation: application and review checklist**” submitted by the applicant, AGRA, has been assessed against the GCF accreditation standards. The result of the entity capacity assessment and the Secretariat's endorsement and recommendations for accreditation under the GCF project-specific assessment approach (PSAA) are as follows:

Overall capacity	High The entity has a well-developed financial management system; a well-functioning control framework, including risk management; and robust systems and policies in place to prevent money laundering and the financing of terrorism and other prohibited practices, such as fraud and corruption. The entity's system has sufficient scope to manage environmental and social and gender risks and impacts with a low likelihood of negative impact on the entity's ability to undertake the proposed project as designed.
Fiduciary functions	Basic fiduciary standards
	Specialized fiduciary standard for project management
Maximum environmental and social risk category	Limited adverse risks (category B)
Conditions	N/A

2. The Secretariat has augmented its capacity by using external experts in undertaking the entity capacity assessment review.¹ A summary is presented below.

I. Entity capacity assessment summary

1.1 Introduction and institutional assessment

1.1.1. Basic information about the project and the entity

3. AGRA (formerly the Alliance for a Green Revolution in Africa), established in 2006 by the Bill & Melinda Gates Foundation and the Rockefeller Foundation, is a pan-African institution headquartered in Nairobi, Kenya. AGRA's vision is to contribute to a food system-inspired inclusive agricultural transformation across Africa, focusing on reducing hunger, improving nutrition and adapting to climate change. AGRA's strategic objectives include empowering and building the resilience of smallholder farmers and supporting the development of inclusive markets and finance to strengthen agricultural systems, as well as strengthening the capability of African countries to sustain agricultural transformation. AGRA collaborates with both public

¹ In line with the PSAA Accreditation Pilot Framework approved by the Board (decision B.31/06), the Secretariat has augmented its capacity by using external experts, third-party organizations and/or service providers to support it in undertaking the entity capacity assessment review.

and private sector partners across numerous African countries to drive agricultural transformation.

4. In 2023 AGRA's total revenues were USD 128.3 million, up from USD 80.3 million in 2022. AGRA is applying for a USD 75.6 million grant from GCF for a five-year programme, RE-GAIN: Scaling solutions for food loss in Africa (hereinafter referred to as the RE-GAIN programme), which falls into the category of medium projects (up to USD 250 million). The annual volume of the proposed programme is less than 15 per cent of the average realized annual revenues in 2022 and 2023. The programme is assigned a category B risk rating as defined in the GCF revised Environmental and Social Policy.

1.2 Methodology

5. AGRA passed the Climate Investment Committee endorsement step (known as CIC2) on 11 October 2023 and submitted the first set of materials for review round #1 (completeness check) on 17 September 2024 using the customized application form and material request. Two further review rounds took place from September to November 2024. The external review team (ERT) conducted a comprehensive review of the submitted materials and assessed AGRA's policies, procedures, capacities, track record and human resources against the GCF accreditation standards and checklist.

6. Reference check interviews were conducted with AGRA's two existing donor partners: the German Development Bank Kreditanstalt für Wiederaufbau (KfW), which has supported AGRA since 2017; and the United Kingdom Foreign & Commonwealth Development Office, which has supported AGRA since 2019. The referees were selected on the basis of their earlier contributions to interventions similar to the RE-GAIN programme under the Partnership for Inclusive Agricultural Transformation in Africa, and their mandate as public government donors. The interviews confirmed that no issues related to the fiduciary role or other capacities of AGRA had been identified during the long-term donor relationship.

7. In addition to the desk review, the ERT conducted a site visit to AGRA's headquarters (HQ) in Nairobi, during which the ERT validated the desk review findings, walked through processes, and engaged in interviews and presentations across all areas of the assessment. A total of 26 representatives from AGRA participated in the discussions, which provided further evidence of the capacity and procedures within AGRA.

8. The applicant was assessed in accordance with the GCF policies and standards to the extent applicable to accreditation below:

- (a) "Updated Strategic Plan for the Green Climate Fund 2024–2027";
- (b) "Matters related to the accreditation framework" regarding the accreditation process (decisions B.24/13 and B.26/01);
- (c) "Guiding Framework and Procedures for Accrediting National, Regional and International Implementing Entities and Intermediaries, Including the Fund's Fiduciary Principles and Standards and Environmental and Social Safeguards" (decision B.07/02);
- (d) "Policy on Prohibited Practices" (decision B.22/19);
- (e) "Anti-Money Laundering and Countering the Financing of Terrorism Policy" (AML/CFT Policy) (decision B.18/10);
- (f) "Policy on the Protection of Whistleblowers and Witnesses" (decision B.BM-2018/21);
- (g) "Environmental and Social Management System: Environmental and Social Policy" (decision B.19/10);

- (h) “Comprehensive Information Disclosure Policy of the Fund” (decision B.12/35) regarding the disclosure of environmental and social (E&S) risk information;
- (i) “Updated Gender Policy and Gender Action Plan 2020–2023” (decision B.24/12);
- (j) “Revised Environmental and Social Policy” (decision B.BM-2021/18);
- (k) “Evaluation Policy” (decision B.BM 2021/07); and
- (l) “Updates to the Accreditation Framework” (decision B.31/06).

1.3 Contribution to the mandate of the GCF

9. The RE-GAIN funding proposal outlines AGRA’s contributions to the GCF focus areas: forests and land use; livelihoods of people and communities; and health, food and water security. The proposed programme aims to enhance climate adaptation by supporting innovative practices in the harvest and post-harvest value chain across seven African countries. It also seeks to improve food security by addressing climate change impacts on post-harvest food losses through the widespread adoption of food loss-reduction solutions (FL-RS) by smallholder farmers. In addition, the funding proposal includes initiatives to raise awareness among farmers about the effects of climate change on post-harvest losses and the benefits of the FL-RS available in the market. Beyond the proposed programme, AGRA’s overall strategy and mission align with the updated Strategic Plan for the GCF 2024–2027 and contributes to the mandates of the GCF.

1.4 Track record and implementation of similar projects

10. As evidence of its track record, AGRA referenced four earlier or ongoing interventions that demonstrate its capacity to manage and deliver similar sized programmes in the proposed regions. AGRA has a track record of both developing and leading continuous activities regionally and of managing country-specific earmarked donor-funded interventions under different business lines with defined timelines, targets, activities and reporting requirements.

11. The four referenced interventions are the multi-country Partnership for Inclusive Agricultural Transformation in Africa (over USD 400 million), the Kaduna Consortium and the Niger Consortium (two separate initiatives aimed at enhancing farmer resilience in Nigeria, each with a funding of USD 1.5 million) and the Strengthening Regenerative Agriculture in Kenya project (USD 4 million). These examples use grants as the primary financial instrument and demonstrate AGRA’s track record of working with programmes involving agricultural productivity, climate resilience and market development. Each intervention incorporates practices aimed at addressing the impacts of climate change and promoting broader resilience in food systems.

12. Globally, AGRA has a network of donor partners ranging from international foundations to government donors, including the founding partners Bill & Melinda Gates Foundation and the Rockefeller Foundation, the MasterCard Foundation, the United States Agency for International Development, German Cooperation (KfW/BMZ) and the United Kingdom Foreign & Commonwealth Development Office, demonstrating a strong funding base.

1.5 Legal status, registration, permits and licences

13. AGRA was registered in 2006 in the United States of America as a non-profit organization under the US Internal Revenue Code (para. 501(c)(3)).

14. AGRA has established country offices in 12 African countries and is physically headquartered in Nairobi, Kenya. In Kenya, AGRA operates and is registered with the Registrar of Companies as a branch of the AGRA US registration under section 366 of the Companies Act, Cap 486 in 2006. This registration grants AGRA legal status and certain rights and obligations as governed by the Kenyan Companies Act. AGRA will be both the accredited entity (AE) as the contractual party with GCF and the executing entity (EE) of the RE-GAIN programme through the programme management unit and the country offices.
15. AGRA has established legal entities in six countries (Ghana, Kenya, Mali, Mozambique, Rwanda and United Republic of Tanzania) to implement its country strategies and operations as well as ensuring legal compliance. The country offices are extensions of AGRA and operate under the supervision of AGRA HQ. In countries where AGRA is not registered, administrative operations are managed through hosting arrangements with partner organizations.
16. In addition, several African governments (Burkina Faso, Ghana, Kenya, Malawi, Mali, Rwanda and United Republic of Tanzania) have granted operational privileges and exemptions through host country agreements.

II. Accreditation assessment

2.1 Basic fiduciary standards

2.1.1. Key administrative and financial capacities

17. AGRA's global board of directors includes members from the Rockefeller Foundation, the Bill & Melinda Gates Foundation and independent directors, supported by board committees dedicated to audit and risk, finance and human resources, nomination and governance, and programme, whose task is to oversee key functions. In addition, AGRA has established five oversight committees governed by their respective policies: procurement, grants, ethics, data protection and safeguarding. Procedures for appointment, termination and remuneration are defined.
18. AGRA's organizational structure is documented, outlining responsibilities and reporting lines. The process for setting objectives involves a five-year strategy cycle and annual reviews, with key performance indicators and a corporate scorecard guiding the tracking of progress.
19. Monitoring and reporting mechanisms are detailed in a quarterly management review, with assigned roles and reporting frequencies. External evaluations are conducted at closure of the larger programmes and at other strategic periods.
20. AGRA's financial statements are prepared as per the United States Generally Accepted Accounting Principles, and include detailed statements of financial position, performance, cash flows, changes in financial position, and notes and disclosures.
21. AGRA uses an enterprise resources planning system for financial management. The system is designed to support compliance with accounting principles and to manage fund accounting. AGRA has a financial operations policy and its Procedures Manual covers financial reporting and donor fund management.
22. The finance team, led by the chief financial officer and consisting of 15 people at the headquarters and eight in-country finance officers, is responsible for maintaining financial reporting systems. The IT infrastructure, which includes the AGRA Management Information System, supports financial tracking and grants management.

23. On the basis of AGRA's track record, this assessment concludes that AGRA has demonstrated that it is capable of managing and reporting financial activities aligned with its strategic framework and has operational capacity to handle international payments.
24. AGRA has an audit and risk committee that oversees both internal and external audit functions. The internal audit function is organized through a four-person internal auditing unit (IAU), which operates under a formally approved charter. The IAU adheres to international standards. The IAU reports functionally to the chair of the audit committee and administratively to the AGRA President. Reports are also shared with senior management, who participate in the process of addressing recommendations by the IAU.
25. AGRA appoints an independent external audit firm for the audit process through a competitive selection process, and audits are conducted in line with International Standards on Auditing. The external auditor issues annual audit opinions. Management letters from the auditor are reviewed by the audit and risk committee, which monitors internal controls and financial reporting.
26. AGRA's control framework is documented through various policies and manuals that govern key operational areas, including governance, human resources and administration, grant management, finance, procurement, risk management and internal audit. Internal policies and procedures support regulatory compliance. The accuracy of financial data and reporting is supported by the built-in controls within the enterprise resources planning system, such as access controls, data validation controls and segregation of duties.
27. AGRA's risk management processes involve a risk register and proactive measures, with updates to the management and the board. Risks are assessed at multiple levels, and actions to enhance mitigation measures are developed and documented. A comprehensive strategic risk report covering seven key risk areas is prepared quarterly for the board.
28. AGRA has detailed procurement guidelines, including a procurement policy and a Procurement Procedure Manual, outlining procurement actions, requirements and responsibilities. AGRA has mechanisms for regular oversight of procurement, including risk assessments, audits, due diligence, key performance indicators and segregation of duties.
29. AGRA uses an e-procurement system, which helps with streamlining workflows and is designed to improve financial control and compliance with procurement policies. In addition, the system offers reporting and analytics to support decision-making.

2.1.2. Transparency and accountability

30. AGRA's transparency and accountability policy framework is built on its ethics policy which hosts various institutional policies including partner and staff codes of conduct, and other targeted policies such as the anti-fraud and AML policies and a policy on sexual harassment. AGRA also has a conflict of interest policy requiring declarations from staff on affiliations. AGRA's ethics committee oversees compliance with the policies.
31. AGRA has policies and measures in place for communicating zero tolerance on fraud and corruption, and for preventing and addressing financial mismanagement. Accountability and transparent organizational culture is promoted in training events and reporting structures with grantees and other partners. Grantees are required to execute a copy of the AGRA Partners' Code of Conduct and safeguarding acknowledgement form, which are annexed to the grant award letter confirming their commitment to AGRA's ethical standards.
32. AGRA has a whistle-blower policy with various reporting channels, but accessibility to reporting forms for different stakeholders could be improved. AGRA is committed to enhancing accessibility of reporting forms on their website. AGRA's audit and risk committee and IAU are responsible for investigating allegations of fraud, financial mismanagement and other forms of

misconduct or prohibited practices. The ERT noted that there is limited documentation of cases escalated to the formal investigation process. AGRA has indicated its commitment to enhancing documentation related to investigations.

33. AGRA has demonstrated that it has policies in place related to AML/CFT and fraud prevention, and provided evidence of its procedures for carrying out adequate due diligence process on partners and fund recipients. The grants unit and the procurement unit conduct organizational capacity assessments on grantees and check the service providers for adverse mentions on sanctions lists using the Refinitiv tool.

2.2 Specialized fiduciary standards

2.2.1. Project management

34. AGRA has structured its operations around thematic areas or business lines to achieve its mission of transforming agriculture in Africa. Under the purview of the Vice-President – Center of Technical Expertise, AGRA has thematic units aligned with the business lines for policy and state capability, seed systems, climate adaptation, sustainable agriculture and resilience, and inclusive markets, trade and finance. Under the purview of the Vice-President – Programs Delivery, AGRA has the regional structures and country offices, as well as units for private sector partnerships, and gender, youth and inclusiveness.

35. AGRA implements projects through a variety of instruments and partners, including grant-making to local non-governmental organizations, farmer organizations, research institutions, extension service providers and so on. All projects and interventions are aligned with broader strategic goals as well as regional and country priorities. Grantees are the primary implementing partners who execute interventions contained in the AGRA strategy. AGRA has a comprehensive Grants Management Policies & Procedures Manual that outlines policies and procedures for managing the entire grant lifecycle.

36. AGRA has a clear monitoring, evaluation and learning (MEL) framework and a comprehensive MEL plan aligned with the AGRA Strategy 3.0. The MEL plan outlines the principles, procedures, tools and responsibilities to execute the MEL framework. AGRA's overall corporate theory of change and results framework expand into detailed business line-specific theories of change. The Director of Strategy & Monitoring & Evaluation and Learning reports directly to the President and oversees the operational units for strategy and analytics, and monitoring and evaluation (M&E). There are regional M&E officers for West Africa and for Southern and East Africa, as well as in-country M&E officers in eight countries.

37. AGRA has demonstrated its experience in project identification, design and management throughout the project cycle. There are fiduciary policies for quality control and oversight mechanisms to monitor budgets and risks in project implementation phase. AGRA has broad programme management and MEL capacity with defined procedures to adhere to established policies and guidelines.

2.2.2. Grant award and/or funding allocation mechanisms

38. AGRA has not applied for accreditation for the specialized fiduciary standard for grant award and/or funding allocation mechanisms. AGRA will not award grants in the proposed RE-GAIN programme, but the field-level implementation partners will be selected and contracted through a procurement process. However, AGRA has provided evidence of its granting process and many of the established grant-making practices and policies will be applied in the RE-GAIN programme.

39. AGRA's approach to awarding and managing grants is documented in its Grants Management Policies & Procedures Manual. Typically, AGRA applies a structured grant-making process based on calls for proposals, with defined eligibility and selection criteria and structured reporting procedures. Country teams develop annual grant pipelines, which are approved by the Vice-President – Country Support, Policy and Delivery. The final grant approvals are handled by the Head of Grants for smaller grants and the grants review committee for larger grants.

2.2.3. On-lending and/or blending for loans and equity

40. AGRA has not applied for accreditation for the specialized fiduciary standard for on-lending and/or blending for loans and equity.

2.3 Environmental and social safeguards

41. AGRA's Environmental and Social Management System (ESMS) is mandated by its highest decision-making body and based on the IFC Performance Standards (PS). The key ESMS document includes an environmental and social policy and separate sections covering (1) identification of E&S risk and impacts, (2) ESMS management and organization, (3) ESMS screening, (4) environmental and social impact assessments, (5) stakeholder engagement (including a grievance mechanism) and (6) collaboration, information sharing and reporting. AGRA's ESMS is comprehensive and is supported by a set of detailed implementation manuals, guidelines and templates.

42. AGRA does not have an Indigenous Peoples policy, but risk screening and monitoring procedures for matters relating to Indigenous Peoples are covered in the ESMS implementation manual/annexes, and sample reports provide evidence of implementation. The project-specific Indigenous Peoples screening and vulnerability assessment annexed to the RE-GAIN funding proposal further indicates AGRA's capacity to comply with PS7 on Indigenous Peoples.

43. AGRA's ESMS requires and guides the identification and assessment of E&S risks and impacts. The process for screening projects is described in the manuals, and the impacts are assessed against PS1–PS8 using the environmental and social screening checklist. An E&S risk categorization guideline provides guidance for assessing the E&S risk category.

44. Most of AGRA's interventions in the agriculture sector are classified as risk category C or B (i.e. low to medium risk). The proposed RE-GAIN programme is classified as category B because of identified potential exposure to social risks related to labour conditions and child labour in post-harvest activities and work safety with agriculture machinery. The environmental risks identified in the RE-GAIN programme are minimal, due to the focus on the post-harvest stage in the agricultural value chain.

45. AGRA has incorporated a system in the delivery model for managing the E&S risks in the grant projects and other activities. All grantees sign an undertaking (comprising a grant award letter and a grant commitment checklist) that commits them to screening and mitigating identified risks. AGRA's grantees are required to document the E&S risks in a project-specific risk register. AGRA is in the process of integrating the ESMS elements in the institution-wide AGRA Management Information System where grant processing, reporting and results monitoring are managed.

46. The roles and responsibilities related to AGRA's ESMS are defined in the ESMS Manual. The information provided by AGRA demonstrates its capacity and expertise on environmental and social matters at the HQ level and at the country level. A senior resilience specialist, sitting in the Adaptation, Sustainable Agriculture and Resilience Unit, oversees the implementation of the ESMS and has been closely engaged in the design of the RE-GAIN programme and its

environmental and social components. Regional environmental and social officers, programme officers and safeguarding officers in country offices support and review AGRA grantees' E&S risk screening. Country offices monitor and report performance to HQ. AGRA's ESMS was developed with an external consultant supported by earmarked KfW funding, and external support for E&S risk matters is contracted as needed. AGRA's ESMS training is mandatory both for staff in environmental and social related positions and for grantees and other implementing partners.

47. AGRA adopted the ESMS in 2020 and it has been rolled out in three West African pilot countries: Burkina Faso, Ghana and Nigeria. AGRA has provided evidence of its track record of implementing the ESMS with local grantees in those countries (E&S risk screening, development of environmental and social action plans, monitoring and performance reporting). However, the ERT assessment noted that, with the exception of Burkina Faso, the ESMS and required capacities have not been fully established in six out of the seven countries targeted by the GCF-funded RE-GAIN programme (Ethiopia, Kenya, Malawi, Uganda, United Republic of Tanzania and Zambia). AGRA has indicated that according to its training plan, all country training scheduled in RE-GAIN countries would be completed by March 2025.

48. AGRA has provided evidence of information disclosure and stakeholder consultation both in terms of past track record and in the context of the current funding proposal. AGRA has a whistle-blowing mechanism and other channels for receiving communications on E&S risk matters. Guidance and templates for setting up a project-level grievance redress mechanism are embedded in the ESMS and the annexed E&S Risk Assessment Toolkit. According to AGRA, complaints related to E&S risk are rarely escalated from the beneficiary level to AGRA; rather, they are typically handled locally by the grant partner.

49. The ERT assesses AGRA's approach to environmental and social risk management as appropriate and sufficient for category B. At the HQ level and in selected countries, AGRA has policies, procedures and capacity to screen, monitor and manage E&S risks and impacts, but the roll-out of the ESMS in all target countries of the RE-GAIN programme has not been completed. AGRA has processes for independent reporting and grievance redress mechanisms at the global and country level and capacity to investigate alleged E&S risk violations.

50. AGRA has zero-tolerance on Sexual Exploitation, Abuse and Harassment (SEAH) which is covered in the human resources and administration policy, ethics policy and safeguarding of vulnerable persons and youth policy. The applicant has identified potential SEAH risks for the programme that include incidents involving staff or programme partners and backlash against women in non-traditional roles. Additionally, subprojects will also be screened for SEAH risks. Mitigation measures to manage these risks include training on AGRA safeguarding policy for staff and programme partners, including regular mandatory training on SEAH/GBV awareness and mitigation, integrating SEAH risk management into staff performance reviews and programme planning as well as SEAH awareness sessions in onboarding for new staff. In addition, the programme will establish community gender dialogues to address power dynamics and promote positive masculinity as well as include men in household financial management training to reduce tensions and offer support to survivors and victims of SEAH. Furthermore, the applicant will ensure that access to grievance mechanisms for SEAH incidents is provided to stakeholders. The applicant's ESMS includes safeguarding officers at the country level for undertaking SEAH assessments and training.

2.4 Gender

51. The AE has provided a gender analysis and action plan and therefore complies with the updated gender policy of the fund. The RE-GAIN programme's gender assessment highlights significant challenges and barriers to gender equality in the agricultural value chains of target

countries. Women face restricted access to critical resources such as land, credit, and climate-resilient technologies, exacerbating disparities in productivity and income compared to men. Sociocultural norms perpetuate unequal responsibilities, with women disproportionately shouldering unpaid care work, limiting their participation in economic and decision-making activities. Women are also underrepresented in leadership roles and formalized agricultural structures, and they lack adequate training and access to extension services tailored to their needs. Climate change further amplifies these gaps by intensifying women's vulnerability due to their reliance on natural resources and exclusion from climate resilience planning and interventions. Institutional barriers, such as inadequate sex-disaggregated data and weak implementation of gender-sensitive policies, continue to impede progress toward equality.

52. The gender and inclusion action plan for the RE-GAIN programme focuses on addressing gender disparities through targeted and actionable measures. The gender action plan defines activities, indicators, and targets for women's participation and benefit with corresponding timelines, budget and gender expertise. During the programme start-up phase country-level gender assessments and action plans will be developed. Key activities include implementing capacity-building activities to enhance women's technical skills, leadership abilities, and access to climate-resilient agricultural practices. To strengthen women's participation in decision-making, the plan includes strategies to establish women-led cooperatives and community-based networks. It also supports awareness campaigns to challenge sociocultural norms that limit women's roles in agriculture and land governance. Additionally, the action plan incorporates provisions for training women in post-harvest handling technologies and facilitating their access to markets through value chain enhancements, as well as finance by identifying relevant credit schemes and financial instruments.

III. Conclusions and recommendations

3.1 Conclusions

53. On the basis of this assessment, AGRA has demonstrated high capacity to implement the proposed RE-GAIN programme and manage the GCF-funded activities in alignment with the relevant GCF policies. In addition, AGRA possesses the relevant legal status to implement the RE-GAIN programme as stipulated in the PSAA pilot framework requirements.

54. AGRA substantially meets the standards for the GCF basic fiduciary standards, specialized fiduciary standards, environmental and social standards, and Gender Policy. The assessment identified some deviations from the requirements as described in the GCF guidance notes, but the ERT does not consider these deviations as material gaps.

- (a) The applicant meets the requirements of the GCF basic fiduciary standards, the GCF Policy on the Protection of Whistleblowers and Witnesses, the GCF Policy on Prohibited Practices and the GCF AML/CFT Policy, and the specialized fiduciary standard for project management; to the extent applicable to PSAA and the funding proposal;
- (b) The applicant meets the GCF revised Environmental and Social Policy, GCF interim environmental and social safeguards (ESS) standards and the GCF Information Disclosure Policy on disclosure of E&S risk information in relation to the E&S risk category B/I-2 (activities with potential mild adverse environmental and/or social risks and/or impacts that are few in number, generally site-specific, largely reversible and readily addressed through mitigation measures) and to the extent applicable to PSAA accreditation;
- (c) The applicant has demonstrated that it has policies, procedures and competencies in order to implement its gender approach, which is found to be consistent with the

Updated GCF Gender Policy to the extent applicable to PSAA and the funding proposal. The applicant has demonstrated that it has experience in gender considerations in the context of climate change.

3.2 Recommendations on project-specific accreditation

55. **Accreditation type:** PSAA

56. **Fiduciary functions**

(a) Basic fiduciary standards; and

(b) Specialized fiduciary standard for project management

57. **Maximum environmental and social risk category:** Category B

58. **Conditions:** No conditions

59. **Remarks:** Roll-out of AGRA's ESMS in all target countries of the proposed RE-GAIN programme has not been completed. The assessment recommends that AGRA ensure that the relevant processes and resources will be in place to appropriately implement the ESMS in Ethiopia, Kenya, Malawi, Uganda, United Republic of Tanzania and Zambia by the start of the RE-GAIN programme activities. AGRA has indicated a timeline to roll out the ESMS and associated training in all RE-GAIN target countries by March 2025.

Section 2. Secretariat's assessment of FP257

Proposal name:	RE-GAIN: Scaling solutions for food loss in Africa
Project-specific assessment approach applicant:	AGRA
Country/(ies):	Burkina Faso, Ethiopia, Kenya, Malawi, Uganda, United Republic of Tanzania, Zambia
Project/programme size:	Medium

I. Overall assessment of the Secretariat

60. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
As the first project targeting food loss mitigation in the current GCF portfolio, this initiative holds significant potential for transformative change by promoting the adoption of food loss-reduction solutions by engaging both the demand and the supply side, with a strong focus on market creation and enhancement, and active private sector participation.	Potential implementation risks given the multi-country nature of the programme, including several contexts affected by fragility and conflict. However, the implementation arrangements, including the legal terms, include adaptive management provisions for mitigating some of these risks.
Six out of the seven targeted countries are least developed countries, with limited public and private investment in food loss reduction. Significant numbers of beneficiaries, with a focus on smallholder farmers and micro, small and medium-sized enterprises (MSMEs), will be supported through capacity development initiatives, the adoption of food loss-reduction solutions and improved access to finance.	A potential risk factor for model 2 is the inability of beneficiaries to access finance or the risk of loan defaults, which could lead to a reduced number of beneficiaries. To address this risk, AGRA has developed food loss-reduction solutions deployment models that have multiple layers of de-risking, including smart subsidies, incentives for the private sector and business development support for MSMEs.
As the first multi-country programme with a direct access entity under the project-specific assessment approach (PSAA), active project Secretariat origination efforts in collaboration with AGRA – leveraging its sectoral expertise, local presence and proven track record – have been deployed in the development of this innovative PSAA initiative.	

61. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the AE and, if considered appropriate, subject to the conditions set out in annex II of document GCF/B.41/02.

II. Summary of the Secretariat's assessment

2.1 Programme background

62. Many African countries face recurrent droughts that affect growing seasons, often shortening the period suitable for agriculture and limiting the viability of farming in marginal areas. Projections indicate that crop yields in some countries could decline by up to 50 per cent by 2030, and net revenues for crops may fall by as much as 90 per cent by 2100, with smallholder farmers being the most affected. Food security, however, is affected not only by reductions in crop production but also by changes throughout the crop value chain, including post-harvest. There is, therefore, a crucial need to address the impacts of climate change on the full agriculture value chain, including production, aggregation, storage, processing and distribution.

63. To address these challenges, the programme aims to build the climate resilience and adaptive capacity of smallholder farmers by promoting the wide-scale adoption of FL-RS, and thereby increasing awareness, capacity and demand for, improving access and the availability of, and enabling environment support for wide-scale adoption of these solutions. These activities will contribute to achieving the targets of the updated Strategic Plan for the GCF 2024–2027: Target 2 (new direct access entities); Target 4 (food systems); Target 9 (adaptation and locally led action); and Target 10 (local private sector early-stage ventures and MSME innovation).

64. The total financing for this proposed programme is USD 105 million with the total GCF funding of USD 75,623,754 in grants, which leverages USD 29.3 million in grants in co-finance by AGRA. The project is rated as ESS category B. The programme will directly benefit 2.6 million vulnerable people and indirectly benefit a further 2.7 million vulnerable people (1.72 per cent of the regional population) across seven countries: Burkina Faso, Ethiopia, Kenya, Malawi, Uganda, United Republic of Tanzania and Zambia.

65. This initiative is the first PSAA multi-country programme in the African region, supporting a direct access entity (regional). The PSAA applicant, AGRA, serving as both the AE and the EE, has over 15 years of experience, operations in 11 countries and a team of over 200 professionals. AGRA's experience and proven track record make it well positioned to lead this large-scale PSAA initiative, with its extensive network including partnerships with governments, local agribusinesses and civil society organizations in the agriculture sector.

66. This funding proposal was developed with support provided by the Project Preparation Facility.

2.2 Component-by-component analysis

Component 1: Increased adoption of FL-RS by smallholder farmers (total cost: USD 43.1 million; GCF cost: USD 22.57 million)

67. Outcome 1 demonstrates several technical strengths, particularly in its comprehensive approach to increasing the adoption of FL-RS by smallholder farmers. By integrating multiple strategies, such as awareness-raising, training, technology transfer and market linkages, the programme effectively addresses the demand side of FL-RS adoption.

68. The extensive engagement strategy further strengthens the programme's potential impact. By leveraging village-based advisers and local institutions, the project aims to reach over a million smallholder farmers, ensuring widespread awareness and participation across seven countries. In addition, the project's commitment to gender responsiveness and youth inclusion is commendable, with specific modules designed to empower women and youth in agricultural value chains.

69. Another key strength lies in its market-oriented approach. By structuring agriculture value chains and fostering linkages between smallholder farmers and cooperatives, MSMEs and potential buyers (retailers, processors and exporters), the programme creates incentives for quality compliance and increased adoption of FL-RS. Practical, on-the-ground demonstrations of FL-RS provide an added layer of support, connecting technical training with real-world application.

Component 2: Enhanced supply and affordability of FL-RS (total cost: USD 43.81 million; GCF cost: USD 40.87 million)

70. Outcome 2 leverages a holistic approach to overcoming supply-side barriers to the adoption of FL-RS by introducing innovative financial mechanisms. Through a dual approach that combines business development and financial innovation, the programme builds the capacity of local MSMEs, creates inclusive opportunities for women and youth, and fosters a sustainable market ecosystem to ensure widespread access to FL-RS for smallholder farmers and MSMEs.

71. The programme strengthens MSMEs by providing targeted training on operations and maintenance, as well as sustainable business models, such as shared equipment use. Engaging FL-RS vendors and incentivizing them to expand distribution channels and after-sales services creates a self-sustaining system that extends the programme's benefits beyond its five-year duration.

72. Innovative financial mechanisms – designed for both smallholder farmers and cooperatives or MSMEs – incorporate smart subsidies, pre-financing partnerships and the integration of FL-RS into agricultural input packages. These approaches help to lower affordability barriers for farmers while de-risking investments for stakeholders. This localized approach ensures interventions are tailored to community needs, with scoping exercises and market intelligence guiding implementation for maximum relevance and impact.

73. The programme's significance lies in its focus on strengthening local markets, reducing dependency on external suppliers and fostering economic growth within target communities. Its scalable design enables replication across regions; meanwhile, empowering women and youth fosters social equity and resilience. By reducing food loss, improving farmer incomes and promoting climate-resilient technologies, outcome 2 aligns with broader development and environmental goals, creating a de-risked investment environment that encourages sustained participation from suppliers, vendors and financial institutions across seven countries.

Component 3: Strengthened enabling environment for the uptake of FL-RS (total cost: USD 11.5 million; GCF cost: USD 6.34 million)

74. Outcome 3 establishes an enabling environment for adopting and scaling FL-RS by addressing policy, institutional and regulatory barriers. This component ensures the sustainability and scalability of interventions under outcomes 1 and 2 while fostering systemic change and long-term impact.

75. Key strengths include a comprehensive policy focus, institutional capacity-building and private sector engagement. By revising policies and promoting quality standards and financial incentives, the programme creates a foundation for the sustainable adoption of FL-RS. Targeted technical assistance equips governments to monitor food loss, link outcomes to national climate

commitments and adapt to changing contexts. Private sector partnerships further enhance investment in post-harvest infrastructure and services.

76. The programme contributes to systemic change by aligning FL-RS initiatives with national and global climate goals, such as nationally determined contributions and the Sustainable Development Goal (SDGs), and creates an environment conducive to large-scale adoption of FL-RS. It ensures sustainability through strengthened institutions, private sector-driven markets and long-term investment potential. Regional knowledge-sharing through AGRA's networks further promotes replication and scale-up across Africa.

Monitoring and evaluation components (total cost: USD 2.76 million; GCF cost: USD 2.05 million)

Project management (total cost: USD 3.79 million; GCF cost: USD 3.79 million)

III. Assessment against investment criteria

3.1 Impact potential

Scale: High

77. The proposed programme delivers transformative adaptation benefits by addressing the vulnerabilities of smallholder farmers and communities across seven African countries. By reducing food losses exacerbated by climate change, the programme strengthens food security and enhances farm-level profitability. It prioritizes vulnerable groups such as women, youth and MSMEs, empowering them through capacity-building, market development and social cohesion. These efforts strengthen community resilience and directly support broader climate adaptation objectives at both the regional and the national level.

78. The programme also complements traditional agricultural adaptation strategies by addressing critical gaps in the food value chain. Its emphasis on preserving harvested crops alleviates pressure on land and water resources, promoting environmental sustainability while mitigating greenhouse gas emissions from compensatory agricultural production.

79. Through catalytic grants and partnerships with the private sector, the programme creates a foundation for sustainable markets and attracts additional investments, ensuring benefits extend beyond the initial funding. Its scalable and replicable design further amplifies its impact, positioning it as a model for regional and global food security initiatives.

3.2 Paradigm shift potential

Scale: High

80. By leveraging GCF grants, the programme catalyses a paradigm shift in post-harvest development pathways through market stimulation, private sector engagement and scalable FL-RS, addressing climate-induced challenges in the agricultural value chain. The programme builds resilient agricultural ecosystems by empowering smallholder farmers with accessible, climate-resilient food loss-reduction systems. It integrates multi-stakeholder engagement and proven last-mile delivery systems, and it aligns with national and regional frameworks to ensure long-term sustainability. These efforts create robust linkages between farmers, markets and financial systems.

81. The programme's design means there is potential for significant scaling and replication. Strengthening local supply chains, fostering private sector innovation and embedding successful models into national flagship programmes bring the potential to extend its reach. Partnerships with regional climate institutions amplify its impact, facilitating adaptation across Africa. Complementing this is a focus on knowledge-sharing, which leverages best practices and innovative financial models to influence broader strategies.

82. By mobilizing private sector engagement, the programme multiplies the impact of GCF grants, unlocking significant additional investment. This approach incentivizes innovation and builds long-term market viability for FL-RS, empowering farmers while strengthening supply chains.
83. Lastly, the programme prioritizes resilience-building for smallholder farmers, with a strong emphasis on inclusivity. Targeting women, youth and MSMEs ensures equitable benefits, fostering economic empowerment and social equity while enhancing food security and climate adaptation.

3.3 Sustainable development potential

Scale: High

84. The proposed programme extends its impact far beyond reducing food losses by delivering a range of co-benefits that advance global goals (SDGs 2, 5, 8, 13), environmental conservation and socioeconomic well-being. Its integrated design fosters systemic change at the local, national and regional level, ensuring long-term sustainability and resilience.
85. The programme generates substantial environmental co-benefits by reducing the need for compensatory agricultural production, which conserves land and water resources, prevents land degradation and protects ecosystems. Efficient agricultural practices decrease dependency on fertilizers and pesticides, reducing the agricultural environmental footprint. In addition, minimizing food waste lowers greenhouse gas emissions and prevents habitat destruction, preserving biodiversity and supporting ecosystem sustainability.
86. The programme's mitigation efforts are significant, with greenhouse gas emission reductions estimated at over 1 million tonnes of carbon dioxide equivalent over 15 years. These reductions stem from decreasing compensatory food production emissions and mitigating land-use changes. This aligns with the climate commitments of target countries, reinforcing their progress towards global climate goals.
87. The programme enhances food quality and safety by reducing contamination risks, improving nutrition and lowering food-borne illnesses, which positively impacts public health systems. It generates employment through FL-RS value chains, creating opportunities for women and youth, and fostering economic inclusion. Stabilizing food markets reduces price volatility, benefiting both producers and consumers, while providing more predictable income and affordable food.
88. Designed with gender sensitivity at its core, the programme ensures women's equal participation and leadership, fostering equity and leveraging their pivotal role in agriculture. Inclusive strategies targeting women-led enterprises and youth integration ensure equitable benefits, strengthening social and economic resilience.

3.4 Needs of the recipient

Scale: Medium to high

89. The target countries rely heavily on agriculture, a sector that employs 40–80 per cent of their populations and contributes 20–40 per cent of their gross domestic product. However, climate change poses severe threats to agricultural productivity, food security and economic stability, exacerbated by low adaptive capacities and limited institutional readiness. The programme directly responds to these challenges by focusing on reducing food losses and building resilience in post-harvest systems – a crucial yet underdeveloped stage in the agricultural value chain.
90. Despite significant investments in improving production systems, post-harvest processes remain neglected, leaving smallholder farmers – the backbone of these economies –

particularly vulnerable to losses from spoilage, contamination and inefficiencies. RE-GAIN addresses this gap by introducing FL-RS that improve storage, processing and market access.

91. The programme is uniquely tailored to local contexts, working with community-level suppliers, agro-dealers and MSMEs to integrate FL-RS sustainably into existing systems. This localized approach not only fosters ownership and capacity but also complements the financial and institutional constraints of national governments, whose limited resources are often concentrated on production rather than post-harvest systems.

92. A core strength of RE-GAIN lies in its focus on empowering vulnerable populations, particularly smallholder farmers, women and youth. These groups face significant barriers to adopting climate-resilient technologies owing to financial constraints and systemic inequalities. The programme addresses these barriers by providing affordable FL-RS, building market infrastructure and creating targeted business development opportunities for marginalized groups. This inclusive approach promotes equitable development and enhances community cohesion.

3.5 Country ownership

Scale: High

93. The RE-GAIN programme exemplifies a model of country-driven, context-specific development, reflecting a strong commitment to aligning interventions with national priorities while fostering local ownership. Through close collaboration with national designated authorities, ministries and stakeholders, AGRA has co-developed project concepts and solutions that address local challenges. This approach involved extensive consultations, including initial government engagements, stakeholder workshops and iterative feedback loops. As a result, the programme integrates with existing initiatives across climate change adaptation, agriculture development, MSME support, financial systems and food security. A participatory design process further strengthens the programme's responsiveness. Extensive consultations with communities, governments and non-governmental entities identified critical gaps and shaped interventions that directly address the realities of smallholder farmers. This process continues throughout implementation, allowing the programme to remain flexible and adaptive to evolving local contexts.

94. The programme also leverages AGRA's expertise and its established presence in the target countries. Drawing from five-year country strategies, comprehensive gap analyses and strategic partnerships, the programme builds on AGRA's foundation to deliver systemic change. Collaboration with governments ensures alignment with national climate strategies and strengthens local capacity through active engagement of agro-dealers, MSMEs and others.

95. This emphasis on government and local ownership is central to the programme's sustainability. By embedding interventions within national systems and fostering collaboration among governments, private sector actors and development organizations, the programme ensures its long-term viability. Local ownership is further reinforced by integrating interventions into existing frameworks and systems, ensuring scalability and institutional commitment.

3.6 Efficiency and effectiveness

Scale: Medium

96. The financial structure exemplifies a strategic and innovative approach to addressing the barriers faced by smallholder farmers in adopting FL-RS while ensuring long-term market sustainability. By integrating catalytic grants, concessional funding and capacity-building initiatives, the programme creates a holistic framework that addresses critical challenges on both the supply and the demand side of the agricultural value chain.

97. At its core, the programme tackles the issues of affordability and access to finance, which are significant obstacles for smallholder farmers. Through concessional funding and grants, private sector manufacturers are incentivized to offer affordable FL-RS while financial institutions are encouraged to develop tailored credit products that mitigate risks. This approach ensures farmers gain access to sustainable solutions while suppliers and financial institutions see reduced risk, fostering a market ecosystem that can thrive beyond the programme's duration.
98. The programme's economic efficiency and resource mobilization are particularly noteworthy. The financial returns are positive across all target countries. Beyond immediate returns, the programme lays a foundation for long-term market viability, unlocking additional private sector investments and amplifying its systemic impact. In addition to financial returns, the programme delivers significant economic and climate benefits.
99. The programme incorporates best practices from AGRA's extensive experience, such as leveraging last-mile delivery systems through village-based advisers and agro-dealers, tailoring financial products to crop cycles and promoting multi-crop food loss-reduction technologies. These practices ensure interventions are relevant, scalable and adaptable to local contexts, fostering systemic change and long-term resilience.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

100. **ESS 1: Assessment and management of environmental and social risks and impacts. Assessment and management of risks.** AGRA has prepared an ESMS that includes policies, procedures and tools for assessing and managing E&S risks and impacts for any activities or interventions that it will finance. Therefore, the assessment and management of E&S risks and impacts will be done during project implementation for each specific activity and grantee in each country. The broad risks identified comprise contextual risks (conflict and child labour); implementing partner risks (labour and working conditions, waste management, road safety); Indigenous Peoples-related risk; soil conservation and management; water management; pesticide management; biodiversity; air emissions; occupational health and safety; community health and safety; and stakeholder engagement.
101. **ESS 2: Labour and working conditions.** AGRA has broadly identified the potential labour risks and impacts of activities, including working conditions, terms of employment, child labour and forced labour, and occupational health and safety. Considering the programme will be implemented in various countries, a country-level analysis on child labour in the agricultural sector should be undertaken.
102. **ESS 3: Resource efficiency and pollution prevention.** AGRA has broadly and generically identified potential impacts to air, water and land, as well as project-associated waste and pesticides. The ESMS details the procedures and tools that will be used to assess interventions, as well as specific pollution risks for grantees during implementation. It further provides guidance and includes procedures and tools for grantees to prepare relevant environmental and social management plans to reduce pollution and promote sustainable use of resources.
103. **ESS 4: Community health, safety and security.** AGRA has identified potential community health and safety risks, including effects on ecosystem services and risk of exposure to pesticides. The ESMS includes procedures and tools for grantees to prepare relevant environmental and social management plans to address community health and safety issues.

104. **ESS 5: Land acquisition and involuntary resettlement.** AGRA has stated that the GCF-funded programme will not be associated with any land acquisition. This should be part of the Exclusion List, stating that no programme activities will result in land acquisition.
105. **ESS 6: Biodiversity conservation and sustainable management of living natural resources.** AGRA broadly identified potential biodiversity risks, including invasive species, ecosystem services and impacts on habitats and protected areas. The ESMS provides procedures, tools and a template that will guide the development of site-specific environmental and social management plans, which will provide additional detail on potential risks/impact and mitigation measure on the programme.
106. **ESS 7: Indigenous Peoples and GCF Indigenous Peoples Policy.** AGRA undertook an assessment on Indigenous Peoples and concluded that the project interventions in five of the seven countries may impact these groups. The assessment concluded that the programme will not result in displacement, loss of access to land or natural resources, or alter the physical or social context of the local areas where Indigenous People live. The ESMS includes due diligence and screening procedures for identifying impacts on Indigenous Peoples and determining whether free, prior and informed consent will be required at an intervention-specific level during project implementation.
107. **ESS 8: Cultural heritage.** AGRA has provided due diligence and screening procedures for identifying impacts on cultural heritage and determining whether risk management measures will be required at an intervention-specific level during project implementation.
108. **GCF Indigenous Peoples Policy and ESS 7 on Indigenous Peoples.** There are pastoralist Indigenous Peoples in most of the countries involved in the proposal. The AE has provided an Indigenous Peoples plan framework that it has adjusted for compliance with the GCF Indigenous Peoples Policy. Project exclusions ensure that no activity results in the infringement of the rights of Indigenous People and that their free, prior and informed consent is secured for the activities that require it; and the activities exclude land acquisition and involuntary resettlement. The AE is recommended to have a sound participative approach and screening for Indigenous Peoples and to engage with the GCF Indigenous Peoples Advisory Group on relevant Indigenous Peoples' issues that may be encountered.
109. **Sexual exploitation, abuse and harassment (SEAH).** AGRA has zero tolerance of SEAH, which is covered in the AGRA human resources and administration policy, its ethics policy and its policy on safeguarding vulnerable persons and youth. The applicant has identified potential SEAH risks for the programme, such as incidents involving staff or programme partners and backlash against women in non-traditional roles. Subprojects will also be screened for SEAH risks. Mitigation measures to manage these risks include training on the AGRA safeguarding policy for staff and programme partners, including regular mandatory training on SEAH and gender-based violence awareness and mitigation; integrating SEAH risk management into staff performance reviews and programme planning; and SEAH awareness sessions in onboarding for new staff. In addition, the programme will establish community gender dialogues to address power dynamics and promote positive masculinity as well as including men in household financial management training to reduce tensions, and offering support to survivors of SEAH. Furthermore, the applicant will ensure that access to grievance mechanisms for SEAH incidents is provided to stakeholders. The applicant's ESMS includes safeguarding officers at the country level for undertaking SEAH assessments and training.

4.2 Gender policy

110. The AE provided a gender assessment and action plan with the funding proposal and therefore complies with the requirements of the GCF Updated Gender Policy.

111. The RE-GAIN programme's gender assessment highlights significant challenges and barriers to gender equality in the agricultural value chains of the target countries. Women face restricted access to critical resources such as land, credit and climate-resilient technologies, exacerbating disparities in productivity and income compared with men. Sociocultural norms perpetuate unequal responsibilities, with women disproportionately shouldering unpaid care work, limiting their participation in economic and decision-making activities. Women are also underrepresented in leadership roles and formalized agricultural structures, and they lack adequate training and access to extension services tailored to their needs. Climate change further amplifies these gaps by intensifying women's vulnerability due to their reliance on natural resources and exclusion from climate resilience planning and interventions. Institutional barriers, such as inadequate sex-disaggregated data and weak implementation of gender-sensitive policies, continue to impede progress toward equality.

112. The gender and inclusion action plan for the RE-GAIN programme focuses on addressing gender disparities through targeted and actionable measures. The gender action plan defines activities, indicators and targets for women's participation and benefit, with corresponding timelines, budget and gender expertise. During the programme start-up phase country-level gender assessments and action plans will be developed. Key activities include implementing capacity-building activities to enhance women's technical skills, leadership abilities and access to climate-resilient agricultural practices. To strengthen women's participation in decision-making, the plan includes strategies to establish women-led cooperatives and community-based networks. It also supports awareness campaigns to challenge sociocultural norms that limit women's roles in agriculture and land governance. In addition, the action plan incorporates provisions for training women in post-harvest handling technologies and facilitating their access to markets through value chain enhancements, as well as finance by identifying relevant credit schemes and financial instruments.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

113. GCF is providing a grant of USD 75.62 million to build smallholder farmers' climate resilience and adaptive capacity by promoting the wide-scale adoption of FL-RS. The programme is co-financed with USD 29.38 million from AGRA, leveraging a total project cost of USD 105 million. This programme covers seven countries in the Africa region. This is the first multi-country programme with a direct access entity under the PSAA modality.

4.3.2. Accredited entity/executing entity capability to execute the current programme (medium risk)

114. The PSAA applicant, AGRA, is carrying out the AE/EE role for this project. AGRA has an established track record in implementing agriculture projects in the region as well as extensive partnerships with governments, local agribusinesses and civil society organizations. The entity capacity assessment of AGRA concludes that AGRA is well positioned to implement the project with no conditions or recommendations

4.3.3. Programme-specific execution risks (medium risk)

115. The proposed procurement models (model 1 and 2) under component 2 have an innovative nature while retain certain risks.

4.3.4. Misalignment risk:

116. There is a misalignment in the timing of the distribution of economic benefits between the local service providers (e.g. manufacturer, agro-dealers) and smallholder farmers. Service providers benefit from the grant immediately upon the completed agreement for the sale of agricultural equipment, while farmers accrue benefits gradually through extended access to the equipment or service provided using the equipment. This disparity creates a risk that GCF concessionality may be eroded owing to:

- (a) **(Model 1) Lack of a monitoring system on pricing:** The memorandum of understanding to be signed between AGRA and the supplier lacks mechanisms to monitor pricing practices by agro-dealers (e.g. margins and sales records). This absence of oversight may dilute GCF concessionality and result in farmers not benefiting from the intended affordability;
- (b) **(Model 2) Default risk:** Youth groups managing the equipment as a business venture may default if they lack sufficient capacity or face economic challenges;
- (c) **(Model 2) Premature exit risk:** Youth groups may choose to exit the business prematurely by disposing of equipment before the economic benefits are fully realized by farmers; and
- (d) **(Model 2) Market risk:** The manufacturer that provides the repurchase assurance may be put a risk if the repurchase assurance does not cover the total outstanding loan, for example, if the resale value has depreciated due to the market situation.

117. To mitigate these risks, the project should incorporate:

- (a) A provision for passing down the GCF concessionality to the relevant smallholder farmers by specifying the monitoring/spot check/reporting mechanisms in models 1 and 2;
- (b) Incentives for youth groups to prioritize long-term capital creation over short-term economic gains;
- (c) Mechanisms ensuring the accessibility and affordability of equipment for local farmers;
- (d) Recovery provisions for the residual value of the grant if the business is terminated early, such as second lien pledges on equipment or negative covenants to redirect funds to similar initiatives; and
- (e) An assurance that the financial institutions and supplier agree on clear depreciation terms for the repurchase value and the risk-sharing ratio and the establishment of regular maintenance requirements to preserve equipment value.

4.3.5. Co-financing risk:

118. The co-financing commitment from AGRA amounts to USD 29.3 million, representing 27.9 per cent of the total project cost. While AGRA's track record and capacity provide confidence in its ability to deliver, delays or shortfalls in co-financing disbursement could affect the timely execution of project activities and reduce the overall impact.

4.3.6. Compliance risk (low risk)

119. The programme's proposed activities are associated with a relatively low inherent risk of money laundering/terrorist financing (ML/TF).

120. The PSAA applicant has assessed ML/TF and other prohibited practices risk as low-medium on the basis of its internal impact and likelihood matrix assessment, which indicated the likelihood to be rare and the potential impact to be major. Mitigating measures include grantee screening and pre-onboarding due diligence as part of organizational capacity

assessments; screening of all contractors, vendors and consultants; staff background checks; and pre-payment reviews by the PSAA applicant's finance team.

121. The PSAA applicant will also act as EE. The Secretariat's capacity assessment found that AGRA has overall high capacity that includes a well-functioning control framework, including relating to risk management, as well as robust policies and systems to prevent ML/TF and other prohibited practices.

122. On the basis of the programme activities and the AE's integrity risk controls to be implemented throughout the programme, the overall compliance risk is determined to be low.

4.3.7. GCF portfolio concentration risk (low risk)

123. If approved, this proposal's impact on the GCF portfolio concentration in terms of results areas and single proposal remains minimal, given the diversified nature of GCF investments in the agriculture sector.

4.3.8. Recommendation

124. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall programme	Medium
Accredited entity (AE)/executing entity (EE) capability	Medium
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance	Low

4.4 Fiduciary

125. The RE-GAIN programme will be implemented by AGRA as both the AE and the EE. AGRA will leverage its extensive network of country offices and partnerships across seven African countries to coordinate programme execution. A dedicated programme implementation unit within AGRA will oversee programme activities, financial management and compliance with GCF policies and standards. The unit will oversee planning and quality assurance; supervise programme monitoring, evaluation and reporting; ensure timely realization of all programme deliverables; provide leadership and technical support to implementing partners; and ensure smooth communication flow across all programme partners. This executing role will be fulfilled through both the Nairobi-based HQ and AGRA's country offices, and the EE will report to the AGRA senior leadership team.

126. A central programme steering committee, co-chaired by AGRA and representatives of the national designated authority, will provide strategic guidance and review project performance annually. The committee will ensure that programme activities are consistent with GCF objectives and national development goals, reinforcing accountability and governance mechanisms.

127. AGRA will establish a robust financial management framework, utilizing its internal financial systems to segregate and track GCF funds. Separate project accounts will be maintained for accurate financial reporting. AGRA's financial management system aligns with international standards, ensuring compliance with GCF fiduciary requirements.

128. Periodic financial reports will be submitted to the GCF, detailing expenditures and progress against budgeted activities. External audits will be conducted annually by an independent auditor to validate financial reports and ensure adherence to established financial controls. AGRA will also provide capacity-building support to local partners on financial management and reporting, to uphold accountability at all implementation levels.

129. Procurement activities under the RE-GAIN programme will adhere to AGRA's procurement policy, which emphasizes transparency, competitiveness and value for money. Procurement plans will be developed annually, specifying timelines and mechanisms for acquiring goods and services. The programme will prioritize local procurement to support regional markets, with AGRA's country offices ensuring compliance with national and international procurement standards.

4.5 Results monitoring and reporting

130. As detailed in the theory of change, three interconnected programme outcomes will contribute to achieving longer-term adaptation outcomes for food and nutrition security and livelihood resilience. The mitigation impact expected from the programme is classified as a co-benefit. The theory of change has appropriately articulated the goal statement to focus on the adoption of food loss-reduction technologies, improving the resilience of farmers, MSMEs and rural communities to climate shocks.

131. The programme's logical framework follows the GCF Integrated Results Management Framework and will measure adequate core and supplementary indicators mapped to two GCF result areas: livelihoods of people and communities; and health, food and water security. The enabling environment and paradigm shift potential indicators selected by the AE are consistent with the requirements and allow for comprehensive results monitoring. The project-specific indicators deal with context-specific results from the output level, ensuring that programme performance is adequately measured. The measurement of a few of the programme-specific indicators will be informed by the baseline survey to be undertaken by the AE at the inception stage.

132. The monitoring and evaluation of the programme will leverage the AE's existing M&E framework and data management systems. The monitoring and evaluation plan submitted as part of the funding proposal package describes how the measurement and reporting against the logical framework will be conducted. The implementation monitoring will be undertaken at the country level, which will be aggregated at the programme level to prepare the annual performance report in accordance with the GCF Monitoring and Accountability Framework. The programme will be commissioning an interim and final evaluation through independent evaluators in line with the GCF Evaluation Policy.

4.6 Legal assessment

133. This funding proposal is being presented under the PSAA. If the funding proposal is approved by the Board, the GCF Secretariat and the applicant will negotiate a funded activity agreement for the proposed programme.

134. The applicant has provided a legal certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the programme.

135. The proposed programme will be implemented in seven host countries. The GCF has signed bilateral agreements on privileges and immunities with two of these host countries, namely Burkina Faso and Zambia. In the case of the remaining five countries, namely Ethiopia, Kenya, Malawi, Uganda and United Republic of Tanzania, GCF is not provided with privileges

and immunities. This means that, among other things, GCF is not protected against litigation or expropriation in these countries, which risks need to be further assessed. Moreover, the ability of GCF to undertake redress activities and/or investigations in such countries may be hindered due to the absence of privileges and immunities for relevant GCF personnel.

136. Therefore, it is recommended that the Board consider whether disbursements of GCF proceeds should be made only after GCF has obtained satisfactory protection against litigation and expropriation in the countries, or has been provided with appropriate privileges and immunities for GCF and its personnel.

137. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and
- (b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

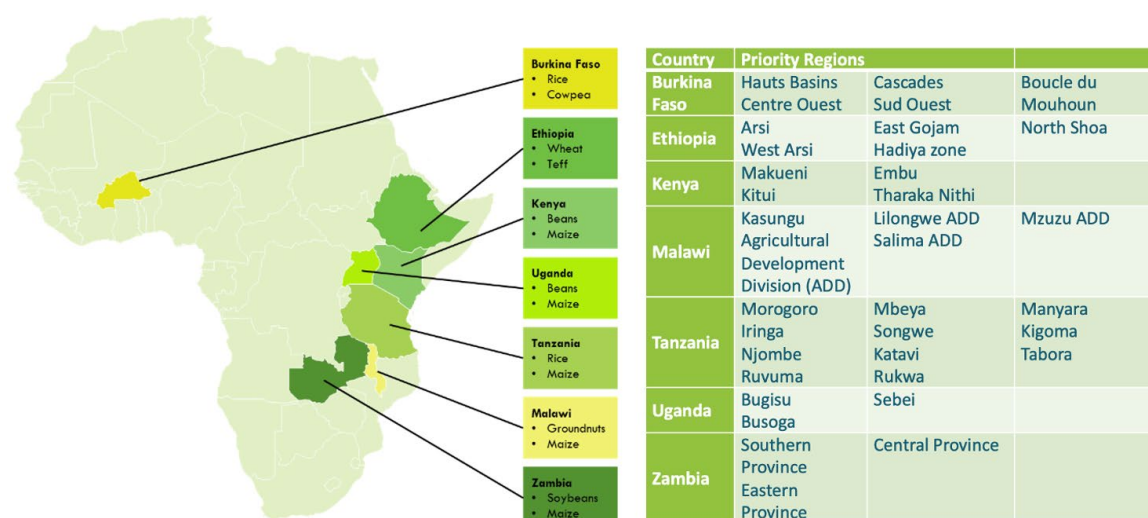
Independent Technical Advisory Panel's assessment of FP257

Proposal name:	RE-GAIN: Scaling solutions for food loss in Africa
Project-specific assessment approach applicant:	AGRA
Country/(ies):	Burkina Faso, Ethiopia, Kenya, Malawi, Uganda, United Republic of Tanzania, Zambia
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1. This funding proposal for a medium-sized, public sector adaptation programme in environmental and social safeguards category B is submitted by AGRA through the GCF project-specific assessment approach (PSAA). The RE-GAIN programme will operate over five years in seven African countries: Burkina Faso and six contiguous countries – Ethiopia, Kenya, Malawi, Uganda, the United Republic of Tanzania and Zambia (see figure 1). AGRA, a continental institution in agriculture and food systems, will be both the PSAA entity and the executing entity for the programme, which will be executed through the AGRA Programme Operational Unit and utilize the AGRA Africa Food Systems Forum and Agribusiness Deal Room, allowing close working relationships with governments, partners and other stakeholders.

Figure 1 Priority climate change-vulnerable regions and target crops for the seven participating countries¹



2. The total cost of the programme is USD 105 million, of which USD 75.6 million is proposed as GCF grant finance and USD 29.4 million as AGRA co-finance. The programme goal is “to build smallholder climate resilience and adaptive capacity by promoting the wide scale

¹ Figure 10 and table 4 from the funding proposal.

adoption of food loss reduction solutions (FL-RS) in the seven participating countries”². The programme will work by increasing demand and absorption capacity for FL-RS by smallholder farmers, enhancing supply and affordability of the solutions, and strengthening enabling environments for scaling them up. This will result in reducing exposure and vulnerability of key grain and legume crops that are grown by smallholders and that are strategic for food security, in regions identified as vulnerable to climate hazards – especially unpredictable rainfall with heavy downpours, high temperatures and flooding (with drought of secondary importance). These hazards are already increasing with climate change and are predicted to intensify, worsening food losses during harvesting, post-harvest handling and storage, and processing, transport and logistics – through mould, rotting, decay, heat damage, pests, rodents and afla/mycotoxin contamination.

3. The programme outputs and intended outcomes, under three components, are as shown in table 1.

Table 1 Programme structure for RE-GAIN: Scaling solutions for food loss in Africa

Component 1: Food Loss-Reduction Solutions (FL-RS) demand side development to increase the adoption of FL-RS by farmers
<i>Outcome 1: Increased adoption of FL-RS by smallholder farmers</i>
Output 1.1: Smallholder farmers supported to adopt FL-RS
Output 1.2: Improved market linkages between agri-value chain actors
Component 2: FL-RS supply side development to increase availability and affordability of FL-RS
<i>Outcome 2: Enhanced supply and affordability of FL-RS</i>
Output 2.1: Business development support provided to local MSMEs to facilitate the improved provision of FL-RS on local markets
Output 2.2: Financial mechanisms established to support smallholders and MSMEs to adopt FL-RS
Component 3: Enabling environment to ensure sustainability of the FL-RS market
<i>Outcome 3: Strengthened enabling environment for the uptake of FL-RS</i>
Output 3.1: Enhanced capacity of national institutions to enable investments in FL-RS

Abbreviation: MSMEs = micro, small and medium enterprises.

4. In addition to extensive work in Component 1 on awareness and training for smallholders and value chain actors to build demand, as well as support to governments in Component 3 to enhance enabling environments, the biggest proportion of the budget (54 per cent of GCF funds) will go to Component 2 to develop the supply side for technical solutions to food loss. Output 2.1 will build partnerships for market linkages and provide training to suppliers and service providers. At the heart of the programme is Output 2.2, in which temporary financial mechanisms will be put in place to support smallholders and MSMEs to adopt FL-RS. According to the detailed programme budget (annex 4 to the funding proposal), a total of USD 20.42 million of GCF funds will be used to subsidize tarpaulins, plastic sheets, metal and plastic silos, and hermetic bags for smallholders farmers to protect harvested crops (in Activity 2.2.1), and USD 12.34 million from GCF will be used to subsidize larger items for which service providers take loans – mechanical multi-crop threshers, moisture meters, communal storage structures, storage protectants and control agents (in Activity 2.2.2). This is to be achieved through two innovative business and financial models, which can be summarized as follows:

² This wording on the project objective is from Slide 5 of the PowerPoint slides presented to iTAP on the funding proposal by the Secretariat on 2 December 2024, and is aligned with the content of paragraph 49 of the funding proposal (which does not include an explicitly stated single-sentence project goal or objective. overall project objective).

- (a) **Model 1** will promote the local supply of FL-RS interventions by using conditional procurements to effectively subsidize interventions at smallholder farmer level, facilitated by 5,350 village-based advisers across the 35 priority regions. This model will enable agrodealers to provide FL-RS to approximately 540,000 smallholder farmers at a discounted rate, with GCF funds used to procure one item for every two items procured and sold by an agrodealer. The subsidy will be passed down to the smallholder farmers as a discount of 33 per cent on the purchase price (see paragraphs 95–105 of the funding proposal for a detailed explanation of the carefully designed model, including use of an escrow account to manage risk), also explained by the PSAA entity as a “Buy two, get one free” promotional offer.³ The aim of this model is to make these interventions more affordable for farmers, and to stimulate production and manufacturer capacity by placing pre-emptive orders.
- (b) **Model 2** will match MSMEs and cooperatives⁴ with suppliers and financial institutions, facilitating the signing of multi-stakeholder agreements to allow the MSMEs to purchase more expensive equipment for post-harvest handling (such as solar-powered multi-crop threshers and shellers, moisture meters, and communal storage structures). The MSMEs will become service providers, operating the equipment on a fee-for-service basis with smallholder customers. Under each agreement, AGRA will commit to an upfront co-payment of 30 per cent of the purchase price, with the MSME taking a loan from the financial institution to pay the other 70 per cent out of its service fee earnings. In the event of a successfully completed loan repayment, the equipment is owned by the MSME. If the MSME defaults, the supplier is obliged in terms of the agreement to buy back the equipment, accounting for depreciation. The establishment of the service operations will be facilitated through the business development support under Output 2.1. The aim of this model is to reduce borrowing costs to MSME service providers, through subsidies on the purchase price, and to reduce credit risk to local financial institutions.
5. The subsidy mechanisms have been carefully designed to achieve these effects, ensuring that the benefits of the subsidies do reach the targeted smallholder farmers without leakages, that systems are in place to ensure agrodealers’ compliance with subsidy conditions, and that contingencies are in place to deal with market price fluctuations or supply chain disruptions. In response to a query from iTAP about the possibility of renting-to-own, AGRA confirmed⁵ that variations on the model will be possible, using the equipment as collateral for the loans (asset finance, buy-back guarantees by the suppliers, rent-to-own, etc.) – to be specified in the four-way agreement. The programme has already engaged the following financial partners: AgLeaseCo in Zambia, Equity for Africa (EFTA in Tanzania and EF-KEN in Kenya), and Coris Bank in Burkina Faso, the last of which AGRA has previously implemented an equipment finance scheme.

1.1 Impact potential

Scale: High

6. The programme will target two of the GCF results areas for adaptation, aiming to enhance the resilience of (a) the most vulnerable people and communities and (b) health and well-being, and food and water security, in the seven participating countries.

³ Responses received by the iTAP on 6 January 2025 from AGRA to questions it received from the iTAP on 18 December 2024.

⁴ The term “youth groups” is used in the funding proposal in the model description “to highlight the priority given to this vulnerable demographic” but it also applies to other cooperatives, including women’s groups and local MSMEs.

⁵ Responses received by the iTAP on 6 January 2025 from AGRA to questions it received from the iTAP on 18 December 2024.

7. **Climate change.** The funding proposal highlights the vulnerability of all seven countries to climate change, with negative impacts experienced and anticipated – not only on crop production and yields but also on volumes of crops that remain of sufficient quality for consumption or sale at harvest and beyond. Historical trends are shown, based on data from the World Bank Climate Change Knowledge Portal, with temperature indicators (such as annual mean surface temperature and the number of days with extreme heat – i.e. temperatures over 35 °C) displaying an increasing trend over time, and the rate of increase accelerating in recent decades. Historical rainfall trends display far greater variability, and for some countries a measure of uncertainty regarding the direction of the trend. The perceived historical increase in the frequency of intense rainfall events is not shown as a key climate variable.

8. Data from the same portal, using two future emissions scenarios, indicate that future projections of rainfall (such as annual mean precipitation, the number of consecutive wet days, and – as a corollary – the number of consecutive dry days) suggest a projected rise in rainfall levels and rainfall days across most of East Africa, likely to increase flooding. The exception is Zambia, where projections display a slight drying signal, consistent with regional trends for southern Africa. Continued trends of increasing variability in rainfall patterns are expected, as are increasing heat extremes. AGRA acknowledges that the impact of climate change on post-harvest food loss has not been well studied, either in terms of observed or projected impacts, but suggests that “[o]f the range of climate hazards that affect harvest and post-harvest value chains of cereal crops and legumes, scholarly publications and country experts pointed to variable, erratic, and unpredictable rainfall (including an increase in heavy rainfall events), high temperatures (extreme heat), and flooding as the priority hazards of concern”.

9. **Adaptation impact.** The programme aims to yield significant adaptation benefits for vulnerable smallholder farmers and their households across the seven countries, through reducing climate change-induced food losses. This will enhance farming communities’ resilience to climate change by improving management at harvest and post-harvest stages, maintaining higher levels of food availability for consumption and sale than would be the case without the GCF–AGRA investment. The intended impacts of the programme are considered complementary to the broader adaptation strategies for agricultural resilience within each target country, which tend to be focused more on primary production. The programme will target vulnerable members of society, aiming to increase the resilience of rural livelihoods through investing in developing local markets and building up farmer cooperatives, youth and women’s groups, and MSMEs, leveraging partnerships with financial institutions and private sector actors to mobilize significant investment beyond the proposed programme.

10. Through these results, the programme aspires to directly benefit over 2.6 million people and indirectly benefit a further 2.7 million people, distributed across the seven target countries as shown in table 7 of the funding proposal (see methodology and detailed calculations in supplementary annex 1 to the funding proposal). The 35 priority regions, shown in detail in maps in annex 16, were selected on the basis of crop profiles and vulnerability to identified climate hazards. Smallholder farmers adopting new climate-resilient practices with FL-RS will be direct beneficiaries, experiencing improved food and nutrition security as well as increased food quality and market connections that improve their incomes. MSMEs and agrodealers will experience improved livelihoods as a result of increased business through the programme subsidy and stimulation of demand. The indirect beneficiaries include local buyers, whose access to affordable and good-quality grains and legumes will improve as a result of the programme interventions, and market players such as suppliers and processors, who will benefit from improved market linkages.

11. At GCF outcome level (see funding proposal section E.3), the programme will measure the number of beneficiaries reached, the number of farmers adopting innovations that strengthen climate change resilience, the number of people with new climate-resilient livelihood options, and the number of vulnerable farm households with improved food security,

using farmer surveys employing the Food Insecurity Experience Scale. Since the innovations and livelihood options in question are predefined and relate to specific FL-RS, these key indicators should deliver credible tracking of results. Although food losses before and after programme interventions will not be quantified, a co-benefit indicator of the percentage of produce meeting quality standards will measure the percentage of grain rejected due to quality concerns associated with afla/mycotoxins, with the aim being to halve the rejection rate.⁶ Overall, it can be anticipated that food loss reductions will be significant, at least during the programme implementation period (i.e. while the subsidies are in operation), improving resilience through better nutrition, greater income from crop sales and enhanced business incomes along the value chains for technical solutions.

12. Overall, impact potential is rated as high.

1.2 Paradigm shift potential

Scale: Medium

13. Paradigm shift relates to the degree to which the proposed programme can successfully catalyse impact beyond the GCF one-off investment of USD 75.6 million over five years, together with the AGRA co-finance. The iTAP recognizes the hard work and careful thought that have gone into designing the programme in a manner that intends to elicit a paradigm shift in the harvest and post-harvest development pathways for the seven target countries and continentally in Africa. A clear focus on achieving scale and sustainability is embedded in the programme design, both through the comprehensive value chain approach to removing barriers to scaling of FL-RS and through the work with governments on country-level enabling environments in Component 3. The design of the business and financial models, however, relying as it does on a large subsidy that will be abruptly withdrawn at programme end, does not seem optimal for achieving sustained demand or long-term market transformation, as discussed further below.

14. **Enabling environment for scaling up.** Component 3 of the programme seeks to create an enabling environment for the uptake of FL-RS. This will involve working with governments on new policies, such as stimulating tax exemptions, subsidies or reduced interest rates for FL-RS investments. RE-GAIN will provide technical assistance to local and national government entities to build their institutional capacity for scaling FL-RS through partnerships, platforms, and support on tools, guidelines and monitoring. RE-GAIN will also work with governments and other development partners to promote additional private sector investment in post-harvest infrastructure, including storage facilities, processing units and cold chains, as well as helping regulatory bodies to develop and implement quality standards and certification processes for FL-RS products and services. Support will also be provided to enhance the capacities of extension services and climate advisory services to monitor food loss in relation to climate conditions and the impact of FL-RS in reducing loss. Although the RE-GAIN investments aim to prolong grain shelf life in the absence of good rural road access and renewable-powered cooling systems, engaging with wider government adaptation and development initiatives will enable complementarity in this regard.

15. **Innovation in institutional markets.** A welcome innovation is the focus on building smallholder farmers' market access by linking them to institutional markets for staple crops – such as schools, universities, armies, hospitals and prisons – as well as to technical and financial partners supporting public sector feeding schemes.⁷ This will be achieved by facilitating purchase agreements with farmer cooperatives or producer groups, and, where no such organizations exist, by using aggregators. Furthermore, the signed purchase agreements can be

⁶ See funding proposal logical framework section E.5.

⁷ For example, the Africa Improved Foods public-private partnership between the government of Rwanda and a number of UK and Netherlands government development partners.

used as collateral to obtain credit from financial institutions.⁸ Technical assistance includes support by village-based advisers to wheat flour millers, enabling them to avoid unnecessary food loss from overprocessing to refined flour, and instead to switch to producing healthier wholegrain bread flour for school-feeding markets. RE-GAIN will leverage the institutional markets through government procurement processes and work with partners such as HarvestPlus, the Partnership for Child Development, Save the Children and the World Food Programme.

16. **Sharing lessons beyond the target countries.** Evidence generated and lessons learned in FL-RS from the target communities and countries will be used to develop capacity, influence policy and promote best practices across Africa. Activity 3.1.4 has budget set aside to work actively to leverage the lessons from the programme in other contexts, building on the extensive networks of AGRA. AGRA will facilitate collaboration with continental and regional bodies and national governments, including African regional climate institutions and regional economic communities. This will include field days and media events bringing together the private sector, government officials, farmers from non-programme areas, and development partners to showcase success stories and encourage replication. Knowledge products will be developed and widely shared, and a business case will be developed for investments in FL-RS manufacturing, supply, service provision and financing.

17. **Exit strategy for subsidies.** The funding proposal states: “Long-term sustainability is at the core of the programme design, with GCF investments seen as catalysers of change within local systems. A key aspect of this GCF intervention is having an exit strategy through the creation of an enabling environment to shift away from the smart-subsidies, catalysing further investments in FL-RS under Component 2” (paragraph 166). Once the programme ends, however, there will be nothing in place to manage this transition away from the subsidy. Prices are likely to go up at least 30 per cent as soon as the subsidy is removed, which may lead to a fall-off in purchasing and thus lower adoption rates for the FL-RS. The iTAP wonders whether a subsidy that was smaller, more gradually phased out, or both, would be less disruptive, and enable continued growth in sales and loans. In sharing examples of successful market stimulation⁹, AGRA made no mention of other similar price subsidy programmes. Arguably, support to local manufacture might be a more appropriate level at which to utilize a subsidy – to stimulate long-term sustainable production of inputs, rather than providing a one-time subsidy of inputs, for which there is no ceiling on imports.

18. **Local manufacturing not prioritized.** The programme does not prioritize direct support to local manufacturing of FLRS, which could have been an alternative focus for subsidization in the form of seed capital or expansion grants to local manufacturers. Although business development support is part of the programme (Activity 2.1.2), this consists of training workshops and provision of market intelligence, and is aimed at businesses that will buy machinery and operate it on a fee-for-service basis. Unlike some of the other AGRA programmes cited, such as the SaYeTech support to artisans in Ghana to manufacture threshers, the RE-GAIN programme does not include any such direct support to local manufacturing.

19. The response to a question from iTAP about import substitution was that while “AGRA fully agrees that long-term sustainable domestic production of inputs is critical for maximizing the impact of the programme,...for the first years of programme implementation, RE-GAIN will prioritize delivering its support to the target groups over import substitution to ensure timely programme delivery and to stimulate and prime the market for local manufacturing capacity.”¹⁰

⁸ This system has been widely used by the United Nations World Food Programme. See, for example, concerning Ethiopia: *WFP Empowers Smallholder Farmers Through Purchase for Progress (P4P)*. Available at <https://reliefweb.int/report/ethiopia/wfp-empowers-smallholder-farmers-through-purchase-progress-p4p>.

⁹ In responses received by the iTAP on 6 January 2025 from AGRA to questions it received from the iTAP on 18 December 2024.

¹⁰ Responses received by the iTAP on 6 January 2025 from AGRA to questions it received from the iTAP on 18 December 2024.

The iTAP notes that AGRA has completed a mapping of local suppliers and manufacturers of the technologies (16 in total – see appendix 9 to annex 2 of the funding proposal) and that the programme “will procure as much domestic input as possible” where this is “comparably priced for the same specifications, or where technical specifications are better for a higher price”.¹¹ The iTAP believes that a more proactive approach could have gone further to promote market transformation.

20. Taking all these factors into account, paradigm shift potential is rated as medium.

1.3 Sustainable development potential

Scale: Medium to high

21. The programme is expected to have a number of social, economic and environmental co-benefits, as outlined below.

22. **Socioeconomic and health co-benefits.** In targeting highly vulnerable smallholder farming communities, the proposed programme will yield multiple socioeconomic co-benefits. Reducing food loss can help stabilize food prices by ensuring a more consistent supply, helping to protect both producers and consumers from price volatility. Improved post-harvest processes that reduce food waste will improve the quality of food reaching consumers, enhancing the nutritional value and reducing the risk of food-related illnesses resulting from contamination by pests, moulds, aflatoxins and other pathogens. Not only will this benefit individuals but also a more stable and safe food supply will reduce public health costs. Working with local suppliers and agrodealers, and supporting business development of MSMEs as service providers, the proposed programme will generate employment opportunities that should extend beyond the initial investment, with the intention for these opportunities to be of particular benefit to vulnerable groups such as women and youth.

23. **Gender and youth co-benefits.** On the customer side, the fee-for-service business model of the programme has significant potential for breaking down barriers to access by women farmers to climate adaptation technology. This model allows small-scale farmers, including female-headed households, to rent equipment, labour for services, or both, that they would otherwise find difficult to access. An interesting example is provided by previous programmes, where AGRA has tested and proven the deployment of “Uber for tractors”¹² digital solutions, resulting in more women getting access to mechanization services. The digital service requests are anonymous, and the service provider deploys the equipment on a first-come, first-served basis, so cannot discriminate against female requesters.

24. On the operator side, although AGRA hopes to prioritize women- and youth-led enterprises in supporting 2,000 MSMEs and cooperatives to take loans and start fee-for-service businesses, inclusivity is more challenging, given the (reasonable) eligibility criteria set out in annex 2 of the funding proposal. The qualifying groups must have a registration certificate and constitution, preferably a documentary track record as a service provider (in extension, aggregation of produce, selling of inputs or provision of mechanized services), and preferably qualified staff or members that have experience in operating, repairing and servicing the machinery. Given that, in many of the rural communities to be engaged in the programme, these are customarily seen as men’s areas of business, it seems unlikely that there will be many women-led businesses or cooperatives that will have sufficient baseline experience to qualify. The funding proposal does not suggest any small steps towards engagement of women’s cooperatives – for example, in providing auxiliary services, such as bookkeeping or keeping machinery clean and oiled. Such options could be explored through Activity 1.1.3 (on training for complementary services).

¹¹ Responses received by the iTAP on 6 January 2025 from AGRA to questions it received from the iTAP on 18 December 2024.

¹² For example, TroTro Tractor in Ghana: <https://www.trotrotractor.com>.

25. On youth groups, however, AGRA has suggested an innovative sub-model which takes into consideration that the eligibility criteria will in practice likely also exclude most youth groups (despite the characterization of the target MSMEs and cooperatives in the funding proposal, somewhat confusingly, as “youth groups”). RE-GAIN will work with formalized MSMEs to partner with local youth groups. The established MSME will take the lead in entering into the multi-stakeholder agreement (including taking a loan to acquire machinery) and taking accountability for the management of the agreement. The MSME will then recruit and work with an informal youth group from the community to operate the equipment, sharing the income and in the process developing the group’s capacities. Through this MSME-supervised approach, the informal youth groups will have the opportunity to build a track record, demonstrating their governance structures and creditworthiness for future opportunities.

26. **Environmental co-benefits.** The RE-GAIN programme interventions to reduce post-harvest food loss will have some environmental co-benefits. Reducing post-harvest food loss will increase the net productivity of agricultural land, improving the efficiency of land and water use and preventing their wasteful use in producing food that is never eaten. The need to compensate for food loss can be a driver of agricultural expansion, and AGRA posits this as a major environmental co-benefit – avoiding land degradation and biodiversity loss (also used in estimating emissions avoided). In the view of the iTAP, however, increased market access and improved grain quality might in fact cause some farmers to expand the land under production to take advantage of additional income-generating opportunities, potentially causing deforestation. So this co-benefit is not a given, and there is nothing in place to test these assumptions about non-expansion. The same applies to the argument that avoiding “regrowing of lost food”¹³ will reduce the environmental impact of fertilizers and pesticides, and related energy demands. A clear benefit is the reduction in air pollutants from burning of waste food, and the reduction of greenhouse gas (GHG) emissions from decomposition.

27. **Mitigation co-benefit.** GHG emission reductions, as a co-benefit of the programme, will be achieved through three pathways (with the emphasis on the first two in funding proposal section D.3): (a) reduction of emissions arising from the production of compensatory food (i.e. food produced to compensate for food lost in harvest and post-harvest without considering land expansion); (b) reduction of food loss as a driver of land-use change linked to compensatory agricultural expansion; and (c) reduction of emissions from wasted food left to rot, causing methane and carbon dioxide emissions. Combined, it is estimated that these pathways could result in a total emission reduction of 111,275 tCO₂e (tonnes of carbon dioxide equivalent) over the programme lifespan of 10 years (see methodology in annex 22 and country-specific breakdown in table 9 of the funding proposal). Given the doubt raised in the previous paragraph about the accuracy of the assumptions on avoided agricultural expansion, which could undermine pathways (a) and (b), it is positive that this is being considered as a co-benefit only.

28. **Mitigation methodology.** The iTAP asked why AGRA chose not to utilize the most obvious choice for estimating emissions reductions, i.e. the Verra VM0046 Methodology for Reducing Food Loss and Waste, choosing instead to base the estimations on a single publication, and queried whether the resultant estimates could be considered as suitably conservative. In a written response,¹⁴ AGRA explained that this methodology was initially considered but that it can only be effective in data-rich environments, which are not present in the programme context of informal and smallholder value chains, while gathering new data would be a big challenge in 14 value chains (two crops in each of the seven countries). The response explained: “In light of these constraints and considerations, the team went back to the first principles and developed an alternative approach that would make optimal use of the data available. The calculations are considered conservative in that they: (i) were focused only on those portions of

¹³ See paragraph 183 of the funding proposal

¹⁴ Responses received by the iTAP on 6 January 2025 from AGRA to questions it received from the iTAP on 18 December 2024.

the crop value of interest to the programme; (ii) did not consider additional emissions (such as those related to transport or storage/packing materials); (iii) made use of estimates for food loss supported by published studies that represent the average rates of loss, rather than very low or very high losses; and (iv) made use of emissions factors derived from a robust global-level study.”

29. In the view of the iTAP, having the anticipated GHG emissions reductions as a co-benefit justifies using a simplified approach to calculating their anticipated values. However, using average figures and not estimates at the lower end, and using only one literature source, is not deemed to constitute a conservative calculation approach. As a result, the figures given for the proposed programme’s anticipated mitigation results could overstate the actual impact that can be achieved.

30. Sustainable development potential is rated as medium to high.

1.4 Needs of the recipient

Scale: High

31. RE-GAIN will target seven African countries, six of which are classified as least developed countries and all of which are recognized as being highly vulnerable to climate change. Of particular concern is the highly affected agriculture sector, which employs between 40 and 80 per cent of the total population and contributes between 20 and 40 per cent of gross domestic product across the target countries. In addition to their sensitivity to climate impacts, all seven of these target countries are recognized as having low adaptive capacity and low readiness to adapt,¹⁵ which highlights their need for support. Most of these countries have a high level of sovereign indebtedness, and need grant funding to help address the challenges faced by poor, small-scale farming households, which are frequently highly exposed and vulnerable to negative impacts of climate change.

32. Climate change-driven (or -exacerbated) food spoilage and contamination in post-production stages of agricultural value chains compound productivity and yield losses already being experienced by these countries during cultivation, as a result of the changing climate.¹⁶ This results in a double blow to farming communities from climatic factors such as rising temperatures, higher air moisture levels, more frequent or intense heavy rainfall events, and more frequent flooding. Increasing post-harvest losses with climate change reduce the output that farmers have left to sell, depress incomes and jeopardize smallholder livelihoods, perpetuating vulnerability to climatic shocks and stresses as well as prolonging the cycle of persistent poverty. Farmers experiencing increasing losses during harvesting, field-drying and on-farm storage may choose to keep less of their reduced output for domestic consumption, worsening household food insecurity in countries that already have markedly high levels of undernourishment¹⁷. All of these reduce adaptive capacity and must be ameliorated to ensure the resilience of smallholder farmers in the seven countries.

¹⁵ According to the Notre Dame Global Adaptation Initiative (ND-GAIN) Index for 2021, where “adaptive capacity” is the ability of society and its supporting sectors to adjust to reduce potential damage and to respond to the negative consequences of climate events, and “readiness” refers to the ability to make quick, effective use of investments for adaptation.

¹⁶ In response to an iTAP question about whether these losses in the target countries have been quantified, the PSAA entity explained (in responses received by the iTAP on 6 January 2025 from AGRA to questions it received from the iTAP on 18 December 2024): “There is currently an absence of robust quantitative/quantified evidence on the current or projected impact of climate change on post-harvest food loss. The climate change linkage in the post-harvest value chain stages is increasingly being investigated, but the body of knowledge on this is relatively nascent and has yet to yield a modeled estimate or quantification that is tied directly to climate projections. For example in Chapter 5 on Food Security of the 2019 IPCC Special Report on Climate Change and Land https://www.ipcc.ch/site/assets/uploads/sites/4/2022/11/SRCCL_Chapter_5.pdf.”

¹⁷ See funding proposal paragraph 27

33. Given the high levels of poverty, weak markets and low adaptive capacity experienced in smallholder farming communities, farmers are frequently unable to adopt climate-resilient FL-RS without initial external support. Moreover, the national governments of each country, while prioritizing climate-resilient agriculture development, lack the financial, institutional and technical capacity to establish sustainable markets for FL-RS, with the majority of the limited resources being directed towards production. The RE-GAIN programme would target this vulnerable stage of the value chain, which, despite the urgent need, has remained relatively underdeveloped at the global scale – particularly when considering smallholder farmers. By targeting local suppliers and agrodealers, while also supporting the business development of MSMEs as service providers in the FL-RS value chain, RE-GAIN will build resilience directly within the target populations, integrating the solutions into local systems that meet the individual needs of each community.
34. The iTAP notes that specific country needs and measures to tailor the interventions to each local context are provided in the country-specific feasibility studies appended to annex 2 of the funding proposal.
35. The needs of the recipient are rated as high.

1.5 Country ownership

Scale: Medium to high

36. The proposed programme arose from discussions between AGRA and GCF, as well as extensive engagements AGRA has conducted in each of the target countries during the development of its five-year country strategies (2023–2027) – engagements resulting in a gap analysis across the agriculture and food system sector of each country. As a continental institution in agriculture and food systems, AGRA has widespread infrastructure and partnerships on the continent, with strong country teams in the target countries. Using its continental forums, the Africa Food Systems Forum and the Agribusiness Deal Room, AGRA will bring on board connections with governments, partners and other stakeholders to enhance ownership and coordinate actors, knowledge and resources that are fundamental to take food loss innovation to scale.
37. **Engagement with governments.** AGRA co-developed the programme together with the seven participating governments. Following initial consultations with the lead ministry (agriculture) and GCF national designated authority for each country, draft concept notes were shared to enable joint development of context-specific solutions. This was followed by multiple local and national stakeholder workshops (detailed in annex 7 of the funding proposal), which were combined with desktop research and experience in the target countries to establish a baseline identifying technical, institutional and market gaps that need to be filled to build the resilience of the target smallholder farmers. This process of stakeholder engagement will continue throughout the programme’s implementation. At the country level, the programme will be implemented under the overall guidance of a programme steering committee, which will meet quarterly. Each such committee will be co-chaired by a representative of the national designated authority and the AGRA country manager, with representatives of other key government departments and agencies (ministries of agriculture and departments for land resources conservation, crop development, agriculture extension services and agriculture planning services), the private sector and civil society organizations.
38. **Alignment with policy.** The focus on post-harvest loss reduction is aligned with key agriculture and adaptation strategies, as outlined in various policy documents and development strategies in each of the seven focus countries, detailed for each country in table 11 of the funding proposal. For example, the Burkina Faso National Economic and Social Development Plan (2016-2020) included a key focus on reducing post-harvest losses through improved storage, processing and distribution techniques to enhance food security and promote sustainable agricultural practices. The ongoing Ethiopian Agricultural Transformation Agenda

focuses on improving productivity and reducing post-harvest losses through infrastructure development and training. The Postharvest Management Strategy in Grains (2018) and the Working Strategy for Strengthening Ethiopia's Teff Value Chain (2015) also both focused on reducing post-harvest losses and enhancing value chains. The Zambian National Agriculture Policy (2012–2030) focuses on reducing post-harvest losses through better storage and handling practices.

39. **Engagement with civil society.** The funding proposal states that “the proposed programme is designed through the participation of a wide array of public and private institutions, and civil society (including national and international and development actors)”.¹⁸ The RE-GAIN programme is closely linked to the broader AGRA strategy, which “works across four strategic priority areas and is implemented through an integrated public-private programme and/or value chain platforms through which 28 million farmers are targeted for yield improving technology through access to extension, improved seeds and markets”.¹⁹ The iTAP notes that some civil society stakeholders favouring an agroecological approach have publicly contested what they perceive as AGRA's focus on sustainable intensification of production through technology-led solutions.

40. In response to a question from the iTAP about public debates on AGRA's approach, AGRA stated that it is “currently prioritizing partnerships with more than 60 civil society and farmers' organizations, including regional farmers' organizations (East African Farmers Federation, Southern Africa Confederation of Agricultural Unions, Association des Grossistes et Détaillants des Intrants Agricoles, Association Burkinabè d'Action Communautaire, Sasakawa Africa Association etc.), to collectively align on key issues to rally around. These efforts are part of AGRA's deliberate strategy and work plan for engaging with African farmers organizations and civil society, and will continue to help mitigate any perceived risks of working with them, as well as aid in the response to occasional criticism from actors who have different priorities and ideologies.”²⁰ It is hoped that as AGRA moves into its next phase, there will be space for a multiplicity of voices from the African continent and a vigorous debate about which approaches are best to achieve the organization's vision “to contribute to a food system-inspired inclusive agricultural transformation across Africa, to reduce hunger, improve nutrition, and adapt to climate change”²¹.

41. **Capacity of AGRA.** Founded in 2006, AGRA is an African-led organization dedicated to sustainably growing the food systems of Africa. The organization operates in 12 countries, employing over 300 staff and managing an average annual operating budget of USD 90 million²². In terms of the due diligence processes for the GCF PSAA, AGRA underwent a two-round capacity assessment that determined its suitability as a PSAA applicant (equivalent to an accredited entity for this funding proposal), the fiduciary functions it can play (basic fiduciary standards and project management) and its maximum environmental and social risk category (category B).

42. In terms of track record with similar initiatives, AGRA has managed the USD 4.6 million YieldWise project, aimed at reducing Tanzanian maize post-harvest losses at scale, as well as related projects funded by the Global Environment Facility and the IKEA Foundation. AGRA currently hosts the Partnership for Inclusive Agricultural Transformation in Africa,²³ a USD 416

¹⁸ Mentioned in paragraph 169, on the exit strategy.

¹⁹ Responses received by the iTAP on 6 January 2025 from AGRA to questions it received from the iTAP on 18 December 2024.

²⁰ Responses received by the iTAP on 6 January 2025 from AGRA to questions it received from the iTAP on 18 December 2024.

²¹ AGRA: About Us. Available at <https://agra.org/who-we-are/about-us>.

²² See funding proposal paragraph 137.

²³ Partnership for Inclusive Agricultural Transformation in Africa, Final Evaluation: Volume I – Final Evaluation Report. Available at <https://agra.org/wp-content/uploads/2023/08/PIATA-Final-Evaluation-Report-Volume-I.pdf>.

million partnership between AGRA and several other major donors²⁴ with AGRA as the implementing partner. The iTAP notes that RE-GAIN will be executed directly by AGRA through its Programme Operational Unit – which falls under the same legal entity as the applicant to the PSAA – and hopes that due care will be taken to maintain a firewall between AGRA in the applicant role of oversight (“the referee”) and AGRA in the executing entity role of execution (“the player”) to avoid any conflict of interest.

43. Country ownership is rated as medium to high.

1.6 Efficiency and effectiveness

Scale: Medium to high

44. The PSAA applicant AGRA, whose permitted fiduciary functions for this GCF programme do not include non-grant instruments, is seeking 100 per cent concessionality for the programme. With a total budget of USD 105 million, AGRA is seeking USD 75.6 million, or 72 per cent of the total, and would itself provide USD 29.4 million, or 28 per cent of the total, in grant co-finance.

45. The AGRA request for full concessionality is argued based on the nature of the investments, which “require targeted capacity development support combined with strategic conditional procurements to catalyse a paradigm shift in the post-harvest value chain using innovative FL-RS in the seven target countries. The two key financial models – 1) catalytic grants for smallholder farmers and 2) catalytic grants for agricultural MSMEs – use concessional funding to incentivize private sector suppliers and financial institutions (FIs) to provide physical solutions that are within the affordability envelope of farmers, while remaining commercially viable to private sector suppliers; and to provide tailored financial products that are affordable to cooperative and MSME service providers, while mitigating credit risk to financial institutions.”²⁵

46. **Concessionality and derisking.** In response to a query from iTAP about the risk of crowding out existing private sector suppliers, AGRA responded that “all interested and eligible suppliers (to be sourced through an open call following the eligibility criteria detailed in annex 2 of the funding proposal) will benefit from the subsidies to build and kick-start the market. The programme seeks to build markets where they are non-existent and incentivizes all suppliers that want to discover new markets and new client segments.”²⁶ The programme addresses the current market failure of FL-RS not being available or affordable in specific targeted geographies or for specific segments because of a range of barriers related to information, capacity, technology and finance. The intention is to grow the market and crowd in private sector players by derisking the rollout of new technologies. RE-GAIN will thus:

- (a) De-risk first-time users investing in new technology that remains unproven in their context. The “pay for two, get one free” type of investment will make sure the concessional funding ends up in farmers’ pockets;
- (b) Reduce the cost of distributing FL-RS to rural areas where demand is not yet confirmed (concessional funding will de-risk agrodealers and distributors from remaining with unsold stock);
- (c) Reduce the risk for FL-RS service providers investing in equipment for which the business case is not yet proven. The manufacturers will be paid, and the service

²⁴ Bayerische Motoren Werke AG (BMW); the Bill and Melinda Gates Foundation; the Mastercard Foundation; the Rockefeller Foundation; the United Kingdom Foreign, Commonwealth and Development Office; and the United States Agency for International Development (USAID).

²⁵ Funding proposal, paragraph 195.

²⁶ Responses received by the iTAP on 6 January 2025 from AGRA to questions it received from the iTAP on 18 December 2024.

providers will not be left with unpaid debts if the equipment is not able to be operated at profitable scale;

- (d) De-risk investments in awareness creation by sellers and distributors of FL-RS (extension, demonstrations, etc.) because awareness creation does not necessarily translate into adoption of FL-RS and profit for the sellers of FL-RS; and
- (e) Incentivize farmers to adopt FL-RS by reducing the cost of the FL-RS and enhancing the availability of FL-RS at the right time.

47. **Financial and economic return.** AGRA expects the RE-GAIN programme's combined approach of addressing supply- and demand-side constraints while improving enabling environments to mobilize USD 144.95 million worth of physical solutions over the next five years, representing a catalytic effect or return of 1.38 times the original investment of USD 105 million by GCF and AGRA. This anticipated financial return on investment excludes the long-term impact of the programme's ability to demonstrate the market viability of FL-RS to both users (farmers) and suppliers (manufacturers and financial institutions), which AGRA hopes will catalyse new private sector markets while addressing food loss and associated climate-related challenges.

48. AGRA undertook an economic analysis for the programme using a cost-benefit analysis approach. Programme costs and benefits were assessed over a 10-year period and using country-specific social discount rates as presented in annex 3 of the funding proposal. The analysis weighed up the total programme costs against the projected benefits associated with the uptake of equipment by smallholder farming households. These benefits included the income gains associated with the value of food that would otherwise have been lost in the absence of the programme interventions, the health benefits from reductions in aflatoxin exposure, the improved nutritional status (particularly of infants) from avoided food losses, and the climate benefits associated with reduced GHG emissions from food production. The results of the economic analysis suggest that, under the current set of assumptions, the total package of interventions is cost-beneficial, with the total programme investment generating an economic net present value of USD 69 million, with a benefit-to-cost ratio of 8:1 and an economic internal rate of return of around 56 per cent. On this basis, and with a sensitivity analysis conducted to show that these statistics are robust to changes in input values, there is a clear economic case for implementing the proposed programme.

49. Efficiency and effectiveness are rated as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

50. This proposed programme – RE-GAIN: Scaling solutions for food loss in Africa (brought to GCF through the PSAA) – is expected to make a significant contribution in helping the smallholder farming sector in Burkina Faso, Ethiopia, Kenya, Malawi, Uganda, the United Republic of Tanzania and Zambia to reduce their exposure and vulnerability to staple grain and legume crop losses that are worsening as a result of intensifying climate change hazards.

51. The iTAP retains a concern that the significant subsidies to be provided to reduce the costs of food loss reduction solutions, derisking the scale-up of these solutions in rural markets, might not have a long-lasting effect, given the subsidies' large and short-term nature. There may be the unintended consequence of a marked drop-off in demand following the subsidies' withdrawal. Nonetheless, the iTAP believes that the programme can have a major impact during the five years of its implementation alone, reducing the vulnerability and enhancing the adaptive capacity of this highly affected sector in vulnerable priority regions of the participating countries.

-
52. The iTAP recommends that AGRA closely track the impacts of the subsidies intended to stimulate market demand in the 35 priority regions of the seven countries – both the subsidy on small-scale food loss reduction solutions being purchased by farmers, and the subsidy on larger technologies for which MSMEs and cooperatives are taking loans in order to set up fee-for-service businesses. Since AGRA will remain engaged in these countries during the programme lifespan of 10 years, it should be feasible to use AGRA resources to track the development of these markets in the five years following programme closure. Although attribution is complex, and other factors will be at play, an evaluation can be made, based on this market tracking, of the extent to which these temporary subsidies, together with the programme’s value chain and enabling environment strengthening activities, have been able to create sustained demand.
53. The iTAP recommends that the Board approve this funding proposal.

Response from the project-specific assessment approach applicant to the independent Technical Advisory Panel's assessment (FP257)

Proposal name:	RE-GAIN: Scaling solutions for food loss in Africa
Project-specific assessment approach applicant:	AGRA
Country/(ies):	Burkina Faso, Ethiopia, Kenya, Malawi, Uganda, United Republic of Tanzania, Zambia
Project/programme size:	Medium

Impact potential
The AE appreciates the overall impact potential of RE-GAIN is assessed as high by iTAP. The programme targets two of the GCF results areas for adaptation, aiming to enhance the resilience of (a) the most vulnerable people and communities and (b) health and well-being, and food and water security, in the seven participating countries.
Paradigm shift potential
The AE acknowledges that the paradigm shift potential is assessed as Medium by iTAP. and agrees with this assessment. The AE also notes the recommendation from the iTAP that the AE closely tracks the impact of the subsidies intended to stimulate market demand in the 35 priority regions of the seven countries. The AE also positively notes iTAP recommendation to take a proactive approach in supporting local suppliers and manufactures.
Sustainable development potential
The AE acknowledges that the sustainability development potential is assessed as Medium-High by iTAP, particularly due to the stated mitigation impact. In the implementation phase RE-GAIN will consult widely to ensure a more balanced approach towards the calculation and tracking of mitigation benefits, with the limitations of the programmes' implementation framework. Further, the AE acknowledges and appreciates iTAP's note of the social, economic and environmental co-benefits of the RE-GAIN Programme.
Needs of the recipient
The AE appreciates that the needs of the recipient are assessed as High by iTAP, and agrees with this rating. RE-GAIN will build resilience directly within the target populations, by integrating the solutions into local systems that meet the individual needs of each community.
Country ownership
The AE acknowledges that the overall country ownership is assessed as Medium- High by iTAP and looks forward to utilising its continental forum, including the Africa Food Systems Forum and the Agribusiness Deal Room to bring onboard connections with governments, partners and other stakeholders to enhance ownership and coordination, sharing knowledge and resources to take food loss innovation to scale. The AE further notes iTAP's observation on maintaining a

firewall in the overseeing and implementation roles of the AGRA. The AE will use its existing programme implementation and overseeing frameworks to enforce and maintain this firewall.

Efficiency and effectiveness

The AE acknowledges that the overall efficiency and effectiveness of the RE-GAIN programme is assessed as **Medium- High** by the iTAP, considering the financial and economic returns from the programme and the overall economic analysis of the programme AE will actively monitor the key assumptions and variables during implementation.

Overall remarks from the independent Technical Advisory Panel:

The AE acknowledges iTAP's overall assessment and recommendation to the Board approval. The AE appreciates the time taken, recommendations given and the lessons learnt through this PSAA process.

ANNEX 8

Gender Assessment and Action Plan

Version 4



2024

RE-GAIN: Scaling Solutions for Food Loss in Africa

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LIST OF ACRONYMS

CCAP	Climate Change Action Plan
ccGAP	Climate Change Gender Action Plan
CSA	Climate Smart Agriculture
E&S	Environmental and Social
ESMS	Environmental and Social Management System
FL-RS	Food Loss Reduction Solutions
GBV	Gender based violence
GCF	Green Climate Fund
GDP	Gross Domestic Product
GGGI	Global Gender Gap Index
GHG	Greenhouse Gas
KNBS	Kenya National Bureau of Statistics
MRV	Monitoring, Reporting and Verification
MSMEs	Micro, Small, and Medium Enterprises
NAP	National Adaptation Plan
NDA	National Designated Authority
NDC	Nationally Determined Contributions
SEP	Stakeholder Engagement Plan
SSA	Sub-Saharan Africa
SWOT	Strengths, Weaknesses, Opportunities, and Threats
WBL	Women, Business and the Law

1 Introduction

1.1 PROGRAMME BACKGROUND

A great deal of attention has been paid in recent decades to the impacts of climate change on crop production, i.e., on growing risks to agricultural productivity. Scholarly investigations and public and private research have invested heavily in identifying and – where feasible – quantifying the ramifications of climate change on crop yields, yield stability over seasons, and in exploring plausible management options for the emerging challenges (CGIAR, 2023). As governments and societies look at how to minimize the risks of climate change, the impact of these changes on food production is increasing, fuelling concerns about food security and livelihoods for current and future generations.

Food security, however, is affected not only by changes in crop production but by changes occurring throughout the crop value chain, including during post-harvest phases (Akoth, 2020). It is therefore crucial to examine the impacts of climate change on a crop's value chain, including production, aggregation, storage, transportation, processing, and distribution. Each stage comprises several sub-processes, and climate change may plausibly affect many or all of the sub-processes too.

With the lion's share of research and resources for resilience interventions in the agricultural sector having been focused on production, the RE-GAIN project is an effort to give dedicated focus to harvest and post-harvest stages of the value chain – specifically, harvesting, post-harvesting handling and storage, processing, transportation, and logistics. As summarized in

Table 1-1, the International Fund for Agricultural Development (IFAD) report highlights a range of climate change concerns in the post-production stages of value chains and potential adaptation interventions that could increase resilience against such climate change concerns (IFAD, 2015).

Table 1-1 - Illustrative climate change risks and climate change risk management interventions in post-production value chain processes (adapted from IFAD, 2015)

Value Chain Components	Climate Risk Issues	Risk Management Interventions
Post-harvest management	Rising losses in harvest volume; declining safety, market quality and nutritional value due to increasing temperatures, humidity, pests and diseases.	Improve knowledge sharing on harvesting techniques to reduce losses. incentivize waste reduction measures and value addition for by-products; provide renewable energy sources to cover changing requirements for cooling, drying, milling, and threshing.
Siting of processing facilities	Extreme climate events (such as, floods, heatwaves, and storms) may damage processing facilities; shifting climatic conditions may render some sites redundant or increase transportation costs. It could create sustainable environment to pests and diseases, affecting both product quality and its suitability for consumption	Use hazard exposure and crop suitability maps to inform the siting of processing facilities; retrofit processing facilities with protective features; insure processing facilities against extreme climate events.
Energy in processing	High dependence on local bioenergy (wood, charcoal, dung, crop residues) has trade-offs with better soil management; rising temperatures require more energy for cooling.	Provide renewable energy sources (such as solar photovoltaic panels for cooling/drying/milling/heating, wind, biogas); equip processing facilities with energy-saving appliances (e.g., solar lighting, solar charging, efficient cook stoves); adopt pollution control measures.
Water in processing	Declining and more irregular water supplies; growing competition with other domestic or industrial users.	Re-site facilities closer to more suitable water sources; increase water storage and distribution capacity (water harvesting, communal ponds, groundwater recharge); introduce demand-side

Value Chain Components	Climate Risk Issues	Risk Management Interventions
		water efficiency measures; support conflict resolution for different water users (e.g., water user groups).
Packaging materials and methods	Rising temperatures and humidity may increase or decrease post-harvest losses and waste, as well as impact food safety, particularly if current packaging materials are impacted by high temperatures leading to produce damage or poor quality.	Design suitable packaging materials in parallel with waste and storage management strategies.
Processing infrastructure	Buildings and roads are exposed to higher peak rainfall, winds, and heat stress.	Introduce protective features and reinforcements into the design of critical infrastructure to handle run-off and higher temperatures; improve ventilation in buildings; harvest surplus water and energy from rooftops and appliances; use early warning systems.
Transport hubs and routes	Routes may become seasonally or permanently impassable (or open up); extreme events will disrupt logistics.	Re-site hubs; develop contingency plans for road, rail, water, and air transport; co-design value addition, storage, and transport components to avoid high-risk transport routes and seasons; upgrade docks, jetties, roads, and railways.
Refrigeration and cold chains	Temperature rises increase requirements for and costs of refrigeration; rising energy requirements increase greenhouse gas emissions.	Conduct cost-benefit analyses of dependency on refrigerated cold chains to assess best routes; introduce renewable energy sources for cooling and ventilation; optimize storage and transport management.
Just-in-time logistics	Extreme climate events (floods, storms, heatwaves) can make it impossible to comply with “just-in time” requirements.	Develop contingency plans for climate shocks and extreme events; create contingency storage opportunities; link into regional markets to avoid over-dependence on high-value export markets.
Demand from retail and consumers	Shifts in quantity and quality requirements and seasonality with climatic trends; disruptions in demand with climate variability, hence higher price fluctuations.	Assess market risks and opportunities before value chain implementation, including likely climatic impacts on high-value markets; strengthen and diversify storage to buffer price fluctuations; diversify into “off- season” crops.
Commodity labelling and certification	Increased consumer awareness as climate change may create new markets for sustainably produced and processed commodities with a low carbon footprint.	Explore opportunities for sustainable procurement, green labelling, and certification.

AGRA is a continental institution working in 15 African countries addressing food systems focussing on smallholder farmers’ production, marketing and nutrition. In the countries where AGRA operates, which are highly diverse in terms of climate, soils, crop choices and institutional capacity, neither all of these climate-related concerns may be applicable, nor all of these potential interventions possible. **Even within the range of what may be applicable, this programme is likely to look at a subset of risks that may be viable to address, and – given resource constraints – only a limited number of high-priority resilience interventions may be feasible to design and deploy.** RE-GAIN is an effort to identify the most salient risks, select the most impactful solutions, and implement the priority interventions through a well-structured, strategic, multi-country programme.

1.2 BRIEF PROGRAMME DESCRIPTION

There is a clear gap in knowledge, data and interventions designed to target the impacts of climate change at the harvest and post-harvest stages of the value chain, despite the mounting evidence of the ramifications on food loss and the impact

this has on land use changes and associated climate change mitigation. The majority of the current programmes designed to tackle climate-induced food loss focus on the pre-harvest stages of the value chain.

To address the pressing need for broader implementation of solutions aimed at reducing climate-related harvest and post-harvest food loss, the proposed programme is designed to raise awareness and build capacity to promote the adoption of Food Loss Reduction Solutions (FL-RS). It will do this by creating institutional capacity, facilitating the uptake of FL-RS by end users and service providers, increasing options of solutions' availability, and enabling practical application through policy interventions. This will include enhanced financial access for farmers and Micro, Small, and Medium Enterprises (MSMEs), empowering them to invest in climate-friendly FL-RS and incentivising vendors, manufacturers, and suppliers of climate-adapted FL-RS, fostering a robust market ecosystem.

A key focus is on strengthening the capabilities of countries to develop climate-resilient post-harvest infrastructure, both through providing physical solutions alongside capacity building along the value chains. This includes investing in strategic frameworks and implementation plans, including a regulated quality-based pricing system and tax exemptions on imports, for reducing food loss. By enhancing access to markets, the programme will encourage farmers to adopt FL-RS products and services, thereby boosting their climate and economic resilience.

Target Countries Overview

During the 2023–2027 period, AGRA plans to target 28 million farmers across 15 Sub-Saharan African countries, 40% of which will be women. The RE-GAIN Programme focuses on AGRA's activities in seven target countries, as shown in Figure 1-1 below. The RE-GAIN Programme is designed to combat food loss during the post-harvest stages and to boost climate resilience by fostering awareness and by building capacity for the adoption of Food Loss Reduction solutions (FL-RS). The programme aims to transfer these solutions to end users and service providers for practical application while facilitating financial access to farmers and Micro, Small, and Medium Enterprises (MSMEs) to invest in climate-resilient FL-RS. The programme plans to incentivize vendors, manufacturers, and suppliers to adopt these solutions and enhance the capacity of countries to develop climate-resilient post-harvest food handling infrastructure.

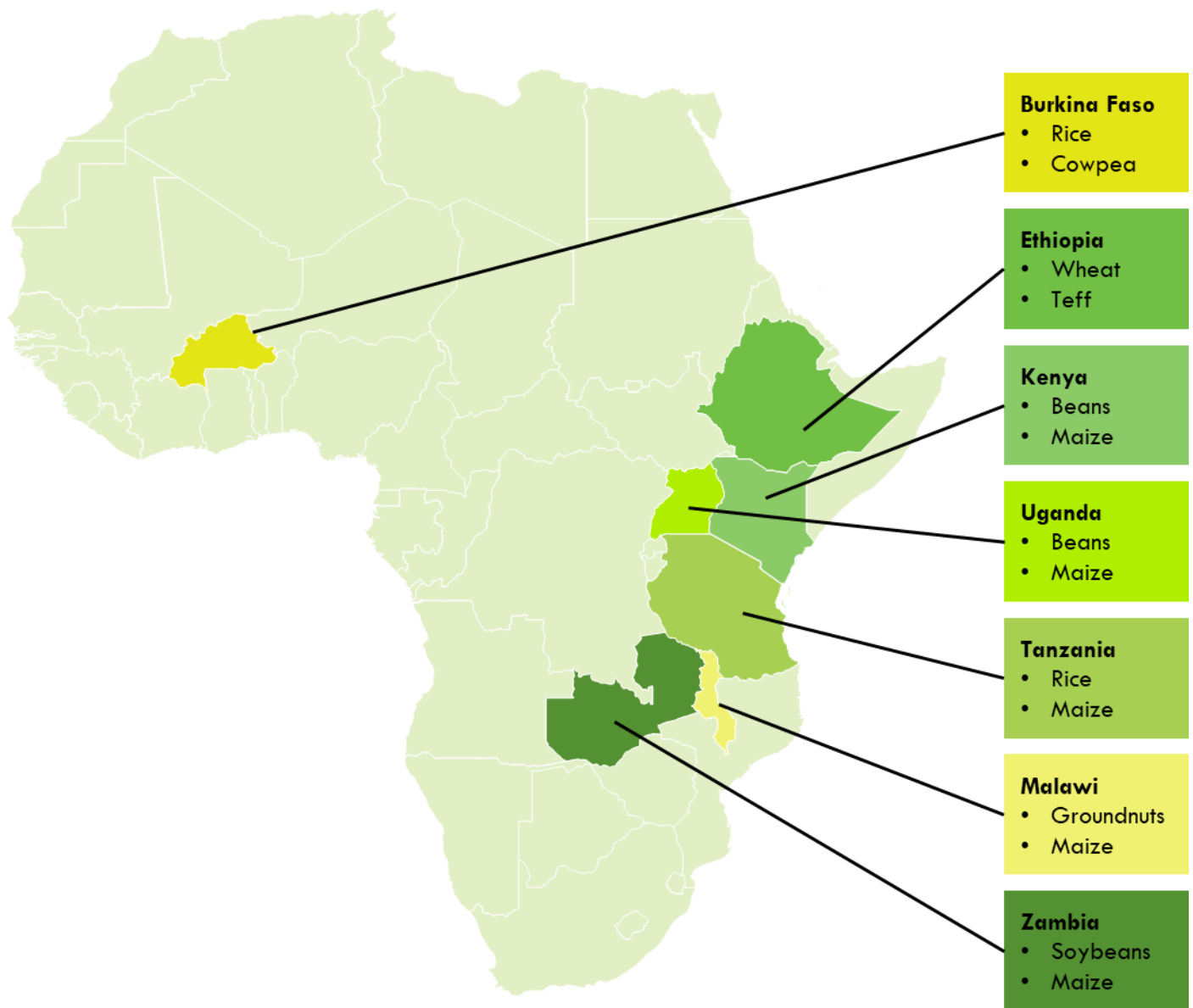


Figure 1-1 Focus Geographies for AGRA (2023-2027)

Crop selection

Key crops were identified by major stakeholders in the respective countries and expert assessments, supported by AGRA and the National Designated Authority (NDA) of each target country. Two major crops per target country were selected, based on area coverage, importance for food security and income, and climate vulnerability, to ensure that sufficient resources would be available for the crafting and execution of targeted solutions. Selected crops are representative of the agricultural dynamics of each country and aligned with the specific needs and strategic agricultural goals of the nation. In addition, these crops hold substantial importance to the country's food security and/or experience particularly high rates of loss within the value chain. Finally, these crops are produced in large parts of the respective countries by a significant number of smallholder farmers. The key crops, therefore, reflect the agronomic and economic realities of each country and provide opportunities for targeted enhancement of food security and sustainable agricultural practices. Additionally, the improved management of these crops is also expected to significantly reduction of GHG emissions contributing to the NDC targets of the countries involved. Figure 1-2 highlights the key crops selected for each of the countries within the programme.

Harvesting and Post Harvesting Definition

For the RE-GAIN programme, the key value chain stages considered are shown in Figure 1-2.



Figure 1-2 Strategic value chain stages included in the RE-GAIN Programme

The harvesting process within this RE-GAIN Programme proposal is defined as the interval between the culmination of agricultural production, marked by the crop reaching its maturity, and the initiation of post-harvest treatment. This process encompasses the identification of the optimal harvesting time and is further delineated into four distinct stages:

1. Removal of contaminated seeds, heads or cobs of matured crops at harvest
2. Reaping, which involves cutting, pulling, or gathering the mature crops.
3. Threshing, the process of separating the grain from the rest of the plant.
4. Cleaning, such as winnowing, to remove chaff and other impurities.
5. Hauling, which entails the transportation of the harvested produce to storage or processing facilities.

The post-harvest handling and storage stage commences once the crop exits the field and is typically conducted on the farm¹.

This stage encompasses several key operations, including:

1. Threshing, which can be performed manually or with mechanical threshing machines.
2. Drying, utilizing cribs, tarpaulins, and similar methods.
3. Cleaning and sorting, such as through winnowing, to remove impurities.
4. On-farm storage, which includes the use of granaries, hermetic bags, ordinary bags, stacks, metal silos, and plastic silos.
5. In some instances, primary processing activities, such as grinding, hulling, pounding, milling, drying, and sieving, are also conducted during this stage.

The processing, transportation, and logistics stage involves farmers selling their harvested crops either directly to traders, who collect the produce from the farm, or to collection centres and processors. These market participants then undertake the tasks of product accumulation, initial processing, quality control, grading, packaging, and transportation to wholesale buyers.

¹ In this instance, a field is where the crops are grown, and a farm consists of the whole small holding including the small aggregation site.

1.3 REASONING FOR REQUESTED FUNDING

Africa's food insecurity challenge has been exacerbated by climate change. Sub-Saharan Africa stands at a crossroads with an unprecedented opportunity for food systems transformation, driven by the demands of a rapidly growing population of 1.5 billion and the pressures of a changing climate (World Bank, 2023) (Worldometer, n.d.). The continent faces significant development challenges including food insecurity, resource degradation, poverty, gender inequality, and social exclusion. The vicious cycle of poverty and environmental degradation in Africa is evident in low crop productivity, deforestation, land degradation, conflict, migration, and vulnerability to climate shocks, which perpetuate persistent food insecurity and poverty. The effects of climate change are expected to be severe in Africa, where the capacity to adapt and respond to a changing climate is weak.

The impacts of climate change have increased over the past decades in Africa, manifesting in more frequent, intense, and prolonged extreme weather events, such as floods, droughts, heatwaves, locust outbreaks, desertification, and sandstorms. These extreme weather events have resulted in increased temperatures and humidity, shifts in precipitation patterns, water stress, and soil erosion. Most African countries already face recurrent droughts that affect growing seasons, often leading to short growing periods reducing the viability of farming in marginal agricultural areas. Projected reductions in crop yields in some countries could reach as much as 50% by 2030, and crop net revenues may fall by up to 90% by 2100, with smallholder farmers being the most affected (IPCC, 2018).

Therefore, the RE-GAIN programme aims to enhance the climate resilience and adaptive capacity of smallholders by promoting the widespread adoption of FL-RS in seven African countries. According to the World Bank estimates, a one percent reduction in post-harvest losses in Sub-Saharan Africa could lead to economic gains of \$40 million each year, and most of the benefits would go directly to smallholder farmers (World Bank, 2011). Moreover, food loss and waste are the result of an extremely inefficient use of resources and account for about 3.3 gigatonnes of greenhouse gas emissions globally (FAO, 2013). Large amounts of water and fertilizer also go into the production of food that never reaches human mouths. Recovering the food that is lost during harvest and post-harvest handling some can help close that calorie gap in Africa while strengthening livelihoods and improving food security— without imposing any additional environmental cost. Therefore, facilitated by the Green Climate Fund (GCF) investment, RE-GAIN will roll out a suite of physical interventions alongside capacity building and enhanced financial and market access. Not only will this benefit the respective countries as whole, but it also has the potential to benefit the region and the wider planet.

1.4 PROGRAMME GOAL STATEMENT

IF the capacity of the target countries and communities to respond to climate-triggered food losses is strengthened through improved and inclusive access to financing, promotion of context-specific and gender-responsive innovations to reduce food losses, and better enabling conditions for public and private investments, **THEN** smallholder farmers will have enhanced food security and livelihood resilience, **BECAUSE** the widespread use of food loss-reduction technologies will reduce food loss and reduce the carbon footprint of food systems, while increasing household income and building the resilience of smallholder farmers, MSMEs and rural communities to climate shocks.

1.5 PURPOSE AND STRUCTURE OF THE ANNEX

This Annex seeks to ensure that the RE-GAIN Programme is designed, financed, implemented and monitored in a gender-responsive and socially inclusive manner. It has been developed in accordance with both the GCF's Gender Policy, and its Gender Assessment and Action Plan Templates. The Annex adopts the GCF's three-pronged approach to mainstreaming gender, by presenting a robust **gender assessment** that considers the various factors (including those exacerbated by climate change) that influence gender inequality in each of RE-GAIN's target countries' agriculture sectors and identifies opportunities for women, men and youth to contribute to inclusive resilience building within post-harvest value chains. These opportunities are translated into **gendered actions** that promote gender equality and youth empowerment, and reduce disparities in the sector, which are captured in a **gender-sensitive monitoring and evaluation (M&E)** framework (i.e. the Gender & Inclusion Action Plan). This Annex is informed by a review of existing literature and secondary databases, extensive stakeholder consultations and expert input.

This Annex sits within the context of the broader funding proposal and should be read as such.

- Chapter 2 provides an overview of the RE-GAIN programme's three key components, outputs and outcomes. This framing is particularly important when reviewing Chapter 5 of this Annex, given the need to ensure that all Gender and Inclusion actions are appropriately embedded within the programme's broader level of ambition and scope of delivery.
- Chapter 3 presents an overview of the status of gender equality in each of RE-GAIN's seven countries. It focuses specifically on two aspects of gender equality, which are of primary relevance to this project – namely, i) gender and agriculture; and ii) gender and climate change. It also outlines the legal, policy and stakeholder landscape as it relates to the intersection of gender, climate and agriculture in each country – providing an important framework within which the Programme-specific gender interventions sit.
- Chapter 4 describes the approach, findings, and recommendations derived from Programme-specific stakeholder engagement and literature review. It identifies preferences, notable barriers and potential opportunities to overcome adoption of physical FL-RS, which in turn informs specific non-physical solutions that should accompany such support.
- Chapter 5 consolidates the recommendations from Chapter 4 into a Gender & Inclusion Action Plan, which contributes to – and aligns with – the Programme's broader Theory of Change.
- Chapter 6 concludes the Annex by outlining the overarching principles and approaches that will guide consistent, gender-responsive and inclusive Programme delivery, all of which are aligned to AGRA's existing institutional commitments and protocols.

2 Programme Overview

The RE-GAIN programme is designed to address food loss in Africa through an integrated approach, taking into account the growing need for climate adaptation in this space and the potential for mitigation of reducing food losses. The RE-GAIN programme's holistic approach to these growing challenges includes strengthening extension services, particularly as it portrays to climate information, demand-side interventions, supply-side development, financial mechanisms, and creating an enabling environment. Below is a comprehensive summary of the programme's three components. Further information on the programme design and process can be uncovered on the Feasibility Study (Annex 2) and the full Funding Proposal.

2.1 COMPONENT 1: FOOD LOSS-REDUCTION SOLUTIONS (FL-RS) DEMAND-SIDE DEVELOPMENT

This component focuses on increasing the demand for and adoption of FL-RS among smallholder farmers to reduce post-harvest food losses, enhance food security, and build resilience to climate change. The activities are centered around raising awareness, providing training, and improving market linkages.

2.1.1 Output 1.1: Support to Smallholder Farmers:

- **Awareness-Raising Campaigns:** Targeted campaigns educate farmers about the impacts of climate change on post-harvest losses and the benefits of adopting climate-resilient FL-RS.
- **Training Programs through Extension Services:** Farmers receive training on best practices for harvesting, post-harvest handling, storage techniques, and the use of FL-RS. Training is tailored to local contexts and emphasizes gender responsiveness and youth engagement.
- **Demonstrations of FL-RS:** Practical demonstrations showcase the effectiveness of FL-RS, helping farmers understand and adopt these solutions in their local environments.
- **Technical Assistance to Food Processors:** Assistance is provided to facilitate a shift to whole-grain flour production, which reduces food waste and improves profitability.

2.1.2 Output 1.2: Improved Market Linkages:

- **Technical Assistance for Value Chain Structuring:** The programme helps establish more structured relationships between farmers, processors, and buyers, ensuring better quality compliance and reduced post-harvest losses.
- **Linkages with Institutional Markets:** Farmers are connected to institutional markets like school feeding programs, which provide reliable demand for high-quality produce, thereby encouraging the adoption of FL-RS.

2.1.3 Outcome for Component 1

The main outcome of the component 1 is increased adoption of FL-RS by smallholder farmers.

2.2 COMPONENT 2: FL-RS SUPPLY-SIDE DEVELOPMENT

This component complements the first by improving the availability and affordability of FL-RS through support for local businesses and the introduction of innovative financial mechanisms to stimulate supply of FL-RS and to improve the

affordability of physical FL-RS solutions that can improve climate resilience for smallholder farmers and micro, small, and medium enterprises (MSMEs).

2.2.1 Output 2.1: Business Development Support for FL-RS Providers:

- **Training for MSMEs and Cooperatives:** Local MSMEs and cooperatives, especially those led by women and youth, are trained to develop sustainable FL-RS service provision operations, with lease-to-own as an example. This includes technical training on equipment use and business management.
- **Market Intelligence and Networking:** AGRA supports the development of market intelligence and networks among local manufacturers, importers, and agro-dealers to boost the FL-RS market.

2.2.2 Output 2.2: Financial Mechanisms and Physical Solutions to Support the Adoption of FL-RS:

To stimulate supply and increase the affordability of FL-RS that can increase smallholder farmers and community-led organisations' climate resilience, the following physical FL-RS solutions have been selected given their impact on food loss and its climate-impacted causes:

- **Hermetic Bags, Metal and Plastic Silos, Tarpaulins, and Plastic Sheets:** These are essential physical solutions aimed at reducing post-harvest losses. They protect crops from pests, moisture, and contamination, thus improving storage and prolonging the shelf life of produce.
- **Harvesting Machinery and Mechanical Threshers/Shellers:** These tools, including solar-powered options, are designed to reduce labor intensity and improve the efficiency of harvesting and post-harvest processing, minimizing losses due to improper handling.
- **Moisture Meters and Storage Protectants:** These devices help maintain optimal conditions for stored crops, preventing spoilage and contamination. Moisture meters are crucial for monitoring and managing crop moisture levels, which is vital in preventing mold growth and toxin production.
- **Communal Storage Structures and Warehouses:** These facilities offer shared storage solutions that help smallholder farmers and aggregators store their produce in controlled environments, reducing the risk of losses due to poor individual storage conditions.

To increase the affordability of these solutions in the market, the following financial models will be leveraged:

Model 1 promotes the local supply of FL-RS interventions by using conditional procurements to effectively subsidise interventions at smallholder farmer level (referred to hereafter as 'smart-subsidies'). At its base, the smart-subsidy model enables agro-dealers to provide FL-RS to smallholder farmers at a discounted rate by using GCF funds to procure one item for every two items² procured and sold by an agro-dealer. The subsidy is passed down to the smallholder farmers as a discount on the purchase price³.

The interventions are typically focused on smallholder farmers, with lower individual ticket sizes to the end-users — i.e. for FL-RS such as tarpaulins and plastic sheets, metal and plastic silos, and hermetic bags. The flow of funds will be facilitated through a trust/escrow account at a local financial institution (FI) and released once proof of offtake by eligible smallholder

² Note: the specific ratios will vary based on context of local markets in each country, with more developed markets receiving a smaller subsidy than less developed markets. A three-for-two model has been used as a base for demonstration purposes.

³ The maximum retail sales price and number of items per customer — as well as the eligibility criteria and geography where the sales at reduced cost can be made — will be included as part of the agreement between RE-GAIN and the suppliers as a condition for participation in the scheme.

farmers has been obtained. However, while funds will pass through an FI, it should be noted that no GCF funds will be transferred to a financial institution as the end beneficiary. The aim of the model is two-fold:

- to stimulate production and manufacturer capacity by placing pre-emptive orders of FL-RS, however, maintaining control over risk through a conditional release of funds to the manufacturer; and
- to reduce the cost of interventions at smallholder farmer level, additional profitability will drive additional demand and facilitate knowledge sharing about the benefits of interventions.

The model has been designed with two variations to allow for administration of the model using the smart-subsidy alone, or with the option of unlocking additional financing from partner financial institutions (FIs), with the latter offering greater potential benefits at the cost of greater complexity.

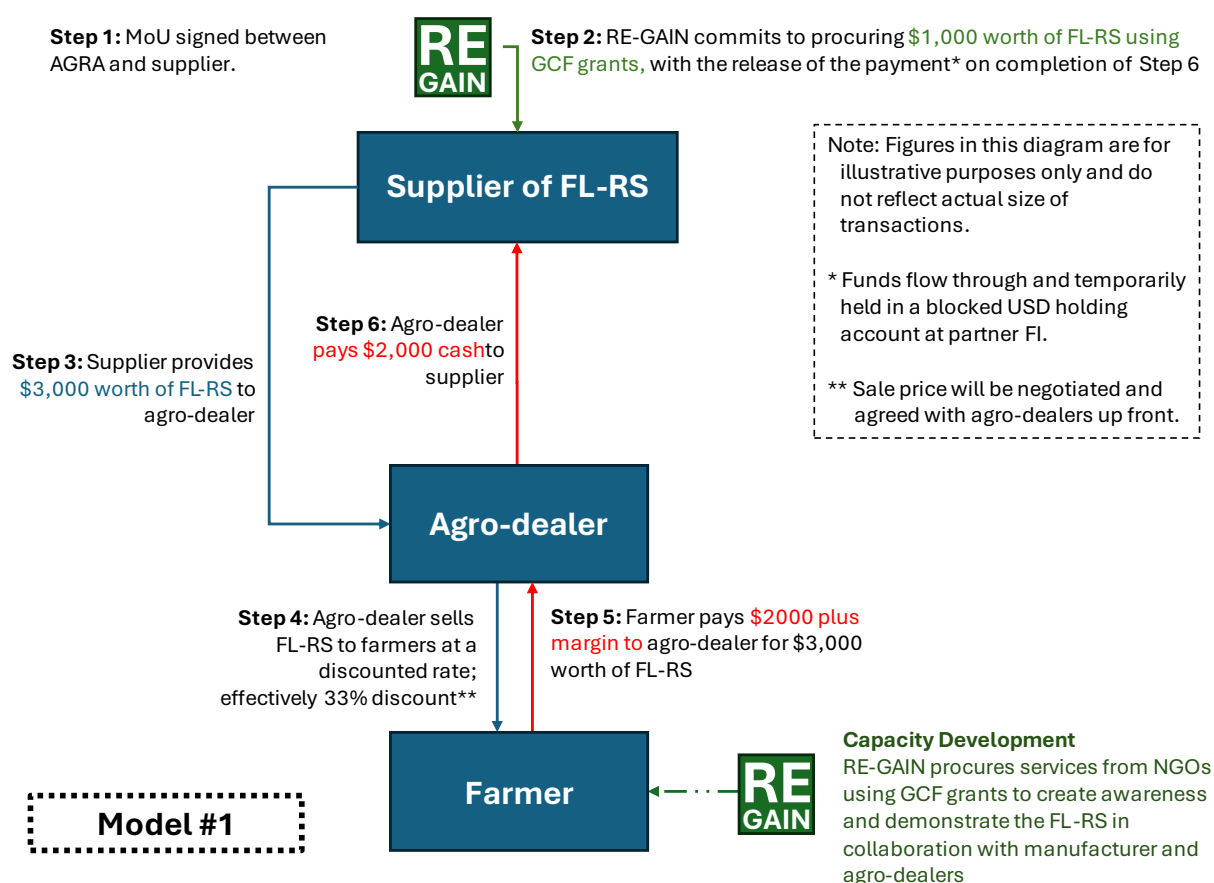


Figure 2-1 Model 1 RE-GAIN Programme

- The above models will be used for smallholder-focussed interventions, which include: tarpaulins and plastic sheets, metal and plastic silos, hermetic bags, and storage protectants and control agents of biological origin.

Youth Group/Cooperative Level:

Model 2 will use GCF grants to provide a unique approach to unlocking larger ticket items and food loss reducing equipment to farmer collectives or small to medium sized enterprises. The term 'youth groups' is used in this model description to highlight the priority given to this vulnerable demographic⁴; however, the model will also apply to other cooperatives, including

⁴ Preferentially targeting youth groups will help build resilience among this vulnerable group of the population, as well as unlock the energy and innovation that youth can bring to the process.

women's groups and local MSMEs. The aim of Model 2 is to create multi-stakeholder agreements with suppliers, youth groups, and financial institutions, with the goal of:

- Enhancing creditworthiness – through repurchase assurances from the suppliers that lower the loss given default.
- Reducing borrowing costs – Through a combination of the lowered credit risk (as per above) and subsidies on the purchase price. The structure will ensure larger-ticket FL-RS become more affordable and thus accessible to youth groups who provide services to smallholder farmers.

At the heart of Model 2 is the engagement of local youth groups that will be supported to act as service providers for FL-RS that require more expensive equipment that can service multiple farmers – such as mechanical multi-crop threshers and shellers (preferably solar-powered), moisture meters and communal storage structures. The establishment of the service operations will be supported through the business development under Output 2.1, ensuring that the youth groups have the necessary foundation to provide a reliable service. This approach leverages several key concepts to unlock the targeted benefits:

- Collectivism provides benefits to smallholder farmers in economies of scale through cost sharing and increased bargaining power with off-takers. These should promote further profitability and therefore additional demand for FL-RS.
- Post-harvest handling will increase the quality and quantity of agricultural produce, allowing smallholder farmers to capture more value, hence increased incomes.

The inclusion of financiers will further unlock access to finance in a typically underserved market. The structure aims to reduce credit risk by providing a partial subsidy that will lead to lower borrowing costs (thanks to smaller loan size and interest payments).

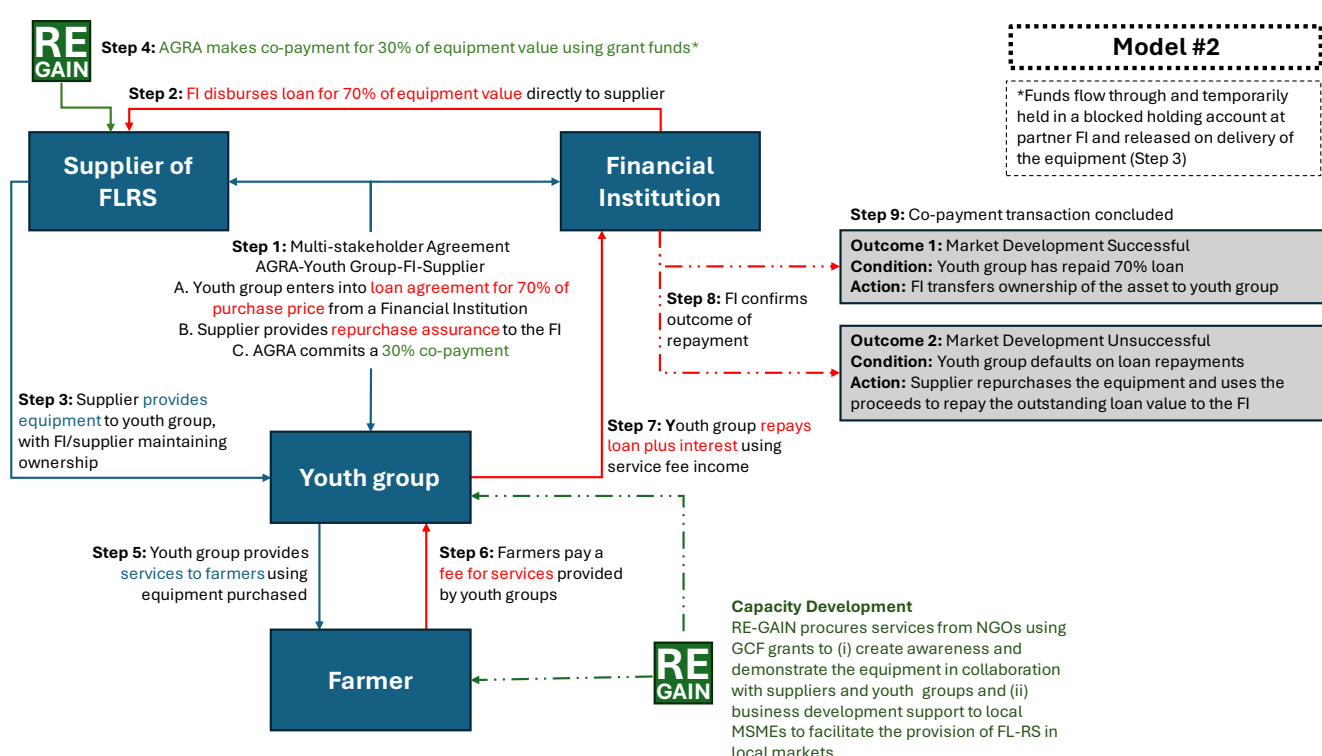


Figure 2-2 Model 2 RE-GAIN Programme

- The above financial model will be used for MSME's - focussed interventions, which include: mechanical multi-crop threshers and shellers (preferably solar-powered), moisture meters and communal storage structures

2.2.3 Outcome of Component 2

The main outcome of the component 2 is enhanced supply and affordability of FL-RS.

2.3 COMPONENT 3: ENABLING ENVIRONMENT FOR FL-RS MARKET SUSTAINABILITY

This component addresses policy-level constraints and builds institutional capacity to ensure the long-term sustainability and scalability of the FL-RS market, ensuring the long-term sustainability of the RE-GAIN programme

2.3.1 Output 3.1: Enhanced Capacity of National Institutions:

- **Policy Reform and Advocacy:** AGRA collaborates with governments to reform policies that currently hinder the adoption of FL-RS, including advocating for tax exemptions, reduced import duties, and the establishment of quality standards. AGRA also supports national governments to formulate, enact and implement new policies and regulations where gaps exist.
- **Institutional Capacity Building:** Technical assistance is provided to local and national government entities to support the scaling of successful FL-RS models and policies. This includes strengthening the analytical capacity of institutions to track and report on food loss and climate change metrics.
- **Development of Business Cases for FL-RS Investments:** Research is conducted, and evidence is gathered to inform solid business cases for investing in FL-RS, which are then used to attract private sector investments and promote successful business models for scaling and replication.
- **Monitoring, Evaluation, and Learning (MEL) System:** A comprehensive MEL framework is implemented to track the progress of the programme, assess the impact of interventions, and ensure continuous improvement and knowledge sharing.

2.3.2 Outcome of Component 3

The main outcome of the component 3 is strengthened enabling environment for the uptake of FL-RS.

3 Country and sector-level analyses

This Chapter describes the status of gender equality in each of RE-GAIN's seven countries. It presents an overview of the progress, commitments, and impediments to closing gender gaps at a national scale, and then presents a high-level summary of two aspects of gender equality that are of primary relevance to this programme – namely, i) gender and agriculture; and ii) the impacts of climate change on gender equality. This follows with an outline of the legal, policy and stakeholder landscape as it relates to the intersection of gender, climate and agriculture in each country – providing an important framework within which programme-specific gender interventions will sit.

3.1 KENYA

3.1.1 National overview: status of gender equality

Kenya is a committed signatory to numerous international agreements that promote gender equality, ensuring the protection and promotion of women's rights. It ratified the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) in 1984, and the Beijing Declaration and Platform for Action (BPfA) in 1995 (UN Women Africa, 2023). Additionally, Kenya is a participant in the Sustainable Development Goals (SDGs), specifically SDG 5, which aims to achieve gender equality and empower all women and girls. Through these treaties and others, such as the Maputo Protocol (the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa), Kenya has integrated international gender standards into its national legal and policy framework, working towards the promotion of women's rights and gender parity (UN Women Africa, 2023).

Box 1 Kenya's National Gender Legislation

Kenya has made significant progress in advancing gender equality, through the enactment of laws, regulations and policies focussed on increasing women's participation in political spheres, their access to finance, and labour force participation (UN Women Africa, 2023). The Constitution incorporates international treaties including CEDAW and BPfA, ensuring equality is embedded in national law with Article 27 guaranteeing equal treatment and opportunities for men and women. This has been complemented by gender-responsive budgeting efforts, where the Kenyan government is considered at the fore of pioneering and implementing such approaches at a federal level (WEF, 2024).

Further to this, there are a number of gender-sensitive policies including the National Land Policy (2009), which recognizes women's equal rights to own property alongside men. Other important policies include the National Policy for the Prevention and Response to Gender-Based Violence (2014) and the National Policy for the Abandonment of Female Genital Mutilation (2019) (UN Women Africa, 2023).

Kenya is ranked above regional averages for the World Bank's Women, Business and the Law (WBL) Index and the World Economic Forum's Global Gender Gap Index (GGGI). With a WBL score of 83.8/100, Kenya attained a perfect score for mobility, workplace pay and marriage indicators. Its primary areas for improvement relate to entrepreneurship and parenthood, where there remains a need to enact legislature that supports women's ability to start and run a business, and provide paid parental leave to allow women to more readily return to the workforce after having a child (World Bank, 2023). This score has increased in recent years, due to reforms that resulted in more stringent laws and policies on domestic violence, prohibition of gender discrimination in employment, and equal remuneration for work of equal value. Contrary to this, Kenya's GGGI rank of 75 out of 146 countries dropped by 2 places between 2023 and 2024 predominantly because of its subindex score for educational attainment, where Kenya ranks in the bottom 15 worldwide. Despite this, Kenya still ranks

15th (of out of 35) in Sub-Saharan Africa (SSA) overall – with a commendable score for its economic participation subindex. (WEF, 2024)

Barriers to effective implementation of policies, legislature and commitments

The differences in measures of progress between the WBL and GGFI can largely be attributed to several key factors, including inadequate implementation of laws, insufficient financing for gender equality measures, poor accountability mechanisms and overriding patriarchal gender norms, all of which have hampered tangible advancements in closing certain gender gaps (UN Women Africa, 2023). In particular, religious and customary laws, attitudes and practices are still considered key impediments to women's empowerment, as illustrated by data from the Kenya National Bureau of Statistics (KNBS) which indicated that only 29% of women were considered to be “empowered” in 2020 (Afrobarometer, 2023) (UN Women Africa, 2023). Notions around this concept of empowerment are linked to several important differences between women and men, with notable gaps in higher education, asset ownership, and financial autonomy, where women trail behind men in each of these areas (Afrobarometer, 2023). As a result, young women in rural areas are a particularly vulnerable group in Kenya. Limited control over, use of and benefit from productive activities, poor access to land markets and other resources, exposure to gender-based violence and the demands of unpaid care work collectively hinder a woman's ability to participate in the economy and perpetuate a cycle of overreliance on men (USAID, 2022). These issues are all pertinent within the context of this Green Climate Fund application. The following subsections therefore explore them in more depth, through the lens of agriculture and climate change.

Women in agriculture

Agriculture is the largest source of income for rural households in Kenya, and women account for 75 percent of the labour force in the sector. This means that, in addition to their reproductive roles relating to water, energy and food security as well as household health and care, they undertake the bulk of agricultural work for home consumption and sale (UNDP, 2020) (IISD, 2018). Despite this, women earn significantly less than male counterparts, and operate largely in the informal agricultural sector (with only 34 percent of women working in the formal sector), and they own a nominal percent of assets – including land. Cultural norms dictate that land is inherited by first-born males, meaning that women generally conduct their agricultural activities on the lands of men and are therefore beholden to the man's decisions on use of income and activities. With women owning as little as 1.62% of agricultural land, and land being an important form of collateral, they face major difficulty in accessing credit (UNDP, 2020). Unfortunately, to date, national climate and environment plans do not place explicit focus on securing women's land ownership rights (IISD, 2018). Women also lack access to information, technologies, extension support, financial services and marketing support – all of which are critical to building resilience and scaling up agricultural operations (Box 2 highlights several additional barriers specific to Kenya's seed system). Collectively, these differences between women and men farmers have resulted in as much as a 30 percent yield gap between male and female managed agricultural enterprises. And poverty is particularly pervasive among low-income earning women and female headed households (UNDP, 2020).

Box 2 Snapshot of findings from AGRA's Gender Gap Analysis in Kenya's National Seed System

AGRA's deep dive into the gendered differences in Kenya's National Seed System was conducted using the SeedSAT. It highlights areas for improvement, requiring effort by both the government and private sector actors – for example (AGRA, 2022):

- There is a need to more explicitly promote gender equality, and the inclusion of women and youth, in national seed systems through reforms to policy, legislature and regulation. Such reforms should be informed by greater engagement with seed producers – focussing specifically on small and micro producer units owned and managed by women.

- Women and youth's accessibility of the government's online and mobile-based platforms used by extension officers (i-Kilimo, National Farmers Information Service (NAFIS) and Plant wise) needs to be improved.
- Data advancements are necessary to ensure that i) seed data is shared widely and accessibly, using gender responsive tools; and ii) seed system data must be gender disaggregated where appropriate.

Climate change and gender equality

Both Kenya's Updated Nationally Determined Contributions (NDCs) and National Adaptation Plan 2015-2030 acknowledge that climate impacts are not gender neutral; women are disproportionately affected by extreme weather events due to their unequal access to key resources and services – such as land, finance, water, energy and education – which has major implications on the economy, particularly in rural areas (Republic of Kenya, 2020) (UNDP, 2020).

As such, the intersection of gender and climate change was explored in depth as part of Kenya's NDC review in 2020, which highlighted critical contextual factors relating to water, agriculture and energy, and the impact that climate change has on women due to their responsibilities and dependence on these sectors. Specifically, women's reliance on the natural resource base – both for productive and reproductive activities – makes them highly vulnerable to climate uncertainty. Prolonged droughts have historically led to high rates of household food insecurity and malnutrition, while extreme flooding destroys crops and infrastructure, increases incidence of waterborne disease and hinders access to key facilities (health, education, markets etc.). All these outcomes affect women more than men, given the traditional domestic and care -related responsibilities that Kenyan women shoulder. And within this broad social grouping, it is understood that pregnant women, women heads of household, lactating mothers, elderly women and schoolgirls are typically most vulnerable (UNDP, 2020). Contrary to this, given men are not tied to domestic responsibilities in the same way that women are, they have increased mobility and often migrate to other areas in search of income in the aftermath of extreme climate events. While this additional income is important for their families, it does mean that women then need to take up roles that were previously allocated to the men in their family (IISD, 2018).

3.1.2 Legal, policy and stakeholder landscape

Kenya's commitments to gender equality are enshrined in its Constitution, legal and policy frameworks, and dedicated gender machinery mandates at national and subnational scales. (USAID, 2022) National instruments and key stakeholders working at the intersection of gender, climate and agriculture in Kenya are captured in the following tables.

National instruments and commitments

Table 3-1 Legal and policy landscape: Kenya

Instrument	Relevant content
National Policy on Gender and Development (2019)	Agriculture is a key thematic area within the Policy, within which several key issues are flagged – relating to women being largely relegated to the informal sector and underpaid, lacking collateral which restricts their credit access, and balancing the constant tension of domestic, social and economic responsibilities. Amongst the policy actions proposed to tackle these challenges, one focuses specifically on facilitating adoption of engendered climate processes to ensure related livelihood practices are climate resilient. Women's role in adaptation and mitigation strategies more broadly is also highlighted within the ENR thematic area.
Updated Nationally Determined Contributions (2020)	Recognises that climate impacts are not gender neutral, and as such Kenya aims to implement its NDCs in a gender responsive manner. As such, four adaptation programmes focussed specifically on gender, youth and vulnerable groups are identified.

National Adaptation Plan (2015 – 2030)	Requires that all adaptation actions be implemented and monitored in a manner that reflects gendered differences, through the collection and analysis of gender disaggregated data. And, within the Social Pillar, a set of short-, medium- and long-term actions are described for Gender, Vulnerable Groups and Youth – aimed at strengthening their adaptive capacity.
National Climate Change Action Plan (NCCAP) 2023-2027	Gender considerations are mainstreamed across all priority action areas within the NCCAP. This includes the promotion of gender-responsive climate technologies and innovation within the private sector, the need for gender disaggregated data, as well as encouraging a gender responsive approach to climate change. It also commits to the development of “Gender-aware agricultural services”, to promote greater inclusion of women in the agricultural sector.
Kenya Climate-Smart Agriculture Strategy 2017–2026	Includes a SWOT analysis of Kenya’s agriculture and climate change sectors, including the degree to which they tackle equity and inclusivity. Key issues highlighted include gender bias, cultural beliefs and practices, and limited training facilities, and financial resources and human resources for women and youth. In response to this, the Strategy identifies several gender -focussed activities in its implementation framework, such as enhancing women and youth’s involvement in the CSA value chain
National Climate Change Response Strategy (2010)	Within the Strategy’s sectoral adaptation and mitigation interventions, there is a focus on expanding existing efforts and funding mechanisms to deliver gender-based response strategies – including the provision of grants to support women led environment and climate efforts; training of gender focal points on environmental management and disseminating climate change information in a gender sensitive manner.
Climate Change Act (Amendment 2023)	Aims to mainstream gender equity in all aspects of climate change responses. It states that the Act “shall be applied in all sectors of the economy by the national and county governments to mainstream intergenerational and gender equity in all aspects of climate change responses.”
Kenya: SDG 5 Strategy 2020-2025	Reflects on the progress that Kenya made in closing gender gaps – predominantly through policy reforms and affirmative action programmes – and highlights areas requiring more targeted attention, such as women’s representation in politics and decision making. It identifies nine broad strategic areas, which cut across sectors and themes – including agriculture and climate change.

Key stakeholders

Table 3-2 Stakeholder map: Kenya

Stakeholder	Mandate
State Department for Gender and Affirmative Action	Responsible for coordinating gender mainstreaming in national development and promoting equitable political and socioeconomic development for all. The functions of the Department are executed through four technical directorates, Country Gender Offices, Gender Focal Points and Administration.
National Gender Equality Commission	Sits at the centre of the gender mainstreaming efforts in Kenya, where it is responsible for providing advice and monitoring gender mainstreaming across various sectors.
National Climate Change Council	Responsible for approving “a national gender and intergenerational responsive public education awareness strategy and implementation programme” in line with the Climate Change Act 2016.

Climate Change Directorate	Responsible for educating on gender and climate change issues, including at national and country government levels.
Ministry of Agriculture and Livestock Development	Within the Ministry, there is a designated Gender Desk that is responsible for ensuring gender considerations are mainstreamed into government-led agricultural programmes.
National Government Affirmative Action Fund	A fund that provides financial support to vulnerable women, youth, people with disability, and the elderly towards their socioeconomic empowerment. It also provides an avenue for promotion of enterprises and value addition initiatives.
Women Enterprise Fund (WEF)	A fund that offers accessible and affordable financial and business support services to Kenyan women, including provision of subsidized credit to women entrepreneurs; capacity building for women led organisations; facilitating linkages between women MSMEs and larger enterprises; and enabling access to marketing products and services.
Kenya Women Finance Trust (also named “Kenya Women Microfinance Bank PLC”)	A microfinance bank regulated by the Central Bank of Kenya, aimed at supporting traditionally “unbankable” women in Kenyan society (who lack collateral) by giving them the opportunity to gain financial access to products, services and personalized financial solutions.
Uwezo Fund	A flagship programme aimed at enabling women, youth and people with disability access finances to promote businesses and enterprises at the constituency level. Since 2013, it has supported 50,842 women’s groups – many of which operate along the agricultural value chain.

3.2 UGANDA

3.2.1 National overview: status of gender equality

The Government of Uganda acknowledges the importance of expanding opportunities for women, both as a fundamental human right and as a pathway to sustainable, inclusive development. This commitment is reflected in Uganda’s gender-responsive legal and policy framework, which aims to reduce gender inequalities and address vulnerabilities across social, political, and economic sectors. At the international level, Uganda aligns with key instruments such as the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW), the Beijing Platform for Action (BPfA), and the Global Agenda 2030 Sustainable Development Goals (SDGs) (Uganda Bureau of Statistics, 2019).

Box 3 Uganda’s National Gender Legislation

Uganda’s National Gender Policy was revised in 2007 to reflect important development imperatives, including the need for increased participation of women in leadership positions, improvement in adult women’s literacy rates, and empowerment of women in business (Government of the Republic of Uganda, 2007). This Policy, alongside other related reforms such as the 2006 National Equal Opportunities Policy, have resulted in some progress in reducing gender inequalities and vulnerabilities, and promoting affirmative action in key sectors (Afrobarometer, 2023) (Uganda Bureau of Statistics, 2019).

Additionally, in 2021, two key pieces of gender legislation were passed (and ascended in 2022). The Succession Amendment Bill aims to address the inequality and marginalization faced by women and girls following the death of a spouse. It promotes gender equality in line with internationally recognized human rights standards. Meanwhile, the Employment Amendment Bill introduces provisions to address sexual violence and harassment in the workplace, ensuring safer and more equitable working conditions for all (WFD, 2021).

Uganda ranks top globally for the WEF's GGGI indicator on health and survival, which is attributed to its high scores for "healthy life expectancy" and "sex ratio at birth" indicators. Conversely, it sits in the bottom 20 countries for educational attainment, largely due to disparities in literacy rates and tertiary education enrolment (which is a stark contrast to their gender parity achievement for primary education enrolments). Overall, the country ranks 83 out of 146 countries. **(WEF, 2024)** According to WBL 2024, Uganda scores 83.8 out of 100, placing it above the SSA regional average. This is an improvement on its 2023 score, in large part due to a recent reform that now grants women the same rights as a man to obtain a divorce. In addition to this, Uganda's laws relating to women's freedom of movement, their decision to work and equal pay, Uganda gets a perfect score. However, there remains significant scope to redefine the country's laws on parenthood particularly as they relate to provision of paid maternity and parental leave. Legal inequalities also place constraints on women starting and running a business and accessing forms of collateral – like land – due the gendered differences in property and inheritance. **(World Bank, 2024)**

The World Bank's Gender Data Portal also sheds light on how Uganda is fairing in closing several key gender gaps, as well as areas requiring intervention. Interestingly, while there remains a gap in labour force participation, female participation has steadily increased since 1990 (67.9% in 2023 compared to 62.2% in 1990), while for males it has declined (from 77.2% in 1990 to 72.4% in 2023). Data also shows that women still spend 1.9 times more time on unpaid domestic and care work than men, and unfortunately the share of women in Uganda who experience intimate partner violence (45%) is significantly higher than the world average of 27%. **(World Bank, 2023)**

Barriers to effective implementation of policies, legislature and commitments

The often-overriding influence of informal institutions, including patriarchy, religion and sociocultural practices, are considered primary drivers of these gendered differences – making it particularly challenging to advance progress in key markers such as asset ownership and employment opportunities **(Uganda Bureau of Statistics, 2019)**. Within more formal institutions, significant obstacles persist in terms of budget constraints, as well as limited human and institutional capacity to implement existing frameworks. There is a need to focus on closing the gap between legislation and practice by enhancing accountability mechanisms to monitor and assess the implementation of these laws and policies effectively **(Uganda Bureau of Statistics, 2019)**.

Women in agriculture

Ugandan women play a critical role in the country's agricultural sector; they make up more than half of the agricultural workforce, and more women are primarily involved in agricultural practices than men (88% versus 78%) (ODI, 2021) (Bowen et al, 2015). However, findings from the Ugandan Bureau of Statistics' Uganda National Panel Survey – which considered differences in male and female managed plots – determined notable disparities between genders: 69% of all women in Uganda do not own land, male managed plots were on average 60% larger than females', and men were 11 percentage points more likely to plant cash crops than women farmers. This means that men dominate high value cash crop value chains and have larger livestock assets. Moreover, while use of improved seeds, fertilizer, herbicides, pesticides, and fungicides is low for women and men alike, female farm managers are less likely to use them than men. (Bowen et al, 2015) (UNDP, 2020)

Other studies have reaffirmed these findings and added further insight into why these disparities persist. Specifically, due to ingrained social norms, women continue to bear the burden of uncompensated time spent collecting water and firewood;

caring for children, the sick and the elderly; and working on their husband's land. These responsibilities result in time poverty, which is compounded by additional factors that disadvantage women, such as less financial control over use of proceeds from sales, limited access to assets / collateral like land, and difficulty in accessing markets or transport in more remote locations. (UNWomen, 2018) (Bowen et al, 2015) (USAID, 2021) Because most of these women are poor and lack access to critical services (including extension support on climate smart practices, and appropriate disaster risk reduction strategies) and infrastructure, they are also more vulnerable to climate-related agricultural shocks, like droughts and floods. And, because of these disasters, their vulnerability is heightened even further because of the increased domestic burdens they face as a result of loss of lives, livelihoods or access to the natural resource base upon which they depend. (USAID, 2021)

If Uganda was able to close its gender gap in the agricultural sector (calculated to be 13% in 2015), it has been estimated that the country could see an increase of 67 million USD in total GDP, whilst simultaneously lifting as many as 119,000 people out of poverty (UNWomen, 2018).

Box 4 Snapshot of findings from AGRA's Gender Gap Analysis in Uganda's National Seed System

AGRA's deep dive into the gendered differences in Uganda's National Seed System was conducted using the SeedSAT. It highlights evidence of inclusivity in the value chain, alongside areas for improvement. For example, women are well represented in agricultural research institutes and laboratories, as well as quality control departments within seed companies (where women in management tend to outnumber men).

However, as with many other countries reviewed using the SeedSAT, there is no evidence that government and private sector seed producers provide open and inclusive communication channels, extension support or training. (AGRA, 2022)

Climate change and gender equality

Uganda's 2019 Human Development Report outlined three rising, interconnected national challenges: climate change, gender inequalities and violent conflicts (Wijenayake et al, 2020). Gendered differences in Ugandan societies mean that women and female headed households are poorer, and more vulnerable to poverty. They also have less agency and ability to participate in decision-making, governance and policy-oriented processes at all levels (from household through to national Government) – all of which contribute to them being less resilient to the impacts of climate change and less empowered (and able) to act and recover in times of crisis. (UNDP, 2020) Given the critical role that women play – not only in Uganda's agricultural sector (described above) but also in its energy sector (where women are the primary providers of fuel) and waste disposal – it is clear that women should be central actors (as leaders and beneficiaries) in these three climate sensitive sectors, if Uganda is going to meet its sustainable development and climate change targets.

As a result, Uganda has made considerable effort to acknowledge the impact of climate change on women and elevate the role that they play in national climate action efforts. The National Climate Change Policy (2018) includes both a policy priority and a policy principle specifically on addressing the needs of, and including, women, girls and other vulnerable groups in Uganda's climate response strategies (Ministry of Water and Environment, 2018), and this has been translated into actionable measures through instruments like the Nationally Determined Contributions (both the original and updated versions) and related implementation plans.

Uganda's Updated NDCs (2022) were informed by a gender analysis which served as a basis for determining how to meaningfully enhance women's resilience and adaptive capacity within the context of existing societal dynamics that influence gendered differences in key sectors (Ministry of Water and Environment, 2022). As a result, the NDCs propose a series of actions to ensure gender-responsive NDC implementation – covering inclusive decision-making processes, specialist gender appointments in key ministries, gender-responsive budgeting, skills and capacity development and enhanced gender

disaggregation within NDC MRV systems⁵. This is complemented by the designation of gender focal points in key NDC sectors who are responsible for integrating gender actions (aligned to the NDCs) into sector planning and budgeting cycles. However, while the institutionalisation of these focal points is present at a national level, this is not consistently reflected in local government and district levels. For example, both Environment and Disaster Management Committees at the district level do not always have dedicated climate and gender personnel, despite being key on-the-ground interfaces with local communities, where implementation takes place (UNDP, 2020).

3.2.2 Legal, policy and stakeholder landscape

National instruments focussed on the intersection of gender, climate and agriculture are described below, along with the stakeholders involved in operationalising or supporting their objectives.

National instruments and commitments

Table 3-3 Legal and policy landscape: Uganda

Instrument	Relevant content
National Gender Policy (2007)	Provides a national framework for redressing gender imbalances, targeted at all levels of planning, resource allocation and implementation. It institutionalises gender mainstreaming across key economic sectors to ensure that gender considerations are integrated, responded to and monitored effectively.
National Equal Opportunities Policy (2006)	Promotes equality of opportunities for all people in Uganda, tackling issues of marginalisation, discrimination, exclusion and inequality in access to resources, services and benefits. The Policy makes mention of the gendered differences in Uganda's agriculture sector, where women play a central role but lack protection of their rights to securing, owning or controlling land.
National Climate Change Policy (2018)	Includes both a policy priority and a policy principle on gender and social inclusion, whereby women, girls and other vulnerable groups in Uganda must be considered central to the country's climate response strategies.
Updated Nationally Determine Contributions (2022)	Gender mainstreaming is an explicit element of NDC implementation, building on the previous NDC's support programme where gender was imbedded in the delivery of adaptation and mitigation measures. Dedicated gender focal points within key ministries (including Agriculture, Energy and Waste) are responsible for ensuring gender is mainstreamed into the implementation of climate related activities.
National Agriculture Policy (2013)	Commits to ensuring all agricultural development services will be offered in a gender equitable manner, and places emphasis on ensuring community empowerment and development is inclusive of all women, men, youth and other vulnerable groups.
National Strategy for Youth Employment in Agriculture (2017)	Considers the specific circumstances and needs of young women in the sector, who shoulder significantly greater burdens – and often make greater personal, economic or educational sacrifices – than male counterparts. Relevant Thematic Strategies focus on their specific needs, including lobbying for young women's access to and utilization of public and private land.

Key stakeholders

Table 3-4 Stakeholder map: Uganda

Stakeholder	Mandate
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⁵ Progress against the Updated NDC Implementation Plan is currently not publicly available, so it is not yet possible to ascertain how well Uganda is fairing against these gender-related commitments.

The Ministry of Gender, Labour and Social Development (MGLSD)	Responsible for the protection and promotion of vulnerable populations, focussed on tackling gender inequality and supporting cultural growth, labour and employment. The Ministry sets the foundation for other sectors to improve their outcomes on gender, social protection and transformation.
Equal Opportunities Commission	Mandated to monitor, evaluate and ensure that all national laws, plans, programmes and practices support equal opportunities and reflect affirmative action commitments, specifically regarding marginalised and vulnerable groups.
National Women's Council	National machinery with a mandate to organize and empower women in Uganda in a unified manner, supporting equality and development through training, networking, lobbying, provision of micro-finance mechanisms, etc.
Climate Change Commission	Sits within the Ministry of Water and Environment, as the designated NDA for the UNFCCC, coordinating all Uganda's climate change activities. A dedicated National Gender Focal Point sits within this Commission, to specifically coordinate gender mainstreaming in climate processes.
Uganda Women's Network	Women's advocacy organisation, with a membership structure that brings together various national organisations and individuals working in support of various thematic programmes. This includes a programme on economic transformation, which strives to overcome negative sociocultural norms, including redistribution of unpaid care work, toward the empowerment of women.
Finance Trust Bank	Microfinance, demand driven bank that focuses on providing low- and middle-income women with greater access to financial services.

3.3 ETHIOPIA

3.3.1 National overview: status of gender equality

Ethiopia is a committed signatory to a wide range of international conventions and protocols that champion gender equality and equity. These include the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW), the Beijing Platform for Action, and the Maputo Protocol (the African Charter on Human and Peoples' Rights on the Rights of Women in Africa), with some reservations related to marriage and inheritance. Ethiopia has also made progress on the Sustainable Development Goals (SDGs), including SDG 5, which focuses on gender equality. Recognizing the transformative potential of these frameworks in eradicating poverty, reducing inequality, and building a sustainable future, Ethiopia has proactively integrated and aligned the SDGs with its Ten-Year Development Plan (TYDP, 2021-2030), ensuring these goals are mainstreamed across all sectors of the economy (UN Women, 2024).

Box 5 Ethiopia's National Gender Legislation

Since the enactment of Ethiopia's National Women Policy in 1993, the country has made consistent progress in advancing gender equality and women's empowerment through the establishment of key institutions (most notably, the Ministry of Women and Social Affairs), participation in international treaties, and the introduction of progressive legislative reforms. To ensure that gender equity is carried through to all levels of decision-making and resource allocation, the National Women Policy requires all federal and regional public institutions to have a department responsible for overseeing gender issues – typically through Women's Affairs Bureaus. Increasingly, dedicated finance and capacities on gender mainstreaming are also being prioritised at the highest levels of government through – for example - the Ministry of Finance and Economic Development's recent adoption of gender-responsive budgeting and reporting for key sectors, like health. (Gender Equity Unit, 2023)

Ethiopia's progress is reflected in both the WBL Index and the GGGI – which each capture critical markers of progress, and highlight where notable disparities persist. WBL scored Ethiopia 76.9 out of 100, which is above the regional average. This is largely due to high scores for mobility, workplace and assets (where, for example, there is no gender difference in property and inheritance laws). Conversely, laws concerning pay and parenthood are particularly low, and to improve on these Ethiopia would need to introduce reforms that: support equal remuneration for work of equal value, allow women to work in the industrial sector, and stipulate paid parental leave to avoid women leaving (or being disadvantaged) in the workplace. (World Bank, 2023) According to the GGGI, Ethiopia ranks 79 out of 146 countries, with its political empowerment subindex scoring particularly high. This is visible through Ethiopia's female head of state (one of only five in SSA) and the fact that the country has almost tripled its score over the past decade (closing 43.1% of its gender gap in this category). In contrast, Ethiopia sits in the bottom ten countries worldwide for educational attainment, largely due to notable gendered differences in literacy rates and tertiary education enrolment (although small improvements are evident year on year). Progress toward parity in economic participation and opportunity is also low; while Ethiopia is one of six countries to have increased the share of women in technical positions by 1 percentage point between 2022 and 2023, in real terms, only 34.3% of technical positions are held by women. (WEF, 2024).

Barriers to effective implementation of policies, legislature and commitments

Beyond these markers of progress and areas for growth, it is evident that gender disparities differ vastly across urban and rural settings, ethnic groups and regions. Despite progressive laws and regulatory reforms, patriarchal social norms and structures continue to perpetuate gender gaps in many contexts which have major economic and social implications. (Gender Equity Unit, 2023) One of the major obstacles to achieving gender equality in Ethiopia remains the lack of reliable data. The absence of accurate information on the status of women hampers efforts to monitor progress and identify areas requiring policies and further intervention. Furthermore, much of the existing data is not disaggregated by gender, making it challenging to gain a full understanding of the situation. Therefore, advancing gender equality in Ethiopia demands a long-term commitment, collective action, and a comprehensive strategy that tackles the complex and interconnected aspects of gender inequality (UN Women, 2024). Additional work is also needed to implement and further develop policies that recognise unpaid care and domestic work. Given that women are disproportionately represented in this area, greater support targeted at reducing this burden on women would have notable impacts on closing the gender gap in employment and earnings. (UN Women, 2018). Gender-based violence is also rife; almost a quarter of all married women suffer physical abuse, and a third of all women experience some form of physical, emotional or sexual violence in their lifetime. (World Bank, n.d.) (USAID, n.d.)

Women in agriculture

In Ethiopia, women farmers perform up to 75% of farm labour, which equates to 70% of the country's household food production (Chandel et al, 2022) (UN Women, 2018). Despite these significant contributions, women farmers face major gendered disparities that have implications on both their personal circumstances and the broader economy. For example, they typically produce 35% less than male farmers and cultivate land that is on average 0.48 ha smaller than the land of their male counterparts. With less access to land, women farmers are inherently at a disadvantage given the critical role that land ownership plays in securing formal credit and other forms of collateral, along with other resources like water supply, improved seeds, fertilizers etc. This is coupled with notable disparities in women's access to extension services, education, transport, information and technologies all of which impact the effectiveness and sustainability of their farming practices, leading to less productive, profitable and diverse outputs. (Chandel et al, 2022) (UN Women, 2018) (World Bank, n.d.)

Poverty in most of Ethiopia's rural areas is also deeply gendered, particularly so in pastoralist areas. Women in these areas are poorer than men, with the poorest groups being those in female-only households. These same households tend to rely on female farm plot managers who are typically older women that are divorced or widowed, with lower education levels and who have a significant number of dependents. And while women dominate the agriculture sector, their participation in farmer forums is still often mediated by men which impedes their voice and agency in the sector. (UNWomen, 2018) (N. Mehary, 2023)

A 2016 study estimated the monetary value of the gender gap in the sector, which amounted to as much as 203.5million USD. Such findings are triggering shifts in agricultural legislative reforms to offer more inclusive and equitable investment in the sector, although noticeable improvements in practice take time due to the aforementioned ingrained societal norms. Some gaps in the country's national seed system are highlighted in Box 6 below.

Box 6 Snapshot of findings from AGRA's Gender Gap Analysis in Ethiopia's National Seed System

AGRA's deep dive into the gendered differences in Ethiopia's National Seed System was conducted using the SeedSAT. It highlights areas for improvement, requiring effort by both the government and private sector actors – for example (AGRA, 2022):

- Women are still not visible in seed quality assurance processes, and efforts are required to ensure greater inclusion of them among seed inspectors, seed importers and exporters, and their participation in national seed trade organizations and national seed growers' organizations.
- Due in part to a lack of guidance on inclusive communication channels, there are significant gender gaps in women's awareness of and participation in farmer groups, as well as their access to extension services.
- There is a need for the Government to consider the provision of incentives to women small-scale producers to effectively participate in the production and commercialisation of seeds.

Climate change and gender equality

The effects of climate change in Ethiopia are not gender neutral. Research shows that women and children are the highest risk groups, most vulnerable to climate shocks and stressors (with sub-factors, like location, age and disability status further compounding this vulnerability). The drivers behind this are attributed to gender-based differences in access to assets and credit, time use and responsibilities at household and community levels, and differential treatment of women and men by formal institutions (which affects their ability to receive information, benefit from support, etc.) (World Bank, 2021) (N. Mehary, 2023). Climate change and environmental degradation have a significant impact on the agriculture sector, which has major implications on farmers' productivity – the majority of whom are women that lack time-saving and climate-responsive technologies, tools and information (UNWomen, 2018). It is evident that climate change is furthering exacerbating inequalities which restrict Ethiopian women's ability to meaningfully participate in all spheres of life (UNDP, 2021).

Acknowledging this, as part of Ethiopia's NDC revision in 2021, a gender analysis was conducted to ensure key issues and entry points to promote gender equality through climate action were effectively integrated into the updated NDCs. The study found that, while Ethiopia's climate change policies and strategies for agriculture, forestry and energy were generally women-responsive, the areas identified for GHG emission reduction in the NDC's agriculture, forest and natural resource measures reflected "limited genderdization". The same was found to be the case for the NDC adaptation measures, where only four out of the 66 performance indicators stipulated the need for gender disaggregated data. This implies that, at the point of integrating these measures into implementation strategies and plans at the sectoral level, more needs to be done to ensure climate action is responsive to the differing needs of women and men. For example, the study recommended that the introduction of green technologies and approaches in the agriculture sector need to focus on reducing existing burdens on women – where the division of labour at the farm level is highly gendered. (UNDP, 2021)

Similarly to the updated NDCs, Ethiopia's Climate Resilient Green Economy National Adaptation Plan (NAP) reflects a clear commitment to ensuring adaptation options are designed and prioritised through a gender-sensitive lens, whereby the needs of women are given due attention so that they benefit equitably from climate action (Federal Democratic Republic of Ethiopia, 2019). However, beyond this principle on gender-sensitivity, little is stated in the NAP on gender-responsive approaches or sex disaggregated data collection that would indicate how this will be achieved in practice.

3.3.2 Legal, policy and stakeholder landscape

National instruments focussed on the intersection of gender, climate and agriculture are described below, along with the stakeholders involved in operationalising or supporting their objectives.

National instruments and commitments

Table 3-5 Legal and policy landscape: Ethiopia

Instrument	Relevant content
National Policy on Women (1993)	Requires that every federal and regional public institution must have a dedicated department responsible for coordinating gender issues, typically in the form of a Gender Directorate / Gender Focal Point.
Climate Resilient Green Economy Facility: Gender Mainstreaming Strategy	Provides strategic direction to guide the CGRE Facility in tackling gender inequalities through its operations, in the form of a Gender Action Plan.
National Adaptation Plan (2019)	Includes a guiding principle focussed specifically on ensuring gender sensitivity informs NAP implementation and prioritisation.
Updated National Determined Contributions (2021)	Informed by a detailed gender gap analysis, the updated NDCs stipulate that gender-responsive approaches must be adopted across all adaptation measures (despite only four performance indicators explicitly requiring sex-disaggregated data).

Key stakeholders

Table 3-6 Stakeholder map: Ethiopia

Stakeholder	Mandate
Ministry of Women and Social Affairs	Responsible for ensuring women's rights and promoting their participation in all spheres of life – social, cultural and economic. The Ministry receives a regular budget from Government for gender-related NDC activity implementation.
Gender Affairs Directorate at the Ministry of Finance	Responsible for gender-responsive budgeting at a national level.
Regional Ministries of Women's Affairs	Bolsters the capacity of, and supports, the Ministry of Women and Social Affairs to deliver its mandate at a decentralised (regional) scale.
Ministry of Agriculture – Gender Focal Point	The Ministry includes a team leader focussed specifically on Women, Children and Youth. As with the Ministry of Women and Social Affairs, this Ministry also receives a regular budget from Government to implement NDC measures – with the understanding that these should be gender-responsive.
The Network of Ethiopian Women's Association	An advocacy group dedicated to advocating for women's voices to be heard, lobbying for their rights and fostering the advancement of gender equality in Ethiopia.
Enat Bank	An inclusive national bank, dedicated to supporting women's economic needs through innovative technologies and services.
Organization for Women in Self Employment (WISE)	A CSO dedicated to supporting low-income self-employed women and girls, to improve their self-reliance and quality of life through targeted programmes focused on

	business skills training, leadership, financial and technical education and the establishment of women-led micro enterprises.
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3.4 BURKINA FASO

3.4.1 National overview: status of gender equality

Burkina Faso has demonstrated its commitment to international gender frameworks, aligning with global efforts to promote gender equality and women's empowerment. As a signatory to key international conventions such as the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) and the Beijing Declaration and Platform for Action, the country has integrated gender equality principles into its national development policies. Burkina Faso has also embraced the Sustainable Development Goals (SDGs), particularly Goal 5, which seeks to achieve gender equality and empower all women and girls. Through these commitments, the government has worked to enhance women's participation in decision-making, improve access to education and healthcare, and address gender-based violence. Despite challenges related to poverty and security, Burkina Faso continues to work towards meeting its international obligations for gender equality (The Global Green Growth Institute, 2019).

Box 7 Burkina Faso's National Gender Legislation

Burkina Faso's Constitution affirms gender equality, and the country has established a national institutional framework to promote the status of women through the Ministry for the Promotion of Women and Children. In 2013, Gender Groups were created in each ministry to promote gender mainstreaming, though their capacity remains limited. The Government adopted the National Gender Policy in 2009, aimed at ensuring equal participation, access to resources, and decision-making for both men and women. That same year, a law was introduced mandating a 30% quota for women's representation on electoral lists, with political parties facing a 50% funding cut for non-compliance (The Global Green Growth Institute, 2019).

Burkina Faso scored 82.5/100 under the World Bank's Women, Business and the Law (WBL) Index, above the regional average, achieving a perfect score in workplace, entrepreneurship, assets and pension. The main areas for improvement include pay, where there is no law that mandates equal remuneration for work of equal value for men and women, and where women are unable to work in jobs deemed hazardous including construction and factories (World Bank, 2023). Under the World Economic Forum's Global Gender Gap Index (GGGI), it was ranked at 120 out of 146 countries, a drop of 11 places since last year. Its position is largely attributed to its low scores for educational attainment and political empowerment, though it was noted that it is one of seven countries in sub-Saharan Africa to have achieved equal gender representation in senior economic roles (WEF, 2024). Nevertheless, it still scored lower than the regional average, with further progress needed.

Barriers to effective implementation of policies, legislature and commitments

Gender inequality remains a significant issue in Burkina Faso, despite efforts to enhance women's participation and representation. Women and girls continue to face disproportionate barriers across education, healthcare and economic opportunities, worsened by humanitarian crises such as the rise of militant extremist groups and the impact of the COVID-19 pandemic. These crises have exacerbated existing inequalities, making it even harder for women to access opportunities and

resources. Furthermore, deeply entrenched patriarchal norms and cultural practices have impeded progress toward gender equality (CEPPS, 2021).

Although the Burkinabe government has introduced measures, including a gender quota for parliamentary candidates, women remain severely underrepresented at the national, regional, and community levels of politics, limiting their influence on decision-making processes (CEPPS, 2021). Additionally, a recent study affirmed that there is a significant gender gap in wages, driven by a notable difference in the number of women who are not paid at all for their work, in part due to drivers like occupational segregation across all sectors, with women typically being employed in lower-productivity sectors. This gender gap is not associated with educational attainment. Instead, it stems from women experiencing lower economic benefits from marriage and the presence of young children in the household, meaning they are often employed as unpaid work in family businesses and farms. This suggests that social norms, which view women's primary role as caregivers, are the underlying cause of this disparity (Aletheia Donald, 2024).

Women in agriculture

Across Burkina Faso, agriculture employs a significant portion of the workforce, accounting for 74% of the overall labour force. In total, 76% of women, compared with 72% of men (International Labour Organization, 2022) are engaged in the agricultural workforce. In rural regions of the country, the primary area for agricultural production, women assume responsibility for domestic, agricultural and non-agricultural tasks. This includes providing the household with food and looking after children, as well as working on the farm doing tasks from weeding to harvesting. Despite their numerous responsibilities, women have significantly less access to the resources and services necessary to enhance their productivity and income and to alleviate their domestic burdens (Diendéré AA, 2023).

In those instances where women own a farm, this reduced access to services translates into barriers when trying access to financial resources and credit. This has led to women primarily concentrating on lower-value crops, as well as having reduced adoption rates of farming technology. Where women are utilized in the form of labour, they are unlikely to be paid at all, largely down to them primarily being recruited to work on family farms (World Bank Group, Africa Region, 2024).

Climate change and gender equality

Burkina Faso submitted its updated Nationally Determined Contributions (NDC) in 2021. It emphasises the need for gender analysis in the priority sectors of the NDC and the development of gender-sensitive NDC sector plans. It also states that any data collected will be disaggregated by sex where possible and to include more women (and young people) in agriculture (Government of Burkina Faso, 2021).

Burkina Faso has submitted a National Adaptation Plan (NAP) in 2015, which acknowledges that women are more vulnerable to the impacts of climate change and there is need to account for this when implementing the activities outlined in the NAP. It also notes the need to integrate gender mainstreaming when considering access to adaptation techniques, and to provide increased capacity building in leadership for women (Burkina Faso Ministry of Environment and Fishery Resources, 2015). It has not submitted a Climate Change Gender Action Plan (ccGAP).

3.4.2 Legal, policy and stakeholder landscape

National instruments and commitments

Table 3-7 Legal and policy landscape: Burkina Faso

Instrument	Relevant content
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National Gender Strategy 2020 – 2024	Emphasizes the integration of gender perspectives in addressing the impacts of climate change on agriculture. It highlights the vulnerability of women in the agricultural sector and the need to empower them through capacity-building and equitable access to resources like land and credit. The strategy advocates for gender-sensitive policies that enhance women's resilience to climate change, ensuring their active participation in climate adaptation strategies
Nationally Determined Contributions (NDC) in 2021	Emphasizes the need for gender-sensitive NDC sector plans, and the need for gender disaggregated data.
National Adaptation Plan (NAP)	The NAP acknowledges both the impacts of climate change on women, along with ways to address this.

Key stakeholders

Table 3-8 Stakeholder map: Burkina Faso

Stakeholder	Mandate
Ministry for the Advancement of Women and Gender	Responsible for ensuring the effective implementation and continuous monitoring of the Government's policy on advancing women and gender equality, with a particular focus on reducing gender inequalities to promote equitable and sustainable human development
Association Burkinabé des Femmes Battantes (ABFB)	Formed in 2008, it was created to improve the development opportunities for women. Many of its programme's centre around agriculture, including building a processing plant for cassava which will serve two purposes; provide a consistent source on income and serve as an education centre for other women who may be interested in processing and selling their own products.

3.5 MALAWI

3.5.1 National overview: status of gender equality

The Government of Malawi has implemented robust measures to ensure that social inclusion and gender equality are integrated into all development frameworks. Internationally, Malawi ratified the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) in 1987 and is a signatory to the Beijing Declaration and Platform for Action (BDPA). Additionally, Malawi ratified the Southern African Development Community (SADC) Protocol on Gender and Development in 2008 and has committed to the UN Sustainable Development Goals (SDGs), including SDG 5, which focuses on achieving gender equality and empowering all women and girls. Progress on SDG 5 in Malawi includes legislative reforms, increased female participation in decision-making, and efforts to reduce gender-based violence. These actions demonstrate the Government's commitment to advancing the legal status and protection of women and vulnerable groups (African Development Bank, 2020).

Box 8 Malawi's National Gender Legislation

The Government of Malawi's efforts to improve the current state of gender equality can be seen through its increasing number of policies on the matter. The Constitution of Malawi explicitly guarantees equal rights for women, prohibiting discrimination based on language, culture, or religion. Section 30 of the Constitution affirms the rights of all individuals and communities, giving special attention to women, children, and people with disabilities.

Additionally, the Government has enacted several gender-related laws to address the challenges faced by women and girls. These include the Prevention of Domestic Violence Act (2006), the Deceased Estates (Wills, Inheritance and Protection) Act (2011), the Gender Equality

Act (2013), the Trafficking in Persons Act (2015), the Child Justice Act (2010), and the Marriage, Divorce and Family Relations Act (2015). In 2017, the Constitution was further amended to align the legal definitions of a child and marriageable age at 18, aiming to eliminate child marriage in all forms. The Land Act was also revised in 2017 to enhance women's access to, control over, and ownership of land, a critical productive asset (African Development Bank, 2020)

Efforts to integrate gender considerations into broader country strategies include the "Malawi 2063 Vision" which acknowledges the negative impact gender inequality has across all sectors and commits to advancing gender equality at all levels through a multi-sectoral approach (Malawi National Planning Commission).

Malawi scored 80.0/100 under the World Bank's Women, Business and the Law (WBL) Index, achieving a perfect score in workplace, pay, marriage and assets. The main areas for improvement include mobility and parenthood, where women are unable to choose where they live or apply for a passport in the same way as a man, and there is no paid leave of at least 14 weeks available to mothers (World Bank, 2023). Malawi was not covered in the 2024 World Economic Forum's Global Gender Gap Index (GGGI) (WEF, 2024). However, it was included in the 2023 report, with a rank of 110 out of 146 countries, and an increase of 22 places from the previous year. This is largely due to a substantial shift in its subindex score for political empowerment, due to the increasing number of women in ministerial positions (WEF, 2023).

Barriers to effective implementation of policies, legislature and commitments

The Ministry responsible for gender issues has limited capacity and funding, weakening coordination efforts. Key structures like gender focal points exist but are under-resourced, hindering effective implementation of gender and empowerment programs. Gender-based violence (GBV) remains prevalent, and laws addressing GBV and early marriage are not widely enforced due to poor dissemination (African Development Bank, 2020). . National data indicates that Malawi has a larger gender gap in secondary enrolment compared to the Sub-Saharan Africa average and lower overall enrolment levels. Girls choosing to drop out of school prematurely, often in favour of starting a family, can have lifelong implications for their health, fertility, skills, economic opportunities, and agency. These factors are interconnected in complex ways, but global evidence indicates that investments in keeping girls in school are particularly crucial. These investments have substantial impacts on their earnings, standard of living, child marriage and early childbearing, fertility, population growth, health and nutrition, agency and decision-making, social capital and institutions, and future per capita investments in human capital (The World Bank, 2022).

Women in agriculture

The Malawi economy is highly dependent on agriculture, contributing over a quarter to the country's GDP, and employing approximately 80% of the population with the majority being women. However, the sector's contribution to GDP is hampered by the large gender productivity gap. On average, plots managed by women are 25% less productive. This is largely attributed to lower levels of inputs (including seeds and fertiliser), reduced access to labour, technology and knowledge as well as constraints on crop choice (The World Bank, 2022).

A recent study in Malawi, looking at Maize fields, found that female farmers were much more likely to use poor quality seeds, reduced amounts of fertiliser and farm on lower quality soil compared to their male counterparts. Additionally, the study found that when accounting for the quality and quantity of these inputs, there was no gender gap in agricultural yield (William J. Burke, 2021). Additionally, extension services are critical for ensuring women have equitable access to agricultural inputs and receive fair returns from those inputs. However, research indicates that female farmers and female-headed

households in Malawi are less likely to receive agricultural advice than men. Additionally, the rate of female-headed households receiving advice decreased between 2016 and 2018. Older women heading households face a particular disadvantage, being the least likely to receive agricultural advice (Ragasa, 2019). This is further compounded by the fact that most women plot managers are widowed, divorced, or separated, leading to fewer household members contributing to agricultural labour. Consequently, women often rely on female family labour, including themselves and their children as they are rarely able to afford to hire male labour. When they do, it's often less productive due to affordability issues, cultural bias, or women's competing domestic responsibilities (The World Bank, 2022).

Box 9 Snapshot of findings from AGRA's Gender Gap Analysis in Malawi's National Seed System

AGRA's deep dive into the gendered differences in Malawi's National Seed System was conducted using the SeedSAT. It highlights areas for improvement, requiring effort by both the government and private sector actors – for example (AGRA, 2022):

- There is no evidence of breeding, variety release, and maintenance activities aimed at promoting gender equality and including women and youth. Additionally, although smallholder farmers, especially women, have extensive knowledge of seed varieties, seeds are still exchanged in agrarian communities marked by gender inequalities, unequal land distribution, and social stigma.
- Trained extension officers are largely men, reinforcing the perception that women are not farmers and should not be consulted on agricultural decisions
- Women's participation in farmer organisations, such as labour-saving cooperatives, producers' organisations, and marketing groups, is limited. This limitation negatively impacts their access to product markets and, more importantly, their potential for resource allocation.

Climate change and gender equality

Malawi submitted its updated Nationally Determined Contributions (NDC) in 2021. As part of its social protection remit it makes mention of *“promotion of gender mainstreaming in policies, programmes and projects”* as well as a commitment to *“actively integrate the mainstreaming of gender considerations within climate change measures and where relevant to track climate change issues and indicators according to gender and vulnerable groups”*. This is echoed throughout, highlighting the need to ensure climate change adaptation measures are planned with a gender lens and tracked to account for gender and other vulnerabilities.

Malawi has not submitted a National Adaptation Plan or a Climate Change Gender Action Plan (ccGAP) thus far. However, it does have a National Adaptation Plan Framework under its Ministry of Natural Resources, Energy and Mining Environmental Affairs Department which includes gender as one of its guiding principles. Within this it calls for a gender-responsive approach to the NAP process including ensuring adaptation actions consider women's access and impact alongside the use of sex-disaggregated data and inputs from gender experts for planning purposes (Government of Malawi, 2020).

3.5.2 Legal, policy and stakeholder landscape

National instruments and commitments

Table 3-9 Legal and policy landscape: Malawi

Instrument	Relevant content
Malawi 2063 Vision	A long-term development blueprint that focuses on agriculture productivity and commercialization, industrialization, and urbanization, through modernization, mechanization, and sustainable practices, aiming for enhanced productivity and food security. The plan emphasizes gender equality and women's empowerment, advocating for gender-responsive policies and women's active participation in various sectors.

National Gender Policy (2015)	Aims to eliminate gender-based violence, enhance women's participation in decision-making, and ensure equal access to education, healthcare, and economic opportunities. Regarding agriculture and climate change, the policy emphasizes the need for gender-responsive strategies to improve women's access to agricultural resources, technology, and training. It also highlights the importance of addressing the specific vulnerabilities of women to climate change impacts and integrating gender considerations into adaptation and mitigation efforts.
National Climate Change Management Policy	Guides efforts to address climate change through adaptation, mitigation, technology, and capacity building. It emphasizes integrating gender considerations, ensuring climate initiatives are inclusive and address women's specific needs, promoting their participation and leadership. In agriculture, it advocates for climate-smart practices, sustainable land management, and improved access to climate information and technology, aiming to boost productivity and resilience, particularly benefiting women who play a significant role in farming.

Key stakeholders

Table 3-10 Stakeholder map: Malawi

Stakeholder	Mandate
Ministry of Gender, Community Development and Social Welfare	The Ministry, amongst other aspects, is responsible for the coordination of all gender equality and women's empowerment measures and programmes.
MicroLoan Foundation Malawi	It has been delivering financial services to women in rural communities since 2002. It provides loans and ongoing business and financial literacy training to clients across its 22 branches and eight satellites. With a large proportion of clients engaging in farming,
National Smallholder Farmers' Association of Malawi (NASFAM)	NAFSAM is the largest independent smallholder-owned membership organization in the country. It provides a number of services to its members including training in business management, marketing, quality control, literacy and basic education and information services. It emphasis a gender-inclusive approach and has developed a gender programme to ensure equitable participation on farming and association leadership.
CISANET	Civil society organization established in 2001 to facilitate the engagement of CSOs working in the agriculture sector with government on policy. They have a strong position on the need to invest in women for agricultural transformation to be possible.

3.6 TANZANIA

3.6.1 National overview: status of gender equality

Tanzania has demonstrated a strong commitment to promoting gender equality through its participation in various international frameworks and agreements. The country is a signatory to key international instruments such as the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW), which it ratified in 1985, reaffirming its pledge to eliminate gender discrimination and promote women's rights. Tanzania is also a member of the Beijing Declaration and Platform for Action (1995), where it committed to taking concrete steps to empower women and promote gender equality in areas such as education, health, and economic participation. Additionally, Tanzania actively engages with the Sustainable Development Goals (SDGs), specifically Goal 5, which aims to achieve gender equality and empower all women and girls by

2030. The country is also part of the African Union's Agenda 2063, which emphasizes the importance of gender equality for sustainable development on the continent (UN Women, 2023).

Box 10 Tanzania's National Gender Legislation

Beginning in 1992, the Government of Tanzania has consistently pushed for the integration of gender equality into its ministries and policies, starting with the creation of the Ministry of Community Development, Women's Affairs and Children (now known as The Ministry of Community Development, Gender, Women and Special Groups) (The World Bank, 2022). A suite of policies and commitments followed this, aimed at increasing women's participation in society and protecting their rights. This includes the 1999 Land and Village Land Acts offer equal land ownership rights to men and women, The 1971 Law of Marriage Act grants women equal property rights within marriage, including the right to contract, sue, and be sued. However, it controversially allows girls aged 14-15 to marry with parental or court consent, despite a 2019 court ruling mandating the marriage age for both genders to be 18. (UN Women, 2023). More recently, gender considerations have been integrated into the Tanzania Development Vision 2025, which commits to redressing gender imbalances and empowering women (The United Republic of Tanzania, 1999).

Due to these efforts, Tanzania is ranked above regional averages for the World Bank's Women, Business and the Law (WBL) Index and the World Economic Forum's Global Gender Gap Index (GGGI). Under the WBL Index, Tanzania scored 81.3/100 and received a perfect score for workplace, pay and mobility (World Bank, 2023). Its main areas for improvement include parenthood and assets, where there is no paid leave of at least 14 weeks available to mothers, and both sons and daughters as well as male and female surviving spouses do not have equal rights to inherit assets (World Bank, 2023). However, for the GGGI, Tanzania is ranked 54 out of 146 countries, down six places from last year (WEF, 2024). This drop is largely due to a decrease in its subindex scores for economic participation and opportunity, as well as educational attainment.

Barriers to effective implementation of policies, legislature and commitments

Whilst the areas for improvement differ between the GGGI and WBL Indexes, they both fundamentally indicate a need to improve the opportunities available to women and girls from education through to work, including the impacts on marriage and parenthood. Many of the legal frameworks perpetuate gender disparities or have significant gaps. One such example is the Law of Marriage Act, which permits marriage for girls as young as 15, whilst the Law of the Child Act does not address the legal age of marriage or prohibit child marriage. Progress was made when Tanzania's high court ruled in 2016 that marriage of girls under 18 is unconstitutional, a finding reaffirmed by the Court of Appeal in 2019. (The World Bank, 2022) However, it is crucial to align statutory laws with this ruling to ensure consistency and clarity. With over one-third of women in Tanzania married before turning 18, and over half married before turning 20, these rates are more than five times higher than those for men (The World Bank, 2022). Further barriers to successful enforcement include the limited representation of women in decision-making roles which hinder their participation in governance. Limited access to resources and institutional bias, along with violence and harassment within governance structures, further discourage women's involvement, undermining their confidence and opportunities for leadership. This is then subsequently reflected through the ongoing systemic challenges, including discriminatory legal frameworks and sociocultural barriers rooted in traditional norms (Lulu, 2024),

Women in agriculture

Across Tanzania, agriculture employs a significant portion of the workforce, accounting for 65% of the overall labour force. In total, 67% of women, compared with 64% of men (International Labour Organization, 2022) are engaged in the agricultural workforce. Despite women making up the majority, female farmers appear to experience a reduction in both quantity and quality of yields compared with their male counterparts. This appears to be because of reduced access to male farm labour and decreased returns from both labour and non-labour inputs, including pesticides and organic fertilizers (Vanya Slavchevska, 2015).

Female plot managers, often widowed, divorced, or separated, lack male household labour and rely more on female family labour, which tends to be less productive. In instances where they can afford to hire male labourers, they tend to be less productive under female management, possibly due to lower-quality labour affordability or cultural biases against working for female managers (The World Bank, 2015). Additionally, women's unequal return from non-labour inputs (i.e. pesticides and fertilizers) is likely related to either using lower quality products, or insufficient knowledge on how and when to apply them (The World Bank, 2022).

Due to the significant role agriculture plays in the Tanzanian economy, in April 2022, President Samia Suluhu announced “Agenda 10/30”, aimed at bolstering agricultural output growth by 10% by 2030. Amongst other measures, the centre of this strategy is the expansion of labour and land, with over 3 million workers expected to be added to the sector by 2025, made up mostly of women and youth. It also aims to ensure each farmer is allocated a minimum of 10 acres of land (UNDP, 2023).

Climate change and gender equality

Tanzania submitted its first Nationally Determined Contributions (NDC) in 2018 and has since submitted an updated version in 2021. In its most recent NDC, under adaptation measures to be pursued, there is a brief mention of gender mainstreaming, highlighting the need for enhanced gender equity in climate change adaptation actions as well as promoting measures to address the negative impacts of climate change on women. Beyond this, there is no further mention of gender issues including anything related to mainstreaming or equality (The United Republic of Tanzania Vice President's Office, 2021).

Tanzania has not submitted a National Adaptation Plan (NAP), though it does have a National Adaptation Programme of Action (NAPA) which makes no mention of gender equality or mainstreaming (United Republic of Tanzania, 2007). It does however have a Climate Change Gender Action Plan (ccGAP), which was issued in 2013. It acknowledges the disproportionate impact climate change will have on women, as well as their role as agents of change. Within this, agriculture was identified as a priority sector, due to its contribution to the Tanzanian economy and employment prospects. As an objective, it largely promotes the need for greater integration and consideration of women in agricultural activities, outlining several broad-scope actions by which to do so (United Republic of Tanzania Vice Presidents Office, 2013).

3.6.2 Legal, policy and stakeholder landscape

National instruments and key stakeholders working at the intersection of gender, climate and agriculture in Tanzania are captured in the following tables.

National instruments and commitments

Table 3-11 Legal and policy landscape: Tanzania

Instrument	Relevant content
The National Guidelines for Mainstreaming Gender into Environment (2014)	A set of guidelines to be used by all actors involved in environmental management.

National Climate Change Response Strategy 2021 - 2026	Provides a set of interventions on adaptation and mitigation, designed to strengthen Tanzania's response to climate change. It references the need for gender mainstreaming, and to do so using the national guidelines (as mentioned above). It also seeks to address gaps including the use of gender-disaggregated data in M&E and gender budgeting. It also places a great significance on agriculture but does not draw any significant connections between the sector and gender issues.
Updated Nationally Determined Contributions (2021)	The 2021 NDC mentions gender mainstreaming briefly, but apart from this there is no further mention of gender.
Climate Change Gender Action Plan (ccGAP)	Addresses the disproportionate impact climate change will have on the livelihoods of women. Within this, agriculture was identified as a priority sector, where it promotes the need for greater integration and consideration of women in agricultural activities, outlining several broad-scope actions by which to do so.
The National Plan of Action for Violence Against Women and Children (NPA-VAWC), and revisions to The Law of Marriage Act	Launched in 2016, the plan aims to lower rates of teen pregnancy and child marriage along with the practice of female genital mutilation.
National Agricultural Policy (2013)	Addresses agricultural challenges and outlines strategies to combat land degradation. It aims to improve climate change adaptation by promoting sustainable environmental practices, enhancing carbon storage through conservation agriculture and agro-forestry, raising awareness of agriculture's potential as a carbon sink, and strengthening the efficient use of renewable natural resources.
The Inheritance Act (1963), The Lands Act (1999), and The Village Lands Act (1999).	Key legislature promoting female land ownership in Tanzania. They aim to boost formal land registration and secure land tenure, addressing gender discrimination in customary land systems. The Village Lands Act invalidates any customary law denying women, children, or persons with disabilities access to land. Additionally, the Inheritance Act protects widows from being disinherited by their in-laws. Despite these policies, implementation challenges persist, and women's land ownership remains low.

Key stakeholders

Table 3-12 Stakeholder map: Tanzania

Stakeholder	Mandate
The Ministry of Community Development, Gender, Women and Special Groups – Department of Gender Development	The Ministry's role includes the integration of gender considerations into national plans and sectoral budgets. As a result, the 2024/25 budget guidelines issued by the government require all ministries and institutions to include gender issues in their budgets. This directive ensures that gender equality is not just a policy on paper but a practical element of the country's development agenda, aligning with the global SDGs on gender equality.
Ministry of Agriculture	The charter of the Ministry of Agriculture covers policy and services aimed at modernising and enabling an effective agricultural system. This includes capacity building, research and development, fostering collaboration and monitoring crops and food reserves. Whilst it doesn't mention gender specifically, it does include within its mandate the need to consider crosscutting issues in agriculture. This is echoed under its "Agricultural Training, Extension Services and Research Division" which also mentions cross cutting issues under training, and specifically identifies gender
Finance2Equal Tanzania (F2E)	A programme launched by the IFC and designed to reduce the gender gap when accessing financial services. Through a three-pronged approach focusing on

	research, peer learning and advisory services, F2E sought to address the underlying constraints and implement firm-level interventions.
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3.7 ZAMBIA

3.7.1 National overview: status of gender equality

Zambia has shown strong dedication to promoting gender equality by engaging in various international agreements. The country ratified the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) in 1985, affirming its commitment to eliminating gender-based discrimination and advancing women's rights. As a participant in the Beijing Declaration and Platform for Action (1995), Zambia has pledged to address critical areas such as women's empowerment, health, and education. Additionally, Zambia is committed to the Sustainable Development Goals (SDGs), with a specific focus on Goal 5, which aims to achieve gender equality and empower all women and girls by 2030. On a regional level, Zambia is a signatory to the African Union's Agenda 2063, which emphasizes gender equality as essential for sustainable development across Africa (The World Bank, 2023).

Box 11 Zambia's National Gender Legislation

In recent years, Zambia has significantly expanded its policy and legal framework to promote and foster gender equality. Legislation on gender equality covers several sections including health and education, as well as economic opportunities and agency. This includes the National Gender Policy which promotes gender equality and addresses gender-based violence through equal opportunity initiatives, enhanced support services for survivors, and community engagement programs involving men and boys. In addition, the Gender Equity and Equality Act (GEEA) No. 22 of 2015 aligns Zambia's legal framework with international and regional gender equality standards, including CEDAW, and prohibits sexual harassment and discriminatory practices. It also mandates the creation of a Gender Equity and Equality Commission to oversee gender issues (The World Bank, 2023).

This commitment to gender equality is mimicked in its development agenda, where as part of its Zambia Vision 2030, the Government of Zambia outline its intention to "eliminate gender inequalities in social economic development" and improve "educational attainment and eliminate gender gaps at all levels of human and social economic development." (Republic of Zambia, 2006). As part of its National Action Plan, it has committed to integrating both climate change and gender into its annual planning and budgeting process, with a set of guidelines and a corresponding checklist to be finalised that will be used to guide government ministries on this matter (Ministry of Green Economy and Environment, 2023).

Zambia is ranked above regional averages for the World Bank's Women, Business and the Law (WBL) Index and the World Economic Forum's Global Gender Gap Index (GGGI). Under the WBL Index, Zambia scored 81.3/100 and received a perfect score for workplace, pay and entrepreneurship (World Bank, 2023). Its main areas for improvement are pensions and mobility, which do not account for periods of absences due to childcare in pension benefits, and insufficient legislation that would allow women to apply for a passport in the same manner as a man (World Bank, 2023). Zambia's GGGI rank currently sits at 92 out of 146 countries, dropping by 2 places between 2023 and 2024, predominantly because of its subindex scores for political empowerment and educational attainment. However, Zambia was one of the 26 countries that came out joint first under the subindex ranking health and survival (WEF, 2024).

Barriers to effective implementation of policies, legislature and commitments

Zambia's progress in closing the gender gap has been challenging, with gender imbalance present in multiple aspects of society. In particular, enforcement of gender equality commitments is significantly hindered by various complex barriers. One of the most pervasive challenges stems from deeply entrenched cultural and traditional norms that uphold patriarchal values. These norms often limit women's rights and perpetuate gender stereotypes, undermining the effectiveness of gender equality policies. In many rural areas, traditional customs, such as early marriage and unequal inheritance practices, take precedence over statutory laws, making it hard to implement legal reforms designed to protect women's rights (Chikatula, 24). Additionally, the capacity of institutions responsible for implementing gender equality policies is often limited by a lack of resources, funding, and trained personnel. Many gender-related programs are underfunded, which leads to weak implementation and monitoring of gender commitments. Even where policies exist, the lack of political will and prioritization of gender issues often leads to slow progress in enforcing them (The World Bank, 2023).

Finally, low levels of public awareness, particularly in rural areas, further complicate efforts to enforce gender equality commitments. Many women and men are unaware of the legal protections and rights guaranteed by international agreements and national policies. This lack of knowledge makes it difficult to build the broad-based support needed to drive change and advocate for gender equality. As a result, enforcement of gender equality commitments remains a challenge, requiring not only legal reforms but also cultural shifts, resource allocation, and increased awareness at all levels of society (The World Bank, 2023).

Women in agriculture

In Zambia, agriculture is a vital sector for the economy, providing food, employment, and income for a significant portion of the population. It is estimated that more than 57% of Zambians rely on agriculture for their livelihoods (International Labour Organization, 2022). This sector plays a critical role in poverty alleviation and economic growth, with various government initiatives in place to support agricultural development and resilience (International Monetary Fund. African Dept., 2023). Of those women in employment, approximately 62% (International Labour Organization, 2022) are engaged in the agricultural sector, making it the primary source of income for the majority of women. Women primarily participate in subsistence and rain-fed agriculture, as many face barriers when it comes to accessing and/or controlling resources that would allow them to expand their capacity. This includes both statutory and customary land tenure being heavily gendered and limited financial inclusion with an estimated 66% of women not having access to any form of financial services (The World Bank, 2023). This is compounded by the comparatively low literacy levels of women (66%), compared to men (82%), which limits their ability to access and engage in formal training and business opportunities (The World Bank, 2023).

Climate change and gender equality

In 2018, the Government of Zambia launched its Climate Change Gender Action Plan (ccGAP) which is designed to ensure that Zambia's climate change processes incorporate gender considerations, ensuring that both women and men have equal access to participate in and benefit from climate change initiatives. It recognises the disproportionate impact climate change will have on the livelihoods of women, as well as the pivotal role women can play in adaptation measures. It advocates for policy reform, education, increased awareness, academic research, sex-disaggregated data and greater capacity-building and training (IUCN, 2018).

The implementation of ccGAP is referenced in both Zambia's National Adaptation Plan (NAP) (Ministry of Green Economy and Environment, 2023) and its updated Nationally Determined Contributions (NDCs) (Government of Zambia, 2021), though the NDC neglects to go into further detail. Zambia's NAP does however raise the need to integrate both climate change and gender into its annual planning and budgeting process. This includes consideration for the different needs and priorities of

men and women, with a set of guidelines and a corresponding checklist to be finalised that will be used to guide government ministries on this matter. It also sets out a training programme for the implementation of the ccGAP, as well as one for the Gender Focal Points in ministries and non-state actors (Ministry of Green Economy and Environment, 2023).

3.7.2 Legal, policy and stakeholder landscape

National instruments and key stakeholders working at the intersection of gender, climate and agriculture in Zambia are captured in the following tables.

National instruments and commitments

Table 3-13 Legal and policy landscape: Zambia

Instrument	Relevant content
National Gender Policy (2023)	Agriculture is identified as a priority sector for fostering sustainable economic growth, and with women supplying 55% of the labour, their involvement is deemed key. It flags key issues including poor economic security as women are often engaged in informal or seasonal employment or work with their families, along with the limited influence women have in decision-making as well as access and control over resources. Measures proposed to tackle this include facilitating women's access to resources and services including land and finance alongside mainstreaming gender into policy and agricultural practices.
Updated Nationally Determined Contributions (2021)	Mentions Zambia's ccGAP, but apart from this there is no further mention of gender.
National Adaptation Plan (2015 – 2030)	Raises the need to integrate both climate and gender into its annual planning and budgeting process, as well as the finalisation of a set of guidelines covering this, and a training programme focused on the implementation of ccGAP.
National Policy on Climate Change	Launched in 2017, the National Policy on Climate Change is designed to stem the impact of climate change, which also includes a focus on reducing crop failure and its impact on economic growth.
Climate Change Gender Action Plan (ccGAP)	Addresses the disproportionate impact climate change will have on the livelihoods of women and advocates for policy reform, education, increased awareness, academic research, sex-disaggregated data and greater capacity-building and training.
The Gender Equity and Equality Act (GEEA) No. 22 of 2015	Incorporates key international and regional human rights instruments on gender, including CEDAW, the Beijing Declaration and Platform for Action, the Maputo Protocol, and the SADC Protocol on Gender and Development. It also prohibits sexual harassment and harmful practices while aiming to eliminate all forms of discrimination against women.

Key stakeholders

Table 3-14 Stakeholder map: Zambia

Stakeholder	Mandate
Gender Division (formerly known as Ministry of Gender)	It is responsible for “ <i>coordinating and monitoring the effective implementation of Gender policies with other government Line Ministries and cooperating partners.</i> ”
Ministry of Agriculture	The mandate of the Ministry of Agriculture covers the adoption of smart and innovative measures to improve productivity as well as supporting agricultural trade and marketing. Under its Programme Planning, Monitoring and Evaluation Unit it lists mainstreaming of gender within the sector as one of its priorities.
Zambia Agricultural Research Institute (ZARI)	ZARI sits under the Ministry of Agriculture and focuses its work on generating agricultural technologies, with a focus on those used by small-scale farmers.

Zambia Women Entrepreneurs Finance Initiative (We-Fi)	Designed to support women entrepreneurs by <i>“scaling up access to financial products and services, building capacity, expanding networks, offering mentors, and providing opportunities to link with domestic and global markets.”</i>
Saving for change	A project funded by the Swiss Agency for Development and Cooperation (SDC) and implemented by Development Aid from People to People (DAPP), a Zambia-based NGO, with the support of World Food Programme (WFP). It helps to form and train groups on financial management and business skills, whilst also allowing farmers to save and borrow from the groups fund. It has a strong footprint in agriculture communities, and actively promotes the inclusions of women.

4 RE-GAIN programme-level gender analysis

Building on the country-level analyses above, this Chapter describes RE-GAIN's approach to understanding the programme-specific gender dynamics within the seven countries, the way in which these dynamics influence key gender domains, and the impact that these domains may have on programme activities. The Chapter then describes the findings from applying this stakeholder-driven approach, serving as a basis for developing the Programme's Gender & Inclusion Action Plan (Chapter 5).

4.1 APPROACH

National and local stakeholder workshops were facilitated in each of the seven RE-GAIN countries, serving as key informants of this programme design process. The first round of workshop discussions (comprising 14 workshops) focussed on the pressing climate challenges faced during harvesting, post-harvesting and processing, transportation and logistics processes – considering the specific regions and crops selected per country – and potential solutions that could be provided by the RE-GAIN programme to overcome these. These workshops were attended by a range of stakeholders, with on average 50% representation of male and female participants.

Building on these workshops, a second round of engagements was facilitated at national and local levels (comprising 14 workshops in total), to garner feedback and recommendations on the solutions and barriers to accessing them, with a particular focus on the gender, environmental and broader social considerations that may influence the sustainability and impact of the proposed solutions on different sub-groups of the target communities. A core focus of these workshops therefore explored:

- The environmental and social risks and impacts of the implementation of solutions, informing the development of Annex 6.
- Potential gender and youth -specific barriers that may impact women and youth's adoption of solutions and identifying opportunities to overcome these, as a guide to determining key recommendations and actions proposed in this Annex 8.

The dedicated gender and inclusion session was structured in three parts:

- The first focussed on describing the roles that women play in harvest and post-harvest value chains – and whether these differ for male counterparts, and youth.
- The second element of discussion considered various domains that influence women, men and youth engagement in agriculture, and specifically the harvest and post-harvest activities for the select country crops. These domains included:
 - Practices and participation;
 - Knowledge, beliefs, and perceptions;
 - Access to assets;
 - Legal rights and status;
 - Gender roles, responsibilities and time use; and

- Power and decision-making.
- The session concluded with group discussions on specific opportunities and barriers that women, men or youth may experience in adopting the identified solutions.

As with the first round of workshops, there was – on average – gender parity in participation in both the national and local workshops, with a diverse range of stakeholders invited, including ministry representatives, development agencies, farmer associations, the private sector and individual farmers. The workshop notes from the gender sessions are contained in APPENDIX A of this document.

4.2 FINDINGS

As with many of the findings presented in Chapter 2 on the disparities (and underlying drivers) between women and men within the context of agriculture and climate change – many clear, common and consistent themes on gender and inclusion emerge from the fourteen workshop reports. This sheds light on the fact that, while the seven RE-GAIN countries differ in terms of gender equality considerations across key sectors and markers / indicators of progress, they experience many similarities in terms of the gender dynamics that influence the involvement of women and youth in agricultural value chains. As such, findings from the workshops are presented in summary format below. Given the inherent linkages between the domains listed below, many findings are interlinked or cut across multiple domains.

Gender domains: the underlying drivers that influence the roles, voice, choice and control that women, men and youth have in agricultural value chains

Patterns in the practices of women, men and youth involved in the agricultural sector are largely reflective of how these groups are viewed more broadly in terms of their contribution to, and position within, society as a whole. During the workshops, most participants coupled discussions on “practices” with the typically assigned “roles and responsibilities” of these different groups. The graphic below presents a snapshot of the most common delineation of roles, indicating differences and areas of overlap in the responsibilities of women, men and youth at various stages along agricultural value chains. For example, while all groups are involved in land preparation activities, the way in which these activities are assigned reflects the underlying patriarchal power dynamic that governs how land is used; men take a lead role and operate at the meta-scale of land preparation activities while women and youth take on supportive, more menial tasks. This delineation is similarly echoed across the post-harvest value chain stages and thus of significance to the planning and design of the RE-GAIN programme, which focuses on these latter stages.

The narrative that follows provides more depth into the implications of these roles in terms of other key gender domains.

	Land preparation 	Harvesting processes 	Post-harvest handling & storage 	Transport & logistics 
Women 	Planting Weeding	Manual reaping, Hand threshing, Winnowing, Cleaning	Hand threshing, Winnowing, Cleaning Drying, Storing, Sorting	Packaging Transport to local markets
Men 	Purchasing inputs Ploughing Seasonal benchmarking & planning Pest control	Mechanised reaping, Hauling (Loading & off loading)	Storage facility construction, Manual/mechanised threshing, Loading & off loading	Transport & logistics to larger markets, Marketing & sales, Financial management
Youth 	General field labour	General field labour	Assistance to older women with post-harvest food preparation tasks	Technology use (marketing / finance / weather)

The land preparation stage is not included in RE-GAIN's remit

Figure 4-1 Typical delineation of roles and responsibilities

Socio-cultural and gender relations were noted as the significant underlying reasons for food value chain inefficiencies, which in turn directly cause food loss. The different productive and social roles of men and women, as prescribed by socio-cultural structures affect their access to and control over assets, knowledge and services, and their participation in productive activities and decision-making. This impacts the efficiency of the food value chain and hence is an underlying reason for food loss. From the findings, it was also evident that women and men have different needs, constraints and preferences when carrying out their activities along the agricultural value chains. These gender concerns are particularly relevant in determining the response of a specific food value chain to food loss reduction approaches and interventions, and consequently determining their effectiveness and impact of FL-RS.

Practices & participation

- Women typically participate in more labour-intensive, menial / tedious tasks like planting, winnowing, weeding, and threshing. This stems from perceptions that women are more patient than men, and also inexperienced / incapable of using machinery, pesticides etc. Often, these tasks are also delineated to specific crop types (for example, beans, grains and teff as well as subsistence fruit and vegetables grown for household consumption) and require traditional methods, while men have full control and responsibility over more profitable “cash crops” including transplanting and harvesting as well as decision-making on use of fertilisers, pricing and marketing of crops, and use of mechanised equipment. Women’s involvement in post-harvest activities is often unpaid and includes processing and food preparation. Typically, a portion of the food prepared during these post-harvest processes is retained for household consumption. Women thus place importance on their involvement in these activities as it provides assurance that certain produce is ringfenced for their families prior to sending goods to the market. Women’s access to and participation in markets is significantly lower than men, which affects their ability to sell produce, earn a fair income, and control the use of these proceeds.

- Both women and men attend agricultural-related groups or meetings, and at times women's level of participation exceeds that of men. Where women's participation has been higher than that of men, this has been due to intentional efforts by development programmes to reach more women. However, the role that they play in discussions is still influenced by patriarchal norms; women and youth are more reserved in meetings, while men take on primary speaking and decision-making roles. Youths' views in particular are often overlooked by elders. However, the creation of dedicated savings groups and farmers associations for women and / or youth shows positive signs of empowerment, where women and youth can express themselves, engage in discussion, and influence decisions. In time, this may lead to greater parity in engagement during mixed meetings as women and youth grow in confidence and men's perspective on the benefits of meaningfully including women and youth evolves.
- Men, and to a lesser extent youth, are still the primary beneficiaries of extension support and agricultural training opportunities. This results in women having less access to agricultural inputs and technologies, and knowledge on new / more efficient or profitable practices, which limits their ability to upscale or diversify farming operations. The gaps women face in access to training, skills, and technology was reported to be due to a combination of gender norms and social biases, lower capital access, mobility limitations, lower levels of education, lack of decision-making power, and limited time due to their high burden of care work. However, where women are included in trainings, it appears that they are generally more receptive to such support.
- While youth provide significant labour across the agricultural value chain, they have limited autonomy and generally participate under the guidance of older family members, including working on labour-intensive activities alongside older men.

Knowledge, beliefs and perceptions

Traditional and cultural beliefs play a significant role in agricultural practices, delineation of roles and responsibilities, and food choices within communities. This perpetuates beliefs that certain tasks are inherently more suitable for men, women or youth which can reinforce stereotypes, patriarchal mindsets and inequalities. In rural settings, across many of the RE-GAIN countries, these beliefs include:

- Women are caretakers, rather than agricultural producers and as such their work on family farms is often unpaid. They are perceived to have less knowledge about advanced farming techniques and have lower literacy rates, and therefore receive less exposure to agricultural education opportunities. Tasks like winnowing, sorting and food processing for household consumption, as well as cleaning tasks in agricultural settings, are performed by women as these activities are associated with the domestic realm.
- Men are viewed as primary decision-makers within household and community spheres – asserting their dominance through ownership / control of assets, higher levels of literacy, seniority within traditional structures, and control over higher value cash crops and livestock. For example, selling produce, even if produced or processed by women and youth, is assumed as a man's duty and the money accrued from these sales is managed to men.
- Youth are considered helpers, rather than primary contributors to agricultural processes and their involvement in decision-making processes is limited. In turn, youth tend to associate farming with poverty, and it is thus not considered a desirable form of employment by many.
- In some communities, polygamy is perceived as a source of labour, with women and children working on family farms which can have implications on girls' literacy levels, school enrolment rates, and early marriages.

Access to assets & legal rights and status

Access to assets is inherently connected to the patriarchal sociocultural norms that dominate many rural communities in each of RE-GAIN's focus countries. This means that the presence of gender equitable legal rights – particularly those relating to ownership or inheritance of land and other productive assets (such laws are present in majority of the RE-GAIN countries)

– does not equate to enforcement or progress. In instances where traditional customs govern decision-making, this means that:

- Property acquired in a household is automatically considered the man's, and thus legally registered in the name of a man. This implies that in most communities and families, women and youth do not own land and other valuable assets (like livestock, infrastructure, etc.) and therefore have limited or no control over how such assets are used. This applies to inheritance as well, where customary laws still tend to favour boy children over women and girls. Such legal bias restricts women's control over family assets.
- Even in matrilineal societies where women do own land, there is still evidence that men control decisions relating to the use of such assets, limiting women's ability to fully participate in and benefit from their own resources. A knock-on effect of this is that women also have limited control over access and use of agricultural inputs like seeds, fertilizers and equipment.
- Land ownership is a primary enabler of accessing credit and other financial services, making it particularly challenging for women who do not have access to such collateral to secure financial support. This undermines their agricultural productivity and economic independence.
- Men and older women generally have more access to land than youth, making it difficult for youth to engage in entrepreneurial activities within the agricultural sector. However, youth tend to have greater access to information through the internet and social media platforms, which is increasingly being viewed by farmers as beneficial and useful.
- Where legal protections exist, women often still struggle to access – or choose not to draw on – legal representation. This extends to issues relating to property disputes, and domestic abuse or gender-based violence. This stems, in part, from very limited representation of women in local government and related legal support services.

Power and decision-making

As evidenced through the above domain descriptions, clear power dynamics exist between women, men and youth within rural contexts. In summary:

- Men hold primary decision-making power within households, on farms and broader community structures. This extends to choices around land use, technology adoption, crop choice, marketing and sale of produce, and related financial matters (budgeting, use of proceeds, etc.). Their role as leaders extends to most local government structures and agricultural organizations (excluding those that are specifically designated for women or youth -only).
- Women make decisions that tie in with their domestic responsibilities, such as food security and nutrition, water and firewood collection and storage, health and hygiene, and small-scale subsistence activities. At times, a husband and wife may jointly make decisions on household budgeting and expenditure. This obviously differs for female-headed households, where all decisions rest with the household's matriarch.
- Youth hold no significant power in domestic or productive spheres, both at household and community levels. This partly stems from a common perception that they lack experience, meaning their opinions are often overlooked purely due to their age. At times, male youth will hold greater power over decisions than women in the household. Youth-targeted agricultural programmes are showing some potential to help empower youth and influence the receptiveness of older relatives to their opinions.

Adopting RE-GAIN's solutions: barriers and opportunities for women and youth

The gender dynamics of these communities, and specifically how they relate to local agriculture sectors, provides an important foundation upon which RE-GAIN's proposed solutions can be explored in terms of uptake and suitability. Gender relations are also highly relevant in determining the effectiveness of food loss reduction strategies and interventions. The findings from

workshop discussions on the various solutions, in terms of potential barriers to women's adoption, and proposed solutions to overcome these, are summarised in the table below (again noting that responses were broadly aligned across all countries):

Table 4-1 Summary of barriers to women's adoption and potential support

Solution in the RE-GAIN Programme ⁶	Women's preference	Barriers to adoption for women	Support necessary to overcome barriers
Mechanical multi-crop thresher	High - Medium	- High upfront capital costs - Low literacy / lack of knowledge & technical skills - Insufficient technical information available - Sociocultural barrier - use of equipment not suitable for women	- Affordable finance options (rent to own / agri leasing loans) - Set up women's savings groups - Technical capacity building / functional literacy - Sensitization of husbands
Metal and plastic silos	High	High cost of materials	- Simplifying the design and using local materials - Capacity building on construction & use / upkeep
Hermetic bags and polypropylene plastic bags	High - Medium	- Lack of local availability - Limited awareness and knowledge, which leads to resistance to change current practices - Delicate materials perish easily	- Set up local distribution facilities - Awareness creation on advantages - Improvements in quality of materials used / durability
Tarpaulins	High	- High cost - Lack of local availability - Usually made with poor quality materials	- Increased accessibility - Flexible payment options - Awareness creation on advantages - Improvements in quality of materials used / durability
Storage structures (storage sheds, huts, baskets)	Medium - Low	- Concerns about durability - Upfront costs for construction - Concerns about security of the crop / susceptibility to infestation - Lack of knowledge on construction / use / benefits	- Training and awareness creation - Community based storage options through aggregation - enables affordability of solution
Storage protectants and control agents	Medium - High	- Concerns about possible effects of chemicals on women & not aware of alternatives - Affordability of solution	Awareness creation and training on safe ways to use and dispose of chemicals

4.3 RECOMMENDATIONS

As per Table 4-1, almost all solutions were considered of medium-to-high preference for women, but several notable barriers were consistently highlighted in discussion – flagging the importance of ensuring targeted gender-responsive support is provided as part of the RE-GAIN programme. Such support includes:

- **Improved, inclusive access to finance for women and youth.** The upfront cost of many of the solutions proposed was highlighted as a primary barrier to adoption by women and youth. It is therefore critical that inclusive and diverse

⁶ Whilst moisture meters are among the final selection of physical solutions, these were not prioritised or discussed in-depth during workshops (with the exception of Kenya) due to a lack of knowledge – both of the existence of such technology and the application thereof.

finance options and services are made available to these groups. Examples of what these could entail include: establishing savings groups / community-based solutions that require investment from farmer groups; provision of subsidies for tools and inputs; rent-to-own schemes; agri-leasing loans with preferential rates for women and youth, etc.

- **Inclusive and accessible capacity building and awareness raising for women and youth.** While women were aware of each of the solutions proposed, they consistently acknowledged the need for capacity building / awareness raising: to ensure that tools, equipment and structures would be used correctly; on the benefits of changing current practices or preferences; and on financial management – including options available, budgeting, saving and repayment processes. Women's typically lower endowment of resources represents one way in which gender may influence FL-RS. Contact with extension agents is seemingly lower among women farmers, and this further gives them less access to the information and skills training that facilitate the adoption of new technologies
- **Sensitization of male community members.** In many communities, women's participation in training workshops, their use of mechanised equipment or chemicals, and ability to invest in – or access – finance options, are controlled by their husbands. It is therefore critical to ensure that men are included in discussions that centre around the importance of such opportunities, to garner their support and understanding of the broader family / community wide benefits of such.
- **Localised distribution of sustainable, high-quality products made using local materials.** Women voiced concerns around the quality and durability of solutions like tarpaulins and hermetic bags. This stems from past experiences, where they have purchased or been provided with such options and have been let down by the delicateness of materials. It is therefore important to both ensure that improved designs made with locally sourced, durable materials are made available, and that information on these improvements is shared widely so that women are suitably assured of the quality of such products and thus encouraged to invest in and utilised them.

Based on the above, and in keeping with RE-GAIN's programmatic approach, and the fact that common gender -related barriers and opportunities exist in each of the seven countries, a set of programme-wide gender and inclusion actions should be prioritised during subsequent design, implementation and M&E processes. These recommendations serve to guide the Gender & Inclusion Action Plan that follows.

5 Gender & Inclusion Action Plan

5.1 KEY ASSUMPTIONS AND CONSIDERATIONS

To ensure that the findings and recommendations highlighted in Chapters 2 and 4 are integrated and implemented into the broader RE-GAIN Programme, a Gender & Inclusion Action Plan (GAP) has been developed. The Gender & Inclusion Action Plan's activities, indicators, implementation timelines and costings will be integrated into the Programme's broader workplan and budget. The Gender & Inclusion Action Plan should be adopted based on the following key assumptions and considerations:

- **Expertise.** The Plan makes an allowance for the appointment of a dedicated full time programme-level Gender Expert who will be responsible for overseeing the planning, delivery and reporting on all indicators; this expert will also serve as the lead technical specialist responsible for all process-related gender activities such as policy revisions, establishment of relevant systems and processes, awareness raising campaigns etc. The Gender Expert will also be responsible for ensuring equitable representation on RE-GAIN's Steering Committee and Programme Advisory Groups and provide the required technical gender expertise for all discussions held at these decision-making levels. AGRA's Gender and Inclusion Director will oversee the Gender Expert's delivery and outputs. The Gender Expert will be based at AGRA's HQ in Kenya. In addition to this fulltime Gender Expert, two parttime Gender Officers have also been budgeted for, who will sit in other RE-GAIN regions and be responsible for coordinating gender efforts and supporting delivery of the Gender & Inclusion Action Plan in their relevant geographies. This staffing structure ensures that the Gender & Inclusion Action Plan has the necessary resources and presence across the continent, so that each region can be readily serviced and risks can be effectively managed and escalated appropriately.
- **Alignment with RE-GAIN's Theory of Change.** Given the contribution of this Gender & Inclusion Action Plan to the Programme's broader Theory of Change (ToC), its Impact Statement, Outcomes and Outputs directly link with those in the ToC. This ensures that all proposed Activities clearly align with the Programme's primary activities and are thus embedded within the workplan – as opposed to being considered “additional” or of peripheral importance.
- **Preliminary processes at Programme Start Up.** As is standard practice for the start-up phase of a multi-country programme of this nature, several preliminary steps will be carried out prior to implementation of this Gender & Inclusion Action Plan. These include:
 - **Finalising RE-GAIN's gender and inclusion institutional arrangements**, which sit within the programme's broader organisational design. This primarily entails the appointment of a Gender Expert, whose responsibilities will be captured in a Terms of Reference, guided by this Gender & Inclusion Action Plan and aligned with AGRA relevant staffing policies. Lines of engagement between the Gender Expert and AGRA's country offices (notably AGRA's embedded gender personnel) will also be established, with roles and responsibilities between each group clearly defined.
 - Develop **country-specific Gender & Inclusion Action Plans** that directly feed into this programme-level Gender & Inclusion Action Plan. These will be informed by baseline gender assessments in each country, to ensure that the nuanced needs and contexts of each region of focus are accurately reflected in activities, targets and timelines. These country-level gender processes and outputs will be delivered during the first six months of the programme, with the country-level GAPs submitted to the GCF for approval in Q3 of Year 1. This is a critical step, which will inform the establishment of more specific disaggregated targets for women, men, youth and other

marginalised groups on a more granular scale. These will be reflected in both the country-level GAPs and the programme-wide GAP contained in this Annex (which will be updated once the country-level GAPs are approved).

- **Guiding principles for all women-only engagements.** Throughout the Gender & Inclusion Action Plan, various activities centre on targeted engagements, extensions services, and capacity-building sessions for women -only. The need for these women-only platforms was raised consistently across all seven countries, along with several important conditions that need to be understood and integrated into the planning and delivery of such events. These include:
 - Selecting a location that is convenient, safe⁷ and hygienic.
 - Selecting a meeting time, day and duration that does not interfere with existing domestic or productive responsibilities, or take place during unsafe hours (early morning or late afternoon, where lighting is poor).
 - Provision of childcare assistance or facilities at the location.
 - Consideration of participants' language preferences and literacy capabilities, when selecting suitable trainers and developing supporting materials.
 - Invitations that explicitly mention women's attendance at trainings and other key interventions (given invitations that are generic tend only to bring men to project gatherings)
 - In contexts where couples training (i.e., a husband and wife are jointly invited to attend) is preferable over women-only engagements, the same principles should be followed.
- **Deliberate and targeted appointment of implementation partners.** Within the eligibility criteria used to select implementation partners, RE-GAIN has included a specific technical competency on the need for a partner to have demonstrated experience of working with women, youth and other marginalised groups. Their applications also need to align with RE-GAIN's E&S policy. This ensures that the on-the-ground support enlisted as part of the programme will be adept at working with the appointed Gender Expert to deliver several activities in the Gender & Inclusion Action Plan below. Given the importance of offering women-to-women extension support, RE-GAIN's selection of extension delivery partners will also ensure that the staff make-up of such delivery partners includes a suitable number of capacitated females who can service and support female beneficiaries.

5.2 GAP BUDGET OVERVIEW

The resource and budget allocations detailed in the Gender & Inclusion Action Plan (section 5.3) comprise the following elements:

Table 5-1 Gender and Inclusion Action Plan costing information

Budget line	Amount	Detail / justification
Programme Start Up	323,956 USD	This line item reflects the gender-related costs that will be incurred in the first six months of the programme, during which time country-level gender assessments and GAPs will be produced. This line item includes: six-month salaries for the Gender Team; travel expenses for field missions to each country; enumerators for qualitative and quantitative data collection.

⁷ In addition to the potential logistical challenges associated with locations that are too far from central communal areas, having to travel too far from homesteads places women at risk of sexual violence.

Gender Experts	3,156,556 USD	This line item covers the salary and all required travel expenses of the Gender team for the duration of the programme <i>excluding the first six months</i> , given this has been captured in the above line item.
Gender and inclusion activities within REGAIN's broader training, awareness raising campaigns and capacity building	259,337 USD per country for the programme duration	This line item represents the budget allocated to gender-related activities associated with training, awareness raising campaigns, capacity building and learning exchange visits across all outputs. It is <i>additional</i> to the Gender Staff costs listed above. It sits within the master budget, and represents 12.5% of the overarching budget's total allocation for all training and awareness campaign expenses.
Additional budget for professional or contractual services required to address unforeseen or country-specific gender activities	479,934 USD per country per annum	Should unforeseen / nuanced / additional country-specific GAP activities be identified that fall outside of the activities currently costed in the overarching GAP, budget will be drawn from this master budget line item.

5.3 GENDER AND INCLUSION ACTION PLAN

Table 5-2 Gender and Inclusion Action Plan

Impact: Enhanced capacity of women and youth to respond to climate-triggered post-harvest food losses through improved and inclusive access to financing, promotion of context specific and gender-responsive innovations to reduce food loss, and better enabling conditions that promote the involvement of women and youth in FL-RS through public and private investments.				
Core outcome indicator	Target (direct)	Target (indirect)	Timeframe	Assumptions & notes
Female beneficiaries with improved food security	87,054 (50% of total direct beneficiaries ⁸)	1,383 (50% of total indirect beneficiaries)	End of programme (Y5)	This indicator represents the expected number of females that will benefit from RE-GAIN, by enabling household access to FL-RS. This differs from the Output level indicators below, which focus on indicators and targets for key sub-groups in the value chain, such as female smallholders, female leaders, and women- and youth- led cooperatives.

Activities & Sub-Activities	Indicator	Target ⁹	Timeframe ¹⁰	Budget allocation / resourcing requirement	Responsible ¹¹
Programme Start-up					

⁸ Assumption is that households, on average across all seven countries, are 50% female.

⁹ Targets to be achieved per country unless otherwise stipulated.

¹⁰ Immediate = first six months; Short-term = Y1; Medium-term = Y2-3; Long-term = Y4-5. These timeframes are subject to revision, based on finalisation of the programme's workplan during the start-up phase.

¹¹ Roles allocated based on likely programme staffing structure. To be finalised during the programme's start-up phase.

Develop country level gender assessments and action plans.	Conduct baseline gender assessments in each country	Country-level gender assessment produced	1 per country	Immediate	Overseen by Gender Expert with data collection and analysis by enumerators	Gender Team
	Develop country-level GAPs that feed into programme level GAP	Country-level GAP produced	1 per country	Immediate		
	Update this programme-wide GAP to reflect new / more detailed disaggregation of targets and indicators.	Programme GAP updated	1 programme-wide update	Short-term		
Outcome 1: Increased adoption of FL-RS by women and youth smallholders						
OUTPUT 1.1: Women and youth smallholders supported to adopt FL-RS						
1.1.1 Provide intentional and on-going access to information and technical advisory services on FL-RS to women and youth through VBAs.	Map out women’s, men’s and youth’s networks and platforms for effective dissemination of FL-RS information.	Dissemination networks mapped	1 per country	Short-term	Overseen by Gender Expert. Cost of training covered in master budget.	Gender Team
	Develop gender-responsive and youth-oriented content on FL-RS for VBAs.	Content developed	1 per country	Short-term		
	Appoint implementation partners with experience in working with, and training, women and youth (individually and through VBAs.	Implementation partners appointed with GESI experience.	100% of partners appointed	Short-term		
	Train VBAs, trainers and women leaders in community structures on targeted gender and youth – focussed support and information sharing.	# trainings for VBAs, trainers and women leaders	1 per year per region ¹²	Annual		Assigned implementation partner together with relevant VBAs
	Establish appropriate and regular forums for VBA engagement, designated for women and youth.	# engagements per community, dedicated to women and youth on FL-RS topics	1 per quarter	On-going		

¹² Noting multiple regions have been identified per country, and within these regions, select communities will receive support.

1.1.2 Identify, prioritise and facilitate women's access to FL-RS that most effectively reduce current workload / drudgery during post-harvest processes and enhance the efficiency and profitability of their efforts.	Facilitate demonstration and training sessions with women-only focus groups on use and benefits of various physical solutions to ascertain preferences.	# trainings per community	2 per year	Short-term	Cost of trainings covered in master budget	Gender Teamt & assigned implementing partner
	Identify community representative / liaison to oversee feedback mechanism whereby women can share preferences or complaints on product quality / needs with vendors to allow for continuous adaption of technologies to suit needs.	Community-vendor liaisons appointed	1 per community	Short-term	Overseen by Gender Expert	Relevant community committee Supported by Gender Team
	Directly link women / women's groups with FL-RS vendors to facilitate their adoption of FL-RS [<i>coupled with activity 2.2.1 on access to financing models</i>]	Proportion of smallholder female farmers benefitting from FL-RS (as a % of total smallholder beneficiaries)	40%	On-going, long-term	Overseen by Gender Expert	Culmination of multiple activities costed in master budget.
1.1.3 Integrate gender and youth empowerment and leadership modules into all elements of RE-GAIN's awareness raising and capacity building initiatives (trainings, awareness activities, mass media campaigns, monitoring and outreach, knowledge products).	Develop appropriate content for the RE-GAIN's awareness raising and capacity building elements – ensuring approaches are gender and socially inclusive with dedicated aspects targeting acceptance / sensitization of male counterparts (including male farmers, vendors, processors, extension officers, etc.).	Gender responsive content developed & integrated	100% of initiatives / products delivered under RE-GAIN's broader Output 1.1 Activity Set.	On-going	Overseen by Gender Expert Cost of trainings and materials covered in master budget	Gender Team
	Advise on all participant lists, ensure selection criteria on trainers includes relevant experience on working with women and youth, and oversee consistent application of this criteria, to ensure	% of participant lists and trainer appointments signed off by Gender Expert	100%	On-going		
		Male: female attendance ratio	60:40			

	equitable representation and expertise at in-person campaigns or trainings.					
1.1.4 Increase women’s decision-making capacities within the context of post-harvest value chains	Deliver targeted training modules to women in FBOs (or otherwise appropriate community farmer models) aimed at enhancing their decision-making and leadership capacities.	% increased of women in leadership roles in FBOs	30% increase from baseline	Long-term	Overseen by Gender Expert Cost of trainings and materials covered in master budget	Gender Team together with assigned implementing partner
Impact Indicator 1.1: An increase in women's appetite for leadership roles in the community as a result of RGAIN’s capacity building.		Measurement method: survey administered on women’s roles within community structures (capturing qualitative and quantitative metrics)		Timeframe: Programme start up and programme closure	Responsible: Overseen by Gender Expert and M&E Team	
OUTPUT 1.2: Improved and inclusive market linkages between agri-value chain actors – prioritising the involvement of women and youth						
1.2.1 Implement measures to overcome mobility, time, terrain and quality of product constraints that impede women’s participation in larger markets.	Conduct gender sensitive value chain (VC) analysis to identify specific nodes within VCs where women are predominant, analyse the challenges they face, and assess opportunities / entry points for women in male-dominated nodes of the VC. (as part of the Programme’s broader Activity on mapping existing actors in local agricultural value chains).	Gender sensitive value chain analysis conducted	1 VC analysis per country	Short-term	Overseen by Gender Expert	Gender Team
	Provide regular training and support to women and youth farmers, cooperatives and networks on meeting market standards, participation in quality control processes, and other marketing / business skills (as identified in VC analysis) – including facilitation of peer-to-peer mentorship support from both successful	# trainings	2 per community	Annual	Cost of training covered in master budget.	VBAs / extension officers Supported by Gender Team

	women and men leaders in relevant VC nodes.					
	Consult with women smallholders to identify off-take service options (incl. collection centres, pooled transport options, etc.) that would overcome their mobility limitations.	# community-level consultations with women <i>on this topic</i>	1 per community	Short-term	Overseen by Gender Expert.	Gender Team
	Facilitate the appointment and capacitation of women as community aggregators responsible for price negotiations, liaison with women farmers and quality control.	Female community aggregators appointed	1 per community / region	Short-term	Overseen by Gender Expert	Relevant community committee Supported by Gender Team
Outcome 2: Enhanced supply and affordability of FL-RS for all						
OUTPUT 2.1: Business development support to women and youth -led initiatives (cooperatives, MSMEs, service providers)						
2.1.1 Identify, incubate and / or support growth of women and youth -led FL-RS cooperatives, MSMEs, & service providers.	Identify women and youth -led initiatives operating in the FL-RS sector, including cooperatives, MSMEs, service providers etc. and assess their current capacity and skills.	Women and youth -led initiatives mapped and analysed	1 mapping per country	Medium-term	Overseen by Gender Expert	Gender Team
	Provide training programmes and bootcamps (delivered under RE-GAIN's broader Output 2.1) on business development, market intelligence, and market development support specifically for women and youth -led FL-RS initiatives – including a digital component that can be accessed post-trainings	# gender-responsive programmes / bootcamps	1 per region per year from Y3	Medium-term	Cost of training covered in master budget.	Business Development Expert Supported from Gender Team

	Arrange exchange visits to successful women and youth -led initiatives operating in nearby villages.	# exchange visits	1 per region per year from Y3	Medium-term	Cost of exchange visits covered in master budget.	Gender Team
	Identify and map out credit / guarantee schemes and financial instruments with favourable lending conditions for women and youth -led agricultural endeavours.	Financial schemes and instruments mapped per country	1 financial mapping per country	Medium-term	Overseen by Gender Expert.	Gender & Finance Experts
	Facilitate women and youth-led FL-RS providers' access to existing credit / guarantee schemes through information sharing and networking platforms.	# community-level information sharing sessions with women and youth <i>on this topic</i>	2 per region per year	Medium-term	Overseen by Gender Expert.	
OUTPUT 2.2: Inclusive and accessible finance mechanisms for women and youth –led initiatives						
2.2.1 Provide enabling support and reduce barriers to access, to ensure women and youth farmers can benefit from RE-GAIN's two financial models ¹³	Facilitate financial literacy training to women and youth -only groups, based on baseline capacity needs assessment.	# trainings	2 per region	Short-to-medium term	Cost of trainings covered in master budget.	Gender & Finance Experts
	Support the establishment or strengthening of women and youth -led cooperatives (or similar entities that prioritise women and youth employment) .	No. of new / strengthened women / youth -led cooperatives	TBC based on country level gender assessments	Medium-to-long term	Overseen by Gender Expert.	
		Proportion of employment opportunities created by RE-GAIN designated for women and youth (as a % of total employment)	40%	Long term		

¹³ Acknowledging that the models have been intentionally designed to offer more inclusive access to finance under terms that are favourable to small holder farmers, cooperatives and agro-dealers who have typically struggled to secure traditional forms of finance.

	Facilitate information sharing sessions on the specific benefits of, differences between, and avenues to access, each model for women and youth -only groups – with agro-dealers and financial institutions present to allow for direct contact between groups.	# of sessions	1 per region	Quarterly, from when models have been established.	Cost of sessions covered in master budget.	
	Engage with FIs and agro-dealers that will be involved in each of RE-GAIN's financial models to determine whether the lending criteria / payment terms respond to the needs of women and youth farmers.	FIs and agro-dealers involved in RE-GAIN's financial models engaged	100%	Short-to-medium term	Overseen by Gender and Finance Experts	
	Provide capacity building and / or advisory support to ensure FIs and agro-dealers involved in each of RE-GAIN's financial models are suitably equipped to engage with and offer financial products and payment terms that suit the needs of women and youth smallholders.	Technical assistance provided to all FI's and agro-dealers identified as needing capacity / advisory support	100% of those identified	Medium-to-long term	Overseen by Gender and Finance Experts	
Impact Indicator 2.2: Enhanced capacity and sustainability of women- and youth-led initiatives (e.g., cooperatives, MSMEs, service providers) as a result of uptake of business development services.		Measurement method: survey administered to women and youth-led initiatives (capturing qualitative and quantitative metrics)		Timeframe: Prior to provision of capacity building support and again at programme closure	Responsible: Overseen by Gender Expert and M&E Team	
Outcome 3: Enabling environment for the uptake of inclusive FL-RS						
OUTPUT 3.1: Enhanced capacity of national institutions to enable investments that prioritise the needs of women and youth in FL-RS						
3.1.1 Strengthen and integrate the voice of women and youth	Where instruments exist that are favourable to women (including related laws on inheritance and ownership of	Local institutions and networks engaged on	1 engagement per country	Medium-to-long term	Overseen by Gender Expert	Gender Team

into FL-RS policy decision-making and implementation.	assets), support their domestication at local levels by working with appropriate local institutions (e.g. gender focal points within relevant agricultural, climate or environment institutions), existing women's networks and NGOs / partners on local dissemination and sensitization efforts.	policy domestication efforts				
	Where instruments do not exist, or are gender blind, provide support to policy development / revision processes, working with suitably mandated gender entities / focal points.	Gender responsive policy reforms / revisions proposed	1 gender responsive policy memo per country	Medium-to-long term		
Impact Indicator 3.1: An increase in the number of commitments and actions by national institutions that prioritize and facilitate investments to address the specific needs of women and youth in FL-RS		Measurement method: baseline and endline assessment of no. of commitments and actions conducted at programme start-up.		Timeframe: Programme start up and programme closure	Responsible: Overseen by Gender Expert and M&E Team	
3.1.2 Ensure that the promotion of successful FL-RS business models includes a gender and socially inclusive lens.	Ensure that within the business case developed for investment in FL-RS, there is an explicit focus on adopting gender-responsive approaches ¹⁴ and guidance on involving and empowering women and youth.	Gender and youth components / case studies integrated into business case	1 gender-responsive business case for the programme	Medium term	Overseen by Gender Expert	Gender Team
	Oversee inclusive awareness creation efforts on the promotion of successful FL-RS business models, through targeted information sharing campaigns to women and youth -led initiatives.	# information sharing campaigns targeted at women and youth <i>on this topic</i>	2 per year	Medium-to-long-term	Cost of campaigns covered in master budget.	Gender Team

¹⁴ Including examples of successful rent-to-buy or lease-to-service schemes (along with other relevant financing mechanisms) that have proven favourable and beneficial to women in comparative contexts

6 Programme application

To ensure consistent application of gender-responsive and socially inclusive planning and implementation, per component and across all countries, several important principles and approaches are outlined below. These comply with AGRA's organisational commitments and globally accepted E&S best practice.

6.1 GENDER INTEGRATION APPROACH

In keeping with AGRA's Five-Year Gender and Inclusion Strategy (2022-2027), the RE-GAIN Programme will adopt the Reach-Benefit-Empower framework across each of its three Outcome areas:

- Women and youth will be **reached** by actively ensuring that they are included as core Programme participants (measured through quotas at meetings and trainings, and number of women / youth -only events) and through targeted effort to identify and deliver benefits that women and youth place explicit value on (be these time savings, workload reduction, skills development, improved well-being, etc.).
- Linked to the above, **benefitting** women and youth will involve ensuring Programme interventions tackle specific needs and constraints to their productivity in the agriculture sector, through tangible actions designed to meet their preferences.
- **Empowering** women and youth will focus on strengthening their voice, choice and control over the various gender domains that they value. This aligns with the Programme's approach to utilising community-driven feedback and evidence-based interventions to effectively tackle drivers of disempowerment and power imbalances between women and men in post-harvest value chains.

These outcome areas should collectively support **transformative change**, alongside the creation of an enabling environment that addresses gender norms, through for example, interaction with powerholders in the community to critically examine gender norms, inequalities and influence change.

6.2 GRIEVANCE MECHANISM, SAFEGUARDING TOOLS AND STAKEHOLDER ENGAGEMENT

6.2.1 Grievance Mechanism

Stakeholder engagement measures will work pro-actively towards identifying and addressing issues before they become grievances. However, when grievances are reported they need to be addressed in a consistent and verifiable manner as part of the implementation of the ESMS (See Annex 6 of this Funding Proposal).

AGRA will ensure through the E&S Due Diligence that the grantee will establish a mechanism to receive and facilitate resolution of stakeholders' concerns, complaints, and grievances about the project's performance. Procedures stipulated by AGRA's Grievance Mechanism (Annex 9 of AGRA's ESMS Manual) will be followed and stakeholders' will be informed about the mechanism. The Grievance Mechanism is designed to be a transparent process that is gender responsive, culturally appropriate, and readily accessible to all segments of the stakeholders at no costs and without retribution. Furthermore, the mechanism will be verified by the AGRA E&S Manager, who will then communicate with the Team of E&S Officers. Those actions are applicable for IFC's risk Category A, B and C projects.

6.2.2 Safeguarding Approach

AGRA is committed to the protection and safeguarding of children and vulnerable adults (together referred to as Vulnerable Persons) and the Youth. AGRA acknowledges that it has a duty to ensure that its programs and operations are conducted in a manner that is safe for Vulnerable Persons and Youth. AGRA recognizes that there is potential for abuse to take place within the context of its programs, which are intended to address the needs of vulnerable communities and groups. AGRA's Safeguarding of Vulnerable Persons and Youth Policy sets out AGRA's commitment to protect the Vulnerable Persons including women and youth we work with either directly or indirectly, from any form of Abuse. It provides mandatory guidelines on preventing any form of Abuse and Exploitation of Vulnerable Persons and sets out codes of conduct that are applicable to AGRA's Staff and Partners in this respect. It also sets out the mechanisms AGRA will follow in the event any abuse is reported.

The policy is applied in conjunction with other AGRA policies, including but not limited to the following: (i) Ethics Policy (ii) Partners Code of Conduct (iii) Whistleblowing policy (iv) Sexual Harassment policy (v) Human Resources Manual, including disciplinary policies & procedures (vi) Data Protection policy. To operationalize AGRA's Safeguarding of Vulnerable Persons and Youth Policy, the HR, Legal and Gender, Youth and Inclusiveness units have developed the following tools for use in program implementation with partners, and are thus all applicable and will be utilised within the context of RE-GAIN:

- Safeguarding Implementation Plan
- Safeguarding and PSHEA Risk Assessment Form
- Travel Risk Assessment Form
- Safeguarding Incident Reporting Form
- Youth travel Information Table

6.2.3 SEAH and GBV Risk Management

AGRA's commitment to the Do-No-Harm principle is embedded in all its programmes. This commitment is operationalized through the abovementioned Safeguarding Policy, designed to prevent and respond to any risks of abuse or exploitation. This includes mandatory gender and safeguarding training for all staff, partners, and community leaders involved in programme implementation. For example, in Rwanda, AGRA has implemented Community-Based Safeguarding Committees composed of both male and female representatives who serve as the first point of contact for reporting SEAH and GBV concerns.

At the programmatic level, AGRA takes a proactive stance by integrating a GBV/SEAH Risk Assessment as part of its Gender Analysis Toolkit. This forward-thinking approach is applied before launching interventions in new contexts, ensuring that potential risks are identified and addressed from the outset. The toolkit includes specific modules on power dynamics, social norms, and potential risks related to GBV/SEAH. For instance, in Malawi, this risk assessment revealed that women engaging in higher-income agricultural roles faced heightened risks of domestic violence. AGRA responded by designing Household Gender Balance Trainings that worked with men and women to reduce tensions around shifting financial roles.

RE-GAIN will therefore adopt **AGRA's SEAH Policy** and **Risk Monitoring and Mitigation Framework**, which entails:

- **Country-Specific Safeguarding Risk Audits:** Conducted annually to identify program-specific risks related to SEAH and GBV.

- **AGRA's Grievance Mechanism**, which as stated above, is a robust system that is further strengthened with gender-sensitive reporting tools. These tools, including anonymous hotlines and online platforms, ensure that survivors of SEAH or GBV can report safely and without fear of retaliation. This effective system is a testament to AGRA's commitment to ensuring the safety and well-being of all individuals involved in its programs.
- **Partnership with Local Women's Rights Organizations:** AGRA collaborates with local organizations specializing in women's rights and GBV prevention in each focus country to ensure that participants can access immediate support and legal resources. For instance, in Tanzania, AGRA's partnership with local NGOs enabled swift referral services for survivors of SEAH during a community-based agricultural training program.

Annex 6's Appendix 1 elaborates on the potential programme risks associated with SEAH, in section 3.3 and Appendix D. The risk mitigation matrix included in Appendix 1 Annex 6 aligns with the instruments in place at AGRA to ensure programme teams are equipped and ready to deal with potential SEAH and GBV risks within the RE-GAIN programme, along with mitigation strategies and responsible parties.

6.2.4 Stakeholder Engagement

Stakeholders are persons or groups who are directly or indirectly affected by a project, as well as those who may have interests in a project and/or the ability to influence its outcome, either positively or negatively. They can include affected communities located near a project, particularly those subject to actual or potential project-related risk and/or adverse impacts on their physical environment, health or livelihoods.

How those stakeholders are engaged can determine the successfulness of a project implementation, as it forms the basis for managing a project's environmental and social impacts. A detailed consultation process is required and all engagement with stakeholders must be conducted in line with AGRA's Environmental & Social Safeguard Policy. A Stakeholder Engagement Plan (SEP) is required to be developed and implemented for each project on an appropriate scale. A SEP should be scaled to the project risks and be tailored to the characteristics of the affected persons or communities. Template of a SEP can be found in Annex 9.

The outcomes of the overall and country level strategic risk assessments should be formally taken into consideration in the updating and planning of AGRA's overall guidelines and strategy for engagement and documented through minutes, and in any engagement guideline or programme plan updates.

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APPENDIX A Workshop Meeting Notes from Gender and Inclusion Sessions

The following subsections summarise the key gender aspects captured in each of the national and local workshop reports. These workshops were rooted in discussions that pertained specifically to the RE-GAIN programme. Each gender session was prefaced by presentations, groupwork and feedback sessions on the targeted agricultural value chains selected per country as well as the specific FL-RS focal areas of the programme, which provided the broader context within which gender specific issues, opportunities and needs were discussed.

While every effort was made to ensure that the gender sessions were delivered consistently across countries, the facilitators also ensured that the nature and direction of discussion focused on issues considered most pertinent to the stakeholders in the room. As such, the focus and granularity of the findings from each workshop differ somewhat.

Key takeaways are combined for each country's national and local workshops.

A 1. KENYA

Table 7-1 Key gender takeaways from Kenya's national and local stakeholder engagements

Discussion topic	Key takeaways
Gender domains	• Women and youth have limited access to land, and no decision-making power, both from a cultural and (presumed) legal perspective. In instances where legal rights do exist for women, they are not necessarily commonly practiced. • Interestingly, most farmer groups are either predominantly composed of women or have a majority of women members. Women tend to attend trainings, accept and seek services than men and youth. • Youth often lack collateral. • Women and youth primarily engage in harvesting, threshing and processing within the value chain. Women also seem to be more focused on nutrition and the overall welfare of the home. Men, traditionally focus on marketing, and transportation, as well as anything that typically involves the use of mechanised equipment. Youth may also be employed alongside men for anything very labour-intensive.
FL-RS preferences, barriers & solutions	• Of the shortlisted solutions for Kenya, tarpaulins and plastic sheets have the highest preference amongst women, followed by hermetic bags, threshers/shellers and moisture meters. Storage structures were included amongst the lowest. • Financial solutions were seen as the main support required to improve opportunities for adoption of these solutions amongst women, be it in the form of credit or improved access. Current credit interest rates are very high (up to 10-12% per month). Second to financial solutions was the need for greater training and capacity building, that would need to be inclusive and accessible.

A 2. UGANDA

Table 7-2 Key gender takeaways from Uganda's national and local stakeholder engagements

Discussion topic	Key takeaways
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Gender domains	• Women are involved in harvesting, drying, storing, threshing beans, cleaning the grain, etc. They are responsible for food security, safety of produce under storage, etc. • Men are involved in transportation and marketing, packaging, threshing maize, processing etc. They make decisions on enterprises, methods of threshing, modes of transport, types of markets where produce is sold. Men have more opportunities to access credit or access to finance loans than women and youth, because they have greater access and ownership of land (most communities and families do not allow women and youth inherit land). They are also given platforms during meetings – overshadowing women and youths views. • Youth are involved in harvesting but have no significant power or decision-making capacity. They have more access to information due accessibility to social media platforms. • Women fear of being ashamed of reporting their husbands to legal structures when husbands do wrongs. • Attendance of meetings and workshops is more by men, who in most cases do not pass on the knowledge to women. In most community meetings, women do not talk or eat if not allowed by men in participation.
FL-RS preferences, barriers & solutions	Highest preference for specific types of support include access to finance for women; empowerment through training (requiring consent from husbands, and their involvement in women trainings); tarpaulins and plastic sheets (which requires access to affordable funding, could be linked to existing programs, and requires improvements in the current poor quality of bags that are available). Mechanical multi-crop threshers are also needed, but these will require subsidies for women farmers due to high upfront costs.

A 3. ETHIOPIA

Table 7-3 Key gender takeaways from Ethiopia's national and local stakeholder engagements

Discussion topic	Key takeaways
Gender domains	• Women are responsible for the food security of the household, as they are the ones that are saving production for household consumption. • Women are quite actively engaged in farming, from harvesting to threshing to processing. They have a significant workload. But most of their employment is informal. • Teff is very labour intensive, and the land levelling is done for women. It's rare for women to be used in ploughing, but there are more than before. • Post-harvest is quite cumbersome (labour and time intensive), and women are the ones primarily doing the activities, as it seems like they have more 'patience'. • Women are often also perceived to being more reliable, so they are often hired for research activities • Women are more involved in the value-addition activities in the smaller scale, while men are more involved in large-scale value addition activities with farmers. • Women are particularly responsible for the building of storage structures. • The more market oriented the production of wheat becomes, the less the women become involved as women are not part of these initiatives or these services. • Extension services are targeted to men.

**FL-RS preferences,
barriers & solutions**

- Primary barriers faced by women farmers span technical issues; financial issues and capacity of the household (given time demands of domestic responsibilities).
- Strategies to empower and include women in agricultural programmes include: provision of temporary childcare to allow women to engage with trainings; get local “gate keepers” to influence the social norms; appoint male gender champions – men that break the social norms; create enabling environment so that women can participate in training and activities; promote different technologies that address women’s needs (fertilizers; biogases (might be able to reduce the workload for women)).

A 4. BURKINA FASO

Table 7-4 Key gender takeaways from Burkina Faso's national and local stakeholder engagements

Discussion topic	Key takeaways
Gender domains	<i>Focus of discussions was on preferences, barriers and solutions only.</i>
FL-RS preferences, barriers & solutions	• Preferred / needed solutions for women farmers include: making airtight bags available, mechanical threshers and gins for several crops, protection and storage control agents, storage structures. • Apart from harvesting machines (deemed unsuitable for women), other tools are possibilities for adoption for women. While many of the proposed solutions have already been adopted, access is poor because of the high prices. • To overcome these obstacles, according to participants, it will be necessary to subsidize the tools, train women on their use, and facilitate women's access to microfinance institutions.

A 5. MALAWI

Table 7-5 Key gender takeaways from Malawi's national and local stakeholder engagements

Discussion topic	Key takeaways
Gender domains	• Women continue to carry the burden of food production and ensure food security in the family. The impacts of climate change make this situation worse. Despite their role in production and post-harvest activities, women are marginalised in several areas: • Most assets are owned by men. In matrilineal societies in some parts of Malawi, women do own land but may have limited decisions related to assets. Most households' assets are passed on through the boy child. • The Constitution of Malawi and the new Land Act provide equal access to land. The reality on the ground is that most of the customary land is owned by men. Even when women do own assets, cultural norms frequently prevent them from having control over these assets. • There is a clear division of labour, with men traditionally handling tasks such as land preparation and threshing, while women are responsible for weeding, transplanting, and harvesting. This division restricts women's opportunities for income generation, as certain tasks are valued more than others. Furthermore, women often have less time for income generating activities due to their childcare and household responsibilities. Women tend to have limited control over some crops such as maize, but

would play a leading role in crops such as beans which are considered as ‘women crops’. In situations where women participate with men, decisions are still made by men. The creation of groups/ cooperatives and training activities tend to enhance equal participation of women and men in farming activities.

- Women tend to have lower literacy levels because some families have lower value in educating women who are sent for early marriages. The lack of education puts them at a lower status in the society. The youth tend to associate farming with poverty. On the other hand, older people feel that young people cannot lead in farming activities or groups.
- There is an unequal distribution of knowledge and training opportunities, with men more likely to attend agricultural training sessions. This disparity restricts women's access to new technologies and farming practices.
- Although both men and women participate in productive labour, women are also involved in reproductive labour as well as welfare for the family, community and society in general. Although most men have spare time, they are rarely available for programmed activities.
- At the household level, power and decision-making are dominated by men. Likewise in agriculture, men typically wield more influence and make the majority of decisions regarding planting, fertilizer use, and selling crops. This power imbalance limits women's control over agricultural practices and income, perpetuating gender disparities. This is not the case for the private and public sectors where qualified individuals have equal opportunities to take senior decision-making roles.

**FL-RS preferences,
barriers & solutions**

Focus of discussions was on gender domains only.

A 6. TANZANIA

Table 7-6 Key gender takeaways from Tanzania's national and local stakeholder engagements

Discussion topic	Key takeaways
Gender domains	• Women tend to participate in labour-intensive activities such as planting, weeding, and harvesting. They have limited participation in decision-making roles or higher-value activities like marketing and financial management. They are also involved in post-harvest processing and food preparation, often unpaid. Women face significant barriers in accessing financial resources, limiting their ability to invest in agricultural inputs and technologies. Access to agricultural technology is also often restricted, with women facing greater challenges in obtaining and using these tools effectively. • Men engage in land preparation, ploughing, and activities that involve machinery. They are more likely to control financial aspects and decision-making in farming operations. And they participate in marketing and selling of produce, typically seen as the face of the farming business. • Youth often provide labour but have limited access to training and capacity-building opportunities. Their participation in farming is usually under the guidance of older family members, with limited autonomy. • Women are perceived as caretakers rather than main agricultural producers. They have limited access to agricultural education and training, perceived as less knowledgeable about advanced farming techniques. Cultural norms may limit their involvement in certain agricultural practices considered “male” tasks.

- Men are viewed as the primary decision-makers and knowledgeable authorities in agriculture. They are more likely to receive formal training and extension services. Cultural beliefs support their dominance in high-value crops and livestock management.
- Youth are seen as helpers rather than primary contributors, limiting their involvement in key decisions. They are often excluded from traditional knowledge-sharing systems.
- Men predominantly have access to post-harvest machinery. This disparity limits women's ability to engage effectively in post-harvest activities, affecting their productivity and economic empowerment.
- Land ownership and access are primarily controlled by men. This gender disparity in land access undermines women's agricultural productivity and economic independence. Without land rights, women face significant barriers to improving their livelihoods and participating fully in agricultural development.
- Women engage in activities perceived as “women’s work” like weeding, harvesting, and food processing.
- Men are primarily responsible for land preparation, market activities, and financial management. They are less burdened by household responsibilities, allowing more focus on agricultural activities.
- Youth assist with both household chores and farming activities. Their time use is largely dictated by school schedules and family needs. They have limited involvement in decision-making due to perceived lack of experience.

**FL-RS preferences,
barriers & solutions**

Focus of discussions was on gender domains only.

A 7. ZAMBIA

Table 7-7 Key gender takeaways from Zambia's national and local stakeholder engagements

Discussion topic	Key takeaways
Gender domains	• Women are responsible for household chores, field labour, harvesting, threshing, and packaging. They hold some household decision making power over how money is spent for domestic purposes. Men make long term decisions for the family. • The general inclination is that men use mechanized equipment, and women use traditional methods for farming and post-harvest processes. The customary thinking is that men understand the equipment better. • Men are also responsible for benchmarking for the next season, constructing storage facilities, operating machinery, and other value addition activities. Men handle the field pesticide spraying, sales, machinery and livestock health monitoring. They are also involved in ox-ploughing the fields, herding cattle, digging water wells, and charcoal burning. • Youths are involved in herding livestock, field labour and household chores. They also help men with transporting and marketing produce. Youth have no power and depend on their parents in this regard. • Access to productive assets is limited to men. Property acquired in the household is automatically the man's. As such, men have more access to assets than women and youths: not only land but also livestock. When a woman gets married, she forfeits the right to family property - although more

	often, some women now have titles due to government's deliberate policies to promote this. However, legal representation remains difficult for a woman to access in a traditional set up. • Most meetings are attended by more women than men, but women have reserved participation in groups with men. They keep their opinions to themselves. • Recently women are taking up leadership positions such as in savings groups. • GBV is still common in Zambia. Men generally do not report domestic violence to avoid feeling embarrassed. • Polygamy is perceived as a source of labour as women and children will work in the farm.
FL-RS preferences, barriers & solutions	• Mechanical multi-crop threshers are a high preference for women but high capital costs, literacy levels and lack of knowledge on use of this equipment are all considered significant barriers to adoption. To overcome these, some example solutions given were the rent to own scheme by Vision Fund; functional literacy training; and engaging with Monze District Women's Association which has some equipment. Agriculture leasing loans, saving groups, and skills development were also listed as support necessary for adoption. • Metal and plastic silos were also noted as a high preference. But materials used are costly and there is a need to simplify designs and use local materials. Learnings should be drawn from Monze, where local famers build a silo using local materials. • The availability of hermetic bags and polypropylene plastic bags is considered a barrier to adoption – learnings should be drawn from the ERDP program which distributed some bags. • Tarpaulin and plastic sheets are available – these solutions are not new, and therefore can be easily adopted but there is a need for increased accessibility, flexible payment options and other financing solutions, and awareness raising of the advantages. • Storage structures (storage sheds, huts, baskets) were of medium preference to women, who stated that there is a need for increased durability of such products. Cost of construction is another barrier, as well as knowledge on construction and use (to avoid infestation). Community based storage solutions through aggregation was a suggested as a potential solution. • Storage protectants and control agents was of high preference to women, but access to these needs to be accompanied by awareness raising on the effects of chemicals on women, alternative options, and safe ways to use and dispose of chemicals. Affordability was also flagged as a potential concern.
